Probability for Finance

Theory, Expanded Proofs, Worked Homework, Past and AI-Generated Exams, and More

These unrevised notes were generated using artificial intelligence, following the student's preferred style and, where possible, the style of Professor Tiziano De Angelis and former tutor Andrea Bovo. Additional proofs and explanations not presented or proved by the professor have been included at the student's discretion. The material has not undergone a complete independent review and may contain errors.

These notes follow the lecturer's order and explain the hypotheses and steps of each proof. Every homework retains its own statement and solution beside the relevant theory. A later homework is anticipated only when the lecturer's own proof takes that exact fact as given; its proof remains at the homework itself. All mathematical displays are unnumbered.

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1. Measure Spaces

From Continuity to Measurability

In classical Calculus the focus is on *continuous* functions. A function f from a (topological) set X into another (topological) set Y (denoted $f : X \to Y$) is said to be continuous if the pre-images of open sets in Y are open sets in X. In formulae

$$f^{-1}(A) = \{x \in X : f(x) \in A\}$$

is open in X whenever A is open in Y.
In Measure Theory instead the focus is on *measurable* functions. A function $f : X \to Y$ is measurable if

$$f^{-1}(A) = \{x \in X : f(x) \in A\}$$

is a *measurable set* in X whenever A is a *measurable set* in Y. So we need to understand the meaning of *measurable set*!
We will learn that in a properly defined probabilistic framework measurable sets are *events* in a random experiment and measurable functions are *random variables*. We start by defining particular collections of sets, called *algebras*.

Definition 1.1 — Algebra

Let $\mathcal{S}$ be a set. A collection Σ_0 of subsets of $\mathcal{S}$ is called an algebra if

(i) $\mathcal{S} \in \Sigma_0$,
(ii) $A \in \Sigma_0 \implies A^c \in \Sigma_0$,
(iii) $A, B \in \Sigma_0 \implies A \cup B \in \Sigma_0$.

If Σ_0 is an algebra on $\mathcal{S}$, then $\varnothing \in \Sigma_0$: the whole space belongs to the algebra and its complement is empty.

HW 1 — Intersection in an algebra

Show that $A, B \in \Sigma_0 \implies A \cap B \in \Sigma_0$.

Solution to HW 1. *We first prove the set equality $A \cap B = (A^c \cup B^c)^c$ by both inclusions.* Let $s_0 \in A \cap B$. Then $s_0 \in A$ and $s_0 \in B$, so it belongs to neither complement. Hence $s_0 \notin A^c \cup B^c$, which means $s_0 \in (A^c \cup B^c)^c$. For the reverse inclusion, let $s_0 \in (A^c \cup B^c)^c$. It belongs to neither A^c nor B^c; therefore it belongs to both A and B, and hence to $A \cap B$. Combining the inclusions proves the equality. Now $A \cap B = (A^c \cup B^c)^c \in \Sigma_0$, where $A^c, B^c \in \Sigma_0$ by the complement axiom, $A^c \cup B^c \in \Sigma_0$ by the two-set union axiom, and the final membership follows by complement closure once more. Only algebra axioms were used; countable closure was not needed.

HW 2 — Finite unions and intersections

Show that $(A_n)_{n=1}^N \subseteq \Sigma_0$ implies

$$\bigcup_{n=1}^{N} A_n \in \Sigma_0 \text{ and } \bigcap_{n=1}^{N} A_n \in \Sigma_0.$$

Solution to HW 2. The union axiom concerns two sets, so we extend it to any finite number by induction. For $N = 1$ the union is A_1, already in the algebra. Suppose $U_N = \bigcup_{i=1}^{N} A_i$ belongs to Σ_0. Since A_{N+1} also belongs to it, the two-set axiom gives $U_N \cup A_{N+1} \in \Sigma_0$. This is exactly U_{N+1}, completing the induction. In

$$U_{N+1} = U_N \cup A_{N+1} \in \Sigma_0,$$

the equality separates the last term of the union and the membership uses the two-set union axiom on the induction hypothesis and $A_{N+1} \in \Sigma_0$.

For the intersection, first complement every A_i. All A_i^c belong to the algebra, so the finite-union result applies to them. Taking one final complement yields

$$\bigcap_{i=1}^{N} A_i = \left(\bigcup_{i=1}^{N} A_i^c \right)^c \in \Sigma_0.$$

The equality is De Morgan: a point belongs to every A_i exactly when it belongs to none of their complements. Each complement belongs to Σ_0, the finite union belongs by the induction just proved, and its complement belongs by the algebra axiom; these justify the final membership. If empty families are allowed, the conventional union is $\varnothing$ and the intersection is S; both also belong to the algebra. The conclusion is finite closure: one cannot infer closure under infinite unions from this induction.

Now we introduce a more abstract collection of sets, called *σ-algebras*.

Definition 1.2 — Sigma-Algebra

Let $\mathcal{S}$ be a set. A collection Σ of subsets of $\mathcal{S}$ is called a σ-algebra if

(i) $\mathcal{S} \in \Sigma$,
(ii) $A \in \Sigma \implies A^c \in \Sigma$,
(iii) $(A_n)_{n=1}^{\infty} \subseteq \Sigma \implies \cup_{n=1}^{\infty} A_n \in \Sigma$.

HW 3 — Countable intersections

Show that $(A_n)_{n=1}^{\infty} \subseteq \Sigma \implies \cap_{n=1}^{\infty} A_n \in \Sigma$.

Solution to HW 3. *We first prove* $\bigcap_{n\geq 1} A_n = (\bigcup_{n\geq 1} A_n^c)^c$. Let s_0 belong to the intersection on the left. Then $s_0 \in A_n$ for every n, so $s_0 \notin A_n^c$ for every n. It is therefore outside their union and inside its complement. This proves the first inclusion. Conversely, let s_0 belong to the complement of that union. No index can satisfy $s_0 \in A_n^c$; thus $s_0 \in A_n$ for every n, giving membership in the intersection. The two inclusions establish the equality. For the measurability conclusion, $\bigcap_{n\geq 1} A_n = \left(\bigcup_{n\geq 1} A_n^c\right)^c \in \Sigma$. Each A_n^c belongs to Σ by complement closure; their union belongs to Σ by the countable-union axiom; the final membership follows from complement closure. These are countable operations; no uncountable closure is being asserted.

In the probabilistic formulation of a random experiment we will use σ-algebras to describe all possible events. However, working with σ-algebras is difficult because it is often difficult to describe ALL sets in a σ-algebra. It is often easier to handle algebras and we will soon learn about π-systems. **Notation**: From now on $\mathcal{S}$ will always denote a set, Σ_0 an algebra of subsets of $\mathcal{S}$ and Σ a σ-algebra of subsets of $\mathcal{S}$.

Definition — Measurable Set

Measurable sets: any $A \in \Sigma$ is said to be a Σ-measurable set.

Definition — Measurable Space

Measurable space: a pair $(\mathcal{S}, \Sigma)$ is called a measurable space.

Definition — Generated Sigma-Algebra

Letting $\mathcal{C}$ be a class of subsets of $\mathcal{S}$ we denote by $\sigma(\mathcal{C})$ the *smallest* σ-algebra that contains all sets in $\mathcal{C}$. We call it the σ-algebra generated by $\mathcal{C}$. Notice that, in particular, if $\mathcal{C} \subseteq \Sigma$, then $\sigma(\mathcal{C}) \subseteq \Sigma$, where the inclusion follows from minimality: Σ is a sigma-algebra containing every member of $\mathcal{C}$, so it contains the smallest such sigma-algebra.

Definition — Borel Sigma-Algebra

One of the most important example of σ-algebra is the so-called Borel σ-algebra. Given a topological space X (i.e., a space in which we define what the open sets are), the Borel σ-algebra $\mathcal{B}(X)$ is the smallest σ-algebra that contains all the open sets of X. That is

$$\mathcal{B}(X) = \sigma(\text{open sets in } X).$$

In particular we will often work with $X = \mathbb{R}$ and $\mathcal{B}(\mathbb{R})$ generated by open sets on the real line. In that context, a very useful class of subsets is the following (example of π-system)

$$\pi(\mathbb{R}) = \{(-\infty, x], x \in \mathbb{R}\}.$$

Indeed it is easy to describe all the sets contained in $\pi(\mathbb{R})$ and the next proposition allows us to connect with $\mathcal{B}(\mathbb{R})$.

Proposition 1.1 — Half-Lines Generate the Borel Sigma-Algebra
It holds $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$.

Expanded proof. We prove both inclusions, following the two steps in the original argument. For an arbitrary $x \in \mathbb{R}$, $(-\infty, x] = \bigcap_{n\geq 1}(-\infty, x + 1/n) \in \mathcal{B}(\mathbb{R})$. For the equality, a number at most x belongs to every open ray on the right. Conversely, if $y > x$, choose n with $1/n < y - x$; then y is outside the nth ray and hence outside the intersection. The final membership follows because the rays are open and a Borel sigma-algebra is closed under countable intersections. Thus $\pi(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R})$. Since $\mathcal{B}(\mathbb{R})$ is a sigma-algebra containing this class, minimality gives $\sigma(\pi(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$.
For the reverse inclusion, recall that an open subset of $\mathbb{R}$ is a countable union of disjoint open intervals. It therefore suffices to show that these intervals belong to $\mathcal{G} = \sigma(\pi(\mathbb{R}))$. For finite $a < b$, first write

$$(a, b) = (-\infty, b) \cap (a, \infty).$$

The equality follows because membership on either side means exactly $a < y < b$, proving both inclusions. Next, $(a, \infty) = (-\infty, a]^c \in \mathcal{G}$, where the equality is the complement identity for a lower closed ray, and the membership uses $(-\infty, a] \in \pi(\mathbb{R}) \subseteq \mathcal{G}$ and closure under complements. Finally, $(-\infty, b) = \bigcup_{n\geq 1}(-\infty, b - 1/n] \in \mathcal{G}$. A point on the right is strictly below b. Conversely, if $y < b$, choose n with $1/n < b - y$; then $y \leq b - 1/n$, so it belongs to the union. These are the two inclusions for the equality. The membership follows from countable-union closure and membership of every closed lower ray in $\pi(\mathbb{R})$. Hence $(a, b) \in \mathcal{G}$ by finite-intersection closure. Unbounded interval components are the open rays just handled; the whole line also belongs to every sigma-algebra.
For an arbitrary open A, use its disjoint open-interval decomposition, say $A = \bigcup_n I_n$. Each I_n belongs to $\mathcal{G}$ by the preceding steps, so $A \in \mathcal{G}$ by countable-union closure. The empty open set is already in $\mathcal{G}$. Thus $\mathcal{G}$ contains all open sets, and minimality of their generated sigma-algebra gives $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{G}$. Combining the two inclusions proves the proposition. QED

Measures, Additivity and Completeness

The next notion we want to introduce is the one of *measure*. We will be particularly concerned with probability measures, which we use to assess "how likely" various events are.

Definition — Set Function

Set functions: A set function is a mapping from an algebra to the positive extended real line $\mu_0 : \Sigma_0 \to [0, \infty]$.

Definition — Additive Set Function

Additive set functions: A set function is additive if $\mu_0(\varnothing) = 0$ and if $A, B \in \Sigma_0$ are such that $A \cap B = \varnothing$ then $\mu_0(A \cup B) = \mu_0(A) + \mu_0(B)$.

Definition — Countably Additive Set Function

Countably additive set functions: A set function is countably additive if $\mu_0(\varnothing) = 0$ and for any $(A_n)_{n=1}^{\infty} \subseteq \Sigma_0$ such that $A_i \cap A_j = \varnothing$ for all $i \neq j$ (pairwise disjoint sets) and $\cup_{n=1}^{\infty} A_n \in \Sigma_0$ it holds

$$\mu_0\big(\cup_{n=1}^{\infty} A_n\big) = \sum_{n=1}^{\infty} \mu_0\big(A_n\big).$$

HW 4 — From two-set to finite additivity

Let μ_0 be an additive set function. Show that for any $(A_n)_{n=1}^N \subseteq \Sigma_0$ such that $A_i \cap A_j = \varnothing$ for all $i \neq j$ (pairwise disjoint sets) it holds

$$\mu_0\big(\cup_{n=1}^N A_n\big) = \sum_{n=1}^N \mu_0\big(A_n\big).$$

Solution to HW 4. We induct on the number of disjoint sets. The assertion for one set is an equality with itself. Assume it holds for N sets and define $U_N = \bigcup_{i=1}^N A_i$. Finite closure of the algebra ensures that U_N is in the domain of μ_0. Pairwise disjointness also implies $U_N \cap A_{N+1} = \varnothing$: membership in that intersection would put a point in $A_i \cap A_{N+1}$ for some $i \leq N$.
We can therefore apply the original two-set additivity rule to U_N and A_{N+1}:

$$\mu_0(U_N \cup A_{N+1}) = \mu_0(U_N) + \mu_0(A_{N+1}) = \sum_{i=1}^N \mu_0(A_i) + \mu_0(A_{N+1}).$$

This is the required formula for $N+1$.
In the displayed chain, the first equality is two-set additivity on the disjoint algebra members U_N, A_{N+1}; the second equality is the induction hypothesis for $\mu_0(U_N)$. All sums are of nonnegative extended numbers; no subtraction of infinities is involved, so infinite values are allowed.

HW 5 — Countable additivity implies additivity

Show that a countably additive set function is also additive.

Solution to HW 5. Take disjoint $A, B \in \Sigma_0$. To use an axiom about infinite sequences for just two sets, define $C_1 = A$, $C_2 = B$ and $C_n = \varnothing$ for every $n \geq 3$. This sequence is pairwise disjoint, and its union is $A \cup B$, which belongs to the algebra. Thus the union meets the domain requirement in the definition of countable additivity on an algebra.

$$\mu_0(A \cup B) = \sum_{n=1}^\infty \mu_0(C_n) = \mu_0(A) + \mu_0(B) + \sum_{n=3}^\infty 0 = \mu_0(A) + \mu_0(B).$$

The equality $\mu_0(\varnothing) = 0$ is part of the definition.
The first displayed equality applies countable additivity to the constructed disjoint sequence. The second separates its first two terms and uses zero mass of the empty set for every remaining term. The last equality sums a series consisting entirely of zeros. It must not be established by subtracting possibly infinite masses from an identity. Together with that zero-mass condition, the displayed equality is precisely additivity.

Definition — Measure and Measure Space

Measure: Given a measurable space $(\mathcal{S}, \Sigma)$ a function $\mu : \Sigma \to [0, \infty]$ is called measure if it is countably additive. The treble $(\mathcal{S}, \Sigma, \mu)$ defines a measure space.
Notice that measures are defined on σ-algebras. Can such functions actually be constructed? We will answer this question shortly.

Definitions — Finite and Sigma-Finite Measures

A measure μ is

- *finite*, if $\mu(\mathcal{S}) < \infty$;
- *σ-finite*, if there exists $(A_n)_{n\in\mathbb{N}} \subset \Sigma$ such that $\cup_{n=1}^\infty A_n = \mathcal{S}$ and $\mu(A_n) < \infty$ for each $n \in \mathbb{N}$.

In particular we will be concerned with probability measures, in which case $\mu(\mathcal{S}) = 1$.

Definition — Null Set

Null sets: a set $A \in \Sigma$ is a null set if $\mu(A) = 0$. Actually, to be fully formal, the collection of null sets is

$$\mathcal{N} := \{A \subset \mathcal{S} : A \subset B \in \Sigma, \mu(B) = 0\}.$$

Notice that in the definition of $\mathcal{N}$ the sets A are not necessarily measurable but they are subsets of measurable sets.

Definition — Complete Measure Space

Complete measure space: a measure space is complete if $\mathcal{N} \subseteq \Sigma$.

Definition — Almost Everywhere

A statement $\mathcal{H}(s)$ about points $s \in \mathcal{S}$ is said to hold almost everywhere (almost surely in case of probability measures) if $\mu(F) = 0$, where

$$F = \{s \in \mathcal{S} : \mathcal{H}(s) \text{ is false}\}.$$

Pi-Systems and Measure Extension

Definition 1.3 — Pi-System

A collection $\mathcal{I}$ of subsets of $\mathcal{S}$ such that

$$I_1, I_2 \in \mathcal{I} \implies I_1 \cap I_2 \in \mathcal{I}$$

is called a π-system.

HW 6 — The lower rays form a pi-system

Prove that $\pi(\mathbb{R})$ is a π-system.

Solution to HW 6. Fix arbitrary $I = (-\infty, a]$ and $J = (-\infty, b]$ in $\pi(\mathbb{R})$. For the first inclusion, take $x_0 \in I \cap J$. Then $x_0 \leq a$ and $x_0 \leq b$, so $x_0 \leq \min\{a, b\}$; hence $x_0 \in (-\infty, \min\{a, b\}]$. For the reverse inclusion, take $x_0 \leq \min\{a, b\}$. Since the minimum is at most each endpoint, $x_0 \leq a, b$. Thus $x_0 \in I \cap J$. Combining the two inclusions gives $I \cap J = (-\infty, \min\{a, b\}] \in \pi(\mathbb{R})$. The final membership follows from $\min\{a, b\} \in \mathbb{R}$ and the definition of $\pi(\mathbb{R})$, not from sigma-algebra closure. As the two chosen members were arbitrary, $\pi(\mathbb{R})$ is a pi-system. The whole real line need not belong to a pi-system and is not a member of this particular generating class.

An important use of π-systems concerns the uniqueness of measures and it is illustrated in the next lemma.

Lemma 1.2 — Uniqueness on a Generating Pi-System

Let $\Sigma = \sigma(\mathcal{I})$ where $\mathcal{I}$ is a π-system on $\mathcal{S}$. Suppose that μ_1 and μ_2 are measures on $(\mathcal{S}, \Sigma)$ such that $\mu_1(\mathcal{S}) = \mu_2(\mathcal{S}) < \infty$ and $\mu_1(I) = \mu_2(I)$ for every $I \in \mathcal{I}$. Then $\mu_1(A) = \mu_2(A)$ for every $A \in \Sigma$.

The next theorem constructs a measure from a premeasure.

Theorem 1.3 — Caratheodory Extension

Let $\mathcal{S}$ be a set, Σ_0 an algebra on $\mathcal{S}$ an $\Sigma = \sigma(\Sigma_0)$. If a set function $\mu_0 : \Sigma_0 \to [0, \infty]$ is countably additive, then there exists a measure $\mu : \Sigma \to [0, \infty]$ such that $\mu(A) = \mu_0(A)$ for any $A \in \Sigma_0$. Moreover, if $\mu_0(\mathcal{S}) < \infty$, then μ is uniquely defined.

We say that μ in the theorem above is an *extension* of μ_0 to the σ-algebra Σ. An important example is that of the *Lebesgue measure*.

Construction — Lebesgue Measure

Lebesgue measure: Let $\mathcal{S} = (0, 1]$, $\Sigma = \mathcal{B}((0, 1])$ and

$$\Sigma_0 = \{F : F = \cup_{i=1}^{n}(a_i, b_i], \text{ for pairwise disjoint sets } (a_i, b_i] \subseteq (0, 1] \text{ and some } n \in \mathbb{N}\}.$$

Then $\sigma(\Sigma_0) = \Sigma$. Define μ_0 for $F \in \Sigma_0$ as

$$\mu_0(F) = \sum_{i=1}^{n}(b_i - a_i).$$

It is possible (but not easy) to prove that μ_0 is countably additive on Σ_0 and clearly $\mu_0(\mathcal{S}) = 1$. Therefore there exists a unique measure μ on Σ that extends μ_0. We call it *Lebesgue measure* and will often denote it by $\mathcal{L}eb$. By the analogous Lebesgue construction one obtains Lebesgue measure on $\mathbb{R}$ and on $\mathbb{R}^d$.

HW 7 — The half-open interval algebra

Prove that Σ_0 defined above is indeed an algebra on $(0,1]$.

Solution to HW 7. The whole space is the single interval $(0,1]$, and the empty union is $\varnothing$. For closure under complement, write a nonempty member as a sorted union of disjoint nonempty intervals, with $0 \le a_1 < b_1 \le a_2 < b_2 \le \cdots \le a_n < b_n \le 1$. Then its relative complement in S is $S \setminus \bigcup_{i=1}^{n}(a_i, b_i] = (0, a_1] \cup \bigcup_{i=1}^{n-1}(b_i, a_{i+1}] \cup (b_n, 1]$. Empty intervals in this expression are simply omitted. Once the two inclusions below establish the equality, the membership of the complement in Σ_0 follows because the displayed right side is a finite disjoint union of intervals of the admitted half-open type.

For the first inclusion in this complement formula, a point outside the sorted original intervals lies before the first interval, in a gap between consecutive intervals, or after the last; these are exactly the displayed pieces. Conversely each piece excludes every original interval, proving the reverse inclusion. When a gap has zero length it is empty. The empty-set representation can be taken as the empty union, or as the degenerate interval $(0,0]$ if a nonempty index family is required. Each remaining interval has the required half-open convention, and the intervals are disjoint. Notice that the left endpoint zero must be allowed: excluding it would prevent representing S.

For closure under union, two initial descriptions might overlap, so one cannot just claim the concatenated intervals are pairwise disjoint. Collect all their endpoints together with zero and one, and sort the distinct numbers as $0 = t_0 < t_1 < \cdots < t_M = 1$. The cells $(t_{j-1}, t_j]$ are disjoint. Every initial interval is a union of these cells, and consequently each of the two sets, and their union, is a union of a selection of cells. There are finitely many cells, giving a valid representation in Σ_0. This checks all three algebra axioms.

HW 8 — Additivity of the interval length function

Prove that μ_0 is (finitely) additive on Σ_0.

Solution to HW 8. First, the definition must not depend on a chosen interval decomposition. Given two finite representations, take the finite sorted list of every endpoint, including zero and one. Each interval splits into consecutive half-open cells. If $a = t_j < b = t_k$, its length satisfies $b - a = \sum_{\ell=j+1}^{k}(t_\ell - t_{\ell-1})$ by telescoping. Thus either representation assigns exactly the sum of the lengths of the same cells contained in F. This proves well-definedness.

The telescoping equality replaces each original interval length by the sum of the consecutive cell lengths. Refinement neither omits nor counts twice a cell contained in the set. Thus the two representations yield the same numerical sum, which is the exact requirement for well-definedness. The empty set contains no cells, so $\mu_0(\varnothing) = 0$. Now let A, B be disjoint algebra members. Use a common endpoint partition for both. No positive-length cell is selected by both sets, since that would contradict $A \cap B = \varnothing$. Hence the cells selected by $A \cup B$ split into the cells selected by A and those selected by B. Summing their lengths gives $\mu_0(A \cup B) = \mu_0(A) + \mu_0(B)$. Repeating this argument, or inducting on the number of sets, yields finite additivity for any finite pairwise disjoint family. It does not yet prove countable additivity; that is a separate, substantially stronger step in constructing Lebesgue measure.

Notation — Increasing, Decreasing and One-Sided Limits

For a sequence of real or extended-real numbers, $x_n \uparrow x$ means both $x_n \le x_{n+1}$ for every n and $\lim_n x_n = x$. Similarly, $x_n \downarrow x$ means $x_n \ge x_{n+1}$ and convergence to x. These arrows assert monotonicity as well as convergence; $x_n \to x$ alone does not assert monotonicity. The inequalities need not be strict. For a continuous positive parameter, $\varepsilon \downarrow 0$ means that $\varepsilon > 0$ approaches zero from above. Thus $\lim_{\varepsilon \downarrow 0} F(z+\varepsilon)$ is the right-hand limit of F at z, whereas $\lim_{\varepsilon \downarrow 0} F(z-\varepsilon)$ is its left-hand limit. The parameter decreases in both expressions; the argument $z - \varepsilon$ increases towards z in the second. The one-sided notation restricts the side of approach, not just one particular sequence such as $\varepsilon = 1/n$.

HW 9 — A singleton is Borel

Prove that $\{x\} \in \mathcal{B}(\mathbb{R})$ for any $x \in \mathbb{R}$.

Solution to HW 9. The Borel sigma-algebra contains all open intervals and is closed under countable intersections. Consider the shrinking intervals $I_n = (x - 1/n, x + 1/n)$. Clearly x belongs to every I_n. Conversely, if $y \neq x$, the distance $|y - x|$ is positive; choose an integer n with $1/n < |y - x|$. Then $y \notin I_n$. Therefore $\{x\} = \bigcap_{n=1}^{\infty}(x - 1/n, x + 1/n)$. Each interval is Borel, so their countable intersection is Borel.
In the displayed equality, the first inclusion was proved by membership of x in every interval; the reverse inclusion was proved by excluding every $y \neq x$. Each interval belongs to $\mathcal{B}(\mathbb{R})$ because it is open; the intersection membership follows from countable-intersection closure. The alternative identity $\bigcap_{n\geq 1}(x - 1/n, x] = \{x\}$ is also valid: the fixed closed right endpoint is retained and every smaller point is eventually excluded.

HW 10 — A singleton has zero Lebesgue measure

Prove that $\mathcal{L}eb(\{x\}) = 0$ for any $x \in \mathbb{R}$.

Solution to HW 10. Let $x \in \mathbb{R}$. HW 9 makes $\{x\}$ Borel, and the open interval $(x - 1/n, x)$ is Borel for every $n \geq 1$. The disjoint decomposition $(x - 1/n, x] = (x - 1/n, x) \cup \{x\}$ holds because a point in the left interval is either strictly below x or equals x, and either alternative gives membership in the left interval. The alternatives are disjoint because a point strictly below x cannot equal x. Hence

$$1/n = \mathcal{L}eb((x - 1/n, x]) = \mathcal{L}eb((x - 1/n, x)) + \mathcal{L}eb(\{x\}) \geq \mathcal{L}eb(\{x\}) \geq 0,$$

where the first equality uses the length of the half-open interval, the second uses finite additivity on the two disjoint Borel pieces, the first inequality uses non-negativity of the open interval's measure, and the last uses non-negativity of the singleton's measure. The bounds hold for every n. If $\mathcal{L}eb(\{x\}) = a > 0$, choosing $n > 1/a$ contradicts $a \leq 1/n$; therefore $\mathcal{L}eb(\{x\}) = 0$. This argument uses no later monotonicity result.

Basic Measure Properties and Monotone Limits

We now turn our attention to basic but important properties of measures.

Lemma 1.4 — Subadditivity and Inclusion–Exclusion

Let $(\mathcal{S}, \Sigma, \mu)$ be a measure space. Then

(a) $\mu(A \cup B) \leq \mu(A) + \mu(B)$ for $A, B \in \Sigma$;
(b) $\mu(\cup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mu(A_i)$ for $(A_i)_{i=1}^{n} \subset \Sigma$;
(c) If $\mu(\mathcal{S}) < \infty$ then $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$ for $A, B \in \Sigma$.

HW 11 — Subadditivity and inclusion-exclusion

Prove Lemma 1.4.

Solution to HW 11. Here $A \setminus B = A \cap B^c$ belongs to the algebra by complement and intersection closure. Decompose the union into disjoint pieces: $A \cup B = (A \setminus B) \cup B$. Also $A = (A \setminus B) \cup (A \cap B)$. Additivity and nonnegativity give

$$\mu(A \cup B) = \mu(A \setminus B) + \mu(B) \leq \mu(A \setminus B) + \mu(A \cap B) + \mu(B) = \mu(A) + \mu(B),$$

proving (a) without subtracting any infinite terms.
The first equality is additivity on the two disjoint pieces of $A \cup B$. The inequality inserts the non-negative term $\mu(A \cap B)$. The last equality is additivity on the two disjoint pieces of A, with the unchanged term $\mu(B)$. Thus every step is valid even when one of the masses is infinite. For (b), apply (a) to $U_n = \bigcup_{i=1}^{n} A_i$ and A_{n+1}, and induct: the base $n = 1$ is equality, and

$$\mu(U_{n+1}) \leq \mu(U_n) + \mu(A_{n+1}) \leq \sum_{i=1}^{n+1} \mu(A_i).$$

In the induction chain, the first inequality follows from (a) applied to U_n, A_{n+1}, the second uses the induction hypothesis and combines the previous finite sum with its next term.

For (c), let $E \in \Sigma$. The disjoint decomposition $S = E \cup (S \setminus E)$ gives

$$\mu(S) = \mu(E) + \mu(S \setminus E) \geq \mu(E).$$

The equality uses additivity and the inequality uses nonnegativity. Since $\mu(S) < \infty$, every measurable set has finite measure; no later monotonicity result is needed. The decompositions used in (a) now give

$$\mu(A) + \mu(B) = \mu(A \setminus B) + \mu(A \cap B) + \mu(B) = \mu(A \cup B) + \mu(A \cap B).$$

The first equality uses the decomposition of A and the second groups the disjoint pieces of $A \cup B$. Subtracting the finite number $\mu(A \cap B)$ proves $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$. This is rearrangement of finite numbers, not an application of the later measure-difference formula. Without finiteness, $\infty - \infty$ may be undefined; the addition-only decompositions remain valid.

HW 12 — Monotonicity of a measure

Prove that if $A, B \in \Sigma$, $A \subset B$, then $\mu(A) \leq \mu(B)$.

Solution to HW 12. Since $A \subseteq B$, the decomposition $B = A \cup (B \setminus A)$ is disjoint. Also $B \setminus A = B \cap A^c \in \Sigma$, where the equality is the definition of set difference and the membership follows from complement and finite-intersection closure of Σ. Therefore

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A),$$

where the equality follows from finite additivity on the two disjoint measurable pieces, and the inequality follows from $\mu(B \setminus A) \geq 0$.

HW 13 — Measure of a difference

The operation of "subtracting" two sets is denoted by $A \backslash B$ and it means $A \cap B^c$. Show that for a finite measure μ, if $B \subset A$ then $\mu(A \backslash B) = \mu(A) - \mu(B)$.

Solution to HW 13. Every point of A either belongs to B or lies in $A \setminus B$, and those alternatives are disjoint. Thus $A = B \cup (A \setminus B)$ and additivity yields $\mu(A) = \mu(B) + \mu(A \setminus B)$. Here the set identity follows from the two alternatives of membership in B and its relative complement; the measure equality follows from finite additivity on these disjoint measurable pieces. Since $\mu(S) < \infty$, the two terms to be subtracted are finite, so rearranging proves the formula.
In fact the weaker assumption $\mu(B) < \infty$ suffices if extended subtraction $\infty - \mu(B) = \infty$ is allowed. Without such a finiteness assumption, subtracting is invalid: for example $A = B = \mathbb{R}$ under Lebesgue measure gives $\mu(A \setminus B) = 0$, while $\mu(A) - \mu(B)$ would read $\infty - \infty$. This explains the role of the hypothesis.

Notation — Increasing and Decreasing Sequences of Sets

For measurable sets, $A_n \uparrow A$ means $A_n \subseteq A_{n+1}$ for every n and $A = \bigcup_{n \geq 1} A_n$. A point belongs to A exactly when it enters the increasing family at some index, after which it remains in all later sets. Similarly, $A_n \downarrow A$ means $A_n \supseteq A_{n+1}$ and $A = \bigcap_{n \geq 1} A_n$: a point of A belongs to every set in the family. Both limiting sets are measurable by the sigma-algebra axioms and HW 3. The set arrow alone makes no claim about the measures; the next lemmas give those conclusions, with a finiteness hypothesis for the decreasing case. For functions, $f_n \uparrow f$ means $f_n(s) \uparrow f(s)$ at every point s, unless the statement explicitly says almost everywhere. It is pointwise monotonicity together with pointwise convergence.

Lemma 1.5 — Continuity from Below

Let $(\mathcal{S}, \Sigma, \mu)$ be a measure space. If $(A_n)_{n \in \mathbb{N}} \subset \Sigma$ such that $A_n \subseteq A_{n+1}$ and $A_n \uparrow A$ as $n \to \infty$ (in the sense that $A = \cup_{n=1}^{\infty} A_n$), then $\mu(A_n) \uparrow \mu(A)$ as $n \to \infty$.

Expanded proof. Countable additivity applies to disjoint sets, whereas the given sets are nested. Following the lecturer, define

$$B_1 = A_1, \; B_n = A_n \setminus \bigcup_{j<n} B_j \; (n \geq 2).$$

These sets are measurable by the algebra properties of Σ. We use their pairwise disjointness, as we will see in HW 14, and the identities

$$\bigcup_{i=1}^{n} B_i = A_n, \ \bigcup_{i\geq 1} B_i = A,$$

as we will see in HW 15. We take precisely these facts as given here. Finite and countable additivity now yield

$$\mu(A_n) = \sum_{i=1}^{n} \mu(B_i) \uparrow \sum_{i=1}^{\infty} \mu(B_i) = \mu(A).$$

The first equality uses the finite-union identity taken as given above and finite additivity on the disjoint B_i. The limit is the definition of a non-negative series. The last equality uses the countable-union identity taken as given above and countable additivity.
The partial sums increase because each summand is non-negative. Their limit defines the sum of this non-negative series. No finiteness assumption is needed: the limit may equal $+\infty$. QED

HW 14 — Disjointifying an increasing family
Prove that $B_i \cap B_j = \varnothing$ for $i \neq j$.
Solution to HW 14. Fix $i < j$. Then

$$B_j \subseteq \left(\bigcup_{k<j} B_k \right)^c \subseteq B_i^c,$$

where the first inclusion follows from the definition of B_j as a difference, and the second follows from $B_i \subseteq \bigcup_{k<j} B_k$ because taking complements reverses inclusion. Hence $B_i \cap B_j = \varnothing$: membership in both sets would contradict $B_j \subseteq B_i^c$. Each B_j is measurable: its defining earlier union is finite and measurable, and its difference from A_j belongs to the sigma-algebra by complement and intersection closure. The disjointness argument itself does not need the A_n to increase; that assumption identifies the partial unions with A_n in the subsequent disjointification argument.

HW 15 — The disjoint pieces have the correct unions
Prove that $\cup_{i=1}^{n} B_i = A_n$ and $\cup_{i=1}^{\infty} B_i = A$.
Solution to HW 15. Put $U_n = \bigcup_{i=1}^{n} B_i$. *We prove $U_n = A_n$ by induction.* For $n = 1$, $U_1 = B_1 = A_1$, where the first equality defines U_1 and the second defines B_1. Assume $U_n = A_n$. Then $U_n \subseteq A_{n+1}$ by the assumed monotonicity of the A_n, and

$$U_{n+1} = U_n \cup B_{n+1} = U_n \cup (A_{n+1} \setminus U_n) = A_{n+1}.$$

The first equality separates the last member of the finite union. The second uses the recursive definition of B_{n+1}. For the final equality, a point of the union is in A_{n+1} because both pieces are subsets of it. Conversely, any point of A_{n+1} either belongs to U_n, or does not and belongs to the difference. These two inclusions prove that last set equality. This completes the induction.
Now let $U = \bigcup_{i\geq 1} B_i$. If $s_0 \in U$, it belongs to some $B_i \subseteq U_i = A_i \subseteq A$; thus $U \subseteq A$. Conversely, if $s_0 \in A = \bigcup_n A_n$, some n satisfies $s_0 \in A_n = U_n \subseteq U$; hence $A \subseteq U$. Therefore $\bigcup_{i\geq 1} B_i = A$. This is a pointwise set identity; it does not require a measure limit.

Lemma 1.6 — Continuity from Above
Let $(\mathcal{S}, \Sigma, \mu)$ be a measure space. If $(A_n)_{n\in\mathbb{N}} \subset \Sigma$ such that $A_n \supseteq A_{n+1}$ and $A_n \downarrow A$ as $n \to \infty$ (in the sense that $A = \cap_{n=1}^{\infty} A_n$), then $\mu(A_n) \downarrow \mu(A)$ as $n \to \infty$ provided that $\mu(A_n) < \infty$ for some $n \in \mathbb{N}$.

HW 16 — Continuity of measures from above

Prove Lemma 1.6.

Hint: take $B_n = A_k \backslash A_{k+n}$ so that $B_n \uparrow A_k \backslash A$ and use HW 13.

Solution to HW 16. Work on the tail after the index k with finite mass; deleting finitely many terms does not change the desired limit. Because A_{k+n} decreases, its complement relative to A_k increases. Moreover a point of A_k eventually leaves some A_{k+n} exactly when it is not in their intersection A. Hence $B_n \uparrow A_k \setminus A$.

Continuity from below, which does not require finite total mass, gives $\mu(B_n) \uparrow \mu(A_k \setminus A)$. All sets here lie in the finite-measure set A_k. We may therefore write

$$\mu(B_n) = \mu(A_k) - \mu(A_{k+n}),\ \mu(A_k \setminus A) = \mu(A_k) - \mu(A).$$

The first displayed equality uses the disjoint decomposition $A_k = B_n \cup A_{k+n}$. The second uses $A_k = (A_k \setminus A) \cup A$. All these masses are bounded by the finite number $\mu(A_k)$, so each subtraction is between finite real numbers.

Passing to the limit and subtracting the same finite number gives $\lim_n \mu(A_{k+n}) = \mu(A)$. Monotonicity ensures this convergence is decreasing, and it also applies to the original sequence. Finiteness is essential: under Lebesgue measure, $A_n = [n, \infty)$ decreases to the empty set but every $\mu(A_n)$ is infinite, while $\mu(\varnothing) = 0$.

Looking at the statement of Lemma 1.6 the question arises as to why do we need the additional condition $\mu(A_n) < \infty$ for some $n \in \mathbb{N}$? The answer is in the next counterexample: take $\mathcal{S} = [0, \infty)$, $\Sigma = \mathcal{B}([0, \infty))$ and $\mu = \mathcal{L}eb$; take $A_n = [n, \infty)$, so that $A_n \downarrow \varnothing$; notice that $\mu(A_n) = \infty$ for all n's and therefore the lemma fails!

Lemma 1.7 — Countable Unions of Null Sets

Countable union of null sets is a null set.

Expanded proof. Let $(N_n)_{n\in\mathbb{N}} \subseteq \Sigma$ satisfy $\mu(N_n) = 0$ for every n. Set $N = \bigcup_{n\geq 1} N_n$ and $N^{(k)} = \bigcup_{n=1}^{k} N_n$. These sets are measurable and $N^{(k)} \uparrow N$. Finite subadditivity gives

$$0 \leq \mu(N^{(k)}) \leq \sum_{n=1}^{k} \mu(N_n) = 0.$$

The first inequality is non-negativity of a measure. The second is finite subadditivity on the first k null sets. The equality substitutes their zero measures and adds finitely many zeros.

Continuity from below therefore gives

$$\mu(N) = \lim_{k\to\infty} \mu(N^{(k)}) = 0.$$

The first equality in the last display is continuity from below on the measurable increasing sets $N^{(k)}$. The second follows because the preceding non-negative upper bound forces every term $\mu(N^{(k)})$ to equal zero.

Countability matters: every singleton has zero Lebesgue measure, whereas an uncountable union of singletons can be an interval of positive measure. QED

HW 17 — Rational points have zero measure

Take the measure space $([0,1], \mathcal{B}([0,1]), \mathcal{L}eb)$. *Prove that* $[0,1]\cap\mathbb{Q} \in \mathcal{B}([0,1])$ *and* $\mathcal{L}eb([0,1]\cap\mathbb{Q}) = 0$.

Solution to HW 17. Rational numbers are countable, so enumerate those in $[0,1]$ as $q_1, q_2, \ldots$. Each singleton $\{q_j\}$ is a Borel subset of $\mathbb{R}$, hence its trace on $[0,1]$ is in $\mathcal{B}([0,1])$. Therefore $[0,1] \cap \mathbb{Q} = \bigcup_{j\geq 1}\{q_j\} \in \mathcal{B}([0,1])$. The singletons are pairwise disjoint and each has zero Lebesgue measure by HW 10. Countable additivity now gives

$$\mathcal{L}eb([0,1] \cap \mathbb{Q}) = \sum_{j\geq 1} \mathcal{L}eb(\{q_j\}) = 0.$$

The first equality uses countable additivity on distinct singletons. The second uses their zero individual masses, so every partial sum is zero and its series limit is zero. The measurable-set

membership is justified before applying the measure. Density of the rationals does not conflict with zero measure: density says every open interval meets the set, whereas measure is determined here by the countable union of zero-mass points.

2. Events

Definition — Probability Space

A mathematical model for an experiment with uncertain (random) outcome is given by a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, where

- Ω is the sample space (the set $\mathcal{S}$ from the previous section). A sample point is a point in Ω, i.e., $\omega \in \Omega$ and it represents (loosely speaking) a possible scenario in the experiment;
- $\mathcal{F}$ is the collection of all events (the σ-algebra Σ from the previous section). An event $E \in \mathcal{F}$ is a collection of sample points, i.e., its a collection of all possible scenarios that may lead to the observed outcome in the experiment;
- P is the probability measure that assigns to each event a probability of occurring. The probability P is a measure on $\mathcal{F}$ with $\mathsf{P}(\Omega) = 1$ (notice that $\mathsf{P} : \mathcal{F} \to [0, 1]$).

Example 1 — Simple Coin-Tossing Models

(a) Toss a coin twice:

$$\Omega = \{HH, HT, TH, TT\},\ \mathcal{F} = \mathcal{P}(\Omega).$$

(b) Toss a coin infinitely many times:

$$\Omega = \{H, T\}^{\mathbb{N}} \implies \omega \in \Omega \text{ is a sequence } \omega = (\omega_1, \omega_2, \omega_3, \dots) \text{ with } \omega_i = H \text{ or } T.$$

In principle one could consider $\mathcal{F} = \mathcal{P}(\Omega)$, where $\mathcal{P}(\Omega)$ is the collection of ALL subsets of Ω and it is called the power set*. It turns out that this is not necessary and we can take the easier σ-algebra*

$$\mathcal{F} = \sigma\big(\{\omega \in \Omega : \omega_n = W\}, n \in \mathbb{N}, W \in \{H, T\}\big).$$

To convince ourselves that this is a good choice we will work towards showing that the set

$$F := \Big\{\omega \in \Omega : \lim_{n\to\infty} \frac{\#(k \le n : \omega_k = H)}{n} = \frac{1}{2}\Big\}$$

belongs to $\mathcal{F}$ and, moreover, $\mathsf{P}(F) = 1$. This is telling us that indeed the Law of Large Numbers (LLN) holds, as it should in any sensible mathematical model for random experiments!

Almost Sure Events and Countable Intersections

Now we prove a result that is in some sense a converse of Lemma 1.7. It tells us that countable intersection of *sure* events give us again a *sure* event. We need a small result first:

HW 18 — A probability-one event and its complement

For any $A \in \mathcal{F}$ show that $\mathsf{P}(A) = 1 \iff \mathsf{P}(A^c) = 0$.

Solution to HW 18. Since A and A^c are disjoint and together cover Ω,

$$1 = \mathsf{P}(\Omega) = \mathsf{P}(A) + \mathsf{P}(A^c).$$

Normalization gives the first equality and additivity gives the second. Both probabilities are finite, so $\mathsf{P}(A^c) = 1 - \mathsf{P}(A)$.
($\Rightarrow$) If $\mathsf{P}(A) = 1$, the established equality gives $\mathsf{P}(A^c) = 0$.
($\Leftarrow$) If $\mathsf{P}(A^c) = 0$, the same equality gives $\mathsf{P}(A) = 1$.
A probability-one event need not be all of Ω: the complement may contain outcomes, but those outcomes form a null event.

Proposition 2.1 — Countable Intersections of Almost Sure Events
If $(A_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ *are such that* $\mathsf{P}(A_n) = 1$ *for all* $n \in \mathbb{N}$*, then*

$$\mathsf{P}\big(\cap_{n=1}^{\infty} A_n\big) = 1.$$

Expanded proof. Each A_n^c is measurable by complement closure. HW 18 gives $\mathsf{P}(A_n^c) = 0$ because $\mathsf{P}(A_n) = 1$. Thus

$$\mathsf{P}\left(\bigcap_n A_n\right) = 1 - \mathsf{P}\left(\left(\bigcap_n A_n\right)^c\right) = 1 - \mathsf{P}\left(\bigcup_n A_n^c\right) = 1,$$

where the first equality follows from HW 18 applied to the measurable intersection, the second follows from De Morgan's identity, and the last follows from Lemma 1.7 applied to the countable family of null events A_n^c. The intersection is measurable by HW 3, so every probability in the chain is defined. QED

Liminf, Limsup and Tail Sets

We are soon going to be using the notions of lim sup and lim inf, so it is good to recall here their meaning in the context of sequences of real numbers. Given a sequence $(x_n)_{n\in\mathbb{N}} \subset \mathbb{R}$ we define

$$\liminf_{n\to\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m \text{ (i.e., the largest minorant in the limit)},$$

$$\limsup_{n\to\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m \text{ (i.e., the smallest majorant in the limit)}.$$

Put $y_n = \sup_{m\geq n} x_m$. Then

$$y_n = \sup_{m\geq n} x_m \geq \sup_{m\geq n+1} x_m = y_{n+1},$$

where the first and last equalities follow from the definition of the tail supremum. The inequality follows because $\{m : m \geq n+1\} \subseteq \{m : m \geq n\}$, so the supremum on the smaller index set cannot be larger. Consequently

$$\limsup_n x_n = \inf_n y_n = \lim_n y_n,$$

where the first equality is the definition of limsup, and the second uses the fact that the limit of a decreasing sequence in the extended real line is its infimum.

HW 19 — Tail infima and the liminf
Show that

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf_{m\geq n} x_m.$$

Solution to HW 19. Define $a_n = \inf_{m\geq n} x_m$. When n increases, the set over which the infimum is taken becomes smaller, so $a_n \leq a_{n+1}$. This monotone sequence has a limit in $[-\infty, \infty]$, and its limit is its supremum. To see this when $a = \sup_n a_n$ is finite, for every $\varepsilon > 0$ some $a_N > a - \varepsilon$; then $a - \varepsilon < a_n \leq a$ for all $n \geq N$. If the supremum is $+\infty$, the same argument with any finite lower bound shows divergence to $+\infty$. If it is $-\infty$, every $a_n = -\infty$.
The definition of the liminf is $\sup_n \inf_{m\geq n} x_m = \sup_n a_n$. Combining this with the monotone limit just proved gives the asserted equality. The original sequence need not converge: the monotonicity belongs to the tail infima, not to (x_n) itself.

Given $(x_n)_{n\in\mathbb{N}} \subset \mathbb{R}$, the values of $\liminf x_n$ and $\limsup x_n$ are always well-defined (not necessarily finite) whereas the value of $\lim x_n$ may not exist. We say that a limit exists if

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n =: \lim_{n\to\infty} x_n.$$

HW 20 — An alternating sequence

Consider the sequence $x_n = (-1)^n$, $n \in \mathbb{N}$. *Calculate its* liminf *and* limsup. *Does the limits exist?*

Solution to HW 20. Every tail contains both an even and an odd index, and indeed infinitely many of each. The smallest value in every tail is -1 and the largest is 1. Thus $\inf_{m\geq n} x_m = -1$ and $\sup_{m\geq n} x_m = 1$ for every n. Taking their respective increasing and decreasing tail limits gives

$$\liminf_n x_n = -1, \ \limsup_n x_n = 1.$$

A sequence has a limit only when its liminf equals its limsup. These two numbers differ, so the limit does not exist. Equivalently the even subsequence is constantly one and the odd subsequence constantly minus one; two subsequences with different limits rule out convergence of the whole sequence.

HW 21 — Threshold properties of liminf and limsup

Prove the following useful properties of liminf *and* limsup

(i) If $z > \limsup_{n\to\infty} x_n$, *then there is* $N \in \mathbb{N}$ *such that* $x_n < z$ *for all* $n \geq N$;
(ii) If $z < \liminf_{n\to\infty} x_n$, *then there is* $N \in \mathbb{N}$ *such that* $x_n > z$ *for all* $n \geq N$;
(iii) If $z < \limsup_{n\to\infty} x_n$, *then there are infinitely many values of* $n \in \mathbb{N}$ *such that* $x_n > z$;
(iv) If $z > \liminf_{n\to\infty} x_n$, *then there are infinitely many values of* $n \in \mathbb{N}$ *such that* $x_n < z$.

Solution to HW 21. Let $b_N = \sup_{n\geq N} x_n$ and $a_N = \inf_{n\geq N} x_n$. Their limits are $b = \limsup x_n = \inf_N b_N$ and $a = \liminf x_n = \sup_N a_N$.

(i) If $z > b$ and no b_N were smaller than z, their infimum would be at least z, a contradiction. Choose N with $b_N < z$. Then $x_n \leq b_N < z$ for every $n \geq N$.

(ii) If $z < a$, some a_N must be larger than z, otherwise their supremum would be at most z. Thus $x_n \geq a_N > z$ for every $n \geq N$.

(iii) Fix any cutoff N. Since $z < b = \inf_M b_M \leq b_N$, we have $z < \sup_{n\geq N} x_n$. The first equality defines b and the inequality is the lower-bound property of the infimum. If every term in this tail were at most z, its supremum would be at most z, a contradiction. Hence some index $n_1 \geq 1$ has $x_{n_1} > z$. Reapply the same argument with cutoff $n_1 + 1$ to choose $n_2 > n_1$ with $x_{n_2} > z$, and continue. The strictly increasing indices give infinitely many exceedances.

(iv) Fix a cutoff N. Since

$$z > a = \sup_M a_M \geq a_N,$$

we have $z > \inf_{n\geq N} x_n$, where the equality defines a and the inequality is the upper-bound property of the supremum. If all terms in the tail were at least z, its infimum would be at least z, a contradiction. Choose a first index below z, then choose another beyond its index, repeating this argument on successive tails. This constructs infinitely many indices with $x_n < z$. In (i), after choosing N with $b_N < z$, write

$$x_n \leq b_N < z \ (n \geq N),$$

where the first inequality follows because b_N is an upper bound for every term in that tail, and the second is the property used to choose N. In (ii), after choosing N with $a_N > z$, write

$$x_n \geq a_N > z \ (n \geq N),$$

where the first inequality follows because a_N is a lower bound for every term in the tail, and the second is the choice of N. These explanations do not assume that a supremum or infimum is attained by one of the terms.

Strict inequalities in the assumptions matter: for example a constant sequence equal to z has limsup z but no terms above z. The proofs use order and tails only, so they continue to work when a limiting bound is infinite and the stated strict comparison is meaningful.

HW 22 — Negation exchanges suprema and infima

Let $(x_\alpha)_{\alpha\in\mathcal{I}}$ be a family of real numbers ($\mathcal{I}$ not necessarily countable, e.g., $\mathcal{I}=[0,1]$). Prove that

$$\sup_{\alpha\in\mathcal{I}} x_\alpha = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha).$$

Deduce that

$$\liminf_{n\to\infty} x_n = -\limsup_{n\to\infty}(-x_n).$$

Solution to HW 22.

For an empty index family the conventional values $\sup\varnothing=-\infty$ and $\inf\varnothing=+\infty$ give the stated identity directly. Assume the family is nonempty and put $M=\inf_\alpha(-x_\alpha)$. If M is finite, $M\le -x_\alpha$ for every α, so $x_\alpha\le -M$ by reversal of inequality under negation. Thus $-M$ is an upper bound. For every $\varepsilon>0$, the greatest-lower-bound property supplies β with $-x_\beta<M+\varepsilon$, and negation gives $x_\beta>-M-\varepsilon$. No number strictly smaller than $-M$ can be an upper bound: choose ε smaller than its gap from $-M$. Hence $\sup_\alpha x_\alpha=-M$. If $M=-\infty$, for every real K some β satisfies $-x_\beta<-K$, hence $x_\beta>K$. The family is unbounded above, so

$$\sup_\alpha x_\alpha=+\infty=-(-\infty)=-M,$$

where the first equality is the definition of unbounded supremum, the second is extended-real negation, and the last substitutes $M=-\infty$. The case $M=+\infty$ is impossible for a nonempty family of finite real numbers, since any selected $-x_\alpha$ is a finite upper bound for its infimum.

The same negation identity holds for a nonempty family of extended-real numbers. The finite-M argument still applies: the greatest-lower-bound property supplies a finite approximating member, while infinite upper members do not affect its bound. For $M=-\infty$, the same arbitrary-threshold argument proves an unbounded supremum, including a possible member $+\infty$. If $M=+\infty$, every $-x_\alpha$ must equal $+\infty$, hence every $x_\alpha=-\infty$ and their supremum is $-\infty=-M$. This extension is needed when a tail infimum or supremum is infinite.

For each tail, this identity gives $\inf_{m\ge n} x_m=-\sup_{m\ge n}(-x_m)$. Taking the supremum over n and applying the identity again,

$$\liminf_n x_n=\sup_n\inf_{m\ge n} x_m=\sup_n\left[-\sup_{m\ge n}(-x_m)\right]=-\inf_n\sup_{m\ge n}(-x_m)=-\limsup_n(-x_n).$$

The first equality in the displayed chain defines liminf. The second applies the already proved negation identity inside each tail. The third applies it to the outer supremum over those tail quantities. The last equality is the definition of the limsup of $(-x_n)$.

The index family may be uncountable in the first assertion because no measure or countable closure is involved, only the completeness property of the extended real order.

Events Occurring Infinitely Often

Now we want to make sense of the notions of liminf and lim sup for sets! Let $(E_n)_{n\in\mathbb{N}}\subset\mathcal{F}$ and define:

$$\begin{aligned}\limsup_{n\to\infty} E_n &:= \{\omega\in\Omega:\omega\in E_n \text{ for infinitely many } n\text{ 's }\}\\ &=\{\omega\in\Omega:\ \text{for any } n\in\mathbb{N} \text{ there is } m\ge n \text{ s.t. } \omega\in E_m\}=\bigcap_{n\ge1}\bigcup_{m\ge n} E_m.\end{aligned}$$

The first equality spells out infinitely many occurrences: after every finite initial segment there is another occurrence. The second equality translates the universal tail condition into an intersection and the existential choice within each tail into a union.

Notice that $\cup_{m\ge n}E_m$ contains those ω's that belong to at least one set E_m for $m\ge n$. Taking the intersection allows to get rid of those ω's that belong to only finitely many sets E_n. Notice also that $\limsup_{n\to\infty}E_n\in\mathcal{F}$ because it is countable intersection of countable unions of sets in $\mathcal{F}$.

HW 23 — Measurability of a limsup event

Prove that $\limsup_{n\to\infty} E_n \in \mathcal{F}$.

Solution to HW 23. Let L be the event of belonging to infinitely many E_n, and let $D = \bigcap_{N\geq 1}\bigcup_{n\geq N} E_n$. For the first inclusion, take $\omega_0 \in L$ and fix N. Infinitely many membership indices cannot all lie in $\{1,\dots,N-1\}$, a finite set. Thus some $n \geq N$ satisfies $\omega_0 \in E_n$, giving $\omega_0 \in \bigcup_{n\geq N} E_n$. As N was arbitrary, $\omega_0 \in D$. For the reverse inclusion, take $\omega_0 \in D$. For every cutoff N, some index $n \geq N$ is a membership index. If there were only finitely many such indices, a cutoff beyond their maximum would contradict this property. Thus $\omega_0 \in L$. Combining the inclusions gives

$$\limsup_n E_n = L = \bigcap_{N\geq 1}\bigcup_{n\geq N} E_n \in \mathcal{F}.$$

The first equality is the definition of the set limsup. The second is the identity just proved. Each inner union belongs to $\mathcal{F}$ by countable-union closure. The final membership follows from countable-intersection closure, already proved in HW 3.

Lemma 2.2 — Reverse Fatou for Sets

Given $(E_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ *we have*

$$\mathsf{P}(\limsup_{n\to\infty} E_n) \geq \limsup_{n\to\infty} \mathsf{P}(E_n).$$

Expanded proof. Define $G_m = \bigcup_{n\geq m} E_n$. We have

$$G_m = E_m \cup G_{m+1} \supseteq G_{m+1},$$

where the equality separates the first index from its tail, and the inclusion follows because a union contains each of its members. The equality has both directions: a point of G_m belongs either to E_m or to some later E_n, and either alternative puts it in G_m. Thus the G_m decrease. Their intersection is the set limsup by its defining tail formula. The events $G_m = \bigcup_{n\geq m} E_n$ are measurable by countable-union closure. They decrease to $\limsup_n E_n$, and $\mathsf{P}(G_1) \leq 1 < \infty$. For each $n \geq m$, the inclusion $E_n \subseteq G_m$ and HW 12 give $\mathsf{P}(E_n) \leq \mathsf{P}(G_m)$. Thus

$$\limsup_n \mathsf{P}(E_n) = \lim_m \sup_{n\geq m} \mathsf{P}(E_n) \leq \lim_m \mathsf{P}(G_m) = \mathsf{P}\left(\bigcap_m G_m\right) = \mathsf{P}(\limsup_n E_n),$$

where the first equality is the numerical tail definition of limsup. The inequality follows by taking the supremum of the preceding bounds over each tail and then passing to the limits, which exist by monotonicity. The second equality follows from Lemma 1.6, whose finite-mass hypothesis was checked above. The final equality follows from the defining tail identity for the set limsup. QED

Lemma 2.3 — First Borel–Cantelli Lemma

Let $(E_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ *be such that*

$$\sum_{n=1}^{\infty} \mathsf{P}(E_n) < \infty,$$

then $\mathsf{P}(\limsup_{n\to\infty} E_n) = 0$ *(i.e., almost every* ω *belongs to only finitely many sets* E_n*'s).*

Expanded proof. Following the lecturer, set $G_m = \bigcup_{n\geq m} E_n$ and

$$G = \bigcap_{m\geq 1} G_m = \limsup_n E_n.$$

Each G_m is measurable by countable-union closure, and G is measurable by countable-intersection closure. Since $G_{m+1} \subseteq G_m$, we have $G_m \downarrow G$. The inclusion holds because every index at least $m+1$ is also at least m; the equality defining G is the definition of the event limsup. Moreover $\mathsf{P}(G_1) \leq 1 < \infty$, so Lemma 1.6 gives

$$\mathsf{P}(G) = \lim_{m\to\infty} \mathsf{P}(G_m).$$

The equality is continuity from above on the decreasing measurable tail unions, with its finite-mass hypothesis just checked.
For a fixed m, define $G_m^k = \bigcup_{n=m}^{m+k} E_n$. These measurable finite unions increase to G_m: each tail index is eventually included, and every finite union is contained in the tail. Lemma 1.5 and finite subadditivity therefore give

$$\mathsf{P}(G_m) = \lim_{k\to\infty} \mathsf{P}(G_m^k) \le \lim_{k\to\infty} \sum_{n=m}^{m+k} \mathsf{P}(E_n) = \sum_{n\ge m} \mathsf{P}(E_n).$$

The first equality is continuity from below; the inequality applies finite subadditivity for every k and preserves the bound on taking limits; the last equality is the definition of a non-negative series. Combining these two calculations yields

$$0 \le \mathsf{P}(G) = \lim_{m\to\infty} \mathsf{P}(G_m) \le \lim_{m\to\infty} \sum_{n\ge m} \mathsf{P}(E_n) = 0.$$

The first inequality is non-negativity of a measure; the first equality is the continuity-from-above calculation. The second inequality takes limits in the tail bound. The final equality holds because the tails of the assumed convergent non-negative series tend to zero. Thus $\mathsf{P}(\limsup_n E_n) = 0$, as required. QED

Let $(E_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ and define:

$$\begin{aligned}\liminf_{n\to\infty} E_n &:= \{\omega \in \Omega : \omega \in E_n \text{ for all but finitely many } n\text{'s}\} \\ &= \{\omega \in \Omega : \exists N_\omega \in \mathbb{N} \text{s.t.} \omega \in E_n \text{for all} n \ge N_\omega\} = \bigcup_{n\ge 1} \bigcap_{m\ge n} E_m.\end{aligned}$$

The first equality spells out all but finitely many occurrences: there is a tail on which every occurrence holds. The second equality translates the choice of that tail into a union and membership throughout that tail into an intersection.
Notice that $\cap_{m\ge n} E_m$ contains those ω's that belong to all sets E_m for $m \ge n$. Taking the union allows to include all those ω's that belong to all but finitely many sets E_n. Notice also that $\liminf_{n\to\infty} E_n \in \mathcal{F}$ because it is countable union of countable intersections of sets in $\mathcal{F}$.

HW 24 — Measurability of a liminf event

Prove that $\liminf_{n\to\infty} E_n \in \mathcal{F}$.

Solution to HW 24. Let L be the event of belonging to all sufficiently late E_n, and put $D = \bigcup_{N\ge 1} \bigcap_{n\ge N} E_n$. If $\omega_0 \in L$, some cutoff N makes $\omega_0 \in E_n$ for every $n \ge N$. It belongs to that tail intersection and therefore to D. Conversely, if $\omega_0 \in D$, the definition of union supplies one N with $\omega_0 \in \bigcap_{n\ge N} E_n$. The definition of intersection then gives membership in every E_n with $n \ge N$, so $\omega_0 \in L$. Both inclusions have been proved, and

$$\liminf_n E_n = L = \bigcup_{N\ge 1} \bigcap_{n\ge N} E_n \in \mathcal{F}.$$

The first equality defines the set liminf, and the second is the two-inclusion identity above. Each tail intersection belongs to $\mathcal{F}$ by HW 3; the final membership uses countable-union closure. The order of the quantifiers is essential: one cutoff works for every later index, rather than a new index being chosen after every cutoff.

HW 25 — Complements of limiting events

Prove that $\liminf_{n\to\infty} E_n = (\limsup_{n\to\infty} E_n^c)^c$.

Solution to HW 25. Start with the defining tail formula for the limsup of the complements, and apply De Morgan's laws twice:

$$\left(\limsup_n E_n^c\right)^c = \left(\bigcap_{N\ge 1} \bigcup_{n\ge N} E_n^c\right)^c = \bigcup_{N\ge 1} \bigcap_{n\ge N} (E_n^c)^c = \bigcup_{N\ge 1} \bigcap_{n\ge N} E_n = \liminf_n E_n.$$

The first equality substitutes the tail definition of limsup. The second uses De Morgan first for the outer intersection and then for each inner union. The third uses double complementation $(E_n^c)^c = E_n$. The final equality is the tail definition of liminf.
In words, failing to be outside E_n infinitely often means being outside only finitely often, which is exactly being inside all sufficiently late E_n. The identity is pointwise and set-theoretic; no probability assumptions are needed.

Lemma 2.4 — Fatou for Sets

Given $(E_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ we have

$$\mathsf{P}(\liminf_{n\to\infty} E_n) \le \liminf_{n\to\infty} \mathsf{P}(E_n).$$

HW 26 — Fatou inequality for events

Prove Lemma 2.4.
Hint: follow the steps in the proof of Lemma 2.2.

Solution to HW 26. By HW 25, $\liminf_n E_n = (\limsup_n E_n^c)^c$. The limiting events are measurable by HW 23–24, and HW 18 gives $\mathsf{P}(A^c) = 1 - \mathsf{P}(A)$ for every measurable A. Apply Lemma 2.2 (Reverse Fatou) to the measurable sequence $(E_n^c)_{n\ge1}$:

$$\mathsf{P}(\liminf_n E_n) = \mathsf{P}((\limsup_n E_n^c)^c) = 1 - \mathsf{P}(\limsup_n E_n^c) \le 1 - \limsup_n \mathsf{P}(E_n^c).$$

The first equality uses HW 25, the second uses the complement formula, and the inequality reverses the Reverse Fatou bound because it is subtracted from one. All probabilities are finite. Using the same complement formula for each E_n gives

$$\begin{aligned} 1 - \limsup_n \mathsf{P}(E_n^c) &= 1 - \limsup_n (1 - \mathsf{P}(E_n)) \\ &= -\limsup_n (-\mathsf{P}(E_n)) = \liminf_n \mathsf{P}(E_n). \end{aligned}$$

The second equality uses that adding the constant one commutes with tail suprema and their limits. The last is HW 22. Combining the chains proves $\mathsf{P}(\liminf_n E_n) \le \liminf_n \mathsf{P}(E_n)$, following the notebook's complement argument.

Definition — Indicator Function

The indicator function of a set $E \in \mathcal{F}$ is defined as

$$1_E(\omega) = \begin{cases} 1, & \text{if } \omega \in E, \\ 0, & \text{if } \omega \notin E. \end{cases}$$

For example, if we take $\Omega = [0,1]$ (i.e., ω's are numbers in $[0,1]$) and $\mathcal{F} = \mathcal{B}([0,1])$, then $1_{[\frac{1}{2},1]}(\omega)$ is a function which is zero when $\omega \in [0, \frac{1}{2})$ and it is equal to one when $\omega \in [\frac{1}{2}, 1]$.

HW 27 — Indicators commute with limsup

(Ex. 2.9 from the book). *Given $(E_n)_{n\in\mathbb{N}} \subset \mathcal{F}$, prove that*

$$1_{\limsup_{n\to\infty} E_n}(\omega) = \limsup_{n\to\infty} 1_{E_n}(\omega).$$

Hint: follow the intermediate steps set out below.

(i) prove that $1_{E_i \cap E_j} = 1_{E_i} \cdot 1_{E_j} = \min\{1_{E_i}, 1_{E_j}\}$;
(ii) prove that $1_{E_i \cup E_j} = 1_{E_i} + 1_{E_j} - 1_{E_i \cap E_j} = \max\{1_{E_i}, 1_{E_j}\}$;
(iii) use (i) and (ii) to conclude.

Solution to HW 27. Fix an outcome ω; every indicator is now either zero or one. For (i), the product and the minimum equal one exactly when both indicators are one, equivalently when ω lies in the intersection. Otherwise both are zero. For (ii), if ω belongs to neither event the displayed sum is zero; if it belongs to exactly one the sum is one; if it belongs to both it is $1 + 1 - 1 = 1$. These are exactly the cases for the indicator of the union and the maximum.
For a countable family, the same pointwise interpretation gives $\mathbf{1}_{\bigcup_{n\ge N} E_n} = \sup_{n\ge N} \mathbf{1}_{E_n}$: either at least one membership occurs, giving one on both sides, or none occurs. Likewise an indicator

of a countable intersection is the infimum of the indicators. Applying these identities to the tail formula gives

$$\mathbf{1}_{\limsup_n E_n} = \mathbf{1}_{\bigcap_{N\geq 1}\bigcup_{n\geq N} E_n} = \inf_{N\geq 1}\sup_{n\geq N}\mathbf{1}_{E_n} = \limsup_n \mathbf{1}_{E_n}.$$

The first displayed equality substitutes the set-limsup formula from HW 23. The second uses the just-verified indicator rule for an intersection outside and a union inside. The third is the numerical tail definition of limsup. All are pointwise equalities for each fixed outcome.
Equivalently, a zero-one sequence has limsup one precisely when ones occur infinitely often. This holds for every outcome, not merely almost surely.

3. Random variables

Pre-images and Measurability
As anticipated at the beginning of the course, random variables are measurable functions. We now make this statement rigorous but we start from a general measurable space for the sake of completeness. Let $(\mathcal{S}, \Sigma)$ be a measurable space and let $h : \mathcal{S} \to \mathbb{R}$ be a function. We define the pre-image of a set $A \subset \mathbb{R}$ as

$$h^{-1}(A) = \{s \in \mathcal{S} : h(s) \in A\}.$$

Definition 3.1 — Measurable Function
A function $h : \mathcal{S} \to \mathbb{R}$ is Σ-measurable (also denoted $h \in m\Sigma$) if $h^{-1}(A) \in \Sigma$ for every $A \in \mathcal{B}(\mathbb{R})$. Equivalently we say that $h \in m\Sigma$ if $h^{-1} : \mathcal{B}(\mathbb{R}) \to \Sigma$.
Notation: We use $(m\Sigma)^+$ for the class of *non-negative* Σ-measurable functions. We use $b\Sigma$ for the class of *bounded* Σ-measurable functions.

Definition — Borel-Measurable Function
Borel-measurable functions: If $\mathcal{S}$ is a topological space and $\Sigma = \mathcal{B}(\mathcal{S})$, then $m\Sigma$ is the class of Borel-measurable functions. For example if $\mathcal{S} = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$ and $h : \mathbb{R} \to \mathbb{R}$ is such that $h \in m\Sigma$, then h is Borel-measurable.

Properties — Pre-images Preserve Set Operations
Each A_α below is a subset of $\mathbb{R}$, so the family is contained in $\mathcal{P}(\mathbb{R})$; this makes explicit the types in the printed shorthand.
Here are some properties of the pre-image mapping: let $h : \mathcal{S} \to \mathbb{R}$ and let $(A_\alpha)_{\alpha\in\mathcal{I}} \subseteq \mathcal{P}(\mathbb{R})$ with $\mathcal{I}$ possibly uncountable (i.e., $\mathcal{I} = [0, 1]$), then

(i) $h^{-1}(\cup_{\alpha\in\mathcal{I}} A_\alpha) = \cup_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$;
(ii) $h^{-1}(A^c) = (h^{-1}(A))^c$;
(iii) $h^{-1}(\cap_{\alpha\in\mathcal{I}} A_\alpha) = \cap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$.

Let us prove (i). If

$$s_0 \in h^{-1}(\cup_{\alpha\in\mathcal{I}} A_\alpha) = \{s \in \mathcal{S} : h(s) \in \cup_{\alpha\in\mathcal{I}} A_\alpha\}$$

then there must be $\alpha_0 \in \mathcal{I}$ for which $h(s_0) \in A_{\alpha_0}$. Thus

$$s_0 \in h^{-1}(A_{\alpha_0}) \subset \cup_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$$

where the membership follows from the definition of pre-image and the inclusion follows because the indicated pre-image is one member of the union. By arbitrariness of s_0 it follows that

$$h^{-1}\left(\cup_{\alpha\in\mathcal{I}} A_\alpha\right) \subseteq \cup_{\alpha\in\mathcal{I}} h^{-1}\left(A_\alpha\right).$$

For the reverse inclusion, notice that if $s_0 \in \cup_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$ then there is $\alpha_0 \in \mathcal{I}$ for which

$$s_0 \in h^{-1}(A_{\alpha_0}) = \{s \in \mathcal{S} : h(s) \in A_{\alpha_0}\} \subset \{s \in \mathcal{S} : h(s) \in \cup_{\alpha\in\mathcal{I}} A_\alpha\} = h^{-1}(\cup_{\alpha\in\mathcal{I}} A_\alpha).$$

The first equality is the definition of the pre-image of A_{α_0}. The inclusion follows from $A_{\alpha_0} \subseteq \bigcup_\alpha A_\alpha$. The last equality is the definition of the pre-image of that union.

Since s_0 was arbitrary, then

$$h^{-1}\left(\cup_{\alpha\in\mathcal{I}}A_\alpha\right)\supseteq\cup_{\alpha\in\mathcal{I}}h^{-1}\left(A_\alpha\right).$$

HW 28 — Preimages preserve complements and intersections
Prove properties (ii) and (iii) of h^{-1}.

Solution to HW 28. *Property (ii): complements.* Let $s_0\in h^{-1}(A^c)$. By definition of pre-image, $h(s_0)\in A^c$, so $h(s_0)\notin A$. Thus $s_0\notin h^{-1}(A)$, and $s_0\in(h^{-1}(A))^c$. This proves the first inclusion. Conversely, let $s_0\in(h^{-1}(A))^c$. Then $s_0\notin h^{-1}(A)$, so $h(s_0)\notin A$, equivalently $h(s_0)\in A^c$. Therefore $s_0\in h^{-1}(A^c)$, proving the reverse inclusion. Combining them gives $h^{-1}(A^c)=(h^{-1}(A))^c$. In the forward inclusion, the first passage uses the definition of pre-image, the second the definition of complement in the codomain, the third the same pre-image definition in reverse, and the fourth the definition of complement in the domain. In the reverse inclusion those definitions are used in the opposite order. Thus the two complements are taken in their respective spaces, rather than silently identified.
Property (iii): intersections. Let $s_0\in h^{-1}(\bigcap_{\alpha\in I}A_\alpha)$. Its image belongs to every A_α, so $s_0\in h^{-1}(A_\alpha)$ for every α. Consequently $s_0\in\bigcap_{\alpha\in I}h^{-1}(A_\alpha)$. For the reverse inclusion, let s_0 belong to that intersection. For every α, pre-image membership gives $h(s_0)\in A_\alpha$. Thus $h(s_0)\in\bigcap_{\alpha\in I}A_\alpha$, giving $s_0\in h^{-1}(\bigcap_{\alpha\in I}A_\alpha)$. Combining the inclusions proves the asserted equality. Every transition was the definition of pre-image, complement or intersection. No measurability was assumed and no countability is needed. For an empty index set, the intersections are the appropriate whole spaces and the formula reduces to $h^{-1}(\mathbb{R})=\mathcal{S}$.

Properties (i), (ii) and (iii) have a very important consequence:

Lemma 3.1 — The Sigma-Algebra of Pre-images
Given $h:\mathcal{S}\to\mathbb{R}$, the class

$$h^{-1}(\mathcal{B}(\mathbb{R})):=\{h^{-1}(A),A\in\mathcal{B}(\mathbb{R})\}$$

is a σ-algebra on $\mathcal{S}$.

Expanded proof. Put $\mathcal{H}=h^{-1}(\mathcal{B}(\mathbb{R}))$. Its members are subsets of the domain $\mathcal{S}$. First,

$$\mathcal{S}=h^{-1}(\mathbb{R})\in\mathcal{H},\ \varnothing=h^{-1}(\varnothing)\in\mathcal{H},$$

where each equality follows from the definition of pre-image and the codomain of h. Each membership follows because the corresponding codomain set is Borel and $\mathcal{H}$ contains every such pre-image. If $H\in\mathcal{H}$, its definition supplies $A\in\mathcal{B}(\mathbb{R})$ with $H=h^{-1}(A)$. Then

$$H^c=(h^{-1}(A))^c=h^{-1}(A^c)\in\mathcal{H},$$

where the first equality substitutes the representation of H, the second follows from HW 28, and the final membership follows because A^c is Borel by complement closure. For $(H_n)_{n\geq1}\subseteq\mathcal{H}$, select $A_n\in\mathcal{B}(\mathbb{R})$ with $H_n=h^{-1}(A_n)$. Then

$$\bigcup_n H_n=\bigcup_n h^{-1}(A_n)=h^{-1}\left(\bigcup_n A_n\right)\in\mathcal{H},$$

where the first equality substitutes these representations, the second is the pre-image identity for unions, and the final membership follows because $\bigcup_n A_n$ is Borel by countable-union closure and $\mathcal{H}$ contains its pre-image. This checks all sigma-algebra axioms. No preliminary measurability of h was assumed. The construction makes h measurable for $\mathcal{H}$. Any sigma-algebra for which h is measurable must contain every Borel pre-image, hence must contain $\mathcal{H}$; this proves its minimality.
QED

It should be clear that h is $h^{-1}(\mathcal{B}(\mathbb{R}))$-measurable! The take-home message is that for any $h:\mathcal{S}\to\mathbb{R}$ we can construct a σ-algebra on $\mathcal{S}$ with respect to which h is measurable. The smallest of such σ-algebras is exactly $h^{-1}(\mathcal{B}(\mathbb{R}))$.

According to Definition 3.1, in order to check whether a function is Σ-measurable or not one needs to verify that pre-images of ALL Borel sets are in Σ. This is virtually an impossible task because Borel sets can take very strange shapes. The next lemma comes to the rescue.

Lemma 3.2 — Testing Measurability on Generators
Let $\mathcal{C} \subset \mathcal{B}(\mathbb{R})$ such that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$ and let $h : \mathcal{S} \to \mathbb{R}$. If $h^{-1} : \mathcal{C} \to \Sigma$, then $h \in m\Sigma$.

Expanded proof. Let $\mathcal{E} = \{B \in \mathcal{B}(\mathbb{R}) : h^{-1}(B) \in \Sigma\}$. By hypothesis $\mathcal{C} \subseteq \mathcal{E}$, while $\mathcal{E} \subseteq \mathcal{B}(\mathbb{R})$ by definition. *We show that $\mathcal{E}$ is a sigma-algebra on $\mathbb{R}$.*
Empty set and whole space. Their pre-images are respectively $\varnothing$ and $\mathcal{S}$, both in Σ. Thus both belong to $\mathcal{E}$.
Complements. Let $B \in \mathcal{E}$. The set B^c is Borel, and the complement identity proved in HW 28 gives $h^{-1}(B^c) = (h^{-1}(B))^c \in \Sigma$. The membership follows from complement closure of Σ. Hence $B^c \in \mathcal{E}$.
Countable unions. Let $(B_n)_{n\in\mathbb{N}} \subseteq \mathcal{E}$. Every B_n is Borel, and every $h^{-1}(B_n)$ lies in Σ. Their union in the codomain is Borel. The union identity proved above gives

$$h^{-1}\Big(\bigcup_{n\geq 1} B_n\Big) = \bigcup_{n\geq 1} h^{-1}(B_n) \in \Sigma.$$

The final membership follows from countable-union closure of Σ; therefore the union belongs to $\mathcal{E}$. Minimality of the generated sigma-algebra now gives

$$\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subseteq \mathcal{E} \subseteq \mathcal{B}(\mathbb{R}).$$

In the final class chain, the equality is the generating hypothesis $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$. The first inclusion is minimality, since $\mathcal{E}$ is now proved to be a sigma-algebra containing $\mathcal{C}$. The last inclusion follows from the defining restriction $B \in \mathcal{B}(\mathbb{R})$ in $\mathcal{E}$.
Thus $\mathcal{E} = \mathcal{B}(\mathbb{R})$. Every Borel set has pre-image in Σ, which is precisely the measurability of h. QED

The following two consequences hold:

(i) A continuous $h : \mathcal{S} \to \mathbb{R}$ on a topological space is Borel measurable.
(ii) On a measurable space $(\mathcal{S}, \Sigma)$, h is measurable if and only if $\{h \leq c\} \in \Sigma$ for every $c \in \mathbb{R}$.

Explanation of (i). Use the open subsets of the codomain as $\mathcal{C}$. They generate $\mathcal{B}(\mathbb{R})$ by definition. Continuity makes each pre-image open in the domain, hence Borel there. Lemma 3.2 applied with domain sigma-algebra $\mathcal{B}(\mathcal{S})$ therefore gives Borel measurability.
Explanation of (ii). For every real c,

$$\{h \leq c\} = \{s \in \mathcal{S} : h(s) \in (-\infty, c]\} = h^{-1}((-\infty, c]).$$

The first equality is the lower-level notation, and the second is the definition of pre-image.
(⇒) If h is measurable, its pre-image of the Borel ray belongs to Σ.
(⇐) Conversely, these memberships for all c are the hypothesis, not a consequence one may presume. Proposition 1.1 gives $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$, so Lemma 3.2 applied to the lower-ray generating class proves measurability. This is the stated practical threshold test.

HW 29 — Measurability of an interval indicator
Let $\mathcal{S} = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$ and $h(s) = 1_{[0,1]}(s)$. Prove that h is Borel-measurable.
Hint: first you need to show that $[0, 1] \in \mathcal{B}(\mathbb{R})$.

Solution to HW 29. The closed interval has the countable-open representation

$$[0, 1] = \textstyle\bigcap_{n\geq 1}(-1/n, 1 + 1/n).$$

A number in $[0, 1]$ belongs to every interval; a number outside is excluded once $1/n$ is smaller than its positive distance from $[0, 1]$. Thus $[0, 1]$ belongs to the Borel sigma-algebra, and so does its complement.

It suffices to check preimages of lower rays $(-\infty, c]$, since these generate $\mathcal{B}(\mathbb{R})$. The indicator takes only the values zero and one, giving

$$\{h \le c\} = \begin{cases} \varnothing, & c < 0, \\ (-\infty, 0) \cup (1, \infty), & 0 \le c < 1, \\ \mathbb{R}, & c \ge 1. \end{cases}$$

For $c < 0$, neither indicator value is at most c, giving the empty set. For $0 \le c < 1$, only zero is at most c, giving exactly the points outside $[0, 1]$. For $c \ge 1$, both values qualify, giving the whole domain. These are the reasons for the three cases, separately from their Borel memberships.
Every set in this list is Borel. Hence h is Borel-measurable.
The empty and full pre-images belong to every sigma-algebra. The middle pre-image is the complement of the Borel interval $[0, 1]$, and therefore belongs to $\mathcal{B}(\mathbb{R})$ by complement closure. Lemma 3.2 applied to the generating closed lower rays proves the conclusion. The middle case is the complement of $[0, 1]$, not the interval itself: the condition $h \le c < 1$ selects points where the indicator is zero.

HW 30 — Equivalent threshold tests for measurability

Prove that $h : \mathcal{S} \to \mathbb{R}$ is Σ-measurable if and only if, any of the following holds:

(i) $\{h < c\} := h^{-1}((-\infty, c)) \in \Sigma$ *for all* $c \in \mathbb{R}$*;*

(ii) $\{h > c\} := h^{-1}((c, \infty)) \in \Sigma$ *for all* $c \in \mathbb{R}$*;*

(iii) $\{h \ge c\} := h^{-1}([c, \infty)) \in \Sigma$ *for all* $c \in \mathbb{R}$*.*

Solution to HW 30. **(⇒)** If h is measurable, the pre-image of every Borel ray belongs to Σ. The open lower and upper rays are Borel because they are open; the closed upper ray is Borel because its complement is an open lower ray. This proves all three forward implications.
(⇐) For the converse in (i), first verify $(-\infty, c] = \bigcap_{n \ge 1} (-\infty, c + 1/n)$ by both inclusions. A number at most c is less than every $c + 1/n$ and hence belongs to the intersection. If a number $y > c$ belonged to that intersection, choosing n with $1/n < y - c$ would exclude it, a contradiction. Thus every point of the intersection is at most c, proving the other inclusion. Assume (i). Using $(-\infty, c] = \bigcap_{n \ge 1} (-\infty, c + 1/n)$, whose identity has been verified, we have

$$\begin{aligned} \{h \le c\} = h^{-1}((-\infty, c]) &= h^{-1}\left(\bigcap_{n \ge 1} (-\infty, c + 1/n) \right) \\ &= \bigcap_{n \ge 1} h^{-1}((-\infty, c + 1/n)) = \bigcap_{n \ge 1} \{h < c + 1/n\} \in \Sigma, \end{aligned}$$

where the first equality follows from the definition of the lower level set. The second substitutes the interval identity, the third uses the pre-image identity for intersections from HW 28, and the fourth uses the definition of strict lower level sets. For the final membership, each strict level set belongs to Σ by assumption (i); their countable intersection belongs to Σ by HW 3. Lemma 3.2 then gives measurability.
Assume (ii). Then

$$\{h \le c\} = h^{-1}((-\infty, c]) = h^{-1}((c, \infty)^c) = (h^{-1}((c, \infty)))^c = \{h > c\}^c \in \Sigma,$$

where the first equality is the lower-level definition, the second is the complement identity for the two rays, the third follows from HW 28, and the fourth is the upper-level definition. The final membership follows from assumption (ii) and complement closure of Σ. Apply the known lower-closed-ray criterion.
Assume (iii). Similarly,

$$\{h < c\} = h^{-1}((-\infty, c)) = h^{-1}([c, \infty)^c) = (h^{-1}([c, \infty)))^c = \{h \ge c\}^c \in \Sigma,$$

where the first equality is the strict lower-level definition, the second uses the complement of the closed upper ray, the third is HW 28, and the fourth is the non-strict upper-level definition. The

final membership follows from assumption (iii) and complement closure. Thus (i) holds, and the already proved implication from (i) gives measurability.

Operations and Limits of Measurable Functions

We start to look at operations between measurable functions.

Lemma 3.3 — Composition of Measurable Functions

Let $h : \mathcal{S} \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}$ be such that $h \in m\Sigma$ and $f \in m\mathcal{B}$. Then $g := f \circ h : \mathcal{S} \to \mathbb{R}$ (i.e., $g(s) = f(h(s))$, $s \in \mathcal{S}$) is a Σ-measurable function.

Expanded proof. Fix an arbitrary $B \in \mathcal{B}(\mathbb{R})$. For the first inclusion, let $s_0 \in (f \circ h)^{-1}(B)$. By definition $f(h(s_0)) \in B$, so $h(s_0) \in f^{-1}(B)$. Applying the pre-image definition again gives $s_0 \in h^{-1}(f^{-1}(B))$. For the reverse inclusion, let $s_0 \in h^{-1}(f^{-1}(B))$. Then $h(s_0) \in f^{-1}(B)$, hence $f(h(s_0)) \in B$. This is precisely $s_0 \in (f \circ h)^{-1}(B)$. Combining the inclusions proves

$$(f \circ h)^{-1}(B) = h^{-1}(f^{-1}(B)) \in \Sigma,$$

where the equality is the identity just proved. For the final membership, first $f^{-1}(B) \in \mathcal{B}(\mathbb{R})$ by Borel measurability of f. Only then can measurability of h be applied to this Borel set, giving $h^{-1}(f^{-1}(B)) \in \Sigma$. Since B was arbitrary, the composition is Σ-measurable. The notation $^{-1}$ denotes pre-images; neither function is assumed invertible. QED

Lemma 3.4 — Operations on Measurable Functions

If $h, h_1, h_2 \in m\Sigma$, then

(i) $h_1 + h_2 \in m\Sigma$;

(ii) $h_1 \cdot h_2 \in m\Sigma$;

(iii) $\lambda h \in m\Sigma$ for any $\lambda \in \mathbb{R}$.

Expanded proof. We prove (i), then (iii), and finally (ii), so that the product proof can use the scalar-multiple result.

Proof of (i). We show that $h_1 + h_2$ is Σ-measurable. By HW 30 it suffices to prove that $\{h_1 + h_2 > c\} \in \Sigma$ for every $c \in \mathbb{R}$. Fix $c \in \mathbb{R}$. If $h_1(s) + h_2(s) > c$, then $c - h_2(s) < h_1(s)$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $q \in \mathbb{Q}$ with $c - h_2(s) < q < h_1(s)$. Conversely, these two strict inequalities imply $h_1(s) + h_2(s) > c$. Therefore

$$\{h_1 + h_2 > c\} = \bigcup_{q \in \mathbb{Q}} \{s \in \mathcal{S} : h_1(s) > q > c - h_2(s)\} = \bigcup_{q \in \mathbb{Q}} \big(\{h_1 > q\} \cap \{h_2 > c - q\}\big) \in \Sigma.$$

The first equality follows from the rational number between the two real values, and the second rewrites $q > c - h_2(s)$ as $h_2(s) > c - q$. For each q, both level sets belong to Σ by HW 30, since h_1, h_2 are measurable. Their intersection belongs to Σ, and the union is measurable because $\mathbb{Q}$ is countable. As c was arbitrary, HW 30 proves $h_1 + h_2 \in m\Sigma$.

Proof of (iii). We show that $\lambda h \in m\Sigma$ for every $\lambda \in \mathbb{R}$. Let $h \in m\Sigma$ and fix $c \in \mathbb{R}$. By HW 30 we need to check $\{\lambda h > c\} \in \Sigma$. We distinguish three cases.

If $\lambda = 0$, then

$$\{\lambda h > c\} = \{s \in \mathcal{S} : 0 > c\} = \begin{cases} \mathcal{S}, & c < 0, \\ \emptyset, & c \geq 0. \end{cases}$$

Both possibilities belong to Σ by the sigma-algebra axioms. If $\lambda > 0$, division preserves the inequality, so

$$\{\lambda h > c\} = \{s \in \mathcal{S} : h(s) > c/\lambda\} = \{h > c/\lambda\} \in \Sigma.$$

If $\lambda < 0$, division reverses the inequality, so

$$\{\lambda h > c\} = \{s \in \mathcal{S} : h(s) < c/\lambda\} = \{h < c/\lambda\} \in \Sigma.$$

The final memberships use HW 30 for the real threshold c/λ. Thus the required level set is measurable in every case, and HW 30 gives $\lambda h \in m\Sigma$.

Proof of (ii). We show that $h_1h_2 \in m\Sigma$. The function $f(x) = x^2$ is continuous, hence Borel-measurable. Lemma 3.3 applied to $f \circ h_1$ and $f \circ h_2$ gives $h_1^2, h_2^2 \in m\Sigma$. Part (i) gives $h_1 + h_2 \in m\Sigma$, and another application of Lemma 3.3 gives $(h_1 + h_2)^2 \in m\Sigma$. Now use the identity

$$h_1h_2 = \frac{1}{2}(h_1 + h_2)^2 - \frac{1}{2}(h_1^2 + h_2^2).$$

Expanding the square cancels the two square terms and leaves h_1h_2, which verifies the equality. The function $h_1^2 + h_2^2$ is measurable by (i). By (iii), both $\frac{1}{2}(h_1 + h_2)^2$ and $-\frac{1}{2}(h_1^2 + h_2^2)$ are measurable. Their sum is measurable by (i). Hence the identity proves $h_1h_2 \in m\Sigma$, using the scalar result already established. QED

The next result concerns sequences of measurable functions.

Lemma 3.5 — Infima, Suprema and Limits

Let $(h_n)_{n\in\mathbb{N}} \subset m\Sigma$ *. Then*

(i) $\inf_{n\in\mathbb{N}} h_n \in m\Sigma$ *and* $\sup_{n\in\mathbb{N}} h_n \in m\Sigma$*;*

(ii) $\liminf_{n\to\infty} h_n \in m\Sigma$ *and* $\limsup_{n\to\infty} h_n \in m\Sigma$*.*

The displayed real-valued memberships require the resulting functions to be finite at every point. Without that extra condition the conclusions mean extended-real measurability, as shown by the level-set proof below. For example $h_n(s) = n$ has supremum $+\infty$ at every point even though each h_n is real-valued.

Expanded proof. For $h = \inf_n h_n$, as we will see in HW 31,

$$\{h \geq c\} = \bigcap_{n\geq 1} \{h_n \geq c\} \in \Sigma,$$

where the equality is precisely the identity taken as given here and deferred to that homework. For the final membership, each set $\{h_n \geq c\}$ belongs to Σ by HW 30, and the intersection belongs to Σ by HW 3. This proves the level-set criterion for the possibly extended-real h. The identity itself is proved in that homework.

For the supremum use $\sup_n h_n = -\inf_n(-h_n)$ from HW 22. Negation preserves level-set measurability because $\{-h \leq c\} = \{h \geq -c\}$, where the equality follows by reversing the inequality under negation and the membership in Σ uses the measurable upper level of h.

Put $\ell_n = \inf_{m\geq n} h_m$. Then

$$\liminf_n h_n = \sup_{n\geq 1} \inf_{m\geq n} h_m = \sup_{n\geq 1} \ell_n,$$

where the first equality is the definition of liminf, and the second substitutes the definition of ℓ_n. Part (i) first proves measurability of each ℓ_n as a countable infimum, and then proves measurability of their supremum by a separate application. The conclusion is extended-real measurability unless pointwise finiteness is additionally known.

Finally $\limsup_n h_n = -\liminf_n(-h_n)$ by HW 22, so the same sign-reversal argument proves the last conclusion. The only deferred identity is the one explicitly indicated above. QED

HW 31 — Superlevel sets of an infimum

For $h = \inf_n h_n$ *prove that* $\{h \geq c\} = \cap_{n=1}^{\infty}\{h_n \geq c\}$.

Solution to HW 31. Proof. We prove equality by showing both inclusions explicitly. First, let $s_0 \in \{h \geq c\}$. By definition, $h(s_0) \geq c$. Since the infimum is a lower bound,

$$h_n(s_0) \geq \inf_{k\geq 1} h_k(s_0) = h(s_0) \geq c \text{ for every } n \in \mathbb{N}.$$

Therefore $s_0 \in \{h_n \geq c\}$ for every n, which means $s_0 \in \bigcap_{n=1}^{\infty}\{h_n \geq c\}$. By arbitrariness of s_0, $\{h \geq c\} \subseteq \bigcap_{n=1}^{\infty}\{h_n \geq c\}$. For the reverse inclusion, let $s_0 \in \bigcap_{n=1}^{\infty}\{h_n \geq c\}$. By definition of intersection, $h_n(s_0) \geq c$ for every n. Thus c is a lower bound for all values $h_n(s_0)$. Since the infimum is their greatest lower bound, $h(s_0) = \inf_{n\geq 1} h_n(s_0) \geq c$. Therefore $s_0 \in \{h \geq c\}$, and by

arbitrariness, $\bigcap_{n=1}^{\infty}\{h_n \geq c\} \subseteq \{h \geq c\}$. Combining the two inclusions gives the stated equality. Therefore, the statement is proved. QED

There is a useful corollary of the lemma above.

Corollary 3.6 — Measurability of the Convergence Set
Let $(h_n)_{n\in\mathbb{N}} \subset m\Sigma$ *. Then* $\{s \in \mathcal{S} : \lim_{n\to\infty} h_n(s)\text{exists}\} \in \Sigma$.

Expanded proof. Let

$$\underline{h} = \liminf_{n\to\infty} h_n,\ \overline{h} = \limsup_{n\to\infty} h_n.$$

By Lemma 3.5 these are extended-real measurable, and $\underline{h} \leq \overline{h}$ pointwise. We follow the lecturer's idea of testing their difference at zero, first restricting to where the subtraction is defined.

Part 1 — Both liminf and limsup are finite real numbers. Let $C = \{s \in \mathcal{S} : \underline{h}(s), \overline{h}(s) \in \mathbb{R}\}$. This set is measurable: since $\underline{h} \leq \overline{h}$, $C = \bigcup_{k\geq 1}(\{\underline{h} \geq -k\} \cap \{\overline{h} \leq k\}) \in \Sigma$. On C both functions are real-valued, so we can use the lecturer's subtraction argument. Define $g(s) := \underline{h}(s) - \overline{h}(s)$ for $s \in C$. By Lemma 3.5 the two functions are measurable, and Lemma 3.4(i),(iii) makes their difference g measurable on C.
We show that finite real convergence is equivalent to $g = 0$. A real sequence converges to a finite limit if and only if its liminf and limsup coincide in $\mathbb{R}$. Since both values are finite on C, this is equivalent to their difference being zero. Therefore

$$E_{\text{fin}} = \{s \in C : \underline{h}(s) - \overline{h}(s) = 0\} = \{s \in C : g(s) = 0\} = g^{-1}(\{0\}) \in \Sigma.$$

The first equality is the finite liminf/limsup convergence criterion, the second substitutes the definition of g, and the third is the definition of pre-image. Since $\{0\} \in \mathcal{B}(\mathbb{R})$ and g is measurable on the measurable set C, its pre-image belongs to Σ.

Part 2 — Both liminf and limsup equal $+\infty$. *We show that convergence to* $+\infty$ *is a measurable event.* The inequality $\underline{h} \leq \overline{h}$ implies that $\underline{h}(s) = +\infty$ forces $\overline{h}(s) = +\infty$. Consequently,

$$E_+ := \{s : \underline{h}(s) = \overline{h}(s) = +\infty\} = \{\underline{h} = +\infty\} = \bigcap_{k\geq 1}\{\underline{h} > k\} \in \Sigma.$$

The first equality uses the inequality between liminf and limsup. For the second, an extended real number equals $+\infty$ exactly when it exceeds every positive integer. Each level set is measurable by Lemma 3.5; closure under countable intersections gives the final membership. If $\underline{h}(s) = +\infty$, the identity $\underline{h}(s) = \sup_N \inf_{n\geq N} h_n(s)$ shows that, for every real M, some tail infimum exceeds M. Hence $h_n(s) > M$ throughout that tail, which is convergence to $+\infty$. Conversely, convergence to $+\infty$ puts $h_n(s) > M + 1$ eventually for every M. The corresponding tail infimum is at least $M + 1$, so $\underline{h}(s) > M$ for every M, and therefore $\underline{h}(s) = +\infty$.

Part 3 — Both liminf and limsup equal $-\infty$. *We show that convergence to* $-\infty$ *is a measurable event.* Again $\underline{h} \leq \overline{h}$, so $\overline{h}(s) = -\infty$ forces $\underline{h}(s) = -\infty$. Thus

$$E_- := \{s : \underline{h}(s) = \overline{h}(s) = -\infty\} = \{\overline{h} = -\infty\} = \bigcap_{k\geq 1}\{\overline{h} < -k\} \in \Sigma.$$

The first equality uses the same order inequality. The second characterizes $-\infty$ by all negative integer thresholds. Each level set is measurable by Lemma 3.5, so its countable intersection belongs to Σ. If $\overline{h}(s) = -\infty$, the identity $\overline{h}(s) = \inf_N \sup_{n\geq N} h_n(s)$ gives, for every real M, a tail supremum below M. Every term of that tail is then below M, which is convergence to $-\infty$. Conversely, that convergence gives $h_n(s) < M - 1$ eventually for every M. The tail supremum is at most $M - 1$, so $\overline{h}(s) < M$ for every M, and therefore $\overline{h}(s) = -\infty$.
Combining the three parts, the event of convergence in $\overline{\mathbb{R}}$ is

$$\{s : \lim_{n\to\infty} h_n(s) \text{ exists in } \overline{\mathbb{R}}\} = E_{\text{fin}} \cup E_+ \cup E_- \in \Sigma.$$

For convergence to a finite real number, the event is E_{fin} alone. No difference of two infinite values is used. QED

Definition — Random Variable

Given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, a random variable X is a $\mathcal{F}$-measurable function, i.e.,

$$X : \Omega \to \mathbb{R} \text{ and } X^{-1} : \mathcal{B}(\mathbb{R}) \to \mathcal{F}.$$

Let us now go back to the infinite coin tossing experiment from Example 1-(b). Defining

$$X_n(\omega) = \begin{cases} 1, & \text{if } \omega_n = H, \\ 0, & \text{if } \omega_n = T, \end{cases}$$

The Frequency Event in Example 1(b)

Expanded proof. Use the head indicator X_n just defined on the infinite coin-tossing space from Example 1(b), and set $S_n = X_1 + \cdots + X_n$. *We first prove that X_n is $\mathcal{F}$-measurable.* By HW 30 it suffices to check its lower-level sets. For $c \in \mathbb{R}$,

$$\{X_n \le c\} = \begin{cases} \varnothing, & c < 0, \\ \{\omega : \omega_n = T\}, & 0 \le c < 1, \\ \Omega, & c \ge 1. \end{cases}$$

The cases follow directly from the indicator values zero and one: neither is below $c < 0$, only zero is at most $0 \le c < 1$, and both are at most $c \ge 1$. The empty and whole sets belong to $\mathcal{F}$; the middle coordinate event is in its defining generating class. Thus all these level sets belong to $\mathcal{F}$, proving $X_n \in m\mathcal{F}$. By Lemma 3.4(i), the finite sum S_n is measurable; by Lemma 3.4(iii), so is S_n/n. Each indicator is between zero and one, so $0 \le S_n/n \le 1$.

Put $U = \limsup_n S_n/n$ and $L = \liminf_n S_n/n$. Lemma 3.5 makes both measurable, and the displayed bounds make both real-valued. The event from Example 1(b) is

$$F = \{\omega : \lim_n S_n(\omega)/n = 1/2\} = \{U = 1/2\} \cap \{L = 1/2\}.$$

The first equality substitutes the head counter S_n into the definition of F. For the second, convergence to $1/2$ gives both limsup and liminf equal to $1/2$; conversely equality of these finite limits gives convergence to $1/2$. These are its two inclusions. Since $\{1/2\}$ is a Borel singleton,

$$\{U = 1/2\} = U^{-1}(\{1/2\}) \in \mathcal{F}, \ \{L = 1/2\} = L^{-1}(\{1/2\}) \in \mathcal{F}.$$

In each chain, the equality is the definition of pre-image, and the membership follows from measurability of the corresponding function. Finite-intersection closure now gives $F \in \mathcal{F}$. Consequently $\mathsf{P}(F)$ is well defined, since P is defined on the events in $\mathcal{F}$. QED

Definition — Law of a Random Variable

We are now ready to introduce the concept of *law* of a random variable. On a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ let X be a real-valued random variable. The law of X is a probability measure on $\mathcal{B}(\mathbb{R})$, i.e., a mapping $\mathcal{L}_X : \mathcal{B}(\mathbb{R}) \to [0, 1]$ defined as $\mathcal{L}_X := \mathsf{P} \circ X^{-1}$. That is, for any $A \in \mathcal{B}(\mathbb{R})$

$$\mathcal{L}_X(A) := \mathsf{P}(X \in A).$$

The diagram below should help your intuition:

$$\mathcal{B}(\mathbb{R}) \xrightarrow{X^{-1}} \mathcal{F} \xrightarrow{\mathsf{P}} [0, 1].$$

In the following countable-additivity proof, the union pre-image identity is used as we will see in HW 32, following the lecturer's explicit deferral.

$\mathcal{L}_X$ is a probability measure MORE DETAIL

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \to \mathbb{R}$ be a r.v. Define $\mathcal{L}_X(A) := \mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A))$, $A \in \mathcal{B}(\mathbb{R})$. We want to prove that $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$.

Proof

Notice that $\mathcal{L}_X$ is well defined. Indeed since X is $\mathcal{F}$-measurable by definition 3.1, $X^{-1}(A) \in \mathcal{F}$ $\forall A \in \mathcal{B}(\mathbb{R})$. Therefore $\mathcal{L}_X(A) = \mathbb{P}(X^{-1}(A)) \in [0,1]$. We now verify the properties of a probability measure.

(i) We first prove that $\mathcal{L}_X(\emptyset) = 0$. By definition of $\mathcal{L}_X$

$$\mathcal{L}_X(\emptyset) = \mathbb{P}(X^{-1}(\emptyset)) = \mathbb{P}(\emptyset) = 0$$

where the first equality follows from the definition of the law X, the second from the fact that pre-image of the empty set is empty, the last from properties of the probability measure $\mathbb{P}$.

(ii) We now prove that $\mathcal{L}_X(\mathbb{R}) = 1$. Again by definition of $\mathcal{L}_X$

$$\mathcal{L}_X(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$$

where the second equality follows the fact that $X : \Omega \to \mathbb{R}$ so $X(\omega) \in \mathbb{R}$ $\forall \omega \in \Omega$ and the last equality follows from the fact that $\mathbb{P}$ is a probability measure.

(iii) Finally we prove countable additivity. Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{B}(\mathbb{R})$ be a sequence of pairwise disjoint sets namely $A_i \cap A_j = \emptyset$ $\forall i \neq j$. *We want to prove that*

$$\mathcal{L}_X\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n).$$

First notice that the pre-images $X^{-1}(A_n)$ are also pairwise disjoint. Indeed for $i \neq j$ by property (iii) of HW 28

$$X^{-1}(A_i) \cap X^{-1}(A_j) = X^{-1}(A_i \cap A_j) = X^{-1}(\emptyset) = \emptyset$$

Moreover since X is $\mathcal{F}$-measurable $X^{-1}(A_n) \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Therefore we can apply the countable additivity of $\mathbb{P}$ to the sequence $(X^{-1}(A_n))_{n\in\mathbb{N}}$. We obtain

$$\mathcal{L}_X\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(X^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right)\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} X^{-1}(A_n)\right) = \sum_{n=1}^{\infty} \mathbb{P}(X^{-1}(A_n)) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n)$$

where the first equality follows from the definition of $\mathcal{L}_X$, the second from HW 32, the third from the countable additivity of $\mathbb{P}$ since the pre-images are pairwise disjoint, and the last from the definition of $\mathcal{L}_X$. Hence $\mathcal{L}_X(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n)$. Therefore $\mathcal{L}_X$ is countably additive. Since $\mathcal{L}_X(\emptyset) = 0$, $\mathcal{L}_X(\mathbb{R}) = 1$ and $\mathcal{L}_X$ is countably additive on $\mathcal{B}(\mathbb{R})$ we conclude that $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$.

HW 32 — Preimage of a countable disjoint union

Prove that if $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{P}(\mathbb{R})$ *are pairwise disjoint sets then*

$$\{\omega \in \Omega : X(\omega) \in \cup_{n=1}^{\infty} A_n\} = \cup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}.$$

Solution to HW 32. Proof. We prove equality by showing both inclusions. For the first inclusion, let $\omega_0 \in \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. By definition of the set and of union, $X(\omega_0) \in \bigcup_{n=1}^{\infty} A_n$ implies that there exists $n_0 \in \mathbb{N}$ such that $X(\omega_0) \in A_{n_0}$. By definition of pre-image, $\omega_0 \in \{\omega : X(\omega) \in A_{n_0}\} \subseteq \bigcup_{n=1}^{\infty} \{\omega : X(\omega) \in A_n\}$. Since ω_0 was arbitrary, $\{\omega : X(\omega) \in \bigcup_n A_n\} \subseteq \bigcup_n \{\omega : X(\omega) \in A_n\}$. For the reverse inclusion, let $\omega_0 \in \bigcup_{n=1}^{\infty} \{\omega : X(\omega) \in A_n\}$. By definition of union, there exists $n_0 \in \mathbb{N}$ such that $\omega_0 \in \{\omega : X(\omega) \in A_{n_0}\}$. By definition of this set, $X(\omega_0) \in A_{n_0}$. Since $A_{n_0} \subseteq \bigcup_{n=1}^{\infty} A_n$, we obtain $\omega_0 \in \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. By arbitrariness of ω_0, the reverse inclusion follows. Combining the two inclusions gives the claimed equality. Therefore, the statement is proved. QED

Remark. The equality does not require disjointness; this assumption becomes necessary in the application to countable additivity of the law. Under disjointness, the index n_0 is unique, but existence alone was sufficient for the argument.

Remark — Why the CDF Determines the Law

Let μ and ν be two probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ satisfying $\mu((-\infty, c]) = \nu((-\infty, c])$ for every $c \in \mathbb{R}$. By HW 6 and Proposition 1.1, the lower closed rays form a pi-system generating $\mathcal{B}(\mathbb{R})$. Also

$$\mu(\mathbb{R}) = \nu(\mathbb{R}) = 1 < \infty.$$

Every assumption of Lemma 1.2 is therefore satisfied, so $\mu(A) = \nu(A)$ for every Borel A. Applying this to the laws of X and Y proves that $F_X = F_Y$ if and only if $\mathcal{L}_X = \mathcal{L}_Y$. This is equality of laws, not pointwise equality of the random variables.

Definition — Cumulative Distribution Function (CDF)

Since $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$ and $\pi(\mathbb{R})$ is a π-system on $\mathbb{R}$, it suffices to specify $\mathcal{L}_X$ on $\pi(\mathbb{R})$ and then apply Lemma 1.2. This is done via the cumulative distribution function (CDF) of X:

$$F_X(c) := \mathcal{L}_X((-\infty, c]) = \mathsf{P}(X^{-1}((-\infty, c])) = \mathsf{P}(X \leq c),\ c \in \mathbb{R}.$$

Here the first equality is the definition of the CDF, the second is the definition of the law, and the third uses the pre-image definition of $\{X \leq c\}$. This event is measurable because X is measurable and the lower closed ray is Borel. The CDF enjoys the following properties:

(a) $F_X : \mathbb{R} \to [0, 1]$ and $F_X(y) \leq F_X(z)$ for $y \leq z$;
(b) If $\mathsf{P}(|X| < \infty) = 1$, then $\lim_{z\to\infty} F_X(z) = 1$ and $\lim_{z\to-\infty} F_X(z) = 0$;
(c) $\lim_{\varepsilon\downarrow 0} F_X(z + \varepsilon) = F_X(z)$ for every $z \in \mathbb{R}$ (i.e., F_X is right-continuous).

Let us prove the claims one by one.

Proof of the CDF properties. (a) We first prove that $F_X : \mathbb{R} \to [0, 1]$. By definition, $F_X(z) = \mathsf{P}(X \leq z)$. Since P is a probability measure, $0 \leq \mathsf{P}(X \leq z) \leq 1$ for every $z \in \mathbb{R}$. Hence $0 \leq F_X(z) \leq 1$.
We now prove that F_X *is non-decreasing.* Let $y, z \in \mathbb{R}$ satisfy $y \leq z$. Notice that $\{X \leq y\} \subseteq \{X \leq z\}$: if $\omega \in \{X \leq y\}$, then $X(\omega) \leq y \leq z$, so $\omega \in \{X \leq z\}$. Both events belong to $\mathcal{F}$ because X is measurable. By monotonicity of probability (HW 12),

$$F_X(y) = \mathsf{P}(X \leq y) \leq \mathsf{P}(X \leq z) = F_X(z),$$

where the first and last equalities follow from the definition of the CDF.
(b) Assume that $\mathsf{P}(|X| < \infty) = 1$. *We first prove that* $\lim_{z\to+\infty} F_X(z) = 1$. For every $n \in \mathbb{N}$, define $A_n := \{\omega \in \Omega : X(\omega) \leq n\}$. Measurability of X gives $A_n \in \mathcal{F}$. Moreover, $A_n \subseteq A_{n+1}$, because $X(\omega) \leq n$ implies $X(\omega) \leq n + 1$. We have $\bigcup_{n=1}^{\infty} A_n = \{X < +\infty\}$: membership in the union means that $X(\omega) \leq n$ for some n, which excludes $+\infty$; conversely, every value less than $+\infty$ is below some positive integer. By almost sure finiteness, this union has probability one. Lemma 1.5 therefore gives

$$\lim_{n\to\infty} F_X(n) = \lim_{n\to\infty} \mathsf{P}(A_n) = \mathsf{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = 1,$$

where the first equality uses the definitions of the CDF and A_n, the second is continuity from below, and the last follows from the union identity and the assumption. For $z \geq n$, monotonicity gives $F_X(n) \leq F_X(z) \leq 1$. Thus the integer limit implies $\lim_{z\to+\infty} F_X(z) = 1$.
We now prove that $\lim_{z\to-\infty} F_X(z) = 0$. For every $n \in \mathbb{N}$, define $B_n := \{\omega \in \Omega : X(\omega) \leq -n\}$. Again $B_n \in \mathcal{F}$. We have $B_n \supseteq B_{n+1}$, because $X(\omega) \leq -(n + 1)$ implies $X(\omega) \leq -n$. Moreover, $\bigcap_{n=1}^{\infty} B_n = \{X = -\infty\}$: $-\infty$ lies below every $-n$, whereas any larger value fails one of these inequalities. The intersection has probability zero by almost sure finiteness. Since the events decrease and $\mathsf{P}(B_1) \leq 1 < \infty$, Lemma 1.6 yields

$$\lim_{n\to\infty} F_X(-n) = \lim_{n\to\infty} \mathsf{P}(B_n) = \mathsf{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = 0,$$

where the first equality uses the definitions, the second is continuity from above, and the last uses the intersection identity and the assumption. For $z \le -n$, $0 \le F_X(z) \le F_X(-n)$ by monotonicity. Consequently, $\lim_{z\to-\infty} F_X(z) = 0$.
(c) We prove that $\lim_{\varepsilon\downarrow 0} F_X(z+\varepsilon) = F_X(z)$ *for every* $z \in \mathbb{R}$. Fix an arbitrary $z \in \mathbb{R}$. For every $n \in \mathbb{N}$, define $A_n := \{\omega \in \Omega : X(\omega) \le z+1/n\}$. Each A_n belongs to $\mathcal{F}$. Since $z+1/(n+1) \le z+1/n$, we have $A_n \supseteq A_{n+1}$.
We now show that $\bigcap_{n=1}^{\infty} A_n = \{X \le z\}$ *by proving both inclusions.* For the first inclusion, let $\omega_0 \in \bigcap_{n=1}^{\infty} A_n$. Then $X(\omega_0) \le z + 1/n$ for every n. Letting $n \to \infty$ gives $X(\omega_0) \le z$, so $\omega_0 \in \{X \le z\}$. For the reverse inclusion, let $\omega_0 \in \{X \le z\}$. Then $X(\omega_0) \le z \le z + 1/n$ for every n, hence $\omega_0 \in A_n$ for every n and therefore $\omega_0 \in \bigcap_{n=1}^{\infty} A_n$. The two inclusions prove the identity. Since (A_n) decreases and $\mathsf{P}(A_1) \le 1 < \infty$, Lemma 1.6 gives

$$\lim_{n\to\infty} F_X(z+1/n) = \lim_{n\to\infty} \mathsf{P}(A_n) = \mathsf{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \mathsf{P}(X \le z) = F_X(z),$$

where the first equality uses the definitions of the CDF and A_n, the second is continuity from above, the third uses the set identity, and the last is the CDF definition. Finally, let $\delta > 0$. The convergence just proved supplies N with $F_X(z+1/N) < F_X(z) + \delta$. For every $0 < \varepsilon \le 1/N$,

$$F_X(z) \le F_X(z+\varepsilon) \le F_X(z+1/N) < F_X(z) + \delta,$$

where the first two inequalities use monotonicity and the last uses the choice of N. Since $\delta > 0$ was arbitrary, $\lim_{\varepsilon\downarrow 0} F_X(z+\varepsilon) = F_X(z)$. As z was arbitrary, F_X is right-continuous. QED

HW 33 — CDF tails with mass at infinity
Prove that if $\mathsf{P}(X = \infty) = p$ *and* $\mathsf{P}(X = -\infty) = q$, *then* $\lim_{z\to\infty} F_X(z) = 1 - p$ *and* $\lim_{z\to-\infty} F_X(z) = q$.
Solution to HW 33. Recall that $F_X(z) = \mathsf{P}(X \le z)$ for $z \in \mathbb{R}$.
We first prove that $\lim_{z\to+\infty} F_X(z) = 1 - p$. For every $n \in \mathbb{N}$, define $A_n := \{X \le n\}$. These events are measurable, and $A_n \subseteq A_{n+1}$ because $X(\omega) \le n$ implies $X(\omega) \le n + 1$. Moreover, $\bigcup_{n=1}^{\infty} A_n = \{X < +\infty\}$. Indeed, membership in the union means $X(\omega) \le n$ for some n, which implies $X(\omega) < +\infty$. Conversely, if $X(\omega) < +\infty$, some positive integer n satisfies $X(\omega) \le n$; this also holds when $X(\omega) = -\infty$. Thus $\omega \in A_n$ and in the union. Lemma 1.5 gives

$$\lim_{n\to\infty} F_X(n) = \lim_{n\to\infty} \mathsf{P}(A_n) = \mathsf{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathsf{P}(X < +\infty) = 1 - \mathsf{P}(X = +\infty) = 1 - p,$$

where the first equality uses the definitions, the second is continuity from below, the third uses the set identity, the fourth is the complement rule, and the last uses the given mass. For $z \ge n$, $F_X(n) \le F_X(z) \le 1 - p$: the first inequality uses monotonicity, and the second follows from $\{X \le z\} \subseteq \{X < +\infty\}$. Therefore $\lim_{z\to+\infty} F_X(z) = 1 - p$.
We now prove that $\lim_{z\to-\infty} F_X(z) = q$. For every $n \in \mathbb{N}$, define $B_n := \{X \le -n\}$. These events are measurable and satisfy $B_n \supseteq B_{n+1}$, because $X(\omega) \le -(n+1)$ implies $X(\omega) \le -n$. Furthermore, $\bigcap_{n=1}^{\infty} B_n = \{X = -\infty\}$: membership means $X(\omega) \le -n$ for every n, which is equivalent to $X(\omega) = -\infty$. Since $\mathsf{P}(B_1) \le 1 < \infty$, Lemma 1.6 gives

$$\lim_{n\to\infty} F_X(-n) = \lim_{n\to\infty} \mathsf{P}(B_n) = \mathsf{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = \mathsf{P}(X = -\infty) = q,$$

where the first equality uses the definitions, the second is continuity from above, the third uses the intersection identity, and the last uses the given mass. For $z \le -n$, $q \le F_X(z) \le F_X(-n)$. The first inequality follows from $\{X = -\infty\} \subseteq \{X \le z\}$, and the second from monotonicity. Hence $\lim_{z\to-\infty} F_X(z) = q$.

HW 34 — The left limit and the atom of a CDF

Prove that $\{X \le z-\varepsilon\} \uparrow \{X < z\}$ *as* $\varepsilon \downarrow 0$. *Then conclude that* $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = F_X(z) - \mathcal{L}_X(\{z\})$.

Solution to HW 34. Fix an arbitrary $z \in \mathbb{R}$. *We first prove that* $\{X \le z-\varepsilon\} \uparrow \{X < z\}$ *as* $\varepsilon \downarrow 0$. For every $n \in \mathbb{N}$, define $A_n := \{\omega \in \Omega : X(\omega) \le z - 1/n\}$. Since X is measurable, $A_n \in \mathcal{F}$. If $\omega_0 \in A_n$, then $X(\omega_0) \le z - 1/n \le z - 1/(n+1)$, so $\omega_0 \in A_{n+1}$. Therefore (A_n) is an increasing sequence of events.

We now prove that $\bigcup_{n=1}^{\infty} A_n = \{X < z\}$ *by both inclusions.* For the first inclusion, let $\omega_0 \in \bigcup_{n=1}^{\infty} A_n$. There exists n_0 with $\omega_0 \in A_{n_0}$. By definition, $X(\omega_0) \le z - 1/n_0 < z$, so $\omega_0 \in \{X < z\}$. For the reverse inclusion, let $\omega_0 \in \{X < z\}$. If $X(\omega_0)$ is finite, then $z - X(\omega_0) > 0$. As $1/n \to 0$, there exists n_0 such that $1/n_0 < z - X(\omega_0)$. Rearranging gives $X(\omega_0) < z - 1/n_0$, hence $\omega_0 \in A_{n_0}$ and in the union. If $X(\omega_0) = -\infty$, it belongs to every A_n, so the same inclusion holds. The two inclusions prove the set identity. The events also increase as the positive parameter ε decreases, and the sequence $1/n$ is cofinal at zero; therefore the asserted increasing limit holds. By Lemma 1.5,

$$\lim_{n\to\infty} F_X(z-1/n) = \lim_{n\to\infty} \mathsf{P}(A_n) = \mathsf{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathsf{P}(X < z),$$

where the first equality uses the definitions, the second is continuity from below, and the third uses the set identity. For $0 < \varepsilon \le 1/n$, monotonicity and the inclusion $\{X \le z-\varepsilon\} \subseteq \{X < z\}$ yield

$$F_X(z-1/n) \le F_X(z-\varepsilon) \le \mathsf{P}(X < z).$$

Since the left side converges to $\mathsf{P}(X < z)$, $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = \mathsf{P}(X < z)$.

Finally, $\{X \le z\} = \{X < z\} \cup \{X = z\}$, with the two events on the right disjoint. Finite additivity gives $\mathsf{P}(X \le z) = \mathsf{P}(X < z) + \mathsf{P}(X = z)$. Rearranging and using the definitions of the CDF and of the law,

$$\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = \mathsf{P}(X < z) = \mathsf{P}(X \le z) - \mathsf{P}(X = z) = F_X(z) - \mathcal{L}_X(\{z\}),$$

where the first equality is the limit just established, the second uses finite additivity, and the last substitutes $F_X(z) = \mathsf{P}(X \le z)$ and $\mathcal{L}_X(\{z\}) = \mathsf{P}(X = z)$. All subtracted probabilities are finite. Since z was arbitrary, the result holds for every $z \in \mathbb{R}$.

Definition — Probability Density Function (PDF)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \to \mathbb{R}$ be a random variable.

Recall that the cumulative distribution function of X is defined by $F_X(z) := \mathbb{P}(X \le z),\ z \in \mathbb{R}$.

A random variable X admits a probability density function (PDF) if there exists a non-negative measurable function $f_X : \mathbb{R} \longrightarrow [0, +\infty)$ such that $\mathbb{P}(X \in A) = \int_A f_X(z)\, dz\ \forall A \in \mathcal{B}(\mathbb{R})$.

In particular, taking $A = (-\infty, z]$, we obtain $F_X(z) = \int_{-\infty}^{z} f_X(t)\, dt$.

The absolute-continuity theorem for Lebesgue integrals identifies the derivative of an integral representation almost everywhere. Applied to a CDF that is absolutely continuous on every bounded interval, it gives a non-negative integrable density of total integral one and

$$f_X(z) = F_X'(z) \text{ for almost every } z \in \mathbb{R}.$$

The hypotheses here are the stated absolute continuity, the CDF limits zero and one, and monotonicity. The theorem gives the integral representation and its almost-everywhere derivative, not a conclusion from mere differentiability.

Properties — Non-negativity and Normalization of the PDF

Statement

Let X be a random variable admitting a probability density function f_X. Then the following properties hold:

(a) Non-negativity. $f_X(z) \ge 0$ for almost every $z \in \mathbb{R}$.

(b) Normalization. $\int_{-\infty}^{+\infty} f_X(z)\, dz = 1$.

We prove the two properties separately.

Proof of (a) — Non-negativity

By definition of a probability density function, f_X is non-negative. We can also see why its non-negativity agrees with the derivative of the CDF.

Recall that F_X is non-decreasing. Therefore, for every $h > 0$, $F_X(z+h) \geq F_X(z)$, and consequently $\frac{F_X(z+h)-F_X(z)}{h} \geq 0$. At every point where F_X is differentiable, letting $h \downarrow 0$ yields

$$F'_X(z) = \lim_{h \downarrow 0} \frac{F_X(z+h) - F_X(z)}{h} \geq 0.$$

When F_X is absolutely continuous, the identity $f_X(z) = F'_X(z)$ holds almost everywhere. Hence, $f_X(z) \geq 0$ for almost every $z \in \mathbb{R}$.

Proof of (b) — Normalization

By the defining property of the PDF, $\mathbb{P}(X \in A) = \int_A f_X(z)\, dz \ \forall A \in \mathcal{B}(\mathbb{R})$. Taking $A = \mathbb{R}$, we obtain

$$\int_{-\infty}^{+\infty} f_X(z)\, dz = \mathbb{P}(X \in \mathbb{R}) = \mathbb{P}(\Omega) = 1,$$

where the first equality follows from the definition of the PDF, the second from the fact that $X : \Omega \to \mathbb{R}$, and the last from the normalization of the probability measure $\mathbb{P}$.

Therefore, $\int_{-\infty}^{+\infty} f_X(z)\, dz = 1$.

Qualification — Absolute Continuity of the CDF

The density criterion uses absolute continuity of the CDF, not mere almost-everywhere differentiability. The definition above specifies the density by its integrals on all Borel sets. Density versions are unique only almost everywhere.

Theorem – Existence of a r.v. with a prescribed CDF

Let $F : \mathbb{R} \to [0,1]$ be a function satisfying the following properties:

(a) Range and monotonicity $F : \mathbb{R} \to [0,1]$, $F(y) \leq F(z)\ \forall y \leq z$
(b) Limits of infinity $\lim_{z \to +\infty} F(z) = 1$ $\lim_{z \to -\infty} F(z) = 0$
(c) Right continuity $\lim_{\varepsilon \downarrow 0} F(z+\varepsilon) = F(z)\ \forall z \in \mathbb{R}$

Then there exist a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a r.v. $X : \Omega \to \mathbb{R}$ s.t. $F_X(c) = F(c)\ \forall c \in \mathbb{R}$

Proof

We want to construct a probability space and a r.v. whose CDF coincide with F.

First we construct the probability space. Let $\Omega = \mathbb{R}$ and $\mathcal{F} = \mathcal{B}(\mathbb{R})$. We now introduce a measure $\mathcal{L}$ by defining $\forall a, b \in \mathbb{R}$ with $a < b$ $\mathcal{L}((a,b]) := F(b) - F(a)$. Notice that, since F is non-decreasing by property (a) we have $F(a) \leq F(b)$ whenever $a < b$. Therefore $\mathcal{L}((a,b]) = F(b) - F(a) \geq 0$. The monotonicity and right-continuity of F allow this assignment on intervals to be extended to a measure $\mathcal{L}$ on $\mathcal{B}(\mathbb{R})$ by construction (is difficult to prove it). *We now verify that $\mathcal{L}$ is a probability measure.* Since $\mathcal{L}$ is a probability measure. Since $\mathcal{L}$ is already a measure on $\mathcal{B}(\mathbb{R})$ it remains to prove that $\mathcal{L}(\mathbb{R}) = 1$. Notice that $(-n, n] \subseteq (-(n+1), n+1]\ \forall n \in \mathbb{N}$ and $\bigcup_{n=1}^{\infty}(-n,n] = \mathbb{R}$. Therefore since $((-n,n])_{n \in \mathbb{N}}$ is an increasing sequence of Borel sets, by continuity of measures from below we obtain

$$\mathcal{L}(\mathbb{R}) = \mathcal{L}\left(\bigcup_{n=1}^{\infty}(-n,n]\right) = \lim_{n \to \infty} \mathcal{L}((-n,n]) = \lim_{n \to \infty}(F(n) - F(-n)) = 1 - 0 = 1$$

Hence $\mathcal{L}$ is a probability measure on $\mathcal{B}(\mathbb{R})$. We can therefore consider the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L})$.

Second we construct the r.v. Define $X : \Omega \to \mathbb{R}$ by $X(\omega) := \omega\ \forall \omega \in \Omega$. Notice that X is identity function on $\mathbb{R}$. *We now verify that X is a r.v. namely that is $\mathcal{F}$-measurable.* Let $A \in \mathcal{B}(\mathbb{R})$. By definition of pre-image

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} = \{\omega \in \mathbb{R} : \omega \in A\} = A$$

where the second equality follows from $\Omega = \mathbb{R}$ and $X(\omega) = \omega$. Since $A \in \mathcal{B}(\mathbb{R}) = \mathcal{F}$ we conclude that $X^{-1}(A) \in \mathcal{F}$. Since $A \in \mathcal{B}(\mathbb{R})$ was arbitrary by definition 3.1 is $\mathcal{F}$-measurable and therefore a r.v.

Finally we prove that the CDF of X coincides with F. Fix an arbitrary $c \in \mathbb{R}$. First notice that $(-n, c] \uparrow (-\infty, c]$ as $n \to \infty$ where we consider sufficiently large n so that $-n < c$. Therefore by continuity of measure from below

$$\mathcal{L}((-\infty, c]) = \lim_{n\to\infty} \mathcal{L}((-n, c]) = \lim_{n\to\infty} (F(c) - F(-n)) = F(c)$$

where the first equality follows from continuity of measure from below, the second from the definition of $\mathcal{L}$ and the third from property (b) since $\lim_{n\to\infty} F(-n) = 0$. Now by definition of the CDF

$$F_X(c) = \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X \le c) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \le c\})$$
$$= \mathbb{P}(\{\omega \in \mathbb{R} : \omega \le c\}) = \mathbb{P}((-\infty, c]) = \mathcal{L}((-\infty, c]) = F(c)$$

where the first and second equalities follow from the definition of CDF and the law of X, the fourth from $X(\omega) = \omega$ and $\Omega = \mathbb{R}$, the sixth from $\mathbb{P} = \mathcal{L}$ and the last from calculation above. Since $c \in \mathbb{R}$ was arbitrary we conclude that $F_X(c) = F(c)$ $\forall c \in \mathbb{R}$. Thus we have constructed a r.v. X whose CDF is precisely F.

Construction — Inverse Transform Method

Another way to construct a random variable with a prescribed CDF F is to start from a uniformly distributed random variable $U \sim \text{Unif}(0, 1)$.

Now, define $X := F^{-1}(U)$, where F^{-1} denotes the inverse function when it exists, or the generalized inverse for a general CDF.

The idea is that applying F^{-1} to a uniform random variable transforms its distribution into the prescribed distribution F.

Thus, the inverse transform method allows us to construct a random variable with any prescribed CDF F.

Information Generated by a Random Variable

We have already seen that the σ-algebra generated by a r.v. Y is given by

$$\sigma(Y) = Y^{-1}\big(\mathcal{B}(\mathbb{R})\big) := \big(\big\{\omega \in \Omega : Y(\omega) \in A\big\}, A \in \mathcal{B}(\mathbb{R})\big).$$

This is the pre-image sigma-algebra proved in Lemma 3.1; its minimality follows because any sigma-algebra making Y measurable must contain every Borel pre-image.

Lemma 3.7 — The σ-Algebra Generated by a Random Variable

Statement

Let $Y : \Omega \to \mathbb{R}$ be a random variable, and consider the class

$$\pi(Y) := \big\{\{\omega \in \Omega : Y(\omega) \le x\} : x \in \mathbb{R}\big\} = Y^{-1}(\pi(\mathbb{R})),$$

where $\pi(\mathbb{R}) := \{(-\infty, x] : x \in \mathbb{R}\}$. Then $\sigma(Y) = \sigma(\pi(Y))$.

Expanded Proof of Lemma 3.7

Recall that $\pi(Y) = Y^{-1}(\pi(\mathbb{R}))$. Moreover, as we will see in HW 35, $\pi(Y)$ is a π-system on Ω.

We want to prove that $\sigma(Y) = \sigma(\pi(Y))$. We establish the equality by proving both inclusions.

First, we prove that $\sigma(Y) \subseteq \sigma(\pi(Y))$. Recall that, by definition, $\pi(\mathbb{R}) = \{(-\infty, x] : x \in \mathbb{R}\}$. Moreover, by definition of the pre-image of a set, for every $x \in \mathbb{R}$,

$$Y^{-1}((-\infty, x]) = \{\omega \in \Omega : Y(\omega) \in (-\infty, x]\} = \{\omega \in \Omega : Y(\omega) \le x\} = \{Y \le x\},$$

where the second equality follows from the fact that $Y(\omega) \in (-\infty, x]$ if and only if $Y(\omega) \le x$.

Now, notice that $\pi(Y) = \{\{Y \le x\} : x \in \mathbb{R}\}$. Therefore, since $\{Y \le x\}$ is precisely one of the sets defining $\pi(Y)$, we obtain

$$Y^{-1}((-\infty, x]) = \{Y \le x\} \in \pi(Y).$$

Since $\sigma(\pi(Y))$ is the smallest σ-algebra containing $\pi(Y)$, we have $\pi(Y) \subseteq \sigma(\pi(Y))$. Consequently, $Y^{-1}((-\infty, x]) \in \sigma(\pi(Y))\ \forall x \in \mathbb{R}$. In other words, the pre-image under Y of every set belonging to $\pi(\mathbb{R})$ belongs to $\sigma(\pi(Y))$.
Moreover, recall that, by Proposition 1.1, $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$. Thus, all the assumptions of Lemma 3.2 are satisfied.
Therefore, by Lemma 3.2, Y is measurable with respect to $\sigma(\pi(Y))$, namely $Y \in m\sigma(\pi(Y))$. Now, recall that $\sigma(Y)$ is the smallest σ-algebra on Ω with respect to which Y is measurable.
Since $\sigma(\pi(Y))$ is a σ-algebra with respect to which Y is measurable, by the minimality of $\sigma(Y)$, we conclude that $\sigma(Y) \subseteq \sigma(\pi(Y))$. *We now prove the reverse inclusion, namely* $\sigma(\pi(Y)) \subseteq \sigma(Y)$.
Recall that $\sigma(Y) = Y^{-1}(\mathcal{B}(\mathbb{R}))$. *We first prove that* $\pi(Y) \subseteq \sigma(Y)$. Let $A \in \pi(Y)$.
By definition of $\pi(Y)$, there exists $x_0 \in \mathbb{R}$ such that $A = \{Y \leq x_0\}$. As shown above, by definition of pre-image,

$$A = \{Y \leq x_0\} = Y^{-1}((-\infty, x_0]).$$

Notice that $(-\infty, x_0]$ is a Borel set, since it is closed in $\mathbb{R}$. Therefore, $(-\infty, x_0] \in \mathcal{B}(\mathbb{R})$. Hence, by definition of $\sigma(Y)$, $A = Y^{-1}((-\infty, x_0]) \in \sigma(Y)$. Since $A \in \pi(Y)$ was arbitrary, we conclude that $\pi(Y) \subseteq \sigma(Y)$. Now, since $\sigma(Y)$ is a σ-algebra containing $\pi(Y)$, and $\sigma(\pi(Y))$ is the smallest σ-algebra containing $\pi(Y)$, we obtain $\sigma(\pi(Y)) \subseteq \sigma(Y)$. Combining the two inclusions, $\sigma(Y) \subseteq \sigma(\pi(Y))$ and $\sigma(\pi(Y)) \subseteq \sigma(Y)$, we conclude that $\sigma(Y) = \sigma(\pi(Y))$. Therefore, the statement is proved.

HW 35 — Prove That $\pi(Y)$ Is a π-System on Ω

Statement

Let $Y : \Omega \to \mathbb{R}$ be a random variable, and consider $\pi(Y) := \{\{Y \leq x\} : x \in \mathbb{R}\}$. Prove that $\pi(Y)$ is a π-system on Ω.

Proof

Recall that a π-system is a non-empty collection of sets closed under finite intersections.
It is trivial that $\pi(Y)$ is non-empty.
We now prove that $\pi(Y)$ *is closed under finite intersections.*
Let $A, B \in \pi(Y)$. By definition, there exist $x, y \in \mathbb{R}$ such that

$$A = \{Y \leq x\},\ B = \{Y \leq y\}.$$

Recall that, as shown in HW 6, $(-\infty, x] \cap (-\infty, y] = (-\infty, \min\{x, y\}]$. Therefore,

$$\begin{aligned} A \cap B &= \{Y \leq x\} \cap \{Y \leq y\} = Y^{-1}((-\infty, x]) \cap Y^{-1}((-\infty, y]) \\ &= Y^{-1}((-\infty, x] \cap (-\infty, y]) = Y^{-1}((-\infty, \min\{x, y\}]) = \{Y \leq \min\{x, y\}\}, \end{aligned}$$

where the third equality follows from HW 28-(iii), and the fourth from HW 6.
Since $\min\{x, y\} \in \mathbb{R}$, by definition of $\pi(Y)$, $\{Y \leq \min\{x, y\}\} \in \pi(Y)$. Hence, $A \cap B \in \pi(Y)$. Since $A, B \in \pi(Y)$ were arbitrary, $\pi(Y)$ is closed under finite intersections.
Therefore, $\pi(Y)$ is a π-system on Ω. The statement is proved.

Properties — Measurable Functions of Random Variables

Here are some useful facts that we state without proof.

(a) Let $Y : \Omega \to \mathbb{R}$ and $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y) \iff \exists f \in m\mathcal{B}(\mathbb{R}) \text{ such that } Z = f(Y),$$

where $f : \mathbb{R} \to \mathbb{R}$.

(b) Let $Y : \Omega \to \mathbb{R}^n$, i.e., $Y(\omega) = (Y_1(\omega), Y_2(\omega), \ldots, Y_n(\omega))$, and let $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y) \iff \exists f \in m\mathcal{B}(\mathbb{R}^n) \text{ such that } Z = f(Y),$$

where $f : \mathbb{R}^n \to \mathbb{R}$.

(c) Let $(Y_\gamma)_{\gamma\in I}$ be a collection of functions $Y_\gamma : \Omega \to \mathbb{R}$, and let $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y_\gamma, \gamma \in I) \iff \text{there exist a countable collection } (\gamma_n)_{n\in\mathbb{N}}$$
$$\text{and a function } f \in m\mathcal{B}(\mathbb{R}^{\mathbb{N}}) \text{ such that } Z = f((Y_{\gamma_n})_{n\in\mathbb{N}}),$$

where $f : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}$.

Financial Interpretation

Item (b) can be interpreted as follows.
Let $(Y_1, \ldots, Y_n)$ be n assets traded on the market, and let Z be the payoff of a derivative contract written on the vector $(Y_1, \ldots, Y_n)$.
Then there must exist a function f such that $Z = f(Y_1, \ldots, Y_n)$, which describes the random payoff Z of the derivative contract.

4. Independence

From Independent Events to Sigma-Algebras

Given $(\Omega, \mathcal{F}, \mathsf{P})$ and events $E, F \in \mathcal{F}$ our familiar concept of "independent events" is expressed by the equation

$$\mathsf{P}(E \cap F) = \mathsf{P}(E)\mathsf{P}(F).$$

(Notice that the notion of independence relies on the particular choice of P!)
For random variables with probability density function independence is also expressed by $f_{X,Y}(x, y) = f_X(x)f_Y(y)$. The two notions above are somewhat consequence of a more general concept of independence that we are now going to explore.

Definition 4.1 — Independent Sigma-Algebras

Let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a collection of sub-σ-algebras of $\mathcal{F}$ (i.e., each $\mathcal{G}_n$ is a σ-algebra on Ω and $\mathcal{G}_n \subset \mathcal{F}$). Then $\mathcal{G}_1, \mathcal{G}_2, \ldots$ are called independent if for any collection of events $(G_n)_{n\in\mathbb{N}}$ with $G_n \in \mathcal{G}_n$ and any finite selection of such events $G_{n_1}, G_{n_2}, \ldots G_{n_k}$ we have

$$\mathsf{P}\big(G_{n_1} \cap G_{n_2} \cap \ldots \cap G_{n_k}\big) = \prod_{i=1}^{k} \mathsf{P}(G_{n_i}).$$

This is a very general definition but effectively impossible to check in practice. We will soon see how to get to a more applicable notion at least for random variables.

Definition 4.2 — Independent Random Variables

Random variables $(X_n)_{n\in\mathbb{N}}$ are called independent if their σ-algebras $\{\sigma(X_n), n \in \mathbb{N}\}$ are independent.
This is still difficult to implement, because it relies on the first definition. One more step...

Definition 4.3 — Independent Events

Events $(E_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ are called independent if, setting $X_n(\omega) = 1_{E_n}(\omega)$, the σ-algebras $\{\sigma(X_n), n \in \mathbb{N}\}$ are independent.
The last definition is actually workable thanks to the next result:

HW 36 — σ-Algebra Generated by an Indicator Function

Statement

Let $E \in \mathcal{F}$. Prove that $\sigma(\mathbf{1}_E) = \{\emptyset, \Omega, E, E^c\}$.

Proof

Recall that, by Lemma 3.7, $\sigma(\mathbf{1}_E) = \sigma(\pi(\mathbf{1}_E))$, where $\pi(\mathbf{1}_E) = \{\{\mathbf{1}_E \le x\} : x \in \mathbb{R}\}$.

Since the indicator function takes only the values 0 and 1, we have

$$\{\mathbf{1}_E \leq x\} = \begin{cases} \emptyset, & x < 0, \\ E^c, & 0 \leq x < 1, \\ \Omega, & x \geq 1. \end{cases}$$

Therefore, $\pi(\mathbf{1}_E) = \{\emptyset, E^c, \Omega\}$.
Since every σ-algebra is closed under complements, we must also include $E = (E^c)^c$.
Notice that the collection $\mathcal{G} := \{\emptyset, \Omega, E, E^c\}$ is already a σ-algebra on Ω, since it contains Ω, is closed under complements and countable unions, and consequently also under intersections.
Therefore, no additional sets are needed, and we obtain $\sigma(\pi(\mathbf{1}_E)) = \{\emptyset, \Omega, E, E^c\}$.
Finally, by Lemma 3.7,

$$\sigma(\mathbf{1}_E) = \sigma(\pi(\mathbf{1}_E)) = \{\emptyset, \Omega, E, E^c\}.$$

Therefore, the statement is proved.

Connection with Independence of Events

This tells us how to calculate all elements in $\sigma(\mathbf{1}_E)$. In particular, given two independent events $E, F \in \mathcal{F}$, according to the definition above we have

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F),\ \mathbb{P}(E^c \cap F) = \mathbb{P}(E^c)\mathbb{P}(F),\ \mathbb{P}(E^c \cap F^c) = \mathbb{P}(E^c)\mathbb{P}(F^c),$$
$$\mathbb{P}(E \cap F^c) = \mathbb{P}(E)\mathbb{P}(F^c).$$

Plus analogous relations involving $\emptyset$ and Ω. So here we recognise the "old" notion of independence!
Next comes a lemma that will further allow us to deepen our understanding of independence for random variables.

Lemma 4.1 — Independence of σ-Algebras Generated by π-Systems

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{G}$ and $\mathcal{H}$ be sub-σ-algebras of $\mathcal{F}$, and let $\mathcal{I}$ and $\mathcal{J}$ be π-systems such that $\sigma(\mathcal{I}) = \mathcal{G},\ \sigma(\mathcal{J}) = \mathcal{H}$.
Then the σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent if and only if

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J)\ \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

Expanded Proof of Lemma 4.1

We prove both implications.
($\Rightarrow$) **Assume that $\mathcal{G}$ and $\mathcal{H}$ are independent.**
By Definition 4.1,

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H)\ \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

Since

$$\mathcal{I} \subseteq \sigma(\mathcal{I}) = \mathcal{G},\ \mathcal{J} \subseteq \sigma(\mathcal{J}) = \mathcal{H},$$

the equality holds, in particular, for every $I \in \mathcal{I}$ and $J \in \mathcal{J}$. Therefore,

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J)\ \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

($\Leftarrow$) **Assume that**

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J)\ \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

We want to prove that $\mathcal{G}$ and $\mathcal{H}$ are independent.
First, fix an arbitrary $I \in \mathcal{I}$, and consider the two set functions defined on $(\Omega, \mathcal{H})$ by

$$\mu_1(H) := \mathbb{P}(I \cap H),\ \mu_2(H) := \mathbb{P}(I)\mathbb{P}(H),\ H \in \mathcal{H}.$$

As we will see in HW 37, these two set functions are measures on $(\Omega, \mathcal{H})$.
Notice that they have the same finite total mass. Indeed,

$$\mu_1(\Omega) = \mathbb{P}(I \cap \Omega) = \mathbb{P}(I) = \mathbb{P}(I)\mathbb{P}(\Omega) = \mu_2(\Omega) < \infty,$$

where the second equality follows from $I \cap \Omega = I$, and the third from $\mathbb{P}(\Omega) = 1$.

Moreover, we prove that the two measures coincide on the π-system $\mathcal{J}$.
Let $J \in \mathcal{J}$. By our initial assumption,

$$\mu_1(J) = \mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) = \mu_2(J),$$

where the second equality follows precisely from the assumption of this implication.
Since $J \in \mathcal{J}$ was arbitrary, we obtain $\mu_1(J) = \mu_2(J)\ \forall J \in \mathcal{J}$.
Thus, the two measures have the same finite total mass and coincide on the π-system $\mathcal{J}$.
Since $\sigma(\mathcal{J}) = \mathcal{H}$, all the assumptions of Lemma 1.2 are satisfied. Therefore, by Lemma 1.2, the two measures coincide on the entire σ-algebra $\mathcal{H}$:

$$\mu_1(H) = \mu_2(H)\ \forall H \in \mathcal{H}.$$

Substituting their definitions, and recalling that $I \in \mathcal{I}$ was arbitrary, we conclude that

$$\mathbb{P}(I \cap H) = \mathbb{P}(I)\mathbb{P}(H)\ \forall I \in \mathcal{I},\ H \in \mathcal{H}.$$

We now fix an arbitrary $H \in \mathcal{H}$ and consider the two measures on $(\Omega, \mathcal{G})$ defined by

$$\nu_1(G) := \mathbb{P}(G \cap H),\ \nu_2(G) := \mathbb{P}(G)\mathbb{P}(H),\ G \in \mathcal{G}.$$

By the same argument as above, these two measures have the same finite total mass:

$$\nu_1(\Omega) = \mathbb{P}(H) = \nu_2(\Omega) < \infty.$$

Moreover, by the equality established in the first part of the proof, they coincide on the π-system $\mathcal{I}$:

$$\nu_1(I) = \nu_2(I)\ \forall I \in \mathcal{I}.$$

Therefore, since $\sigma(\mathcal{I}) = \mathcal{G}$, by Lemma 1.2 the two measures also coincide on the entire σ-algebra $\mathcal{G}$:

$$\nu_1(G) = \nu_2(G)\ \forall G \in \mathcal{G}.$$

Substituting their definitions, and recalling that $H \in \mathcal{H}$ was arbitrary, we conclude that

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H)\ \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent.
Therefore, the statement is proved.

HW 37 — Prove That Two Set Functions Define Measures

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\mathcal{H} \subseteq \mathcal{F}$ be a σ-algebra, and let $I \in \mathcal{F}$.
Prove that the two set functions $H \longmapsto \mathbb{P}(I \cap H),\ H \longmapsto \mathbb{P}(I)\mathbb{P}(H)$ define two measures on $(\Omega, \mathcal{H})$.

Expanded Proof of HW 37

Let $I \in \mathcal{F}$ be fixed, and define the two set functions

$$\mu_1(H) := \mathbb{P}(I \cap H),\ \mu_2(H) := \mathbb{P}(I)\mathbb{P}(H),\ H \in \mathcal{H}.$$

We prove that both set functions are measures on $(\Omega, \mathcal{H})$.
First, we prove that μ_1 is a measure.
Let $H \in \mathcal{H}$. Since $\mathcal{H} \subseteq \mathcal{F}$, we have $H \in \mathcal{F}$. Moreover, since $I \in \mathcal{F}$ and $\mathcal{F}$ is closed under intersections, $I \cap H \in \mathcal{F}$.
Therefore, $\mu_1(H) = \mathbb{P}(I \cap H)$ is well-defined and non-negative: $\mu_1(H) \geq 0\ \forall H \in \mathcal{H}$.
Furthermore,

$$\mu_1(\emptyset) = \mathbb{P}(I \cap \emptyset) = \mathbb{P}(\emptyset) = 0.$$

We now prove countable additivity.
Let $(H_n)_{n \in \mathbb{N}} \subseteq \mathcal{H}$ be a sequence of pairwise disjoint sets, namely $H_i \cap H_j = \emptyset\ \forall i \neq j$.
Notice that the sets $I \cap H_n$ are also pairwise disjoint. Indeed, for every $i \neq j$,

$$(I \cap H_i) \cap (I \cap H_j) = I \cap (H_i \cap H_j) = I \cap \emptyset = \emptyset.$$

Moreover, since $I \in \mathcal{F}$ and $H_n \in \mathcal{H} \subseteq \mathcal{F}$, we have $I \cap H_n \in \mathcal{F}$ for every $n \in \mathbb{N}$.

Therefore,

$$\mu_1\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}\left(I \cap \bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty}(I \cap H_n)\right) = \sum_{n=1}^{\infty}\mathbb{P}(I \cap H_n) = \sum_{n=1}^{\infty}\mu_1(H_n),$$

where the second equality follows from the distributive property of intersections over unions, and the third from the countable additivity of $\mathbb{P}$, since the sets $I \cap H_n$ are measurable and pairwise disjoint.
Thus, μ_1 is a measure on $(\Omega, \mathcal{H})$.
We now prove that μ_2 is also a measure.
Recall that $\mu_2(H) = \mathbb{P}(I)\mathbb{P}(H),\ H \in \mathcal{H}$.
Since $\mathbb{P}(I) \geq 0$ and $\mathbb{P}$ is a probability measure, we immediately have $\mu_2(H) \geq 0\ \forall H \in \mathcal{H}$, and $\mu_2(\emptyset) = \mathbb{P}(I)\mathbb{P}(\emptyset) = 0$.
Moreover, let $(H_n)_{n\in\mathbb{N}} \subseteq \mathcal{H}$ be a sequence of pairwise disjoint sets. By the countable additivity of $\mathbb{P}$,

$$\mu_2\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}(I)\mathbb{P}\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}(I)\sum_{n=1}^{\infty}\mathbb{P}(H_n) = \sum_{n=1}^{\infty}\mathbb{P}(I)\mathbb{P}(H_n) = \sum_{n=1}^{\infty}\mu_2(H_n).$$

Therefore, μ_2 is also a measure on $(\Omega, \mathcal{H})$.
We conclude that both set functions

$$\begin{aligned}\mu_1(H) &= \mathbb{P}(I \cap H),\\ \mu_2(H) &= \mathbb{P}(I)\mathbb{P}(H),\end{aligned} \quad H \in \mathcal{H}$$

define measures on $(\Omega, \mathcal{H})$.
Therefore, the statement is proved.

Corollary — Characterisation of Independence via Threshold Events
Let $X, Y : \Omega \to \mathbb{R}$ be two random variables.
Then X and Y are independent if and only if

$$\mathbb{P}(X \leq x,\ Y \leq y) = \mathbb{P}(X \leq x)\,\mathbb{P}(Y \leq y)\ \forall x, y \in \mathbb{R}.$$

Indeed, this is a direct application of Lemma 4.1 with

$$\mathcal{I} := \pi(X) = \big\{\{\omega \in \Omega : X(\omega) \leq x\} : x \in \mathbb{R}\big\},$$

and

$$\mathcal{J} := \pi(Y) = \big\{\{\omega \in \Omega : Y(\omega) \leq y\} : y \in \mathbb{R}\big\}.$$

Recall that, by Lemma 3.7,

$$\sigma(\pi(X)) = \sigma(X),\ \sigma(\pi(Y)) = \sigma(Y).$$

Therefore, the independence of the random variables X and Y is equivalent to the factorisation of the probabilities of the events $\{X \leq x\}$ and $\{Y \leq y\}$ for every $x, y \in \mathbb{R}$.

Generalisation — Independence of Finitely Many Random Variables
More generally, random variables $X_1, \ldots, X_n$ are independent if and only if, for every $x_1, \ldots, x_n \in \mathbb{R}$,

$$\mathbb{P}(X_1 \leq x_1, \ldots, X_n \leq x_n) = \prod_{i=1}^{n}\mathbb{P}(X_i \leq x_i).$$

This is the analogous finite-dimensional characterisation of independence, obtained by applying the same idea to the π-systems generated by the threshold events of the variables $X_1, \ldots, X_n$. (Williams, E4.1, p. 225).

HW 38 — Independence of Three σ-Algebras Generated by π-Systems

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3$ be three π-systems such that $\mathcal{I}_k \subseteq \mathcal{F},\ \Omega \in \mathcal{I}_k,\ k = 1, 2, 3$.

Prove that $\sigma(\mathcal{I}_1)$, $\sigma(\mathcal{I}_2)$ and $\sigma(\mathcal{I}_3)$ are independent if and only if

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k)$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.

Why is it necessary to assume that $\Omega \in \mathcal{I}_k$, $k = 1, 2, 3$?

Part I — Expanded Proof of the Equivalence

Let $\mathcal{G}_k := \sigma(\mathcal{I}_k),\ k = 1, 2, 3$.

We prove both implications.

($\Rightarrow$) Assume that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.

By Definition 4.1, for every $G_k \in \mathcal{G}_k$, $k = 1, 2, 3$,

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \prod_{k=1}^{3} \mathbb{P}(G_k).$$

Since

$$\mathcal{I}_k \subseteq \sigma(\mathcal{I}_k) = \mathcal{G}_k,\ k = 1, 2, 3,$$

the equality holds, in particular, for every $I_k \in \mathcal{I}_k$. Therefore,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k).$$

($\Leftarrow$) Assume that

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k)$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.

We want to prove that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.

First, without loss of generality, we consider $\mathcal{G}_1$ and $\mathcal{G}_2$, since the three indices play symmetric roles. In particular, since $\Omega \in \mathcal{I}_3$, we can choose $I_3 = \Omega$ in our assumption.

Therefore, for every $I_1 \in \mathcal{I}_1$ and $I_2 \in \mathcal{I}_2$,

$$\mathbb{P}(I_1 \cap I_2) = \mathbb{P}(I_1 \cap I_2 \cap \Omega) = \mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(\Omega) = \mathbb{P}(I_1)\mathbb{P}(I_2),$$

where the second equality follows from our assumption, and the third from $\mathbb{P}(\Omega) = 1$.

Since $\mathcal{I}_1$ and $\mathcal{I}_2$ are π-systems generating $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively, by Lemma 4.1 we conclude that

$$\mathcal{G}_1 \text{ and } \mathcal{G}_2 \text{ are independent.}$$

We now introduce the collection

$$\mathcal{I}_{1,2} := \{I_1 \cap I_2 : I_1 \in \mathcal{I}_1,\ I_2 \in \mathcal{I}_2\}.$$

We first prove that $\mathcal{I}_{1,2}$ is a π-system.

Let $A, B \in \mathcal{I}_{1,2}$. By definition, there exist $I_1, J_1 \in \mathcal{I}_1$ and $I_2, J_2 \in \mathcal{I}_2$ such that $A = I_1 \cap I_2,\ B = J_1 \cap J_2$.

Therefore,

$$A \cap B = (I_1 \cap I_2) \cap (J_1 \cap J_2) = (I_1 \cap J_1) \cap (I_2 \cap J_2),$$

where the second equality follows from the associative and commutative properties of intersections.

Since $\mathcal{I}_1$ and $\mathcal{I}_2$ are π-systems, $I_1 \cap J_1 \in \mathcal{I}_1,\ I_2 \cap J_2 \in \mathcal{I}_2$.

Hence, $A \cap B \in \mathcal{I}_{1,2}$.

Moreover, $\Omega = \Omega \cap \Omega \in \mathcal{I}_{1,2}$, since $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$.

Thus, $\mathcal{I}_{1,2}$ is a non-empty collection closed under intersections, and therefore a π-system.
Notice also that, since $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$, $I_1 = I_1 \cap \Omega \in \mathcal{I}_{1,2}\ \forall I_1 \in \mathcal{I}_1$, and $I_2 = \Omega \cap I_2 \in \mathcal{I}_{1,2}\ \forall I_2 \in \mathcal{I}_2$.
Therefore, $\mathcal{I}_1 \subseteq \mathcal{I}_{1,2},\ \mathcal{I}_2 \subseteq \mathcal{I}_{1,2}$.
Since $\sigma(\mathcal{I}_{1,2})$ is a σ-algebra containing both $\mathcal{I}_1$ and $\mathcal{I}_2$, by the minimality of generated σ-algebras we obtain

$$\mathcal{G}_1 \subseteq \sigma(\mathcal{I}_{1,2}),\ \mathcal{G}_2 \subseteq \sigma(\mathcal{I}_{1,2}).$$

We now prove that $\sigma(\mathcal{I}_{1,2})$ and $\mathcal{G}_3$ are independent.
Let $A \in \mathcal{I}_{1,2}$ and $I_3 \in \mathcal{I}_3$. By definition, there exist $I_1 \in \mathcal{I}_1$ and $I_2 \in \mathcal{I}_2$ such that $A = I_1 \cap I_2$.
Therefore,

$$\mathbb{P}(A \cap I_3) = \mathbb{P}(I_1 \cap I_2 \cap I_3) = \mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(I_3) = \mathbb{P}(I_1 \cap I_2)\mathbb{P}(I_3) = \mathbb{P}(A)\mathbb{P}(I_3),$$

where the second equality follows from our initial assumption, and the third from the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$, established above.
Since $A \in \mathcal{I}_{1,2}$ and $I_3 \in \mathcal{I}_3$ were arbitrary,

$$\mathbb{P}(A \cap I_3) = \mathbb{P}(A)\mathbb{P}(I_3), \forall A \in \mathcal{I}_{1,2},\ I_3 \in \mathcal{I}_3.$$

Since $\mathcal{I}_{1,2}$ and $\mathcal{I}_3$ are π-systems, by Lemma 4.1 we conclude that

$$\sigma(\mathcal{I}_{1,2}) \text{ and } \mathcal{G}_3 \text{ are independent.}$$

Finally, let $G_k \in \mathcal{G}_k,\ k = 1, 2, 3$.
Since $\mathcal{G}_1, \mathcal{G}_2 \subseteq \sigma(\mathcal{I}_{1,2})$, and $\sigma(\mathcal{I}_{1,2})$ is closed under intersections, we have $G_1 \cap G_2 \in \sigma(\mathcal{I}_{1,2})$.
Therefore,

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \mathbb{P}((G_1 \cap G_2) \cap G_3) = \mathbb{P}(G_1 \cap G_2)\mathbb{P}(G_3) = \mathbb{P}(G_1)\mathbb{P}(G_2)\mathbb{P}(G_3),$$

where the second equality follows from the independence of $\sigma(\mathcal{I}_{1,2})$ and $\mathcal{G}_3$, and the third from the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$.
Since $G_k \in \mathcal{G}_k$, $k = 1, 2, 3$, were arbitrary, we conclude that

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \prod_{k=1}^{3} \mathbb{P}(G_k), \forall G_k \in \mathcal{G}_k.$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.
Therefore, the equivalence is proved.

HW 38 — Part II: Why Is the Assumption $\Omega \in \mathcal{I}_k$ Necessary?
In Part I, the assumption $\Omega \in \mathcal{I}_3$ allowed us to choose $I_3 = \Omega$ and establish the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$.
However, pairwise independence alone does not imply joint independence. We first illustrate this fact and then construct a counterexample showing why the assumption $\Omega \in \mathcal{I}_k$ is needed.

Counterexample 1 — Pairwise Independence Does Not Imply Joint Independence
Let X_1 and X_2 be two independent and identically distributed random variables such that

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2},\ i = 1, 2.$$

Define $X_3 := X_1 X_2$.
We first prove that X_1, X_2, X_3 are pairwise independent.
By assumption, X_1 and X_2 are independent. Therefore, it remains to consider the pairs (X_1, X_3) and (X_2, X_3).
Notice that $X_3 = 1$ if and only if X_1 and X_2 have the same sign. Thus,

$$\{X_3 = 1\} = \big(\{X_1 = 1\} \cap \{X_2 = 1\}\big) \cup \big(\{X_1 = -1\} \cap \{X_2 = -1\}\big).$$

Since these two events are disjoint, by additivity and the independence of X_1 and X_2,

$$\begin{aligned}\mathbb{P}(X_3 = 1) &= \mathbb{P}(X_1 = 1, X_2 = 1) \ + \mathbb{P}(X_1 = -1, X_2 = -1) \\ &= \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) \ + \mathbb{P}(X_1 = -1)\mathbb{P}(X_2 = -1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.\end{aligned}$$

We now prove that X_1 and X_3 are independent.
Since each X_i takes only the values -1 and 1, we have

$$\sigma(X_i) = \{\emptyset, \Omega, \{X_i = 1\}, \{X_i = -1\}\}, \ i = 1, 2, 3.$$

Therefore, the collection $\mathcal{J}_i := \big\{\{X_i = 1\}\big\}$ is a π-system generating $\sigma(X_i)$. Indeed, it is closed under intersections, and its generated σ-algebra also contains $\{X_i = 1\}^c = \{X_i = -1\}$.
Thus, by Lemma 4.1, it is sufficient to prove that

$$\mathbb{P}(X_1 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1)\mathbb{P}(X_3 = 1).$$

Notice that, since $X_3 = X_1X_2$,

$$\{X_1 = 1, X_3 = 1\} = \{X_1 = 1, X_2 = 1\}.$$

Indeed, when $X_1 = 1$, the equality $X_3 = 1$ holds if and only if $X_2 = 1$.
Therefore,

$$\mathbb{P}(X_1 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_1 = 1)\mathbb{P}(X_3 = 1),$$

where the second equality follows from independence, and the last from $\mathbb{P}(X_3 = 1) = 1/2$, established above.
Hence, by Lemma 4.1, $\sigma(X_1)$ and $\sigma(X_3)$ are independent.
Similarly, exchanging X_1 and X_2, we obtain

$$\mathbb{P}(X_2 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_2 = 1)\mathbb{P}(X_3 = 1).$$

Therefore, again by Lemma 4.1, $\sigma(X_2)$ and $\sigma(X_3)$ are independent.
We conclude that

$$X_1, X_2, X_3 \text{ are pairwise independent.}$$

We now prove that they are not jointly independent.
Since $X_3 = X_1X_2$, we have

$$\{X_1 = 1, X_2 = 1, X_3 = 1\} = \{X_1 = 1, X_2 = 1\}.$$

Therefore,

$$\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4}.$$

On the other hand,

$$\prod_{i=1}^{3} \mathbb{P}(X_i = 1) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $\frac{1}{4} \neq \frac{1}{8}$, we obtain

$$\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 1) \neq \prod_{i=1}^{3} \mathbb{P}(X_i = 1).$$

Thus, by Definition 4.1, the three σ-algebras $\sigma(X_1)$, $\sigma(X_2)$ and $\sigma(X_3)$ are not jointly independent.
Therefore, this counterexample shows that pairwise independence does not imply joint independence.
In particular, it explains why, in Part I of HW 38, establishing the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$ was not sufficient.

Counterexample 2 — Necessity of $\Omega \in \mathcal{I}_k$

We now prove that, without the assumption $\Omega \in \mathcal{I}_k$, the factorisation on the three π-systems does not necessarily imply the independence of the generated σ-algebras.

Consider the probability space $\Omega = \{0, 1\},\ \mathcal{F} = \mathcal{P}(\Omega)$,
with $\mathbb{P}(\{0\}) = \mathbb{P}(\{1\}) = \frac{1}{2}$.
Let $A := \{1\}$, so that $\mathbb{P}(A) = \frac{1}{2}$.
Define

$$\mathcal{I}_1 = \mathcal{I}_2 = \{\Omega, A\},\ \mathcal{I}_3 = \{\emptyset\}.$$

First, notice that these collections are π-systems.
Indeed, $\{\Omega, A\}$ is closed under intersections, since $\Omega \cap A = A,\ A \cap A = A,\ \Omega \cap \Omega = \Omega$.
Moreover, $\{\emptyset\}$ is a non-empty collection closed under intersections, because $\emptyset \cap \emptyset = \emptyset$.
However, although $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$, $\Omega \notin \mathcal{I}_3$.
We first prove that the factorisation in HW 38 holds on these three π-systems.
Let $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
Since $\mathcal{I}_3 = \{\emptyset\}$, necessarily $I_3 = \emptyset$.
Therefore,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \mathbb{P}(I_1 \cap I_2 \cap \emptyset) = \mathbb{P}(\emptyset) = 0,$$

where the second equality follows from the fact that the intersection of any set with the empty set is empty.
On the other hand,

$$\mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(I_3) = \mathbb{P}(I_1)\mathbb{P}(I_2) \cdot 0 = 0.$$

Consequently,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k).$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
We now prove that the generated σ-algebras are not independent.
Let $\mathcal{G}_k := \sigma(\mathcal{I}_k),\ k = 1, 2, 3$.
Since $A \in \mathcal{I}_1$ and $A \in \mathcal{I}_2$, we have $A \in \mathcal{G}_1,\ A \in \mathcal{G}_2$.
Moreover, although $\Omega \notin \mathcal{I}_3$, we necessarily have $\Omega \in \mathcal{G}_3$, because every σ-algebra on Ω contains Ω.
Therefore, we can choose $G_1 = A,\ G_2 = A,\ G_3 = \Omega$.
If the three σ-algebras were independent, by Definition 4.1 we would have

$$\mathbb{P}(A \cap A \cap \Omega) = \mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega).$$

However,

$$\mathbb{P}(A \cap A \cap \Omega) = \mathbb{P}(A) = \frac{1}{2},$$

where the first equality follows from $A \cap A \cap \Omega = A$.
On the other hand,

$$\mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega) = \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{1}{4}.$$

Since $\frac{1}{2} \neq \frac{1}{4}$, we obtain $\mathbb{P}(A \cap A \cap \Omega) \neq \mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega)$.
Thus, by Definition 4.1, the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are not independent.
Notice that this does not contradict the factorisation established above. Indeed, that factorisation holds only for events belonging to the original π-systems. Since $\Omega \notin \mathcal{I}_3$, we could not choose $I_3 = \Omega$ there, whereas Ω necessarily belongs to the generated σ-algebra $\mathcal{G}_3$.
Therefore, the factorisation holds on the three π-systems but fails on their generated σ-algebras.
Hence, the assumption $\Omega \in \mathcal{I}_k$, $k = 1, 2, 3$, is necessary to guarantee the equivalence in HW 38.
Therefore, the statement is proved.

Borel–Cantelli and Independent Repetition

Now that we have a proper notion of independence we can state a converse to the first Borel-Cantelli's lemma.

Lemma 4.2 — Second Borel–Cantelli's Lemma

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be a sequence of independent events such that

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) = +\infty.$$

Then

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 1.$$

In other words, with probability one, infinitely many of the events E_n occur.

Expanded Proof of Lemma 4.2

Set $E := \limsup_{n\to\infty} E_n$.

By definition of the limit superior of a sequence of sets and De Morgan's laws, we have

$$E^c = \left(\bigcap_{n=1}^{\infty} \bigcup_{m\geq n} E_m\right)^c = \bigcup_{n=1}^{\infty} \left(\bigcup_{m\geq n} E_m\right)^c = \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} E_m^c = \liminf_{n\to\infty} E_n^c,$$

where the second and third equalities follow from De Morgan's laws, and the last from the definition of the limit inferior.

We want to prove that $\mathbb{P}(E^c) = 0$, *which will imply* $\mathbb{P}(E) = 1$.

Set $p_n := \mathbb{P}(E_n),\ n \in \mathbb{N}$.

By assumption, $\sum_{n=1}^{\infty} p_n = +\infty$.

Fix an arbitrary $n \in \mathbb{N}$. *We first calculate* $\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right)$.

For every $k \geq 0$, define $A_k := \bigcap_{m=n}^{n+k} E_m^c$.

Notice that $(A_k)_{k\geq 0}$ is a decreasing sequence of measurable sets. Indeed,

$$A_{k+1} = A_k \cap E_{n+k+1}^c \subseteq A_k.$$

Moreover, $\bigcap_{k=0}^{\infty} A_k = \bigcap_{m\geq n} E_m^c$.

Since $\mathbb{P}$ is a probability measure, we also have $\mathbb{P}(A_0) \leq \mathbb{P}(\Omega) = 1 < \infty$.

Thus, all the assumptions of Lemma 1.6 are satisfied. Therefore, by continuity of measures from above,

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) = \lim_{k\to\infty} \mathbb{P}\left(\bigcap_{m=n}^{n+k} E_m^c\right).$$

Now, since the events $(E_n)_{n\in\mathbb{N}}$ are independent, their complements are also independent. Indeed, by Definition 4.3 and HW 36, $E_m^c \in \sigma(\mathbf{1}_{E_m})\ \forall m \in \mathbb{N}$, and the σ-algebras generated by the indicator functions of the independent events are themselves independent.

Consequently, for every finite $k \geq 0$,

$$\mathbb{P}\left(\bigcap_{m=n}^{n+k} E_m^c\right) = \prod_{m=n}^{n+k} \mathbb{P}(E_m^c) = \prod_{m=n}^{n+k} (1 - p_m),$$

where the first equality follows from the independence of the complementary events, and the second from $\mathbb{P}(E_m^c) = 1 - \mathbb{P}(E_m) = 1 - p_m$.

Combining this equality with continuity from above, we obtain

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) = \lim_{k\to\infty} \prod_{m=n}^{n+k} (1 - p_m).$$

We now use the inequality $1 - x \leq e^{-x},\ x \geq 0$.
Since $p_m = \mathbb{P}(E_m) \in [0,1]$, applying this inequality with $x = p_m$ gives

$$0 \leq 1 - p_m \leq e^{-p_m}.$$

Therefore, multiplying these inequalities for $m = n, \ldots, n+k$, we obtain

$$\prod_{m=n}^{n+k} (1 - p_m) \leq \prod_{m=n}^{n+k} e^{-p_m} = \exp\left(- \sum_{m=n}^{n+k} p_m \right),$$

where the last equality follows from the properties of the exponential function.
Consequently,

$$0 \leq \mathbb{P}\left(\bigcap_{m \geq n} E_m^c \right) \leq \lim_{k \to \infty} \exp\left(- \sum_{m=n}^{n+k} p_m \right).$$

Notice that, since $\sum_{m=1}^{\infty} p_m = +\infty$ and each $p_m \in [0,1]$, removing finitely many terms does not change the divergence of the series. Hence, for every fixed $n \in \mathbb{N}$, $\sum_{m=n}^{\infty} p_m = +\infty$.
Therefore,

$$\lim_{k \to \infty} \exp\left(- \sum_{m=n}^{n+k} p_m \right) = 0,$$

since $e^{-t} \to 0$ as $t \to +\infty$.
Thus, by the inequality established above,

$$0 \leq \mathbb{P}\left(\bigcap_{m \geq n} E_m^c \right) \leq 0.$$

We conclude that

$$\mathbb{P}\left(\bigcap_{m \geq n} E_m^c \right) = 0 \ \forall n \in \mathbb{N}.$$

Finally, recall that we established at the beginning of the proof that $E^c = \bigcup_{n=1}^{\infty} \bigcap_{m \geq n} E_m^c$.
Since each event in this countable union has probability zero, by Lemma 1.7 (a countable union of null sets is a null set), we obtain $\mathbb{P}(E^c) = 0$.
Therefore,

$$\mathbb{P}(E) = 1 - \mathbb{P}(E^c) = 1.$$

Recalling that $E = \limsup_{n \to \infty} E_n$, we conclude that

$$\mathbb{P}\left(\limsup_{n \to \infty} E_n \right) = 1.$$

Therefore, the statement is proved.
We are now going to see an application of Borel-Cantelli's two lemmas.

Example 2 (Williams, Example 4.4) — Application of the Borel–Cantelli Lemmas
Statement
Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of independent and identically distributed random variables such that

$$X_n \sim \text{Exp}(1),\ n \in \mathbb{N}.$$

For $\alpha > 0$, determine the probability of the event

$$E_\alpha := \{\omega \in \Omega : X_n(\omega) > \alpha \log n,\ \text{i.o.}\},$$

where *i.o.* means *infinitely often*. Then study the almost sure behaviour of

$$L(\omega) := \limsup_{n \to \infty} \frac{X_n(\omega)}{\log n}.$$

Expanded Solution of Example 2

Fix $\alpha > 0$, and define

$$E_n^\alpha := \{\omega \in \Omega : X_n(\omega) > \alpha \log n\},\ n \in \mathbb{N}.$$

The survival function of X_n satisfies

$$\mathbb{P}(X_n > t) = e^{-t},\ t \geq 0.$$

This follows from $X_n \sim \text{Exp}(1)$. Consequently,

$$\mathbb{P}(E_n^\alpha) = \mathbb{P}(X_n > \alpha \log n) = e^{-\alpha \log n} = n^{-\alpha}.$$

The first equality uses the definition of E_n^α, the second the exponential survival function (since $\alpha \log n \geq 0$), and the last the identity $e^{a \log n} = n^a$.

Moreover,

$$E_n^\alpha = X_n^{-1}((\alpha \log n, \infty)) \in \sigma(X_n) \subseteq \mathcal{F}.$$

The equality is the definition of a pre-image; the membership follows from the definition of $\sigma(X_n)$, and the final inclusion from the $\mathcal{F}$-measurability of X_n. Since the X_n are independent, the events $(E_n^\alpha)_{n\in\mathbb{N}}$ are independent by Definition 4.2.

We now express E_α as a limit superior:

$$\begin{aligned} E_\alpha &= \{\omega \in \Omega : X_n(\omega) > \alpha \log n \text{ for infinitely many } n\} \\ &= \{\omega \in \Omega : \forall N \in \mathbb{N},\ \exists m \geq N \text{ such that } X_m(\omega) > \alpha \log m\} \\ &= \bigcap_{N=1}^{\infty} \bigcup_{m \geq N} \{\omega \in \Omega : X_m(\omega) > \alpha \log m\} = \bigcap_{N=1}^{\infty} \bigcup_{m \geq N} E_m^\alpha = \limsup_{n\to\infty} E_n^\alpha. \end{aligned}$$

The first equality states the definition of E_α. The second expresses *infinitely often*: after every N there must be some $m \geq N$ for which the inequality holds. The third translates "for every N" into an intersection and "there exists $m \geq N$" into a union. The fourth substitutes the definition of E_m^α, and the last is the definition of the limit superior of sets.

We will use formula (4.3):

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

As we will prove in HW 39, this follows from the two Borel–Cantelli lemmas and the convergence criterion for the p-series $\sum_{n=1}^{\infty} n^{-\alpha}$.

We now consider

$$L(\omega) := \limsup_{n\to\infty} \frac{X_n(\omega)}{\log n},$$

where the ratios are defined for $n \geq 2$, because $\log 1 = 0$. For each $n \geq 2$, the function $X_n/\log n$ is measurable, since X_n is measurable and $\log n$ is a positive constant. Moreover,

$$L = \inf_{N\geq 2} \sup_{n \geq N} \frac{X_n}{\log n}.$$

This is the definition of the numerical limit superior; since countable suprema and infima of measurable functions are measurable, L is an extended-real-valued measurable random variable.

We first prove that $\mathbb{P}(L \geq 1) = 1$.

Let $\omega_0 \in E_1$. We have

$$\frac{X_n(\omega_0)}{\log n} > 1 \text{ for infinitely many } n \geq 2.$$

Indeed, the definition of E_1 gives $X_n(\omega_0) > \log n$ infinitely often, and division by $\log n > 0$ preserves the inequality. Consequently,

$$L(\omega_0) \geq 1.$$

This follows from the definition of the limit superior: a sequence exceeding 1 infinitely often cannot have limit superior below 1. Since $\omega_0 \in E_1$ was arbitrary,

$$E_1 \subseteq \{L \geq 1\}.$$

We therefore obtain

$$1 \geq \mathbb{P}(L \geq 1) \geq \mathbb{P}(E_1) = 1.$$

The first bound holds for every probability, the second inequality follows from the inclusion just proved and monotonicity of probability, and the last equality uses formula (4.3) with $\alpha = 1$. Hence $\mathbb{P}(L \geq 1) = 1$.
We now prove that $\mathbb{P}(L > 1) = 0$.
First, notice that

$$\{L > 1\} = \bigcup_{k=1}^{\infty} \left\{ L \geq 1 + \frac{2}{k} \right\} = \bigcup_{k=1}^{\infty} A_k,$$

where, for each fixed $k \in \mathbb{N}$, we define

$$A_k := \left\{ L \geq 1 + \frac{2}{k} \right\}.$$

The first equality follows from $2/k \to 0$ and the Archimedean property of the real numbers (including the case $L = +\infty$); the second follows from the definition of A_k.
Now fix $k \in \mathbb{N}$, and define

$$B_k := \left\{ X_n > \left(1 + \frac{1}{k}\right) \log n, \text{ i.o.} \right\}.$$

We want to show that

$$A_k \subseteq B_k \iff A_k \cap B_k^c = \emptyset.$$

The equivalence holds because an inclusion $A_k \subseteq B_k$ means that no outcome can belong to A_k without belonging to B_k.
Suppose, for a contradiction, that there exists

$$\omega_0 \in A_k \cap B_k^c.$$

From the membership $\omega_0 \in A_k$, we obtain

$$L(\omega_0) \geq 1 + \frac{2}{k}.$$

This is the definition of A_k. On the other hand,

$$X_n(\omega_0) > \left(1 + \frac{1}{k}\right) \log n \text{ only finitely often.}$$

This follows from $\omega_0 \in B_k^c$: the outcome does not belong to the event where the inequality holds infinitely often. Thus, for some $N \geq 2$,

$$X_n(\omega_0) \leq \left(1 + \frac{1}{k}\right) \log n \; \forall n \geq N.$$

Indeed, after the last index (if any) at which the strict inequality holds, its negation must hold for every subsequent index. Dividing yields

$$\frac{X_n(\omega_0)}{\log n} \leq 1 + \frac{1}{k} \; \forall n \geq N.$$

The inequality is preserved because $\log n > 0$ for $n \geq 2$. Therefore,

$$L(\omega_0) \leq 1 + \frac{1}{k}.$$

This follows from the definition of limit superior, since every sufficiently late term is bounded by $1 + 1/k$, and contradicts the membership $\omega_0 \in A_k$ established above. Thus,

$$A_k \cap B_k^c = \emptyset, \; A_k \subseteq B_k.$$

The contradiction rules out any outcome in $A_k \cap B_k^c$, which is equivalent to the desired inclusion. Consequently,

$$\mathbb{P}(A_k) \leq \mathbb{P}(B_k) = \mathbb{P}(E_{1+1/k}) = 0.$$

The inequality follows from $A_k \subseteq B_k$ and monotonicity of probability, the first equality from the definition of E_α, and the last from formula (4.3), since $1 + 1/k > 1$.
Since k was arbitrary, we obtain

$$\mathbb{P}(L > 1) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = 0.$$

The first equality uses the identity for $\{L > 1\}$, and the second follows from Lemma 1.7 because each A_k has probability zero.
Finally,

$$\mathbb{P}(L = 1) = \mathbb{P}(L \geq 1) - \mathbb{P}(L > 1) = 1 - 0 = 1.$$

The first equality follows from the disjoint decomposition $\{L \geq 1\} = \{L = 1\} \cup \{L > 1\}$ and additivity of probability; the other equalities use the two probabilities already established. Recalling the definition of L, we conclude that

$$\mathbb{P}\left(\limsup_{n\to\infty} \frac{X_n}{\log n} = 1\right) = 1.$$

Therefore, the statement is proved.

HW 39 — Proof of Formula (4.3)

Statement

Let $(X_n)_{n\in\mathbb{N}}$ be independent and identically distributed random variables with $X_n \sim \text{Exp}(1)$. For $\alpha > 0$, let $E_\alpha := \{X_n > \alpha \log n, \text{ i.o.}\}$. Prove formula (4.3):

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

Expanded Proof of HW 39

Fix $\alpha > 0$, and consider

$$E_n^\alpha := \{X_n > \alpha \log n\}, \ n \in \mathbb{N}.$$

As established in Example 2, these events are measurable and independent, and

$$\mathbb{P}(E_n^\alpha) = n^{-\alpha}, \ E_\alpha = \limsup_{n\to\infty} E_n^\alpha.$$

The first identity was obtained from the exponential survival function; the second from the meaning of *infinitely often* and the definition of the limit superior. Hence,

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = \sum_{n=1}^{\infty} \frac{1}{n^\alpha}.$$

The equality follows by substituting the probability of each E_n^α.
If $\alpha > 1$, we have

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = \sum_{n=1}^{\infty} \frac{1}{n^\alpha} < \infty.$$

The equality was established above; the inequality follows from the p-series convergence criterion for $\alpha > 1$. Consequently,

$$\mathbb{P}(E_\alpha) = \mathbb{P}\left(\limsup_{n\to\infty} E_n^\alpha\right) = 0.$$

The first equality uses the representation of E_α, and the second follows from Lemma 2.3 (First Borel–Cantelli), whose assumptions hold because the events are measurable and their probabilities have finite sum. Independence is not required here.

If $0 < \alpha \leq 1$, we have

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = \sum_{n=1}^{\infty} \frac{1}{n^\alpha} = +\infty.$$

The first equality is the formula already established, while the second follows from the divergence of the p-series for $0 < \alpha \leq 1$; in the boundary case $\alpha = 1$ this is the harmonic series. Therefore,

$$\mathbb{P}(E_\alpha) = \mathbb{P}\left(\limsup_{n\to\infty} E_n^\alpha\right) = 1.$$

The first equality uses the same representation of E_α, and the second follows from Lemma 4.2 (Second Borel–Cantelli), because the events are independent and the sum of their probabilities diverges.
Combining the two cases, we conclude that

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

Therefore, formula (4.3) is proved.

Fundamental Question for Modelling

So far, we have considered sequences of random variables with arbitrary distributions. A natural question is whether we can actually construct a sequence of independent random variables $(X_n)_{n\in\mathbb{N}}$, each with a prescribed CDF F_n.
The answer is yes. Indeed, let $(U_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that

$$U_n \sim \text{Unif}(0, 1).$$

Define

$$X_n := F_n^{-1}(U_n),\ n \in \mathbb{N},$$

where F_n^{-1} denotes the generalised inverse of the CDF F_n.
Then

$$X_n \sim F_n,$$

by the *inverse transform method*. Moreover, the random variables $(X_n)_{n\in\mathbb{N}}$ are independent, since each X_n is a measurable function of U_n, and measurable transformations of independent random variables preserve independence.

Stochastic Processes

A *stochastic process* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a family of random variables $(Y_t)_{t\in\mathcal{I}}$, indexed by a parameter t, usually interpreted as time.
For example, the index set $\mathcal{I}$ may be $\mathbb{N}$, corresponding to discrete time, or $[0, \infty)$, corresponding to continuous time.

Existence of Stochastic Processes

A natural question is whether a stochastic process can be constructed with a prescribed distributional structure.
Under suitable consistency conditions, the answer is yes, as established by the *Kolmogorov Existence Theorem*.
The technical details of this theorem are beyond the scope of the present discussion.

Example 3 — Markov Chains

Statement

Let E be a finite or countable set, and let $P = (p_{ij})_{i,j\in E}$ be a square matrix with positive entries satisfying

$$\sum_{j\in E} p_{ij} = 1,\ \forall i \in E.$$

Such a matrix is called a *stochastic matrix*.

Let μ be a probability measure on E, and set

$$\mu_i := \mu(\{i\}),\ i \in E.$$

A *time-homogeneous Markov chain* $Z = (Z_n)_{n\in\mathbb{N}\cup\{0\}}$ on E, with initial distribution μ and transition matrix P, is a stochastic process satisfying

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}$$

for every $n \in \mathbb{N}$ and every choice of states $i_0, \ldots, i_n \in E$.
We want to construct such a process using independent random variables.

Expanded Construction and Proof of Example 3

First, let Z_0 be an E-valued random variable with distribution μ, so that

$$\mathbb{P}(Z_0 = i) = \mu_i,\ \forall i \in E.$$

Next, consider a family of E-valued random variables $(Y_{i,n})_{i\in E,\ n\in\mathbb{N}}$ such that

$$\mathbb{P}(Y_{i,n} = j) = p_{ij},\ \forall i, j \in E,\ n \in \mathbb{N}.$$

We assume that all the random variables $Y_{i,n}$ are mutually independent and also independent of Z_0.
Notice that this independence is an assumption of our initial construction: **the random variables $Y_{i,n}$ are chosen independently before defining the process Z.**
For each fixed state i and time n, the random variable $Y_{i,n}$ takes values in E, according to the probabilities p_{ij}. Thus, the index i identifies which random variable we are considering, whereas j represents one of its possible values.
We now define the process recursively by setting

$$Z_n(\omega) := Y_{Z_{n-1}(\omega),n}(\omega),\ n \geq 1.$$

This means that the value of $Z_{n-1}(\omega)$ determines which random variable $Y_{i,n}$ is selected to obtain $Z_n(\omega)$.
In particular, if $Z_0(\omega) = i_0$, then

$$Z_1(\omega) = Y_{i_0,1}(\omega).$$

The equality follows from the recursive definition with $n = 1$, after substituting $Z_0(\omega) = i_0$.
Notice that knowing $Z_0(\omega) = i_0$ identifies the random variable $Y_{i_0,1}$, **but does not determine its realised value**. We only know that

$$\mathbb{P}(Y_{i_0,1} = j) = p_{i_0 j},\ j \in E.$$

If we additionally consider an outcome ω for which $Z_1(\omega) = i_1$, then

$$Z_2(\omega) = Y_{i_1,2}(\omega).$$

This follows from the same recursive definition with $n = 2$. Again, the condition $Z_1(\omega) = i_1$ identifies the variable used at the second step, without specifying its realised value.
Notice also that, although the variables $Y_{i,n}$ are independent, the resulting variables Z_n are generally not independent, since the selection of the variable used at time n depends on the previous state Z_{n-1}.
Nevertheless, the construction defines measurable random variables. Indeed, for every $j \in E$ and $n \geq 1$,

$$\{Z_n = j\} = \bigcup_{i\in E} \left(\{Z_{n-1} = i\} \cap \{Y_{i,n} = j\}\right).$$

The equality follows from the recursive definition: $Z_n = j$ precisely when the previous state equals some $i \in E$ and the corresponding variable $Y_{i,n}$ takes the value j. Since E is finite or countable, the right-hand side is a countable union of measurable events. Starting from Z_0, this proves the measurability of each Z_n by induction.
We now verify that the constructed process has the required joint distribution.
We first consider the first two transitions.

Fix arbitrary states $i_0, i_1, i_2 \in E$. We obtain

$$\{Z_0 = i_0, Z_1 = i_1, Z_2 = i_2\} = \{Z_0 = i_0, Y_{i_0,1} = i_1, Z_2 = i_2\} = \{Z_0 = i_0, Y_{i_0,1} = i_1, Y_{i_1,2} = i_2\}.$$

The first equality follows from the recursive definition: on the event $\{Z_0 = i_0\}$, we have $Z_1 = Y_{i_0,1}$. Therefore, requiring $Z_1 = i_1$ is equivalent to requiring $Y_{i_0,1} = i_1$.
The second equality follows by applying the same reasoning to the next step. Once the first two conditions hold, we have $Z_1 = i_1$, and therefore $Z_2 = Y_{i_1,2}$. Hence, requiring $Z_2 = i_2$ is equivalent to requiring $Y_{i_1,2} = i_2$.
The important point is that i_0, i_1, i_2 are fixed states. We are considering the event in which the process follows the particular path

$$i_0 \longrightarrow i_1 \longrightarrow i_2.$$

Consequently, the event on the right-hand side involves the three specific random variables

$$Z_0,\ Y_{i_0,1},\ Y_{i_1,2},$$

which are independent by our initial assumption. We are not using independence of Z_0, Z_1, Z_2. Therefore,

$$\begin{aligned}\mathbb{P}(Z_0 = i_0, Z_1 = i_1, Z_2 = i_2) &= \mathbb{P}(Z_0 = i_0, Y_{i_0,1} = i_1, Y_{i_1,2} = i_2) \\ &= \mathbb{P}(Z_0 = i_0)\mathbb{P}(Y_{i_0,1} = i_1)\mathbb{P}(Y_{i_1,2} = i_2) = \mu_{i_0} p_{i_0 i_1} p_{i_1 i_2}.\end{aligned}$$

The first equality follows from the equality of events established above. The second follows from the independence of $Z_0, Y_{i_0,1}, Y_{i_1,2}$, and the last from their prescribed distributions.
We now extend the argument to an arbitrary number of transitions.
Fix $n \in \mathbb{N}$ and arbitrary states $i_0, \ldots, i_n \in E$.
Repeatedly applying the recursive definition gives

$$\{Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n\} = \{Z_0 = i_0\} \cap \bigcap_{r=1}^{n} \{Y_{i_{r-1},r} = i_r\}.$$

This follows from the same argument used for the first two transitions. For a fixed path, the condition $Z_{r-1} = i_{r-1}$ implies that the variable selected at time r is $Y_{i_{r-1},r}$. Therefore, requiring $Z_r = i_r$ is equivalent to requiring $Y_{i_{r-1},r} = i_r$.
Since the states $i_0, \ldots, i_n$ are fixed, the event on the right-hand side involves the random variables

$$Z_0,\ Y_{i_0,1},\ Y_{i_1,2},\ \ldots,\ Y_{i_{n-1},n}.$$

These are independent by construction. In particular, the selected Y-variables are distinct because their time indices are different, even if some of the states coincide.
Consequently,

$$\begin{aligned}\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n) &= \mathbb{P}\left(\{Z_0 = i_0\} \cap \bigcap_{r=1}^{n} \{Y_{i_{r-1},r} = i_r\}\right) \\ = \mathbb{P}(Z_0 = i_0) \prod_{r=1}^{n} \mathbb{P}(Y_{i_{r-1},r} = i_r) &= \mu_{i_0} \prod_{r=1}^{n} p_{i_{r-1} i_r} = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.\end{aligned}$$

The first equality follows from the representation of the event just established. The second follows from the independence of the selected random variables, the third from their prescribed distributions, and the last from expanding the finite product.
Since $n \in \mathbb{N}$ and the states $i_0, \ldots, i_n \in E$ were arbitrary, we have proved that

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.$$

Thus, the constructed process satisfies the required joint distribution and is a time-homogeneous Markov chain with initial distribution μ and transition matrix P. Therefore, the statement is proved.

A Monkey Types Shakespeare

Statement and construction

Consider a fixed text, such as the complete works of William Shakespeare (WS), consisting of $N \in \mathbb{N}$ symbols in a specified order. Write the target sequence as

$$(w_1, w_2, \ldots, w_N).$$

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables, where X_n is the symbol typed by the monkey at the n-th keystroke. We assume that each symbol needed to type WS has a probability uniformly bounded below by a constant $\varepsilon > 0$ at every keystroke:

$$\mathbb{P}(X_n = x) \geq \varepsilon > 0 \text{ for every } n \in \mathbb{N} \text{ and every symbol } x \text{ appearing in WS.}$$

This makes explicit the uniform-in-time meaning of the positive lower bound used in the lecture notes. No assumption that the X_n have identical distributions is needed.
For $s \in \mathbb{N} \cup \{0\}$, let

$$C_s := \bigcap_{r=1}^{N} \{X_{s+r} = w_r\}.$$

The event C_s means that a complete copy of WS starts at keystroke $s+1$ and ends at keystroke $s+N$. Different occurrences may overlap; each occurrence is identified by its starting position.
Following the lecture notes, introduce the events

$$\begin{aligned} H &:= \{\text{the monkey types infinitely many copies of WS}\},\\ H_k &:= \{\text{the monkey types at least } k \text{ copies of WS}\},\\ H_{m,k} &:= \{\text{the monkey types at least } k \text{ copies of WS by time } m\},\\ H^{(m)} &:= \{\text{the monkey types infinitely many copies of WS after time } m\}. \end{aligned}$$

Here $H_{m,k}$ concerns copies completed no later than the m-th keystroke, whereas $H^{(m)}$ concerns copies whose starting positions are after that keystroke. In particular, $H_{m,k}$ depends only on $X_1, \ldots, X_m$, while $H^{(m)}$ depends only on $X_{m+1}, X_{m+2}, \ldots$. These are measurable events: each occurrence C_s is a finite intersection of events determined by random variables, and the conditions involving finitely or infinitely many occurrences can be expressed through countable unions and intersections.
We want to prove

$$\mathbb{P}(H) = 1.$$

Expanded proof

We first show that $\mathbb{P}(H)$ *is either* 0 *or* 1.
Fix $m, k \in \mathbb{N}$. We obtain

$$\mathbb{P}\big(H_{m,k} \cap H^{(m)}\big) = \mathbb{P}(H_{m,k})\mathbb{P}\big(H^{(m)}\big).$$

The equality follows from independence: $H_{m,k}$ is determined by the first m symbols, whereas $H^{(m)}$ is determined by symbols typed strictly after time m; the corresponding families of random variables are independent.
As we will see in HW 40, deleting finitely many possible starting positions does not change whether infinitely many copies occur, so $H^{(m)} = H$. Consequently,

$$\mathbb{P}(H_{m,k} \cap H) = \mathbb{P}(H_{m,k})\mathbb{P}(H).$$

This equality follows by substituting $H^{(m)} = H$ into the preceding identity.
As we will also see in HW 40,

$$H_{m,k} \uparrow H_k \text{ as } m \to \infty,\ H_k \cap H = H.$$

Therefore,

$$\mathbb{P}(H) = \mathbb{P}(H_k \cap H) = \lim_{m\to\infty} \mathbb{P}(H_{m,k} \cap H) = \lim_{m\to\infty} \big[\mathbb{P}(H_{m,k})\mathbb{P}(H)\big] = \mathbb{P}(H_k)\mathbb{P}(H).$$

The first equality uses $H_k \cap H = H$. The second follows from Lemma 1.5 (continuity from below), since $H_{m,k} \cap H \uparrow H_k \cap H$. The third uses the independence identity already established, and the last follows from continuity from below applied to $H_{m,k} \uparrow H_k$.
As we will prove in HW 40,

$$H_k \downarrow H \text{ as } k \to \infty.$$

Consequently,

$$\mathbb{P}(H) = \mathbb{P}(H)^2.$$

The equality follows by taking $k \to \infty$ in $\mathbb{P}(H) = \mathbb{P}(H_k)\mathbb{P}(H)$ and applying Lemma 1.6 (continuity from above). Since $\mathbb{P}(H) \in [0,1]$, we conclude that

$$\mathbb{P}(H) \in \{0,1\}.$$

Indeed, the equation $p = p^2$ is equivalent to $p(1-p) = 0$.
We now prove that $\mathbb{P}(H) = 1$.
Divide the successive keystrokes into disjoint blocks of length N. For every $k \in \mathbb{N}$, define

$$E_k := \bigcap_{r=1}^{N} \{X_{(k-1)N+r} = w_r\}.$$

Thus E_k is the event that the monkey types an entire copy of WS in the k-th block, consisting precisely of keystrokes $(k-1)N+1, \ldots, kN$.
We obtain

$$\mathbb{P}(E_k) = \prod_{r=1}^{N} \mathbb{P}(X_{(k-1)N+r} = w_r) \geq \varepsilon^N > 0.$$

The equality follows from independence of the N random variables used in the block, while the inequality follows from the uniform lower bound on the probability of each required symbol. In particular, the same positive bound holds for every k.
Consequently,

$$\sum_{k=1}^{\infty} \mathbb{P}(E_k) \geq \sum_{k=1}^{\infty} \varepsilon^N = +\infty.$$

The inequality follows from the preceding bound, and the equality holds because the sum contains infinitely many copies of the fixed positive constant ε^N.
Moreover, the events $(E_k)_{k\in\mathbb{N}}$ are independent, because distinct blocks involve disjoint sets of the independent random variables $(X_n)_{n\in\mathbb{N}}$. Thus all the assumptions of Lemma 4.2, the Second Borel–Cantelli Lemma, are satisfied, and we obtain

$$\mathbb{P}\left(\limsup_{k\to\infty} E_k\right) = 1.$$

As we will see in HW 40,

$$\limsup_{k\to\infty} E_k \subseteq H.$$

Therefore,

$$1 = \mathbb{P}\left(\limsup_{k\to\infty} E_k\right) \leq \mathbb{P}(H) \leq 1.$$

The first equality is the conclusion of the Second Borel–Cantelli Lemma, the first inequality follows from the inclusion and monotonicity of probability, and the last inequality follows from $H \in \mathcal{F}$.
Hence,

$$\mathbb{P}(H) = 1.$$

Therefore, the statement is proved.

HW 40 — Set-Theoretic Verification of the Events in the Monkey Example

Statement

With the notation introduced in *A Monkey Types Shakespeare*, prove that

(i) $H = H^{(m)}$;
(ii) $H_k \cap H = H$;
(iii) $\limsup_{k\to\infty} E_k \subseteq H$;
(iv) $H_k = \bigcup_{m=1}^{\infty} H_{m,k}$;
(v) $H = \bigcap_{k=1}^{\infty} H_k$.

Expanded proof of HW 40

Recall that $C_s := \bigcap_{r=1}^{N} \{X_{s+r} = w_r\},\ s \in \mathbb{N}\cup\{0\}$, is the event that the monkey types one complete copy of WS starting at keystroke $s+1$.

(i) *We first prove that $H = H^{(m)}$.*

Recall that H is the event of infinitely many occurrences of WS in the entire sequence, while $H^{(m)}$ concerns infinitely many occurrences whose first keystrokes are strictly after time m. We also allow $m = 0$.

For every fixed threshold $q \geq 0$, consider the event $\bigcup_{s\geq q} C_s$. This is the event that at least one complete copy of WS starts at a keystroke $s+1$ with $s \geq q$. Here s labels a possible starting position and q is a variable threshold; all indices are nonnegative integers.

For fixed $m \geq 0$, we have

$$H^{(m)} = \bigcap_{q=m}^{\infty} \bigcup_{s\geq q} C_s,$$

$$\omega \in H^{(m)} \iff \forall q \geq m,\ \exists s \geq q :\ \omega \in C_s.$$

The outer intersection requires the condition just described to hold for every threshold $q \geq m$, so occurrences can be found beyond every such threshold. This is precisely the meaning of infinitely many occurrences after time m; the second line writes the same condition with quantifiers.

In particular,

$$H = H^{(0)} = \limsup_{s\to\infty} C_s.$$

The first equality uses the case $m = 0$, when the entire sequence is considered; the second is the definition of the limit superior of events.

Fix $m \in \mathbb{N}$. *First, we prove that $H^{(m)} \subseteq H$.*

For every $0 \leq q < m$, we have

$$H^{(m)} \subseteq \bigcup_{s\geq m} C_s \subseteq \bigcup_{s\geq q} C_s.$$

The first inclusion follows by taking the threshold m in the intersection defining $H^{(m)}$; the second follows because $s \geq m$ implies $s \geq q$. Hence,

$$H^{(m)} \subseteq H^{(0)} = H.$$

The inclusion holds because the conditions with $q < m$ follow from the preceding inclusions, while those with $q \geq m$ already enter the definition of $H^{(m)}$; the equality was established above.

Conversely, we prove that $H \subseteq H^{(m)}$ by contradiction.

Recall that

$$H \subseteq H^{(m)} \iff H \cap \left(H^{(m)}\right)^c = \emptyset.$$

The equivalence expresses that no outcome belongs to H without belonging to $H^{(m)}$.

Suppose, for a contradiction, that $\omega_0 \in H \cap \left(H^{(m)}\right)^c$. We have

$$\left(H^{(m)}\right)^c = \bigcup_{q=m}^{\infty} \bigcap_{s\geq q} C_s^c.$$

The equality follows by applying De Morgan's laws to the representation of $H^{(m)}$ above. Consequently, there exists a threshold $q_0 \geq m$ such that

$$\omega_0 \notin C_s \qquad \text{for every } s \geq q_0.$$

This follows from $\omega_0 \in \left(H^{(m)}\right)^c$ and the preceding representation, using the same interpretation of unions and intersections.
On the other hand, we obtain

$$\omega_0 \in C_{s_0} \qquad \text{for some } s_0 \geq q_0.$$

This follows from $\omega_0 \in H = H^{(0)}$ by taking $q = q_0$ in the membership characterisation, which is permitted since $q_0 \geq m \geq 0$. It contradicts the preceding exclusion of every occurrence with $s \geq q_0$.
Hence,

$$H \cap \left(H^{(m)}\right)^c = \emptyset, \qquad H \subseteq H^{(m)}.$$

The empty intersection follows from the contradiction, and the inclusion follows from the equivalence established above. Finally,

$$H = H^{(m)}.$$

The equality follows from the two inclusions just proved.
(iv) *We now prove that* $H_k = \bigcup_{m=1}^{\infty} H_{m,k}$.
We first express the two events in terms of the C_s:

$$H_k = \bigcup_{0 \leq s_1 < \cdots < s_k} \bigcap_{j=1}^{k} C_{s_j}$$

and

$$H_{m,k} = \bigcup_{\substack{0 \leq s_1 < \cdots < s_k \\ s_k + N \leq m}} \bigcap_{j=1}^{k} C_{s_j}.$$

In the first expression, the intersection requires k distinct occurrences of WS, while the union considers every possible selection of their starting positions. In the second, the additional restriction $s_k + N \leq m$ requires all selected copies to be completed by time m.
Consequently,

$$\bigcup_{m=1}^{\infty} H_{m,k} = \bigcup_{m=1}^{\infty} \bigcup_{\substack{0 \leq s_1 < \cdots < s_k \\ s_k + N \leq m}} \bigcap_{j=1}^{k} C_{s_j} = \bigcup_{0 \leq s_1 < \cdots < s_k} \bigcap_{j=1}^{k} C_{s_j} = H_k.$$

The first equality substitutes the definition of $H_{m,k}$. The second follows because, for every finite selection of starting positions, we can choose $m \geq s_k + N$, so that selection eventually appears in the union. The last equality uses the representation of H_k.
Moreover, $H_{m,k} \subseteq H_{m+1,k}$, because increasing the deadline from m to $m+1$ does not exclude any previously completed copies.
Therefore,

$$H_{m,k} \uparrow H_k \text{ as } m \to \infty.$$

(v) *We next prove that* $H = \bigcap_{k=1}^{\infty} H_k$.
Using complements, we obtain

$$\left(\bigcap_{k=1}^{\infty} H_k\right)^c = \bigcup_{k=1}^{\infty} H_k^c = \bigcup_{q=0}^{\infty} \bigcap_{s \geq q} C_s^c = \left(\bigcap_{q=0}^{\infty} \bigcup_{s \geq q} C_s\right)^c = H^c.$$

The first equality follows from De Morgan's law. The second expresses the equivalence between having fewer than k occurrences for some k and having only finitely many occurrences, which means that from some position q onward no further C_s occurs. The third equality follows again from De Morgan's law, while the last uses the representation of H established above.

Taking complements, we conclude that

$$H = \bigcap_{k=1}^{\infty} H_k.$$

Moreover, $H_{k+1} \subseteq H_k$, because having at least $k+1$ copies implies having at least k copies. Hence,

$$H_k \downarrow H \text{ as } k \to \infty.$$

(ii) In particular,

$$H \subseteq H_k,\ H_k \cap H = H.$$

The inclusion follows because an intersection is contained in each of its constituent sets, while the equality follows from the identity $A \cap B = B$ whenever $B \subseteq A$.
(iii) *Finally, we prove that* $\limsup_{k\to\infty} E_k \subseteq H$.
Recall that $E_k = C_{(k-1)N}$. The equality follows because E_k requires one complete copy of WS in the k-th block of N keystrokes.
For every $K \in \mathbb{N}$,

$$\bigcup_{k\geq K} E_k = \bigcup_{k\geq K} C_{(k-1)N} \subseteq \bigcup_{s\geq (K-1)N} C_s.$$

The equality substitutes the definition of E_k, while the inclusion follows because the positions $(k-1)N$, for $k \geq K$, form a subset of all starting positions $s \geq (K-1)N$.
Therefore,

$$\limsup_{k\to\infty} E_k = \bigcap_{K=1}^{\infty} \bigcup_{k\geq K} E_k \subseteq \bigcap_{K=1}^{\infty} \bigcup_{s\geq (K-1)N} C_s = \bigcap_{q=0}^{\infty} \bigcup_{s\geq q} C_s = H.$$

The first equality is the definition of the limit superior, and the inclusion follows from the preceding inclusion of unions. The next equality holds because the thresholds $(K-1)N$ tend to infinity, so restricting the intersection to these thresholds does not change it. The last equality follows from the representation of H.
Therefore, the statement is proved.

HW 41 — Independence of Products of Random Variables
Statement
Let $(Y_n)_{n\in\mathbb{N}\cup\{0\}}$ be a sequence of i.i.d. random variables such that

$$\mathbb{P}(Y_n = 1) = \mathbb{P}(Y_n = -1) = \frac{1}{2}.$$

Define

$$X_n := \prod_{i=0}^{n} Y_i,\ n \in \mathbb{N} \cup \{0\}.$$

Prove that $(X_n)_{n\in\mathbb{N}\cup\{0\}}$ is a sequence of i.i.d. random variables.

Expanded Proof of HW 41
We first notice that

$$X_0 = Y_0,\ X_n = X_{n-1}Y_n,\ n \geq 1.$$

The first equality follows from the definition of X_0, while the second follows by separating the last factor in the product defining X_n.
Moreover, $X_n \in \{-1, 1\}$ for every n, since each Y_i takes values in $\{-1, 1\}$.
We now use the same idea as in Example 3 on Markov chains: we fix a possible path of the process and calculate its probability.
Fix $n \in \mathbb{N} \cup \{0\}$ and arbitrary states $x_0, \ldots, x_n \in \{-1, 1\}$. We obtain

$$\{X_0 = x_0, \ldots, X_n = x_n\} = \{Y_0 = x_0, Y_1 = x_0x_1, \ldots, Y_n = x_{n-1}x_n\}.$$

The equality follows from the recursive relation $X_k = X_{k-1}Y_k$. Indeed, since $X_{k-1}^2 = 1$, the conditions $X_{k-1} = x_{k-1}$ and $X_k = x_k$ are equivalent to requiring $Y_k = x_{k-1}x_k$, once the previous states have been fixed.
Therefore,

$$\begin{aligned}\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) &= \mathbb{P}(Y_0 = x_0, Y_1 = x_0x_1, \dots, Y_n = x_{n-1}x_n) \\ &= \mathbb{P}(Y_0 = x_0) \prod_{k=1}^{n} \mathbb{P}(Y_k = x_{k-1}x_k) = \left(\frac{1}{2}\right)^{n+1}.\end{aligned}$$

The first equality follows from the identity of events just established, the second from independence of the Y_k, and the last from their common distribution, since every $x_{k-1}x_k$ belongs to $\{-1, 1\}$.
We now relate this result to Example 3.
Consider the state space $E = \{-1, 1\}$, with initial distribution and transition matrix given by

$$\mu_{-1} = \mu_1 = \frac{1}{2},\ P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The joint distribution obtained above can be written as

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mu_{x_0} \prod_{k=1}^{n} p_{x_{k-1}x_k}.$$

This is precisely the joint-distribution formula for a Markov chain introduced in Example 3, with $\mu_i = 1/2$ and $p_{ij} = 1/2$ for every $i, j \in E$.
In this particular case, all rows of the transition matrix are identical: the probability of the next state does not depend on the current state.
We now determine the marginal distributions. For every $m \geq 0$ and $a \in \{-1, 1\}$,

$$\mathbb{P}(X_m = a) = \sum_{x_0, \dots, x_{m-1} \in E} \mathbb{P}(X_0 = x_0, \dots, X_{m-1} = x_{m-1}, X_m = a) = 2^m \left(\frac{1}{2}\right)^{m+1} = \frac{1}{2}.$$

The first equality follows by summing over all possible values of the preceding variables, while the second uses the joint distribution already established and the fact that there are 2^m possible choices of $x_0, \dots, x_{m-1}$.
Consequently,

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \left(\frac{1}{2}\right)^{n+1} = \prod_{k=0}^{n} \mathbb{P}(X_k = x_k).$$

The first equality was proved above, and the second follows from the marginal distributions. The factorisation also holds for any finite selection of the X_n, by summing over the remaining coordinates. Since the variables take values only in $\{-1, 1\}$, this proves their independence according to Definition 4.2.
Moreover, all the X_n have the same distribution. Therefore,

$$(X_n)_{n \in \mathbb{N} \cup \{0\}} \text{ is an i.i.d. sequence.}$$

Therefore, the statement is proved.

5. Integration

A key quantity of interest in probability theory is the expected value $\mathsf{E}[X]$ of a random variable X. But what is the real mathematical meaning of that symbol? If X takes discrete values $x_1, x_2, \dots$, then

$$\mathsf{E}[X] = \sum_{i=1}^{\infty} x_i \mathsf{P}(X = x_i).$$

If X takes values in $\mathbb{R}$ and it admits a pdf, then

$$\mathsf{E}[X] = \int_{-\infty}^{\infty} z f_X(z) \mathrm{d}z.$$

What about all the other random variables?
We want to make sense of the expected value in the form

$$\mathsf{E}[X] = \int_{\Omega} X(\omega) \mathrm{d}\mathsf{P}(\omega).$$

Quick recap on Riemann integral: if $g \in C([a,b])$ then $\int_a^b g(z)\mathrm{d}z := \lim_{n\to\infty} \sum_{i=0}^{2^n-1} g(z_i^n)(z_{i+1}^n - z_i^n)$ where

$$z_i^n = a + \frac{i}{2^n}(b-a),\ i = 0, 1, 2, \dots 2^n.$$

We only know that $\omega \mapsto X(\omega)$ is $\mathcal{F}$-measurable but not continuous, therefore Riemann integral doesn't work here (plus, what is the meaning of $\mathrm{d}\mathsf{P}(\omega)$?). We need a new notion of integration that works for measurable function and that coincides with the Riemann integral in the special case of continuous functions. (Recall that any continuous function is Borel-measurable.)
Let $(\mathcal{S}, \Sigma, \mu)$ be a measure space. We will define the *Lebesgue integral* by making sense of the expression

$$\mu(f) := \int_{\mathcal{S}} f(s) \mathrm{d}\mu(s) = \int_{\mathcal{S}} f \mathrm{d}\mu$$

for $f \in m\Sigma$. For $A \in \Sigma$, sometimes we will use the notation

$$\mu(f; A) = \mu(f 1_A) := \int_{\mathcal{S}} f 1_A \mathrm{d}\mu = \int_A f \mathrm{d}\mu.$$

A special case is the $\mathcal{S} = \mathbb{N}$, $\Sigma = \mathcal{P}(\mathbb{N})$ and $\mu = \#$ the counting measure (i.e., $\mu(\{k\}) = 1$ for any $k \in \mathbb{N}$). A function $f : \mathbb{N} \to \mathbb{R}$ is of the form $f(k) = a_k \in \mathbb{R}$ for $k \in \mathbb{N}$ an the integral with respect to the counting measure reads

$$\int_{\mathcal{S}} f \mathrm{d}\mu = \sum_{k=1}^{\infty} f(k)\mu(\{k\}) = \sum_{k=1}^{\infty} a_k.$$

The first equality is integration against counting measure, which assigns mass one to every singleton. The second substitutes $f(k) = a_k$ term by term. For non-negative terms the values may be infinite; for signed terms the equality is used when the integral or series is defined.
So we must make sure that our definition of integral tallies with the formula above.

Definition — Integral of a Non-negative Simple Function
Integrals of positive simple functions. We start by defining the integral of indicator functions. For $A \in \Sigma$ we define

$$\mu_0(1_A) = \int_{\mathcal{S}} 1_A \mathrm{d}\mu := \mu(A)$$

where we use μ_0 to emphasise that this is an integral that holds *only* for indicator functions.
A function $f \in (m\Sigma)^+$ is called "simple" if

$$f(s) = \sum_{k=1}^{m} a_k 1_{A_k}(s)$$

where $a_k \in [0, \infty)$ and $A_k \in \Sigma$.

Simple Functions and Their Representations
HW 42 — A simple function is measurable
Check that $f \in m\Sigma$.

Solution to HW 42. An indicator of a measurable set is measurable: for each threshold its lower-ray preimage is either $\varnothing$, the complement of the set, or S. Multiplication by a finite constant and finite sums preserve measurability, so the stated representation already proves the claim.

One can also see directly why overlapping sets do not matter. For each pattern $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_m) \in \{0,1\}^m$, form the measurable cell

$$C_\varepsilon = \bigcap_{k=1}^{m} A_k^{(\varepsilon_k)},\ A_k^{(1)} = A_k,\ A_k^{(0)} = S \setminus A_k.$$

These finitely many cells partition S after empty ones are discarded. On C_ε, the value of f is the constant $c_\varepsilon = \sum_k a_k \varepsilon_k$. For any Borel set $B \subseteq \mathbb{R}$, $f^{-1}(B) = \bigcup_{\varepsilon : c_\varepsilon \in B} C_\varepsilon$. This is a finite measurable union.

For the first inclusion in the pre-image formula, a point with $f(s) \in B$ belongs to its unique partition cell C_ε, whose value $c_\varepsilon = f(s)$ lies in B; it is therefore in the displayed union. Conversely a point in a selected cell has $f(s) = c_\varepsilon \in B$, so is in the pre-image. Each cell is measurable by finite intersections and complements, and the final membership follows by finite-union closure. Thus every Borel preimage is measurable. This description also shows that the function takes only finitely many values, even when the original sets overlap.

There may be different ways of writing the same simple function (i.e., different choices of a_k and A_k). For example, take $\mathcal{S} = [0,1]$ and $\Sigma = \mathcal{B}([0,1])$, then

$$f(s) = 1_{[\frac{1}{2},1]}(s) = 1_{[\frac{1}{2},\frac{2}{3})}(s) + 1_{[\frac{2}{3},1]}(s).$$

The first equality defines f. The second follows because the two intervals on the right are disjoint and their union is $[1/2, 1]$: a point contributes one to exactly one of the two indicators when it lies in that interval, and zero otherwise.

We say that a simple function is expressed in *canonical form* if $a_1 < a_2 < \ldots < a_m$ and $A_k = f^{-1}(\{a_k\})$ for $k = 1, \ldots m$.

We denote the collection of simple functions by SF^+ and we define the Lebesgue integral for $f \in SF^+$ by

$$\mu_0(f) = \int_{\mathcal{S}} f \mathrm{d}\mu := \sum_{k=1}^{m} a_k \mu(A_k).$$

Observation: If $\mathcal{S} = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$, $\mu = \mathcal{L}eb$ and $f = 0$, then

$$\int_{\mathcal{S}} f \mathrm{d}\mu = 0 \cdot \mu(\mathbb{R}) = 0 \cdot \infty.$$

The first equality is the simple-function integral of the zero function, represented as $0\mathbf{1}_{\mathbb{R}}$. The second substitutes $\mathcal{L}eb(\mathbb{R}) = \infty$. The expression is assigned value zero by the non-negative integration convention specified next.

It stands to reason that any sensible integral of the zero function should take the value zero. Therefore we adopt the convention that $0 \cdot \infty = 0$ when we define the Lebesgue integral. From the definition of $\mu_0(f)$ (with $f \in SF^+$) and the observation that there are many different representations of a simple function, a question naturally arises: does the integral $\mu_0(f)$ depend on the chosen representation of f? The answer is NO and you can try to prove it in the next piece of homework.

HW 43 — The simple-function integral is well defined

Fix $(\mathcal{S}, \Sigma, \mu)$ and let $f \in SF^+$ have the following two representations:

$$f = \sum_{i=1}^{n} a_i 1_{A_i} \ \text{ and } \ f = \sum_{j=1}^{m} b_j 1_{B_j},$$

for numbers $(a_i)_{i=1}^n, (b_j)_{j=1}^m \subset [0, \infty)$ and sets $(A_i)_{i=1}^n, (B_j)_{j=1}^m \subset \Sigma$. Show that

$$\mu_0(f) = \sum_{i=1}^{n} a_i \mu(A_i) = \sum_{j=1}^{m} b_j \mu(B_j).$$

Solution to HW 43. The issue is that a formula for the integral must depend on the function, not on how its summands were chosen. A common refinement makes the two descriptions comparable.

Partition S according to membership in every A_i and every B_j: each cell is an intersection which chooses either the set or its complement for each of the $n+m$ sets. There are at most 2^{n+m} cells; discard the empty ones and call the remaining finite measurable partition $\mathcal{C}$.
For each $C \in \mathcal{C}$, membership in any A_i or B_j is constant. Hence the common function takes on that cell the value

$$c_C = \sum_{i:C\subseteq A_i} a_i = \sum_{j:C\subseteq B_j} b_j.$$

This equality follows from the pointwise equality of the two representations, evaluated at any point of the nonempty cell. Also each A_i is the disjoint union of the cells contained in it, so finite additivity gives $\mu(A_i) = \sum_{C\subseteq A_i} \mu(C)$. Substituting and rearranging only finite sums of nonnegative terms,

$$\sum_i a_i\mu(A_i) = \sum_i a_i \sum_{C\subseteq A_i} \mu(C) = \sum_{C\in\mathcal{C}} \left(\sum_{i:C\subseteq A_i} a_i \right) \mu(C)$$

$$= \sum_{C\in\mathcal{C}} c_C\mu(C) = \sum_{C\in\mathcal{C}} \left(\sum_{j:C\subseteq B_j} b_j \right) \mu(C) = \sum_j b_j\mu(B_j).$$

The first equality replaces each set mass by the masses of its disjoint refinement cells. The second groups a finite non-negative double sum by cells. The third uses the first pointwise representation of c_C, and the fourth uses the second representation. The last equality reverses the refinement for the B_j. These reasons account for every equality in the displayed chain.
No cancellation or subtraction is performed, so infinite measures of cells cause no difficulty. If $c_C = 0$, every included nonnegative coefficient is zero and that cell contributes zero under the stated convention. If a positive coefficient multiplies an infinite cell mass, both common-refinement formulas take value infinity consistently. Thus the definition is independent of the representation, even in the presence of overlaps or infinite total measure.

HW 44 — Properties of simple-function integration

(**Properties of integrals of simple functions).** *Prove the following properties:*

(i) If $f, g \in SF^+$ and $\mu(\{s \in \mathcal{S} : f(s) \neq g(s)\}) = 0$, then $\mu_0(f) = \mu_0(g)$.

(ii) If $f, g \in SF^+$ and $c \geq 0$, then $f + g \in SF^+$ and $c \cdot f \in SF^+$; moreover

$$\mu_0(f+g) = \mu_0(f) + \mu_0(g) \quad \text{and} \quad \mu_0(c\cdot f) = c\mu_0(f).$$

(iii) If $f, g \in SF^+$ and $f \leq g$, then $\mu_0(f) \leq \mu_0(g)$.

(iv) If $f, g \in SF^+$, then $f \vee g := \max\{f, g\}$ and $f \wedge g := \min\{f, g\}$ are also in SF^+.

Solution to HW 44. Choose a common finite measurable partition $C_1, \ldots, C_M$ on which both functions are constant: $f = \sum_k a_k \mathbf{1}_{C_k}$ and $g = \sum_k b_k \mathbf{1}_{C_k}$. Such a partition is obtained by intersecting the level sets of f with those of g; it covers S, including any zero-value cells. Representation independence of the simple integral lets us use this convenient partition.

(i) If $a_k \neq b_k$ on a nonempty cell, that whole cell lies in $\{f \neq g\}$, so its measure is zero. Its contributions to either integral are therefore zero. On every other cell the values coincide. Summing over cells gives equal integrals. This argument does not need the functions to agree at each individual point, only outside a null set.

(ii) On C_k the sum has value $a_k + b_k$ and the scaled function value ca_k. There are only finitely many such values, and all cells are measurable, so both functions are nonnegative simple. Summing gives

$$\mu_0(f+g) = \sum_k (a_k + b_k)\mu(C_k) = \sum_k a_k\mu(C_k) + \sum_k b_k\mu(C_k),$$

and

$$\mu_0(cf) = \sum_k ca_k\mu(C_k) = c\mu_0(f).$$

The first equality in the sum formula is the definition of the simple integral on the common partition. The second distributes multiplication and rearranges a finite sum. Replacing each resulting sum by its defining integral yields $\mu_0(f) + \mu_0(g)$.
For $c = 0$ the function is identically zero and its integral is zero; this is the reason for the convention $0 \cdot \infty = 0$.

(iii) If $f \leq g$ pointwise, $a_k \leq b_k$ on every nonempty cell. Multiplication by nonnegative cell measures and addition preserve the inequality, yielding $\sum a_k \mu(C_k) \leq \sum b_k \mu(C_k)$.

(iv) The maximum and minimum are constant on each cell, with values $\max(a_k, b_k)$ and $\min(a_k, b_k)$. These are finite nonnegative numbers drawn from a finite list. Thus both functions are nonnegative simple and measurable.

In the stated example both graphs are zero to the left of $1/2$ and one to its right. They differ only at $s = 1/2$, where f has the upper value one and g has the lower value zero. Since this singleton has measure zero, both integrals equal the length of the right half-interval, $1/2$. Changing the endpoint value does not change the area measured by the integral.

An example of property (i) is the following (plot f and g): Take $([0,1], \mathcal{B}([0,1]), \mathcal{L}eb)$ and

$$f(s) = 1_{[\frac{1}{2},1]}(s) \quad \text{and} \quad g(s) = 1_{(\frac{1}{2},1]}(s);$$

The set $\{s \in \mathcal{S} : f(s) \neq g(s)\} = \{\frac{1}{2}\}$ has zero Lebesgue measure, $\mathcal{L}eb(\{\frac{1}{2}\}) = 0$ and indeed

$$\mathcal{L}eb(f) = \mathcal{L}eb([\tfrac{1}{2}, 1]) = \mathcal{L}eb((\tfrac{1}{2}, 1]) = \mathcal{L}eb(g).$$

The first equality is the integral of the indicator defining f. The second removes the singleton $\{1/2\}$, whose Lebesgue measure is zero; additivity gives the same interval measure. The third is the indicator-integral definition applied to g.

Definition — Non-negative Lebesgue Integral

Now we extend the definition of the Lebesgue integral to positive measurable functions: for any $f \in (m\Sigma)^+$ we define

$$\mu(f) := \sup\{\mu_0(h) : h \in SF^+, h \leq f\}.$$

Notice that if $f \in SF^+$ clearly $\mu(f) = \mu_0(f)$. Indeed f itself is an admissible simple minorant, giving $\mu(f) \geq \mu_0(f)$. Every other admissible simple $h \leq f$ has $\mu_0(h) \leq \mu_0(f)$ by the simple-integral monotonicity already proved in HW 44. Taking the defining supremum gives the reverse inequality. Combining them proves agreement of the two definitions.

HW 45 — Monotonicity of the nonnegative integral

Prove that if $f, g \in (m\Sigma)^+$ are such that $f \leq g$, then $\mu(f) \leq \mu(g)$.

Solution to HW 45. At this point the integral of a general nonnegative measurable function is defined from simple functions: $\mu(f) = \sup\{\mu_0(h) : h \text{ nonnegative simple}, \ h \leq f\}$. If $h \leq f$ and $f \leq g$, then $h \leq g$. Thus every simple function admitted in the supremum for f is also admitted in the supremum for g. Taking a supremum over a smaller collection cannot give a larger value. Therefore

$$\sup_{0 \leq h \leq f, \ h \text{ simple}} \mu_0(h) \leq \sup_{0 \leq h \leq g, \ h \text{ simple}} \mu_0(h).$$

This is precisely $\mu(f) \leq \mu(g)$. Both admissible collections contain the zero function, so the suprema are well defined in $[0, \infty]$. No convergence theorem or additivity of the general integral is needed, avoiding circular use of results that have not yet been proved.

Lemma 5.1 — Zero Integral of a Non-negative Function

Let $f \in (m\Sigma)^+$ be such that $\mu(f) = 0$. Then

$$\mu(\{f > 0\}) = \mu(\{s \in \mathcal{S} : f(s) > 0\}) = 0.$$

Expanded proof. Suppose for contradiction that $\mu(\{f > 0\}) > 0$. We use the identity $\{f > 0\} = \bigcup_{n \geq 1} \{f > 1/n\}$, as we will see in HW 46. If every set on the right were null, countable

subadditivity would make their union null. Thus some n satisfies $\mu(\{f > 1/n\}) > 0$. The simple function $n^{-1}\mathbf{1}_{\{f>1/n\}}$ is dominated by f; the definition of the non-negative integral gives

$$\mu(f) \geq n^{-1}\mu(\{f > 1/n\}) > 0.$$

This contradicts $\mu(f) = 0$. Non-negativity rules out cancellation between positive and negative parts. QED

HW 46 — Positive superlevel sets exhaust the positive set
Prove that

$$\{f > 0\} = \cup_{n=1}^{\infty}\{f > \tfrac{1}{n}\} = \lim_{n\to\infty}\{f > \tfrac{1}{n}\}.$$

Solution to HW 46. Write $A_n = \{s : f(s) > 1/n\}$. The thresholds decrease, so $A_n \subseteq A_{n+1}$: exceeding the larger threshold implies exceeding the smaller one. Every A_n is contained in $\{f > 0\}$ because $1/n > 0$.
For the reverse inclusion, take a point s with $f(s) > 0$. If $f(s)$ is finite, the Archimedean property provides an integer $n > 1/f(s)$, so $1/n < f(s)$ and $s \in A_n$. If $f(s) = +\infty$, it belongs to every A_n immediately. Hence $\{f > 0\} \subseteq \bigcup_n A_n$, proving equality.
An increasing set sequence has limit equal to its union: once a point enters some A_n, it belongs to every later set, while a point never entering is absent throughout. Thus its liminf and limsup both equal that union. This proves the last equality as a set identity; applying measure continuity from below would additionally yield $\mu(\{f > 0\}) = \lim_n \mu(A_n)$. The strict inequality must be preserved: a point with $f(s) = 0$ should not enter any of these sets.
A simple application of the lemma above is $([0,1], \mathcal{B}([0,1]), \mathcal{L}eb)$ and $f(s) = 1_{\{\frac{1}{2}\}}(s)$.
The next theorem is probably the most useful single tool in the whole course.

Theorem 5.2 — Monotone Convergence (MON)
If $(f_n)_{n\in\mathbb{N}} \subset (m\Sigma)^+$ is such that $f_n(s) \uparrow f(s)$ for (μ a.e.) $s \in \mathcal{S}$, then

$$\int_{\mathcal{S}} f_n \mathrm{d}\mu \uparrow \int_{\mathcal{S}} f \mathrm{d}\mu.$$

Expanded proof. First assume $f_n \uparrow f$ everywhere. Monotonicity gives an increasing sequence $\mu(f_n)$ with $I := \lim_n \mu(f_n) \leq \mu(f)$. *We prove the reverse inequality.* Fix $0 < c < 1$ and a simple function $0 \leq h \leq f$, and let $E_n = \{ch \leq f_n\}$. Measurability follows from the level-set criteria and the elementary operations already proved. Since $f_n \leq f_{n+1}$, we have $E_n \subseteq E_{n+1}$. For any fixed s, if $h(s) = 0$, then $s \in E_n$ for every n. If $h(s) > 0$, then $ch(s) < h(s) \leq f(s)$; convergence of $f_n(s)$ therefore gives $f_n(s) \geq ch(s)$ for all sufficiently large n. Thus $\bigcup_n E_n = \mathcal{S}$, proving $E_n \uparrow \mathcal{S}$. This argument is part of the lecturer's proof and is established here.
Write $h = \sum_{j=1}^{r} a_j \mathbf{1}_{A_j}$, with measurable disjoint A_j and $a_j \geq 0$. Since $f_n \geq ch\mathbf{1}_{E_n}$,

$$\mu(f_n) \geq c\sum_{j=1}^{r} a_j \mu(A_j \cap E_n) \longrightarrow c\sum_{j=1}^{r} a_j \mu(A_j) = c\mu(h).$$

The limit uses continuity from below for the measure and a finite non-negative sum, so infinite values cause no subtraction. It follows that $I \geq c\mu(h)$. Let $c \uparrow 1$, then take the supremum over all simple $h \leq f$. The definition of the integral yields $I \geq \mu(f)$.
If convergence and monotonicity hold only almost everywhere, choose one measurable null set outside which all these assertions hold, and set every function equal to zero on that set. This preserves measurability and integrals: it holds for simple functions by HW 44 and then follows for non-negative functions from the defining supremum. The everywhere case applies to the modified functions, without using the present theorem circularly. QED

HW 47 — The increasing sets in the monotone convergence proof
Let

$$E_n = \{s \in \mathcal{S} : ch \leq f_n\} \in \Sigma$$

and prove that $E_n \subseteq E_{n+1}$ *with* $E_n \uparrow \mathcal{S}$.

Solution to HW 47. The inequality defining E_n compares measurable functions. Since a simple function is finite-valued, subtraction of ch from the nonnegative extended-valued f_n is well-defined and measurable. Therefore $E_n = \{f_n - ch \geq 0\}$ is measurable.
If $s \in E_n$, then $ch(s) \leq f_n(s) \leq f_{n+1}(s)$, so $s \in E_{n+1}$. This proves that the sequence increases. To identify its union, fix a point s. If $h(s) = 0$, nonnegativity of every f_n gives $ch(s) = 0 \leq f_n(s)$, so the point belongs to every E_n. This case must be separated: the strict inequality $f(s) > ch(s)$ need not hold when both values are zero.
If $h(s) > 0$, the strict choice $c < 1$ gives $ch(s) < h(s) \leq f(s)$. Suppose first that $f(s) < \infty$. The positive number $\delta = (f(s) - ch(s))/2$ has the property that $f(s) - \delta > ch(s)$. Convergence supplies N_s such that $f_n(s) > f(s) - \delta > ch(s)$ for every $n \geq N_s$. If $f(s) = \infty$, convergence to infinity instead ensures eventually $f_n(s) > ch(s)$, since $ch(s)$ is finite. Thus in either case the point belongs to some E_{N_s}. As s was arbitrary, $\bigcup_n E_n = S$, proving $E_n \uparrow S$.
If convergence holds only outside a measurable null set N, the same proof gives $S \setminus N \subset \bigcup_n E_n$; it does not necessarily give equality with S. The complement of the union is then a measurable subset of N and has measure zero. Consequently, for every measurable A, continuity from below still gives $\mu(A \cap E_n) \uparrow \mu(A)$, which is the exact conclusion needed for the integrals in the monotone convergence proof. No claim of literal coverage of exceptional points is necessary.

From MON the next question arises: given $f \in (m\Sigma)^+$ how can we construct an increasing sequence $(f_n) \subset SF^+$ that approximates f? Here is how: let

$$A_k^r := \{s \in \mathcal{S} : \tfrac{k-1}{2^r} \leq f(s) < \tfrac{k}{2^r}\}$$

for $k, r \in \mathbb{N}$. Set

$$f^{(r)}(s) = \sum_{k=1}^{r2^r} \tfrac{k-1}{2^r} 1_{A_k^r}(s) + r 1_{\{f \geq r\}}(s)$$

and notice that by construction $f^{(r)} \leq f^{(r+1)} \leq f$ and $f^{(r)}(s) \uparrow f(s)$, $s \in \mathcal{S}$.
We will use the construction above and MON to prove numerous results.

Lemma 5.3 — Almost-Everywhere Equality
If $f, g \in (m\Sigma)^+$ *and* $\mu(\{f \neq g\}) = 0$, *then* $\mu(f) = \mu(g)$.

Expanded proof. Choose the same truncated dyadic approximation rule for both functions, as in the preceding construction. Then $f^{(r)} \uparrow f$ and $g^{(r)} \uparrow g$, and equality of $f(s)$ and $g(s)$ forces equality of their approximations at s. Thus the simple approximations differ only on the given null set. HW 44 gives $\mu(f^{(r)}) = \mu(g^{(r)})$. Applying MON to the two increasing sequences proves

$$\mu(f) = \lim_r \mu(f^{(r)}) = \lim_r \mu(g^{(r)}) = \mu(g).$$

The first and last equalities are MON on their respective simple approximations. The middle equality holds term by term by HW 44, so their numerical limits agree.
The common approximation rule matters: arbitrary independent choices would not ensure equality of the simple functions outside the null set. QED

Lemma 5.4 — Fatou for Functions
For $(f_n)_{n \in \mathbb{N}} \subset (m\Sigma)^+$ *it holds*

$$\liminf_{n \to \infty} \int_{\mathcal{S}} f_n \mathrm{d}\mu \geq \int_{\mathcal{S}} \liminf_{n \to \infty} f_n \mathrm{d}\mu.$$

Expanded proof. Let $g_n = \inf_{m\geq n} f_m$. These functions are measurable and non-negative, and $g_n \uparrow \liminf_n f_n$. For every $m \geq n$, $g_n \leq f_m$, so monotonicity gives $\mu(g_n) \leq \inf_{m\geq n} \mu(f_m)$. MON now yields

$$\mu(\liminf_n f_n) = \lim_n \mu(g_n) \leq \lim_n \inf_{m\geq n} \mu(f_m) = \liminf_n \mu(f_n).$$

The first equality is MON for g_n. The inequality passes the previously proved numerical bounds to their limits. The final equality is the tail definition of the numerical liminf.
The inequality is obtained before taking the limit; we have not interchanged an arbitrary infimum and an integral. Non-negativity makes all expressions meaningful even for infinite integrals. QED

Lemma 5.5 — Reverse Fatou for Functions
If $(f_n)_{n\in\mathbb{N}} \subset (m\Sigma)^+$ and there is $g \in (m\Sigma)^+$ such that $f_n \leq g$ for all but finitely many n's and $\mu(g) < \infty$, then

$$\limsup_{n\to\infty} \int_{\mathcal{S}} f_n \mathrm{d}\mu \leq \int_{\mathcal{S}} \limsup_{n\to\infty} f_n \mathrm{d}\mu.$$

HW 48 — Reverse Fatou under an integrable ceiling
Prove the lemma by applying Fatou's lemma to $g - f_n$.

Solution to HW 48. Removing finitely many terms changes neither lower nor upper limits, so we work with the tail for which $0 \leq f_n \leq g$. Integrability of g implies that g is finite almost everywhere: if $\mu(g = \infty) > 0$, the inequality $g \geq M\mathbf{1}_{\{g=\infty\}}$ for every M would force an infinite integral. We may redefine all the functions to be zero on the measurable null exceptional set. Hence every subtraction below is between finite pointwise values, and the integrals of f_n are bounded by the finite number $G = \int g\, d\mu$.
The additivity needed below follows from results already available. For two non-negative measurable functions, choose increasing simple approximations of each. Their sums increase to the sum of the limits. Additivity on simple functions is HW 44, and MON passes that finite additivity identity to the two limits. Applying it to $g = (g - f_n) + f_n$ gives $\int(g - f_n) = G - a_n$; applying it to $g = (g - \limsup f_n) + \limsup f_n$ gives $G - L$. Both rearrangements subtract finite numbers bounded by G. Thus no later general-linearity theorem is assumed.
Put $u_n = g - f_n \geq 0$. Fatou's lemma states that for a nonnegative measurable sequence,

$$\int \liminf_n u_n \, d\mu \leq \liminf_n \int u_n \, d\mu.$$

At each point, subtracting from the fixed value $g(s)$ reverses order, so

$$\liminf_n [g(s) - f_n(s)] = g(s) - \limsup_n f_n(s).$$

If $a_n = \int f_n \, d\mu$ and $L = \int \limsup_n f_n \, d\mu$, both a_n and L lie in $[0, G]$. Substitution into Fatou gives

$$G - L \leq \liminf_n (G - a_n) = G - \limsup_n a_n.$$

The last equality follows directly from tail suprema: $\inf_{n\geq k}(G - a_n) = G - \sup_{n\geq k} a_n$, and then take the limit in k. Cancelling the finite number G and reversing the sign yields $L \geq \limsup_n a_n$, exactly the desired inequality.
The finite ceiling is what makes this argument valid. Without $G < \infty$, the expression $G - L$ could be $\infty - \infty$, so one could not derive a reversed inequality by subtraction.

As one would expect, Lebesgue integrals are linear:

Lemma 5.6 — Non-negative Linearity
For $\alpha, \beta \in [0, \infty)$ and $f, g \in (m\Sigma)^+$ we have

$$\int_{\mathcal{S}} (\alpha f + \beta g) \mathrm{d}\mu = \alpha \int_{\mathcal{S}} f \mathrm{d}\mu + \beta \int_{\mathcal{S}} g \mathrm{d}\mu.$$

Expanded proof. Choose non-negative simple $f^{(r)} \uparrow f$ and $g^{(r)} \uparrow g$. For $\alpha, \beta \geq 0$, $\alpha f^{(r)} + \beta g^{(r)} \uparrow \alpha f + \beta g$. Linearity for simple functions was proved in HW 44. MON gives

$$\mu(\alpha f + \beta g) = \lim_r \big(\alpha\mu(f^{(r)}) + \beta\mu(g^{(r)})\big) = \alpha\mu(f) + \beta\mu(g).$$

The first equality combines MON for the approximating sum with HW 44 at each simple-function stage. The second uses MON separately for the two summands and the limit rule for a finite sum of non-negative increasing quantities.

A finite sum of non-negative increasing terms converges to the sum of its limits. There is no subtraction of infinite quantities. A zero coefficient contributes zero under the usual $0 \cdot \infty = 0$ convention for non-negative integration. QED

Signed Functions and Integrability

Of course we would like to extend our results to functions that can change their sign. Given $f \in m\Sigma$ we can decompose it as

$$f = \max\{f, 0\} - \max\{-f, 0\} =: f^+ - f^-$$

so that $f^+, f^- \in (m\Sigma)^+$ and $|f| = f^+ + f^-$.

HW 49 — Positive and negative parts of a measurable function

Check that $f^+, f^- \in (m\Sigma)^+$ and $|f| = f^+ + f^-$.

Solution to HW 49. The functions $p(x) = \max(x, 0)$ and $n(x) = \max(-x, 0)$ are continuous on $\mathbb{R}$. For example $p(x) = (x + |x|)/2$ and $n(x) = (|x| - x)/2$. A continuous function is Borel-measurable, and composing a Borel-measurable function with a Σ-measurable one is measurable: for a Borel set B,

$$(p \circ f)^{-1}(B) = f^{-1}(p^{-1}(B)) \in \Sigma.$$

Thus $f^+ = p \circ f$ and $f^- = n \circ f$ are measurable.

The pre-image equality is the composition identity. The inner pre-image $p^{-1}(B)$ is Borel because p is continuous; its pre-image under f belongs to Σ because f is measurable. These are separate memberships in different spaces. Both are nonnegative by their definitions, placing them in $(m\Sigma)^+$. The identities are pointwise. At a point where $f(s) \geq 0$, $f^+(s) = f(s)$ and $f^-(s) = 0$, so the sum is $f(s) = |f(s)|$. At a point where $f(s) < 0$, $f^+(s) = 0$ and $f^-(s) = -f(s)$, so the sum is again $|f(s)|$. These two cases cover the domain, including $f(s) = 0$. They also show $f = f^+ - f^-$ and $f^+ f^- = 0$: positive and negative contributions never occur simultaneously at one point. This decomposition is the basis for defining a signed integral by subtracting the two nonnegative integrals when that subtraction is meaningful.

If $\min\{\mu(f^+), \mu(f^-)\} < \infty$ the Lebesgue integral is well-defined as

$$\mu(f) = \int_{\mathcal{S}} f^+ \mathrm{d}\mu - \int_{\mathcal{S}} f^- \mathrm{d}\mu.$$

Definition 5.1 — The L1 Space

We denote by $\mathcal{L}^1(\mathcal{S}, \Sigma, \mu)$ the collection of functions $f \in m\Sigma$ such that

$$\mu(|f|) = \int_{\mathcal{S}} f^+ \mathrm{d}\mu + \int_{\mathcal{S}} f^- \mathrm{d}\mu < \infty.$$

HW 50 — Absolute value of an integral

For $f \in \mathcal{L}^1(\mathcal{S}, \Sigma, \mu)$ verify that $|\mu(f)| \leq \mu(|f|)$.

Solution to HW 50. Write $f = f^+ - f^-$ and $|f| = f^+ + f^-$, where the two parts are nonnegative. Both are bounded above by $|f|$, so $a = \int f^+ \, d\mu$ and $b = \int f^- \, d\mu$ are finite nonnegative numbers. This finiteness justifies the signed integral $\int f \, d\mu = a - b$; there is no subtraction of infinite quantities.

For any finite $a, b \geq 0$, the triangle inequality for real numbers gives $|a - b| \leq |a| + |-b| = a + b$. Applying it to the two part-integrals yields

$$\left| \int f \, d\mu \right| = |a - b| \leq a + b = \int (f^+ + f^-) \, d\mu = \int |f| \, d\mu.$$

The first equality in the displayed chain defines the signed integral; the inequality is the triangle inequality for the finite real numbers a, b. The next equality is non-negative linearity of the integral, and the last substitutes the pointwise identity $f^+ + f^- = |f|$.
Thus cancellation between positive and negative contributions can only reduce the absolute value of a signed integral. Equivalently, one can integrate the pointwise bounds $-|f| \leq f \leq |f|$, obtaining

$$-\int |f| \leq \int f \leq \int |f|;$$

the displayed conclusion is precisely these two inequalities combined.

HW 51 — Linear combinations remain integrable
For $f, g \in \mathcal{L}^1(\mathcal{S}, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$ verify that $\alpha f + \beta g \in \mathcal{L}^1(\mathcal{S}, \Sigma, \mu)$.

Solution to HW 51. Membership in L^1 requires two separate checks: measurability and absolute integrability. The function $h = \alpha f + \beta g$ is measurable by the addition and scalar-multiplication conclusions of Lemma 3.4. This uses only previously established real-function measurability, without invoking a later product-space theorem.
The ordinary triangle inequality applied at each point gives

$$|h(s)| = |\alpha f(s) + \beta g(s)| \leq |\alpha| \, |f(s)| + |\beta| \, |g(s)|.$$

The upper bound is nonnegative measurable. Its integral is finite, because nonnegative linearity gives

$$\int \left(|\alpha| \, |f| + |\beta| \, |g| \right) d\mu = |\alpha| \int |f| \, d\mu + |\beta| \int |g| \, d\mu < \infty.$$

By monotonicity, the integral of $|h|$ is bounded by this finite number. Thus h is absolutely integrable and belongs to L^1. This proves the closure needed for L^1 to be a real vector space; it does not assume signed linearity before knowing that the relevant signed integrals exist.
Linearity of the integral is also preserved in the sense that if $f, g \in \mathcal{L}^1(\mathcal{S}, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$, then

$$\int_{\mathcal{S}} (\alpha f + \beta g) \mathrm{d}\mu = \alpha \int_{\mathcal{S}} f \mathrm{d}\mu + \beta \int_{\mathcal{S}} g \mathrm{d}\mu.$$

We will often need to swap the order of limits and integrals. One result in this direction is MON. The next one is also a very useful result of a similar kind.

Theorem 5.7 — Dominated Convergence (DOM)
Suppose $(f_n)_{n \in \mathbb{N}} \subset m\Sigma$ and $f \in m\Sigma$ be such that $f_n(s) \to f(s)$ for (μ-a.e.) $s \in \mathcal{S}$ as $n \to \infty$. If $|f_n| \leq g$ for all $n \in \mathbb{N}$ and some $g \in \mathcal{L}^1(\mathcal{S}, \Sigma, \mu)$, then

$$\lim_{n \to \infty} \int_{\mathcal{S}} |f_n - f| \mathrm{d}\mu = 0$$

and, in particular

$$\lim_{n \to \infty} \int_{\mathcal{S}} f_n \mathrm{d}\mu = \int_{\mathcal{S}} f \mathrm{d}\mu.$$

Expanded proof. The almost-everywhere limit and $|f_n| \leq g$ imply $|f| \leq g$ almost everywhere; hence f is integrable. Let $h_n = |f_n - f|$. Then $0 \leq h_n \leq 2g$ and $h_n \to 0$ almost everywhere. The reverse Fatou inequality already proved in HW 48 therefore gives

$$0 \leq \limsup_n \mu(h_n) \leq \mu(\limsup_n h_n) = 0.$$

Thus $\mu(|f_n - f|) \to 0$. By linearity of the integrable integral and its triangle inequality,

$$|\mu(f_n) - \mu(f)| = |\mu(f_n - f)| \leq \mu(|f_n - f|) \longrightarrow 0.$$

In the last chain the equality is linearity, justified because f_n, f are integrable. The inequality is the integral triangle inequality, and the limit is the convergence of $\mu(h_n)$ already proved.
The domination is by a single integrable function independent of n; separate integrability of each term would not supply this bound. QED

HW 52 — A shrinking support and failure of domination
Let $(\mathcal{S}, \Sigma, \mu) = ([0,1], \mathcal{B}([0,1]), \mathcal{L}eb)$ and define $f_n(s) = n1_{(0,\frac{1}{n})}(s)$. Show that DOM fails.

Solution to HW 52. For $s = 0$, the open support never contains s, so $f_n(0) = 0$ for every n. For a fixed $s > 0$, choose an integer $N > 1/s$. Whenever $n \geq N$, $1/n < s$, so $s \notin (0, 1/n)$ and again $f_n(s) = 0$. Hence $f_n \to 0$ at every point of $[0,1]$, not merely almost everywhere.
Each function is nonnegative measurable and has integral equal to height times support length:

$$\int_0^1 f_n(s)\, ds = n\mathcal{L}eb((0,1/n)) = n\frac{1}{n} = 1.$$

In the preceding integral chain, the first equality is the definition of the integral of the simple function $n\mathbf{1}_{(0,1/n)}$; the second uses that Lebesgue measure of the interval is its length; the last is ordinary multiplication $n(1/n) = 1$.
Consequently

$$\lim_n \int_0^1 f_n\, ds = 1 \text{ but } \int_0^1 \lim_n f_n\, ds = 0.$$

The narrow support tends to zero in length, but its increasing height preserves unit mass. Pointwise convergence does not control the total mass on those shrinking intervals.
The dominated convergence theorem requires a *single* integrable function g with $|f_n| \leq g$ for every n. It does not suffice that each f_n separately be integrable. To see directly that no such common g exists, take $0 < s < 1/4$ and let $n = \lfloor 1/(2s) \rfloor + 1$. Then $n > 1/(2s)$ and $n \leq 1/(2s) + 1 < 1/s$, so $s < 1/n$ and $f_n(s) = n > 1/(2s)$. Every common dominator would thus satisfy $g(s) \geq 1/(2s)$ throughout this interval, up to a null set if the original bounds are almost-everywhere bounds. Because there are countably many n, their exceptional sets can first be combined into one null set. But $\int_0^{1/4} (2s)^{-1} ds = \infty$, contradicting integrability of g. The theorem is therefore not contradicted: its domination hypothesis fails.

The Standard Approximation Procedure
Many of the things we are able to prove in this course rely on a 4-step procedure, which we call *the standard machinery* and which we outline here so that we don't need to repeat it every time it is used.
The standard machinery: Imagine we want to prove a specific property related to Lebesgue integrals. Then we can proceed as follows:

(1) Show that the property holds for indicator functions of measurable sets;
(2) Use 1. and linearity to extend the property to simple functions;
(3) Use MON to pass from simple functions to positive measurable functions;
(4) Use $f = f^+ - f^-$ to prove the property for general measurable functions.

Change of Measure and Densities
As we have seen already, on a measurable space $(\mathcal{S}, \Sigma)$ we can put many different measures. The corresponding Lebesgue integrals are going to be different. Is there a way of transforming a Lebesgue integral with respect to a measure μ_1 into a Lebesgue integral with respect to another measure μ_2?
Fix $f \in (m\Sigma)^+$. For any $A \in \Sigma$ let us denote

$$\lambda(A) := \int_{\mathcal{S}} f 1_A \mathrm{d}\mu.$$

Then $\lambda : \Sigma \to [0, \infty]$ defines a measure. Notice that if $\mu(N) = 0$ then $\lambda(N) = 0$. We say that λ is *absolutely continuous* with respect to μ and use the symbol $\lambda << \mu$.

HW 53 — A nonnegative density defines a measure

Check that λ is countably additive and therefore it is a measure on $(\mathcal{S}, \Sigma)$.

Solution to HW 53. A nonnegative integral is well-defined with values in $[0, \infty]$, so this rule assigns an admissible mass to every measurable set. The convention is that $f\mathbf{1}_A$ vanishes outside A, including when $f = \infty$ there; equivalently we integrate f over A. The empty set gives the zero function, and therefore $\lambda(\varnothing) = 0$.

Let $A_1, A_2, \ldots$ be pairwise disjoint members of Σ. At any point, at most one of their indicators is one. Thus $\mathbf{1}_{\bigcup_{j\geq 1} A_j} = \sum_{j\geq 1} \mathbf{1}_{A_j}$, $f\sum_{j=1}^{N} \mathbf{1}_{A_j} \uparrow f\mathbf{1}_{\bigcup_{j\geq 1} A_j}$. All functions in this increasing sequence are nonnegative measurable. The monotone convergence theorem, which permits infinite limiting integrals, consequently gives

$$\lambda\Big(\bigcup_{j\geq 1} A_j\Big) = \lim_{N\to\infty} \int_S f \sum_{j=1}^{N} \mathbf{1}_{A_j}\, d\mu = \lim_{N\to\infty} \sum_{j=1}^{N} \int_S f\mathbf{1}_{A_j}\, d\mu = \sum_{j\geq 1} \lambda(A_j).$$

The first equality in the displayed calculation uses the definition of λ and MON for the increasing indicator sums. The second is linearity for a finite sum of non-negative functions. The last equality is the definition of the non-negative series together with $\lambda(A_j) = \int f\mathbf{1}_{A_j}\, d\mu$.

Only a finite sum is moved through the integral before taking its monotone limit, so no unjustified interchange of a signed series occurs. Nonnegativity, zero mass of the empty set, and this disjoint countable-additivity identity are all the measure axioms.

This construction also implies absolute continuity: if $\mu(A) = 0$, every nonnegative measurable function supported on A has integral zero, so $\lambda(A) = 0$. Finiteness of λ is not claimed; it holds when $\int f\, d\mu < \infty$.

The construction shown above can be made systematic:

Theorem 5.8 — Radon–Nikodym

If μ and λ are σ-finite measures on $(\mathcal{S}, \Sigma)$ such that $\lambda << \mu$, then there is $f \in (m\Sigma)^+$ such that

$$\lambda(A) = \int_{\mathcal{S}} f 1_A \mathrm{d}\mu \; \forall A \in \Sigma.$$

We say that $f =: \frac{\mathrm{d}\lambda}{\mathrm{d}\mu}$ is the Radon-Nikodym derivative of λ with respect to μ.

For its finite-measure proof we use the Hahn sign-decomposition theorem: a finite signed measure γ has a measurable partition into a negative set N and a positive set P, meaning that every measurable subset has the corresponding non-positive or non-negative value. Here the signed measures are differences of finite non-negative measures, so their values are finite and the theorem applies. The precise special case and its full proof are given in the sign-decomposition appendix. The Radon–Nikodym argument itself follows here in full.

Proof of Theorem 5.8

Proof. Finite measures. Assume first that $\mu(\mathcal{S}), \lambda(\mathcal{S}) < \infty$. Let $\mathcal{C}$ consist of non-negative measurable functions g such that $\int_A g\, d\mu \leq \lambda(A)$ for every $A \in \Sigma$. The zero function belongs. Each candidate has integral at most $\lambda(\mathcal{S})$, so $a = \sup_{g\in\mathcal{C}} \int g\, d\mu$ is a finite non-negative number. The maximum of two candidates is a candidate: split A into the measurable pieces $A \cap \{g \geq h\}$ and $A \cap \{g < h\}$, integrate g on the first and h on the second, and add the two bounds for λ. This also proves the maximum is integrable.

Choose $g_n \in \mathcal{C}$ with $\int g_n\, d\mu > a - 1/n$ and let $f_n = \max(g_1, \ldots, g_n)$. Finite maxima preserve membership, and $f_n \uparrow f$ for a non-negative measurable extended function. For each measurable A, MON gives

$$\int_A f\, d\mu = \lim_n \int_A f_n\, d\mu \leq \lambda(A), \int f\, d\mu = a.$$

The first equality is MON on $f_n\mathbf{1}_A$, and the inequality passes their common bound to the limit. For the last equality, the upper bound is the definition of a applied to each f_n; the lower bound is

$$\int f_n \geq \int g_n > a - 1/n.$$

Taking limits gives both inequalities. Since this integral is finite, f is finite μ-almost everywhere: on $E = \{f = \infty\}$ it dominates $k\mathbf{1}_E$ for every k, implying $k\mu(E) \le a$ and hence $\mu(E) = 0$. Set $f = 0$ on this measurable null set if a finite representative is desired.

The set function $\rho(A) = \lambda(A) - \int_A f\,d\mu$ is a finite non-negative measure. Non-negativity is the preceding test inequality. For disjoint A_n, countable additivity of λ and the density measure from HW 53 give countable additivity of the difference: both non-negative series have finite sums, so subtraction commutes with their partial-sum limits. Moreover $\rho \ll \mu$, since both terms are zero on a μ-null set.

Suppose $\rho(\mathcal{S}) > 0$. Then $\mu(\mathcal{S}) > 0$, by absolute continuity. Put $c = \rho(\mathcal{S})/(2\mu(\mathcal{S})) > 0$ and apply the Hahn sign-decomposition theorem to $\gamma = \rho - c\mu$. Its total is $\rho(\mathcal{S})/2 > 0$, by substituting c and cancelling. For its positive set P, $\gamma(P) > 0$: the complementary negative set has non-positive value, and the two values add to the positive total. Absolute continuity implies $\mu(P) > 0$, since otherwise $\rho(P) = 0$ and $\gamma(P) = 0$. For every $A \in \Sigma$, the sign property on $A \cap P$ gives $c\mu(A \cap P) \le \rho(A \cap P) \le \rho(A)$; the last inequality is monotonicity of the measure ρ. Therefore

$$\int_A (f + c\mathbf{1}_P)\,d\mu = \int_A f\,d\mu + c\mu(A \cap P) \le \int_A f\,d\mu + \rho(A) = \lambda(A).$$

The first equality is non-negative linearity and the indicator integral; the inequality is the residual bound; the last equality is the definition of ρ. Thus $f + c\mathbf{1}_P \in \mathcal{C}$, but its total integral is $a + c\mu(P) > a$, contradicting the supremum. Hence $\rho(\mathcal{S}) = 0$, and non-negativity implies $\rho(A) = 0$ for every A. This is precisely $\lambda(A) = \int_A f\,d\mu$. If $\mu(\mathcal{S}) = 0$, absolute continuity gives $\lambda = 0$ and the zero density proves the conclusion without this contradiction argument.

Sigma-finite measures. Choose increasing measurable covers U_n, V_n with $\mu(U_n), \lambda(V_n) < \infty$; finite unions turn any finite-measure covers into increasing ones. Then $C_n = U_n \cap V_n$ increases to $\mathcal{S}$ and has finite mass for both measures. Let $E_1 = C_1$ and $E_n = C_n \setminus C_{n-1}$ for $n > 1$. The pieces are measurable, disjoint, and cover $\mathcal{S}$ by the least-index argument. The finite case on each trace space E_n supplies f_n, extended by zero outside its piece. Such an extension is measurable because its positive level sets are trace-measurable subsets of E_n, and the other level sets additionally contain its measurable complement. The non-negative measurable function $f = \sum_n f_n \mathbf{1}_{E_n}$ satisfies

$$\int_A f\,d\mu = \sum_n \int_{A \cap E_n} f_n\,d\mu = \sum_n \lambda(A \cap E_n) = \lambda(A).$$

The first equality uses MON on the non-negative finite sums and finite linearity. The second is the finite-case identity on each piece. The third is countable additivity on the partition of A.

For uniqueness, let g be another density. If $\{f > g\}$ has positive μ-measure, some $A = E_j \cap \{f > g + 1/k\}$ has positive measure: the union over j, k is $\{f > g\}$ outside the null sets where a density is infinite. On A both density integrals equal the finite number $\lambda(A)$, whereas

$$\int_A f \ge \int_A g + \mu(A)/k > \int_A g,$$

a contradiction. The first inequality integrates the pointwise level bound; the strict one uses $\mu(A) > 0$ and $k > 0$. Interchange f, g to exclude the other strict inequality. Thus densities agree almost everywhere. QED

6. Expectation

Definition — Expectation

Thanks to the hard work in the previous section we can now define expectations of random variables. Taking $(\mathcal{S}, \Sigma, \mu) = (\Omega, \mathcal{F}, \mathsf{P})$ we now know the meaning of the expression

$$\mathsf{E}[X] = \int_\Omega X(\omega)\mathrm{d}\mathsf{P}(\omega)$$

for $X \in m\mathcal{F}$. We denote

$$\mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P}) = \{X \in m\mathcal{F} : \mathsf{E}[|X|] < \infty\}.$$

Here is a list of properties that we have already proven in the previous section and that we only recap with the new notation:

- If $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ and $\alpha, \beta \in \mathbb{R}$ then $\mathsf{E}[\alpha X + \beta Y] = \alpha \mathsf{E}[X] + \beta \mathsf{E}[Y]$.
- If $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ and $X \leq Y$, then $\mathsf{E}[X] \leq \mathsf{E}[Y]$.

Suppose $(X_n)_{n\in\mathbb{N}} \subset m\mathcal{F}$.

- (MON) If $0 \leq X_n \uparrow X$, then $\mathsf{E}[X_n] \uparrow \mathsf{E}[X]$.
- (FATOU) If $X_n \geq 0$ for all $n \in \mathbb{N}$, then $\liminf_{n\to\infty} \mathsf{E}[X_n] \geq \mathsf{E}[\liminf_{n\to\infty} X_n]$.
- (DOM) If $|X_n(\omega)| \leq Y(\omega)$ for all $(n, \omega) \in \mathbb{N} \times \Omega$ with $\mathsf{E}[Y] < \infty$ and $\mathsf{P}(X_n \to X) = 1$ for some $X \in m\mathcal{F}$, then $\lim_{n\to\infty} \mathsf{E}[|X_n - X|] = 0$.

Lemma 6.1 — Markov Inequality
Let $Z \in m\mathcal{F}$ and $g : \mathbb{R} \to [0, \infty]$ be Borel-measurable and non-decreasing. Then $g(Z) \in m\mathcal{F}$ and

$$\mathsf{E}[g(Z)] \geq \mathsf{E}[g(Z)1_{\{Z\geq c\}}] \geq g(c)\mathsf{P}(Z \geq c).$$

Expanded proof. Composition makes $g(Z)$ measurable. Since $g \geq 0$, deleting its contribution off $\{Z \geq c\}$ can only decrease the integral. On that event, monotonicity of g gives $g(Z) \geq g(c)$. Integrating these pointwise comparisons yields

$$\mathsf{E}[g(Z)] \geq \mathsf{E}[g(Z)\mathbf{1}_{\{Z\geq c\}}] \geq g(c)\mathsf{E}[\mathbf{1}_{\{Z\geq c\}}] = g(c)\mathsf{P}(Z \geq c).$$

The first inequality deletes a non-negative contribution. The second integrates the pointwise lower bound $g(Z) \geq g(c)$ on $\{Z \geq c\}$. The equality uses the definition of an indicator integral, $\mathbb{E}\mathbf{1}_A = \mathbb{P}(A)$.
Division by $g(c)$ requires $0 < g(c) < \infty$. The undivided inequality also permits an infinite non-negative expectation. QED

Example 4 — Applications of Markov Inequality
(1) If $Z \in (m\mathcal{F})^+$, then $\mathsf{P}(Z \geq c) \leq \frac{1}{c}\mathsf{E}[Z]$;
(2) If $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ then $\mathsf{P}(|X| \geq c) \leq \frac{1}{c}\mathsf{E}[|X|]$;
(3) If $g(z) = \mathrm{e}^{\theta z}$ for a fixed $\theta > 0$, then $\mathsf{P}(Z \geq c) \leq \mathrm{e}^{-\theta c}\mathsf{E}[\mathrm{e}^{\theta Z}]$.
For non-negative random variables there are some additional useful properties, which we discuss below.

(a) If $X \in (m\mathcal{F})^+$ and $\mathsf{E}[X] < \infty$, then $\mathsf{P}(X < \infty) = 1$.
(b) If $(Z_k)_{k\in\mathbb{N}} \subset (m\mathcal{F})^+$, then

$$\mathsf{E}\Big[\sum_{k=1}^{\infty} Z_k\Big] = \sum_{k=1}^{\infty} \mathsf{E}\big[Z_k\big].$$

(c) Combining (a) and (b) we see that if $(Z_k)_{k\in\mathbb{N}} \subset (m\mathcal{F})^+$ is such that $\sum_{k=1}^{\infty} \mathsf{E}\big[Z_k\big] < \infty$, then

$$\sum_{k=1}^{\infty} Z_k < \infty, \mathsf{P} - a.s.$$

The latter implies $Z_k \to 0$, $\mathsf{P} - a.s.$ as $k \to \infty$.
(d) A particular case of (c) is when $Z_k = 1_{F_k}$ for some $(F_n)_{n\in\mathbb{N}} \subset \mathcal{F}$. The condition $\sum_k \mathsf{E}\big[Z_k\big] < \infty$ reads $\sum_k \mathsf{P}(F_k) < \infty$ and the conclusion is

$$\sum_{k=1}^{\infty} 1_{F_k} < \infty, \mathsf{P} - a.s.$$

For each $\omega \in \Omega$ the sum counts how many are the sets F_k such that $\omega \in F_k$. There are only finitely many, for every $\omega \in \Omega$. This corresponds to BC1, i.e.,

$$\mathsf{P}(\limsup_{n\to\infty} F_n) = 0.$$

HW 54 — Nonnegative expectations and random series

Prove (a)–(c) above.

Solution to HW 54. For (a), a finite expectation must prevent a positive chance of an infinite value. For every integer $m \geq 1$, $X \geq m\mathbf{1}_{\{X\geq m\}}$, so integrating gives $\mathsf{P}(X \geq m) \leq \mathsf{E}X/m$. The events $\{X \geq m\}$ decrease to $\{X = \infty\}$. Continuity from above is applicable because the underlying measure is a probability measure and hence finite. Therefore

$$\mathsf{P}(X = \infty) = \lim_{m\to\infty} \mathsf{P}(X \geq m) \leq \lim_{m\to\infty} \frac{\mathsf{E}X}{m} = 0.$$

Its complement has probability one, proving (a).

For (b), define the partial sums $S_N = \sum_{k=1}^{N} Z_k$. They are nonnegative measurable, and $S_N \uparrow S := \sum_{k\geq 1} Z_k$ by the definition of a nonnegative series. In particular S is an extended-valued measurable variable. Monotone convergence and finite linearity yield

$$\mathsf{E}S = \lim_N \mathsf{E}S_N = \lim_N \sum_{k=1}^{N} \mathsf{E}Z_k = \sum_{k\geq 1} \mathsf{E}Z_k.$$

Neither independence nor a dominating integrable variable is needed.

The first equality in the expectation chain is MON. The second applies finite non-negative linearity to S_N. The last equality defines the sum of the non-negative numerical series of expectations. The reason the interchange works is the monotone increase of the nonnegative partial sums.

For (c), the right side is now finite, so (b) gives $\mathsf{E}S < \infty$. Applying (a) to S proves $S < \infty$ almost surely. On a path where this series is finite, S_N converges to a finite real number, and

$$Z_N = S_N - S_{N-1} \longrightarrow S - S = 0.$$

The first equality is the difference of two successive partial sums. The convergence follows since both partial sums tend to the same finite S. The last equality is subtraction of that finite number from itself.

This pathwise argument proves the final almost-sure convergence on the same probability-one event. Notice that the implication starts from a finite *sum of expectations*; finiteness of each individual $\mathsf{E}Z_k$ would not be enough to make the series converge.

Theorem 6.2 — Jensen Inequality

Let $c : \mathbb{R} \to \mathbb{R}$ be a convex function. Let $X \in \mathcal{L}^1$ be such that $\mathsf{E}[|c(X)|] < \infty$. Then $\mathsf{E}[c(X)] \geq c(\mathsf{E}[X])$.

Expanded proof. We first establish the elementary convexity facts needed here and for conditional Jensen: a finite convex function on $\mathbb{R}$ is continuous, has a supporting affine line at every point, and is the supremum of the supporting lines chosen at rational points.

For $x < y < z$, express y as the convex combination $y = ((z-y)/(z-x))x + ((y-x)/(z-x))z$. The defining convexity inequality, rearranged by multiplication and division by the positive differences, gives $\frac{c(y)-c(x)}{y-x} \leq \frac{c(z)-c(x)}{z-x} \leq \frac{c(z)-c(y)}{z-y}$. Both inequalities are the same convexity bound, rearranged to isolate the left and right subinterval slope, respectively. Fixing an interior point m therefore gives

$$L := \sup_{x<m} \frac{c(m)-c(x)}{m-x} \leq R := \inf_{y>m} \frac{c(y)-c(m)}{y-m}.$$

The left supremum is at least the finite chord slope at $x = m-1$ and at most the finite right chord at $y = m+1$. The right infimum is bounded by the same two finite numbers in the opposite directions. Thus L, R are finite. Choose $a \in [L, R]$. For $x < m$, the bound on the left slope rearranges to $c(x) \geq c(m) + a(x-m)$; for $x > m$, use the right slope bound. Equality holds at

$x = m$, so this line supports c everywhere. The same secant inequalities show continuity at m: for $0 < |h| < 1$, the slope between m and $m + h$ lies between the two fixed finite chord bounds from $m - 1, m, m + 1$. Multiplying by $|h|$ bounds $|c(m + h) - c(m)|$ by a fixed constant times $|h|$, which tends to zero.

Supporting slopes near a fixed m are uniformly bounded. Indeed, convexity bounds c above on $[m - 2, m + 2]$ by the maximum of its endpoint values. On $[m - 1, m + 1]$, reflection about m and $2c(m) \le c(q) + c(2m - q)$ also give a finite uniform lower bound. For a supporting slope at $q \in [m - 1, m + 1]$, compare with the chords from $m - 2$ and to $m + 2$. Their denominators are at least one and their numerators are uniformly bounded by the preceding value bounds. This proves the claimed bound on the slopes. Choose one supporting line $a_q x + b_q$ at each rational q. Each is at most c everywhere. For rational $q \to m$, $c(q) + a_q(m - q) \to c(m)$, since continuity handles the first term and the bounded-slope estimate makes the second vanish. Thus the supremum of this countable family equals $c(m)$ at every m, proving the third fact.

Now let $m = \mathsf{E}X$, which is finite by $X \in L^1$, and use a supporting line at m. Substitute $x = X(\omega)$ and integrate:

$$\mathsf{E}[c(X)] \ge c(m) + a\mathsf{E}[X - m] = c(\mathsf{E}X).$$

The inequality integrates the pointwise supporting-line bound and uses linearity on its integrable affine terms. The equality uses $\mathsf{E}[X - m] = \mathsf{E}X - m = 0$ and then substitutes $m = \mathsf{E}X$. The integrability of $c(X)$ is the hypothesis of the theorem. This proves Jensen without assuming differentiability of c. QED

Definition — Lp Space and Norm

$\mathcal{L}^p$-spaces of random variables. For $1 \le p < \infty$ we say that $X \in \mathcal{L}^p(\Omega, \mathcal{F}, \mathsf{P})$ if $X \in m\mathcal{F}$ and

$$\|X\|_p := \big(\mathsf{E}[|X|^p]\big)^{\frac{1}{p}} < \infty.$$

We refer to $\|\cdot\|_p$ as $\mathcal{L}^p$-norm, although proving that it is a norm requires a bit of work and a clever trick.

Lp norms are defined on almost sure equivalence classes: $\|X\|_p = 0$ means $X = 0$ almost surely, not necessarily at every outcome.

HW 55 — Statement

(Tricky!). *Show that $\|\cdot\|_p$ is a norm, i.e., $\|X\|_p \ge 0$, $\|X\|_p = 0$ if and only if $X = 0$, $\|\alpha X\|_p = |\alpha|\|X\|_p$ and $\|X + Y\|_p \le \|X\|_p + \|Y\|_p$.*

Solution to HW 55. Identification modulo almost-sure equality is necessary. Changing a measurable function at a point of probability zero does not change any of these moments, so on pointwise functions the expression is only a seminorm. It is well-defined on equivalence classes because almost-sure equality preserves $|X|^p$ and hence its expectation.

Nonnegativity follows from $|X|^p \ge 0$. If $X = 0$ almost surely, its norm is zero. Conversely, if $\mathsf{E}|X|^p = 0$, then for every integer $m \ge 1$,

$$|X|^p \ge m^{-p}\mathbf{1}_{\{|X|>1/m\}},\ \mathsf{P}(|X| > 1/m) \le m^p\mathsf{E}|X|^p = 0.$$

Since $\{|X| > 0\} = \bigcup_{m\ge1}\{|X| > 1/m\}$, countable subadditivity gives $\mathsf{P}(|X| > 0) = 0$. Thus the equivalence class is precisely zero. For a scalar a, $|aX|^p = |a|^p|X|^p$; integrating and taking the pth root gives absolute homogeneity, including $a = 0$.

The triangle inequality requires more work because taking a nonlinear power does not commute with addition. First ensure that the proposed sum has finite moment, without using the result being proved. For nonnegative u, v, convexity of t^p gives

$$\left(\frac{u + v}{2}\right)^p \le \frac{u^p + v^p}{2},\ (u + v)^p \le 2^{p-1}(u^p + v^p).$$

Together with $|X + Y| \le |X| + |Y|$, this shows $\mathsf{E}|X + Y|^p \le 2^{p-1}(\mathsf{E}|X|^p + \mathsf{E}|Y|^p) < \infty$. All the norms subsequently divided by are therefore finite.

If $p = 1$, integrating the ordinary absolute-value triangle inequality already proves

$$\|X + Y\|_1 \le \|X\|_1 + \|Y\|_1.$$

For $p > 1$, put $q = p/(p-1)$, so $1/p + 1/q = 1$. We will derive the Hölder bound used below rather than assume a norm triangle inequality circularly. For $u, v \ge 0$, minimizing $u^p/p - uv$ over u gives derivative $u^{p-1} - v$ and minimum at $u = v^{1/(p-1)}$. The minimum value is $-v^q/q$, so $uv \le \frac{u^p}{p} + \frac{v^q}{q}$. For variables $A \in L^p$, $B \in L^q$ with nonzero norms, apply this pointwise to $u = |A|/\|A\|_p$, $v = |B|/\|B\|_q$. Its nonnegative right side has expectation $1/p + 1/q = 1$. Consequently the product is integrable and $\mathsf{E}|AB| \le \|A\|_p \|B\|_q$. If either norm is zero, the corresponding variable is zero almost surely by the definiteness proof and the same estimate is immediate. This proves Hölder in all needed cases. Now let $W = X + Y$ and $B = |W|^{p-1}$. The moment calculation already made gives

$$\mathsf{E}|B|^q = \mathsf{E}|W|^{(p-1)q} = \mathsf{E}|W|^p < \infty, \ \|B\|_q = \|W\|_p^{p-1}.$$

The first equality substitutes $|B| = |W|^{p-1}$ and takes the qth power. The second uses $(p-1)q = p$. Finiteness holds because $W \in L^p$ as already established. The norm equality takes the qth root and uses $p/q = p - 1$.
Multiply $|W| \le |X| + |Y|$ by $|W|^{p-1}$ and integrate. Applying Hölder separately to the two resulting products yields

$$\|W\|_p^p \le \mathsf{E}[|X||W|^{p-1}] + \mathsf{E}[|Y||W|^{p-1}] \le (\|X\|_p + \|Y\|_p)\|W\|_p^{p-1}.$$

If $\|W\|_p = 0$, the desired inequality is automatic. Otherwise divide by the positive finite number $\|W\|_p^{p-1}$ to obtain $\|X + Y\|_p \le \|X\|_p + \|Y\|_p$. This is Minkowski's inequality. We have checked every norm axiom, the finiteness of the sum, and the zero-norm cases that would otherwise make the normalisations invalid.

Lemma 6.3 — Monotonicity of Lp Norms
Fix $1 \le p < r < \infty$. If $Y \in \mathcal{L}^p$ and $Y \in \mathcal{L}^r$, then $\|Y\|_p \le \|Y\|_r$.

Expanded proof. The probability measure has total mass one. Put $V_n = |Y| \wedge n$ to avoid presupposing the integrability we want to prove. Since $t \mapsto t^{r/p}$ is convex on $[0, \infty)$, Jensen gives

$$(\mathsf{E}V_n^p)^{r/p} \le \mathsf{E}V_n^r \le \mathsf{E}|Y|^r.$$

The first inequality is Jensen for the bounded variable V_n^p; the second follows from $V_n^r \le |Y|^r$ and monotonicity. All terms involving V_n are finite, so taking roots is legitimate.
Taking the r-th root yields $\|V_n\|_p \le \|Y\|_r$. MON applied to $V_n^p \uparrow |Y|^p$ proves both $Y \in L^p$ and $\|Y\|_p \le \|Y\|_r$. The total-mass-one assumption is essential to this constant-one comparison. QED

HW 56 — Statement
For $X, Y \in \mathcal{L}^p$ and $\alpha, \beta \in \mathbb{R}$ prove that $\alpha X + \beta Y \in \mathcal{L}^p$.

Solution to HW 56. The linear combination is measurable because sums and constant multiples of measurable real variables are measurable. *We must prove finiteness of its pth absolute moment.* A proof by an elementary convexity bound avoids assuming the norm triangle inequality in order to prove the vector-space property needed for that norm.
For $u, v \ge 0$, convexity of $t \mapsto t^p$ gives $((u + v)/2)^p \le (u^p + v^p)/2$. Multiplying by 2^p yields $(u + v)^p \le 2^{p-1}(u^p + v^p)$. Apply it with $u = |a||X|$ and $v = |b||Y|$, after the ordinary triangle inequality:

$$|aX + bY|^p \le (|a||X| + |b||Y|)^p \le 2^{p-1}\big(|a|^p|X|^p + |b|^p|Y|^p\big).$$

The upper bound is integrable by the assumptions. Taking expectations gives

$$\mathsf{E}|aX + bY|^p \le 2^{p-1}\left(|a|^p \mathsf{E}|X|^p + |b|^p \mathsf{E}|Y|^p\right) < \infty.$$

Thus the combination belongs to L^p. This argument includes $p = 1$ (where the prefactor equals one) and zero coefficients. On almost-sure equivalence classes, changing representatives only changes this combination on the union of their two null exceptional sets, so the vector operation is also well-defined.

The result from the homework tells us that $\mathcal{L}^p$ is a *vector space*. The next result instead gives us an important inequality.

Theorem 6.4 — Cauchy–Schwarz Inequality
If $X, Y \in \mathcal{L}^2$, then $X \cdot Y \in \mathcal{L}^1$ and

$$\mathsf{E}[|XY|] \le \|X\|_2 \cdot \|Y\|_2.$$

Expanded proof. Let $U = |X| \wedge n$, $V = |Y| \wedge n$, so all products are integrable. For every $t \in \mathbb{R}$, $0 \le \mathsf{E}[(U - tV)^2] = \mathsf{E}U^2 - 2t\mathsf{E}(UV) + t^2\mathsf{E}V^2$. If $\mathsf{E}V^2 = 0$, then $V = 0$ almost surely and the inequality is immediate. Otherwise choose $t = \mathsf{E}(UV)/\mathsf{E}V^2$ to obtain $(\mathsf{E}UV)^2 \le \mathsf{E}U^2\mathsf{E}V^2$. The truncated products increase to $|XY|$. MON gives $\mathsf{E}|XY| \le (\mathsf{E}X^2)^{1/2}(\mathsf{E}Y^2)^{1/2} < \infty$. Only after this integrability check do we use $|\mathsf{E}(XY)| \le \mathsf{E}|XY|$ to obtain the stated form. QED

HW 57 — Statement
Use Cauchy-Schwartz inequality to prove

$$\|X + Y\|_2 \le \|X\|_2 + \|Y\|_2.$$

Solution to HW 57. First, $(X + Y)^2 \le 2X^2 + 2Y^2$ shows that the sum is square-integrable. Cauchy–Schwarz states that the product of two square-integrable variables is integrable and $|\mathsf{E}[XY]| \le \mathsf{E}|XY| \le (\mathsf{E}X^2)^{1/2}(\mathsf{E}Y^2)^{1/2}$. All terms in the following expansion are therefore finite:

$$\begin{aligned}\|X + Y\|_2^2 &= \mathsf{E}[(X + Y)^2] = \mathsf{E}X^2 + 2\mathsf{E}[XY] + \mathsf{E}Y^2 \\ &\le \|X\|_2^2 + 2|\mathsf{E}[XY]| + \|Y\|_2^2 \le \|X\|_2^2 + 2\|X\|_2\|Y\|_2 + \|Y\|_2^2 = (\|X\|_2 + \|Y\|_2)^2.\end{aligned}$$

Both sides are squares of nonnegative finite numbers.
The first equality is the definition of the squared norm. The next expands the square and uses linearity for three integrable terms. The first inequality replaces the mixed expectation by its absolute value; the second applies Cauchy–Schwarz. The final equality expands the real square $(\|X\|_2 + \|Y\|_2)^2$. The square-root function is increasing on $[0, \infty)$, so taking square roots preserves the inequality and gives the result. No independence or zero-mean assumption on X, Y is required: the cross term is bounded by Cauchy–Schwarz rather than discarded.

Variance, Covariance and Orthogonality
Definitions — Variance and Covariance
Variance and covariance. If $X, Y \in \mathcal{L}^2$ then $XY \in \mathcal{L}^1$ by CS and $X, Y \in \mathcal{L}^1$ by monotonicity of norms. In particular $\mu_X := \mathsf{E}[X]$ and $\mu_Y := \mathsf{E}[Y]$ are well-defined. Defining $\tilde{X}(\omega) := X(\omega) - \mu_X$ and $\tilde{Y}(\omega) := Y(\omega) - \mu_Y$ we have $\mathsf{E}[\tilde{X}] = \mathsf{E}[\tilde{Y}] = 0$. Moreover $\mathrm{Cov}(X, Y) = \mathsf{E}[\tilde{X}\tilde{Y}]$ and by CS we get $\mathsf{E}[\tilde{X}\tilde{Y}] \le \mathsf{E}[\tilde{X}^2]^{1/2}\mathsf{E}[\tilde{Y}^2]^{1/2}$ and rewriting everything in terms of X and Y we get

$$|\,\mathrm{Cov}(X, Y)| \le \mathrm{Var}(X)^{1/2}\,\mathrm{Var}(Y)^{1/2},$$

which is a well-known property that relates to the correlation coefficient

$$|\rho_{X,Y}| := \frac{|\mathrm{Cov}(X, Y)|}{\mathrm{Var}(X)^{1/2}\,\mathrm{Var}(Y)^{1/2}} \le 1.$$

Correlation requires both variances to be positive. The corresponding signed ratio belongs to $[-1, 1]$.

Definition — Scalar Product and Orthogonality
Scalar product in $\mathcal{L}^2$. You may have realised that $\mathcal{L}^2$ is a vector space. We can equip it with a scalar product

$$\langle X, Y\rangle := \mathsf{E}[XY],\ X, Y \in \mathcal{L}^2.$$

Two random variables $X, Y \in \mathcal{L}^2$ are said to be orthogonal if $\langle X, Y\rangle = 0$.
Pythagora's theorem holds:

$$\|X + Y\|_2^2 = \|X\|_2^2 + \|Y\|_2^2 \iff \langle X, Y\rangle = 0.$$

HW 58 — Statement

Prove Pythagora's theorem above.

Solution to HW 58. Cauchy–Schwarz ensures $XY \in L^1$, so we may expand the square and integrate each term:

$$\|X+Y\|_2^2 = \mathsf{E}[X^2 + 2XY + Y^2] = \|X\|_2^2 + \|Y\|_2^2 + 2\langle X, Y\rangle.$$

The first equality in that chain expands $(X+Y)^2$. The second uses linearity of expectation and the definitions of the squared norms and the inner product. Integrability was verified before these operations, so neither equality hides an undefined term.

If the inner product is zero, the last term vanishes and the Pythagorean identity follows. Conversely, if that identity holds, subtract the two finite squared norms from both sides of this expansion; it gives $2\langle X, Y\rangle = 0$, hence orthogonality.

The hypothesis is orthogonality, not necessarily independence. For example, if X is symmetric with finite fourth moment and $Y = X^2$, then $\mathsf{E}[XY] = \mathsf{E}X^3 = 0$ although Y is a function of X. Also, zero covariance alone is different unless the means are handled: $\mathsf{E}[XY] = \operatorname{Cov}(X, Y) + \mathsf{E}X\,\mathsf{E}Y$. The identity concerns the uncentred L^2 inner product exactly as stated.

Completeness and Cauchy Sequences

Definitions — Cauchy Sequence and Completeness

Completeness. Let $p \in [1, \infty)$.

- We say that a sequence $(X_n)_{n\in\mathbb{N}} \subset \mathcal{L}^p$ is a *Cauchy sequence* if for any $\varepsilon > 0$ there is $n_\varepsilon \in \mathbb{N}$ s.t. $\|X_n - X_j\|_p \le \varepsilon$ for all $n, j \ge n_\varepsilon$.
- We say that $\mathcal{L}^p$ is *complete* if any Cauchy sequence in $\mathcal{L}^p$ converges to an element in $\mathcal{L}^p$.

Lemma 6.5 — Completeness of Lp

The space $\mathcal{L}^p(\Omega, \mathcal{F}, \mathsf{P})$ is complete.

Expanded proof. Let $(X_n)_{n\in\mathbb{N}} \subseteq L^p$ be Cauchy, where $1 \le p < \infty$. Choose strictly increasing k_j such that $\|X_r - X_s\|_p \le 2^{-j}$ whenever $r, s \ge k_j$. Monotonicity of norms gives

$$\sum_j \mathsf{E}|X_{k_{j+1}} - X_{k_j}| \le \sum_j 2^{-j} < \infty,$$

where the geometric-series bound is the fact used by the lecturer, as we will see in HW 59. By the summability result in the already completed HW 54, the series of absolute increments is finite almost surely. The telescoping sums therefore define a finite limit $X = \lim_j X_{k_j}$ almost surely. Set $X = 0$ on the measurable exceptional null set to obtain a measurable real-valued representative.

For $r \ge k_j$, Fatou along this subsequence gives

$$\mathsf{E}|X_r - X|^p \le \liminf_{m\to\infty} \mathsf{E}|X_r - X_{k_m}|^p \le 2^{-jp}.$$

In particular $X - X_{k_j} \in L^p$. The elementary convexity bound $|u+v|^p \le 2^{p-1}(|u|^p + |v|^p)$ then gives $X \in L^p$. The estimate holds for every $r \ge k_j$, not only the chosen subsequence, so $X_r \to X$ in L^p. It is not necessary, and is generally false, that the entire original sequence converge pointwise. QED

HW 59 — The geometric series used in completeness

Calculate $\sum_{n=0}^{\infty} \frac{1}{2^n}$.

Solution to HW 59. Work first with the finite sum $s_N = 1 + 1/2 + \cdots + 2^{-N}$. Multiplication by one half gives $s_N/2 = 1/2 + 1/4 + \cdots + 2^{-(N+1)}$. Subtracting these finite sums cancels every middle term, leaving

$$s_N - \frac{s_N}{2} = 1 - 2^{-(N+1)},\ s_N = 2\big(1 - 2^{-(N+1)}\big) = 2 - 2^{-N}.$$

The first displayed equality follows by cancelling the overlapping terms of the two finite geometric sums. In the next chain, the first equality multiplies that identity by two, and the second distributes the factor and uses $2 \cdot 2^{-(N+1)} = 2^{-N}$.
As $N \to \infty$, $2^{-N} \to 0$, and the definition of an infinite series as the limit of its partial sums gives $\sum_{n=0}^{\infty} 2^{-n} = 2$. Starting instead at $n = 1$ removes the first term one and gives a sum of one. The index matters when the series is used as an explicit bound.

How does all the abstract maths we have seen so far with the classical notion of expectation? Recall that $\mathcal{L}_X = \mathsf{P} \circ X^{-1}$ defines the law of a random variable.

Lemma 6.6 — Integration with Respect to the Law
Let $h : \mathbb{R} \to \mathbb{R}$ be Borel-measurable and $X \in m\mathcal{F}$. Then $h(X) \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ if and only if $h \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$ and

$$\mathsf{E}[h(X)] = \int_{\mathbb{R}} h(x)\mathcal{L}_X(\mathrm{d}x).$$

Expanded proof. For $h = \mathbf{1}_B$, B Borel, the assertion is the definition of the law:

$$\mathsf{E}[\mathbf{1}_B(X)] = \mathsf{P}(X \in B) = \mathcal{L}_X(B).$$

For a non-negative simple $h = \sum_j a_j \mathbf{1}_{B_j}$, finite linearity adds these identities. For a non-negative Borel h, choose simple $h_n \uparrow h$. Then $h_n(X) \uparrow h(X)$, and MON on the two spaces gives $\mathsf{E}[h(X)] = \int_{\mathbb{R}} h(x)\,\mathcal{L}_X(\mathrm{d}x)$. Apply this identity to $|h|$: the two absolute integrals are finite in exactly the same cases. When finite, subtract the identities for h^+ and h^-. This proves both the signed formula and the stated equivalence of integrability. QED

In the special case of X admitting a p.d.f. $f_X(x)$ the formula above reduces to $\mathcal{L}_X(\mathrm{d}x) = f_X(x)\mathrm{d}x$ and

$$\mathsf{E}[h(X)] = \int_{\mathbb{R}} h(x)\mathcal{L}_X(\mathrm{d}x) = \int_{\mathbb{R}} h(x)f_X(x)\mathrm{d}x.$$

The first equality is the transfer formula just proved, under its non-negativity or integrability hypotheses. The second uses that the law has density f_X with respect to Lebesgue measure, so integration against the law becomes integration against that density.
In particular f_X is the Radon-Nykodim derivative of $\mathcal{L}_X$ with respect to the Lebesgue measure

$$f_X = \frac{\mathrm{d}\mathcal{L}_X}{\mathrm{d}x} = \frac{\mathrm{d}\mathcal{L}_X}{\mathrm{d}\mathcal{L}eb}.$$

The first equality identifies the density as a Radon–Nikodym derivative. The second makes explicit that dx denotes Lebesgue measure; it is the same derivative with the measure named in full.
There are three important inequalities that we state before closing this section.

Lemma 6.7 — Holder and Minkowski Inequalities
Suppose $f, g \in \mathcal{L}^p(\mathcal{S}, \Sigma, \mu)$, $h \in \mathcal{L}^q(\mathcal{S}, \Sigma, \mu)$ with

$$\frac{1}{p} + \frac{1}{q} = 1,\ p > 1.$$

Then we have

- *Hölder inequality:*

$$|\mu(f \cdot h)| \le \mu(|fh|) \le \|f\|_p \|h\|_q.$$

- *Minkowski inequality:*

$$\|f + g\|_p \le \|f\|_p + \|g\|_p.$$

Expanded proof. Let $1 < p < \infty$ and $1/p + 1/q = 1$. Young's inequality $uv \le u^p/p + v^q/q$, $u, v \ge 0$, follows by minimizing $u^p/p - uv$ over $u \ge 0$: its minimum at $u = v^{1/(p-1)}$ is $-v^q/q$. If either

norm is zero, the product is zero almost everywhere. Otherwise apply Young to $u = |f|/\|f\|_p$, $v = |h|/\|h\|_q$ and integrate:

$$\int \frac{|fh|}{\|f\|_p \|h\|_q} \, \mathrm{d}\mu \le \frac{1}{p} + \frac{1}{q} = 1.$$

This proves Hölder and integrability of the product.
For Minkowski, first $|f+g|^p \le 2^{p-1}(|f|^p + |g|^p)$ ensures that $u = f + g$ has finite p-norm. Hölder gives

$$\|u\|_p^p \le \int (|f| + |g|)|u|^{p-1} \, \mathrm{d}\mu \le (\|f\|_p + \|g\|_p)\big\||u|^{p-1}\big\|_q = (\|f\|_p + \|g\|_p)\|u\|_p^{p-1}.$$

The first inequality integrates the pointwise triangle inequality after multiplication by the non-negative factor $|u|^{p-1}$. The second applies HÃ¶lder separately to the two summands. The last equality uses $(p-1)q = p$, which gives $\||u|^{p-1}\|_q = \|u\|_p^{p-1}$.
If $\|u\|_p = 0$, the assertion is immediate; otherwise divide by its positive $(p-1)$-st power. For $p = 1$, integrate the pointwise triangle inequality directly. QED

Lemma 6.8 — Chebyshev Inequality
For $c > 0$ and $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathsf{P})$ we have

$$\mathsf{P}(|X - \mathsf{E}(X)| \ge c) \le \frac{Var(X)}{c^2}.$$

HW 60 — Chebyshev's inequality
Prove Chebyshev's inequality.

Solution to HW 60. Square integrability implies integrability because $\mathsf{E}|X| \le (\mathsf{E}X^2)^{1/2}$ by Cauchy–Schwarz with the constant one. Thus $m = \mathsf{E}X$ is a finite real number, and $Z = (X-m)^2$ is a nonnegative integrable variable with $\mathsf{E}Z = \mathrm{Var}(X)$.
Let $A = \{|X - m| \ge c\}$. On A, $Z \ge c^2$; on A^c, $Z \ge 0$. These two observations combine into the pointwise bound $Z \ge c^2 \mathbf{1}_A$. Integrating and using $\mathsf{E}\mathbf{1}_A = \mathsf{P}(A)$ yields $\mathrm{Var}(X) = \mathsf{E}Z \ge c^2 \mathsf{P}(A)$. Division by the positive number c^2 proves the statement. This is Markov's inequality applied to the squared deviation, so no distributional shape, independence, or density is assumed. If the variance is zero, then $X = m$ almost surely and the same inequality correctly gives zero probability for any positive deviation threshold.

The next theorem shows how independence impacts the covariance of random variables.

Theorem 6.9 — Products of Independent Variables
Assume X and Y are independent r.v.'s in $\mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$. Then $X \cdot Y \in \mathcal{L}^1$ and $\mathsf{E}[XY] = \mathsf{E}[X]\mathsf{E}[Y]$. If in addition $X, Y \in \mathcal{L}^2$, then $Cov(X,Y) = 0$ and

$$Var(X+Y) = Var(X) + Var(Y).$$

Expanded proof. First suppose $X, Y \ge 0$. Choose simple approximations $X_n \uparrow X$, $Y_n \uparrow Y$ depending respectively only on X and Y. Write $X_n = \sum_i a_i \mathbf{1}_{A_i}$, $Y_n = \sum_j b_j \mathbf{1}_{B_j}$, where $A_i \in \sigma(X)$, $B_j \in \sigma(Y)$. Independence gives

$$\mathsf{E}[X_n Y_n] = \sum_{i,j} a_i b_j \mathsf{P}(A_i \cap B_j) = \sum_{i,j} a_i b_j \mathsf{P}(A_i)\mathsf{P}(B_j) = \mathsf{E}X_n \mathsf{E}Y_n.$$

The first equality expands the two simple functions and uses finite linearity and indicator intersections. The second uses independence of A_i and B_j. The last factors a finite double sum into its two marginal simple integrals.
The products increase to XY, so MON proves the non-negative formula. Apply it to $|X|, |Y|$, which remain independent: $\mathsf{E}|XY| = \mathsf{E}|X|\mathsf{E}|Y| < \infty$. We may therefore expand $XY = (X^+ - X^-)(Y^+ - Y^-)$, apply the non-negative formula to all four integrable terms, and recombine to obtain $\mathsf{E}(XY) = \mathsf{E}X\,\mathsf{E}Y$. QED

Remark: if $X, Y \in \mathcal{L}^1$ are not independent, then $XY \notin \mathcal{L}^1$ in general.
Now we want to prove an easy version of the Strong Law of Large Numbers, under more restrictive assumptions (boundedness of the fourth moment) than what is actually needed. The more restrictive assumptions allow for an easy proof.

Theorem 6.10 — A Strong Law of Large Numbers
Let $(X_n)_{n\in\mathbb{N}} \subset \mathcal{L}^4(\Omega, \mathcal{F}, \mathsf{P})$ *be independent and such that*

$$\mathsf{E}[X_n] = 0 \text{ and } \sup_{n\in\mathbb{N}} \mathsf{E}[X_n^4] \leq K$$

for some $K > 0$. *Setting* $S_n = \sum_{i=1}^n X_i$ *we have*

$$\mathsf{P}\left(\lim_{n\to\infty} n^{-1} S_n = 0\right) = 1.$$

Expanded proof. Let $S_n = \sum_{j=1}^n X_j$. In the expansion of S_n^4, a term containing an index exactly once has expectation zero: independence separates its zero-mean factor. Only four equal indices and two repeated pairs remain. If $\mathsf{E}X_j^4 \leq K$, Cauchy–Schwarz gives $\mathsf{E}X_j^2 \leq K^{1/2}$. Thus

$$\mathsf{E}S_n^4 = \sum_{j=1}^n \mathsf{E}X_j^4 + 6\sum_{i<j} \mathsf{E}X_i^2\,\mathsf{E}X_j^2 \leq nK + 3n(n-1)K \leq 3Kn^2.$$

The first equality in the fourth-moment display groups the surviving index patterns. There are six orderings for two pairs, accounting for the coefficient. The first inequality bounds each fourth moment by K and each product of second moments by K; the second uses $n + 3n(n-1) \leq 3n^2$. For fixed $\varepsilon > 0$, Markov gives $\mathsf{P}(|S_n/n| > \varepsilon) \leq \frac{3K}{\varepsilon^4 n^2}$. The bound is summable. The first Borel–Cantelli lemma implies that the threshold is exceeded only finitely often almost surely. Apply this to the countable thresholds $\varepsilon = 1/m$ and intersect the resulting probability-one events. On this intersection, every positive threshold eventually bounds $|S_n/n|$, proving convergence to zero.
QED

HW 61 — A strong law with nonzero common mean
Prove that if in the theorem above we replace $\mathsf{E}[X_n] = 0$ *with* $\mathsf{E}[X_n] = \mu$, *for some* $\mu \in \mathbb{R}$, *then* $n^{-1}S_n \to \mu$, P*-a.s. as* $n \to \infty$.

Solution to HW 61. The natural reduction is to remove the common mean, but the hypotheses of the centred statement must still be verified. Set $Y_n = X_n - \mu$. Subtracting a deterministic constant preserves independence, and $\mathsf{E}Y_n = \mathsf{E}X_n - \mu = 0$. For real x, the elementary convexity bound $(u+v)^4 \leq 8(u^4+v^4)$ for $u, v \geq 0$ gives

$$|x-\mu|^4 \leq (|x|+|\mu|)^4 \leq 8(|x|^4 + |\mu|^4).$$

Consequently

$$\sup_n \mathsf{E}|Y_n|^4 \leq 8(K + |\mu|^4) =: K' < \infty.$$

The centred fourth-moment law therefore applies to the independent Y_n.
To make its conclusion self-contained, let $T_n = \sum_{j=1}^n Y_j$. On expanding T_n^4, any term containing a variable to the first power and independent of all remaining factors has expectation zero, since that factor has mean zero. The surviving terms are the individual fourth powers and the pairs of squares: $\mathsf{E}T_n^4 = \sum_{i=1}^n \mathsf{E}Y_i^4 + 6\sum_{1\leq i<j\leq n} \mathsf{E}Y_i^2\,\mathsf{E}Y_j^2$. The factor six counts the arrangements of two copies of each of two indices. Cauchy–Schwarz gives $\mathsf{E}Y_i^2 \leq (\mathsf{E}Y_i^4)^{1/2} \leq \sqrt{K'}$. Hence $\mathsf{E}T_n^4 \leq nK' + 6\frac{n(n-1)}{2}K' \leq 3K'n^2$. It follows that $\sum_{n=1}^\infty \mathsf{E}\left|\frac{T_n}{n}\right|^4 \leq 3K'\sum_{n=1}^\infty n^{-2} < \infty$. By monotone convergence, the expectation of the nonnegative series $\sum_n |T_n/n|^4$ is this finite sum. A nonnegative variable with finite expectation is finite almost surely, so the series converges almost surely. Its terms must then tend to zero; thus $|T_n/n|^4 \to 0$ and $T_n/n \to 0$ almost surely.
Finally $S_n = T_n + n\mu$, so $S_n/n = \mu + T_n/n \to \mu$ on that same probability-one event. This verifies the reduction and shows explicitly why the uniform fourth-moment assumption survives centering.

Application — The Frequency Assertion in Example 1(b)

Proof. Use the fair-coin model of Example 1(b). In this model the head indicators X_j are independent fair Bernoulli variables. Put $Y_j = X_j - 1/2$. Then $\mathsf{E}Y_j = 0$ and $\mathsf{E}Y_j^4 = 1/16$: the first identity averages the two values $-1/2, 1/2$ with equal probabilities, and the second averages their common fourth power. Theorem 6.10 now applies and gives

$$\frac{1}{n}\sum_{j=1}^{n} X_j = \frac{1}{2} + \frac{1}{n}\sum_{j=1}^{n} Y_j \longrightarrow \frac{1}{2} \text{ almost surely.}$$

The equality substitutes $X_j = Y_j + 1/2$, sums the n constants and divides by n. The convergence uses the proved strong law on Y_j. Its convergence event is exactly the measurable event F already identified after Lemma 3.5. Therefore $\mathsf{P}(F) = 1$. QED

7. Product measure

Product Spaces and Measurable Rectangles

In this section we review rather superficially the theory of product measure. Our main objective is to state Fubini's theorem, which is a very useful tool.

Construction — Product Measurable Space

Let $(\mathcal{S}_1, \Sigma_1)$ and $(\mathcal{S}_2, \Sigma_2)$ be two measurable spaces. We denote $\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2$ meaning that $s \in \mathcal{S}$ if $s = (s_1, s_2)$ with $s_1 \in \mathcal{S}_1$ and $s_2 \in \mathcal{S}_2$. We further define

$$\mathcal{I} := \{B : B = B_1 \times B_2, B_1 \in \Sigma_1, B_2 \in \Sigma_2\}.$$

HW 62 — Measurable rectangles form a pi-system

Show that $\mathcal{I}$ is a π-system on $\mathcal{S}$.

Solution to HW 62. Let $A_1 \times A_2$ and $B_1 \times B_2$ be arbitrary members of $\mathcal{I}$. *We prove*

$$(A_1 \times A_2) \cap (B_1 \times B_2) = (A_1 \cap B_1) \times (A_2 \cap B_2)$$

by both inclusions. For the first, let (s_1, s_2) belong to the left-hand side. Membership in the first rectangle gives $s_i \in A_i$, and membership in the second gives $s_i \in B_i$, for $i = 1, 2$. Thus $s_i \in A_i \cap B_i$, so the pair belongs to the right-hand side. Conversely, let the pair belong to the right-hand side. Then each coordinate belongs to both corresponding sets. Consequently the pair belongs to both original rectangles, and hence to their intersection. Combining the inclusions proves the equality. Now $A_i \cap B_i \in \Sigma_i$ by the finite-intersection closure of Σ_i. Therefore the right-hand side belongs to $\mathcal{I}$ by its definition. The equality and this final membership establish the pi-system property, since the original two members were arbitrary. The whole product also belongs to $\mathcal{I}$, since $S_i \in \Sigma_i$. However, measurable rectangles alone need not form a sigma-algebra: their unions need not be single rectangles. The product sigma-algebra is the generated class $\sigma(\mathcal{I})$.

Definition — Product Sigma-Algebra

We define the product σ-algebra $\Sigma_1 \times \Sigma_2$ as

$$\Sigma_1 \times \Sigma_2 = \sigma(\mathcal{I})$$

and the product measurable space $(\mathcal{S}, \Sigma) = (\mathcal{S}_1 \times \mathcal{S}_2, \Sigma_1 \times \Sigma_2)$. We say that $f : \mathcal{S}_1 \times \mathcal{S}_2 \to \mathbb{R}$ is $\Sigma_1 \times \Sigma_2$-measurable if $f^{-1}(A) \in \Sigma_1 \times \Sigma_2$ for every $A \in \mathcal{B}(\mathbb{R})$.

Lemma 7.1 — Measurability of Sections

Let $(\mathcal{S}, \Sigma) = (\mathcal{S}_1 \times \mathcal{S}_2, \Sigma_1 \times \Sigma_2)$ and let $\mathcal{H}$ be the class of functions $f \in b\Sigma$ that satisfy

(i) $s_2 \mapsto f(s_1, s_2)$ is Σ_2-measurable for each $s_1 \in \mathcal{S}_1$;

(ii) $s_1 \mapsto f(s_1, s_2)$ is Σ_1-measurable for each $s_2 \in \mathcal{S}_2$.

Then $\mathcal{H} = b\Sigma$.

Expanded proof. Fix $s_1 \in \mathcal{S}_1$. The section of a rectangle $B_1 \times B_2$ is B_2 if $s_1 \in B_1$, and empty otherwise; both belong to Σ_2. Let $\mathcal{D} = \{A \in \Sigma_1 \times \Sigma_2 : A_{s_1} \in \Sigma_2\}$. Sections commute with complements and countable unions: for a point s_2, these assertions follow respectively by negating $(s_1, s_2) \in A$, or by asking whether this membership holds for at least one of the sets in the union. Consequently $\mathcal{D}$ is a sigma-algebra containing all rectangles, and hence the product sigma-algebra. For a Borel set B, $\{s_2 : f(s_1, s_2) \in B\} = \left(f^{-1}(B)\right)_{s_1} \in \Sigma_2$. Thus the section function is measurable. Exchange the two coordinates for the other section. This direct sigma-algebra argument proves the source's claim without requiring the functional monotone-class theorem here. Joint measurability implies section measurability; separate measurability alone does not imply joint measurability in general. QED

The take-home message of the lemma above is that if a function is measurable with respect to the product σ-algebra, then it is also measurable for each individual σ-algebra once all but one variables are fixed.
For $f \in b\Sigma$, given two finite measures for the bounded signed formulation μ_1 and μ_2 defined on $(\mathcal{S}_1, \Sigma_1)$ and $(\mathcal{S}_2, \Sigma_2)$, respectively, let us introduce

$$I_1^f(s_1) := \int_{\mathcal{S}_2} f(s_1, s_2)\mu_2(\mathrm{d}s_2) \quad \text{and} \quad I_2^f(s_2) := \int_{\mathcal{S}_1} f(s_1, s_2)\mu_1(\mathrm{d}s_1).$$

Qualification — Finite Measures in the Bounded Formulation
In the next lemma assume $\mu_1(\mathcal{S}_1), \mu_2(\mathcal{S}_2) < \infty$. For merely sigma-finite measures, the extension concerns non-negative or integrable functions, as explained in the proof and the following theorem.

Lemma 7.2 — Integration of Sections
Let $\mathcal{H}$ be the class of functions $f \in b\Sigma$ such that
(i) $I_1^f \in b\Sigma_1$ and $I_2^f \in b\Sigma_2$;
(ii)

$$\int_{\mathcal{S}_1} I_1^f(s_1)\mu_1(\mathrm{d}s_1) = \int_{\mathcal{S}_2} I_2^f(s_2)\mu_2(\mathrm{d}s_2).$$

Then $\mathcal{H} = b\Sigma$.

Expanded proof. Assume first that both measures are finite. Let $\mathcal{D}$ be the class of product-measurable sets A for which the two section-integral functions of $\mathbf{1}_A$ are measurable and their iterated integrals agree. The functions are bounded by the finite masses of the opposite spaces. For a rectangle $A_1 \times A_2$, its section-integral functions are $\mathbf{1}_{A_1}\mu_2(A_2)$ and $\mathbf{1}_{A_2}\mu_1(A_1)$. Their outer integrals both equal $\mu_1(A_1)\mu_2(A_2)$ by the indicator-integral definition. Thus rectangles, including the whole space, belong to $\mathcal{D}$.
If $A \in \mathcal{D}$, its complementary section integrals are $\mu_2(\mathcal{S}_2) - I_1^{\mathbf{1}_A}$ and $\mu_1(\mathcal{S}_1) - I_2^{\mathbf{1}_A}$. These are measurable bounded functions, and finite linearity makes their outer integrals equal: each is the product total mass minus the corresponding integral for A. No infinite subtraction occurs. For disjoint $(A_n) \subseteq \mathcal{D}$, the indicator of the union is the increasing limit of the finite indicator sums. To verify the identity, a point in one of the sets contributes one to exactly one summand, while a point outside the union contributes zero to all. MON in each section gives measurable section integrals for the union; MON again in the outer integrals gives equality of their increasing limits, since equality holds for each finite sum. Thus $\mathcal{D}$ is a Dynkin class containing the rectangle pi-system. The pi–d theorem in the lecturer's technical appendix gives all sets in the product sigma-algebra. This establishes the assertion for every indicator.
Finite linearity now gives it for every non-negative measurable simple function. For bounded non-negative f, choose the increasing dyadic simple approximation from Chapter 5 and apply MON first to each section and then to the outer integrals. The section functions remain bounded by $\|f\|_\infty$ times the opposite finite total mass. For bounded signed f, apply the non-negative assertion to f^+ and f^- and subtract their finite section integrals and finite outer integrals. These steps

prove both properties of the lemma for every bounded product-measurable f, without using a later course theorem.
For sigma-finite measures choose increasing finite-measure sets E_n and F_n exhausting the two spaces. For non-negative f, apply the finite result on $E_n \times F_n$ to $f \wedge n$. Equivalently, extend this bounded restriction by zero on the full space. These functions increase to f. Section MON and outer MON therefore give measurable possibly infinite section integrals and the same iterated-integral identity. When the common iterated integral of $|f|$ is finite, apply the non-negative result to f^+ and f^-; finiteness of the integral of $|f|$ justifies the subtractions wherever the signed section is defined, with arbitrary values on the resulting marginal null sets. Boundedness alone cannot replace finiteness of the measures: the constant one on $\mathbb{R}^2$ has infinite section integrals for Lebesgue measure. This explains the qualification on the original bounded signed statement. QED

Thanks to the lemma above we can define a measure on the product σ-algebra, because the lemma holds for any 1_F. Then we define a set function on $\Sigma = \Sigma_1 \times \Sigma_2$ as

$$\mu(F) = \int_{\mathcal{S}_2} \int_{\mathcal{S}_1} 1_F(s_1, s_2)\mu_1(\mathrm{d}s_1)\mu_2(\mathrm{d}s_2) = \int_{\mathcal{S}_1} \int_{\mathcal{S}_2} 1_F(s_1, s_2)\mu_2(\mathrm{d}s_2)\mu_1(\mathrm{d}s_1).$$

Theorem 7.3 — Fubini Theorem
The set function μ defined above is a measure on $\Sigma_1 \times \Sigma_2$. It is called product measure *ad it is denoted by $\mu = \mu_1 \times \mu_2$. That is the unique measure on Σ such that (recall μ_1, μ_2 are σ-finite)*

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2), \text{ for } A_1 \in \Sigma_1, A_2 \in \Sigma_2.$$

If $f \in (m\Sigma)^+$ then

$$\mu(f) = \int_{\mathcal{S}_1 \times \mathcal{S}_2} f\mathrm{d}\mu = \int_{\mathcal{S}_2} \int_{\mathcal{S}_1} f\mathrm{d}\mu_1\mathrm{d}\mu_2 = \int_{\mathcal{S}_1} \int_{\mathcal{S}_2} f\mathrm{d}\mu_2\mathrm{d}\mu_1.$$

If $f \in m\Sigma$ and $\mu(|f|) < \infty$, then (source formula 7.1) *continues to hold.*

Expanded proof. Define $\mu(F)$ by the iterated integral of $\mathbf{1}_F$. The preceding lemma with sigma-finite localization ensures measurability of the section integrals and equality of their two integration orders. We use countable additivity of μ, as we will see in HW 63. Also $\mu(\varnothing) = 0$, since the indicator of the empty set is zero. Thus μ is a measure. For rectangles, direct integration gives $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$, with $0 \cdot \infty = 0$. For uniqueness, first restrict two candidate measures to an exhausting sequence of finite-mass product rectangles. Their restrictions agree on the rectangle pi-system and have equal finite total masses. Lemma 1.2 makes them equal on each restriction. Continuity from below then makes them equal on the whole product space.
The iterated-integral formula holds for indicators and hence, by finite linearity, for non-negative simple functions. Approximate any non-negative measurable f by increasing simple functions and apply MON to each integral. Infinite non-negative integrals are allowed; this is the Tonelli form. When $\mu(|f|) < \infty$, both f^+ and f^- have finite integrals. Subtract their identities to obtain Fubini for signed f, without subtracting two infinities. QED

HW 63 — Countable additivity of the product measure
Verify that μ is countably additive.

Solution to HW 63. The section integral is a nonnegative measurable function of s_1, so the displayed mass is defined in $[0, \infty]$. For the empty set its indicator is zero everywhere, giving $\mu(\varnothing) = 0$. To establish countable additivity, take pairwise disjoint product-measurable sets $F_1, F_2, \ldots$ and put $F = \bigcup_n F_n$.
For every pair (s_1, s_2), disjointness gives $\mathbf{1}_F = \sum_{n \geq 1} \mathbf{1}_{F_n}$. Fix s_1 temporarily. The partial sums of these indicators increase, so monotone convergence for μ_2 gives

$$\int_{S_2} \mathbf{1}_F(s_1, s_2)\, \mu_2(ds_2) = \sum_{n \geq 1} \int_{S_2} \mathbf{1}_{F_n}(s_1, s_2)\, \mu_2(ds_2).$$

Write the nth section integral as $g_n(s_1) \geq 0$. Each is measurable, and $\sum_{n=1}^{N} g_n$ increases to $\sum_n g_n$. Applying monotone convergence a second time, now for μ_1, yields

$$\mu(F) = \int_{S_1} \sum_{n\geq 1} g_n \, d\mu_1 = \lim_N \sum_{n=1}^{N} \int_{S_1} g_n \, d\mu_1 = \sum_{n\geq 1} \int_{S_1} \int_{S_2} \mathbf{1}_{F_n} \, d\mu_2 \, d\mu_1 = \sum_{n\geq 1} \mu(F_n).$$

The order of integration has been kept fixed throughout.

In the displayed mass chain, the first equality inserts the section identity proved just above. The second uses MON on the outer integral and finite non-negative linearity for each partial sum. The third defines the infinite series and substitutes the definition of g_n. The last equality is the definition of $\mu(F_n)$. Thus the proof of additivity does not assume an interchange theorem for a measure whose measure axioms are still being checked. Both monotone-convergence applications allow infinite masses, so no finiteness of the union is needed.

On rectangles, the section integral is $\mathbf{1}_{A_1}(s_1)\mu_2(A_2)$, which gives the expected product mass $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$ with the usual zero-mass convention. Sigma-finiteness underlies the product-measure theorem and its uniqueness; the additivity calculation itself relies on nonnegative monotone limits.

If we drop the assumption that μ_1 and μ_2 are σ-finite Fubini's theorem may fail. An example is provided in the next homework.

HW 64 — Why sigma-finiteness matters for iterated integrals

Let $\mathcal{S}_1 = [0,1]$, $\mathcal{S}_2 = [0,1]$, $\Sigma_1 = \mathcal{B}([0,1])$ and $\Sigma_2 = \mathcal{B}([0,1])$. Let $\mu_1 = \mathcal{L}eb$ and $\mu_2 = \#$ (i.e., the counting measure). Take

$$F = \{(x,y) \in \mathcal{S}_1 \times \mathcal{S}_2 : x = y\}$$

and check that $F \in \Sigma_1 \times \Sigma_2$ but

$$\int_{\mathcal{S}_1} \int_{\mathcal{S}_2} 1_F \mu_2(\mathrm{d}s_2)\mu_1(\mathrm{d}s_1) \neq \int_{\mathcal{S}_2} \int_{\mathcal{S}_1} 1_F \mu_1(\mathrm{d}s_1)\mu_2(\mathrm{d}s_2).$$

Solution to HW 64. The subtraction map $(x,y) \mapsto x - y$ is continuous on the square, so its zero-level set D is closed and Borel. On $[0,1]^2$, the Borel sigma-algebra is the product of the coordinate Borel sigma-algebras: products of open intervals give a countable base, so open sets are countable unions of Borel rectangles. Thus this closed diagonal is measurable for the product sigma-algebra required in the question.

For the first order, fix x. The section in the y coordinate is the singleton $\{x\}$, containing exactly one point. Therefore

$$\int_{[0,1]} \mathbf{1}_D(x,y) \, \#(dy) = \#(\{x\}) = 1, \int_{[0,1]} 1 \, dx = 1.$$

The first equality integrates an indicator against counting measure on its singleton section. The second uses that the singleton has exactly one element. The separate integral of one is the Lebesgue length of $[0,1]$, which is one.

For the opposite order, fix y. The section in x is the singleton $\{y\}$, which has Lebesgue measure zero. Hence

$$\int_{[0,1]} \mathbf{1}_D(x,y) \, dx = \mathcal{L}eb(\{y\}) = 0, \int_{[0,1]} 0 \, \#(dy) = 0.$$

The first equality integrates the indicator of the singleton section against Lebesgue measure. The second uses that singletons have zero Lebesgue measure. The separate zero integral is zero by the definition of the non-negative integral.

The zero function has integral zero even on a space with infinite mass; it is not an undefined product $0 \cdot \infty$. Thus the two iterated values really are one and zero.

The missing hypothesis is sigma-finiteness of counting measure on the uncountable interval. A set of finite counting mass must contain only finitely many points. A countable union of such sets contains at most countably many points and cannot cover $[0,1]$. Consequently this counting

measure is not sigma-finite. Nonnegativity and measurability alone do not warrant the usual Tonelli interchange without the measure hypotheses; the direct section computations explain its failure here.

Tail Integrals and Joint Laws

Here is a useful application of Fubini's theorem. Take the product space

$$\big(\Omega \times [0,\infty), \mathcal{F} \times \mathcal{B}([0,\infty)), \mathsf{P} \times \mathcal{L}eb\big).$$

Let $X \in (m\mathcal{F})^+$ and consider the function $h = 1_A$ for the set

$$A = \{(\omega, x) : 0 \le x \le X(\omega)\}.$$

Denoting the product measure $\mu = \mathsf{P} \times \mathcal{L}eb$, we can compute $\mu(h)$ in two equivalent ways:

HW 65 — Expectation as the area under a survival function

Show that h is measurable on the product σ-algebra. Use Fubini's theorem to show that

$$\mu(h) = \mathsf{E}[X] = \int_0^\infty \mathsf{P}(X > x)\mathcal{L}eb(\mathrm{d}x).$$

Solution to HW 65. The lift $(\omega, x) \mapsto X(\omega)$ is product-measurable: its sublevel sets are rectangles $\{X \le c\} \times [0,\infty)$. The map $(\omega, x) \mapsto x$ is likewise measurable, since its sublevel sets are $\Omega \times [0, c]$ for $c \ge 0$ and empty for $c < 0$. Their comparison set $A = \{X(\omega) - x \ge 0\}$ is measurable; subtraction is legitimate because x is finite even when X is infinite. Thus its indicator h is nonnegative measurable.

The measures are sigma-finite: probability is finite, and the half-line is covered by the finite-length intervals $[0, m]$. Tonelli's theorem therefore identifies the product integral with both iterated nonnegative integrals, without first requiring a finite expectation. For a fixed ω, the section of A is $[0, X(\omega)]$ if $X(\omega) < \infty$, and the whole half-line otherwise. Its length equals $X(\omega)$ in either case. Consequently

$$\int h \, d\nu = \int_\Omega \left(\int_0^\infty h(\omega, x)\, dx \right) \mathsf{P}(d\omega) = \mathsf{E}X.$$

The first equality is Tonelli for the non-negative measurable h. The second uses that the inner section has length $X(\omega)$, also when this value is infinite, and then applies the definition of expectation.

In the opposite order, for each $x \ge 0$ the path section is $\{X \ge x\}$. Hence the same theorem gives $\int h \, d\nu = \int_0^\infty \mathsf{P}(X \ge x)\, dx$.

The question uses $>$ instead of $\ge$, which must be justified if X has atoms. Define $k(\omega, x) = \mathbf{1}_{\{X(\omega) = x\}}$, another nonnegative measurable function. For each fixed ω with finite $X(\omega)$ its support in x is a singleton of length zero; if $X(\omega) = \infty$, it is empty on the finite half-line. Tonelli thus gives

$$\int_0^\infty \mathsf{P}(X = x)\, dx = \int_\Omega \int_0^\infty k(\omega, x)\, dx\, \mathsf{P}(d\omega) = 0.$$

The first equality is Tonelli applied to k, identifying its section integral as $\mathsf{P}(X = x)$. The second uses that each section in the real coordinate is either a singleton or empty, and therefore has Lebesgue integral zero.

Since $\mathsf{P}(X \ge x) = \mathsf{P}(X > x) + \mathsf{P}(X = x)$, the two survival integrals coincide by nonnegative linearity. This proves the stated identity, including infinite expectations, without subtracting infinite integrals or assuming a continuous distribution.

Joint laws. Given two random variables X, Y on $(\Omega, \mathcal{F}, \mathsf{P})$, their joint law is a mapping

$$\mathcal{L}_{X,Y} : \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R}) \to [0, 1]$$

defined by $\mathcal{L}_{X,Y}(\Gamma) = \mathsf{P}((X, Y) \in \Gamma)$ for any $\Gamma \in \mathcal{B}(\mathbb{R}) \times \mathcal{B}(\mathbb{R})$. Noticing that

$$\pi(\mathbb{R}^2) = \{(-\infty, x] \times (-\infty, y], x, y \in \mathbb{R}\}$$

is a π-system on $\mathbb{R}^2$, it is enough to define $\mathcal{L}_{X,Y}$ on $\pi(\mathbb{R}^2)$ (which is a much easier task): the joint distribution of (X, Y) is

$$F_{X,Y}(x, y) = \mathsf{P}(X \le x, Y \le y).$$

We say that (X, Y) have a joint p.d.f. if there is $f_{X,Y} \ge 0$ such that

$$\mathsf{P}((X, Y) \in \Gamma) = \int_{\mathbb{R}^2} 1_\Gamma(x, y) f_{X,Y}(x, y) \mathrm{d}x\mathrm{d}y.$$

When the p.d.f. exists Fubini's theorem gives us the marginals:

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)\mathrm{d}y, \; f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\mathrm{d}x.$$

Product space and independence. Let X and Y be two random variables on $(\Omega, \mathcal{F}, \mathsf{P})$ with laws $\mathcal{L}_X$ and $\mathcal{L}_Y$ and CDFs F_X and F_Y. The following statements are equivalent:

(i) X and Y are independent;
(ii) $\mathcal{L}_{X,Y} = \mathcal{L}_X \times \mathcal{L}_Y$;
(iii) $F_{X,Y}(x, y) = F_X(x)F_Y(y)$;

Moreover, if (X, Y) admits joint p.d.f. then (i)–(iii) are also equivalent to

(iv) $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ for Lebesgue-almost every (x, y).

HW 66 — A pair of measurable coordinates is measurable

Let $(\Omega, \mathcal{F})$, $(\mathcal{S}_1, \Sigma_1)$, $(\mathcal{S}_2, \Sigma_2)$ be measurable spaces. For $i = 1, 2$ let $X_i : \Omega \to \mathcal{S}_i$ with $X^{-1} : \Sigma_i \to \mathcal{F}$. Define $\mathcal{S} := \mathcal{S}_1 \times \mathcal{S}_2$, $\Sigma := \Sigma_1 \times \Sigma_2$ and $X(\omega) := (X_1(\omega), X_2(\omega))$. Show that $X^{-1} : \Sigma \to \mathcal{F}$.

Solution to HW 66. Measurability means that the inverse image of every product-measurable set belongs to $\mathcal{F}$. The product sigma-algebra is generated by rectangles, so begin with one rectangle $A_1 \times A_2$, where $A_i \in \Sigma_i$. Directly from the coordinate definition,

$$X^{-1}(A_1 \times A_2) = \{X_1 \in A_1, X_2 \in A_2\} = X_1^{-1}(A_1) \cap X_2^{-1}(A_2) \in \mathcal{F}.$$

For the first inclusion in the rectangle pre-image formula, let $\omega_0 \in X^{-1}(A_1 \times A_2)$. By definition, $X(\omega_0) = (X_1(\omega_0), X_2(\omega_0))$ belongs to the rectangle, so $X_i(\omega_0) \in A_i$ for both coordinates. Thus $\omega_0 \in X_1^{-1}(A_1) \cap X_2^{-1}(A_2)$. Conversely, membership in this intersection gives both coordinate memberships and hence membership of the pair in $A_1 \times A_2$, which means $\omega_0 \in X^{-1}(A_1 \times A_2)$. The two inclusions justify the equality; the intermediate event notation names exactly these two simultaneous conditions. Each coordinate pre-image is in $\mathcal{F}$ by the assumed coordinate measurability, and the final membership follows from finite-intersection closure.

This verifies the generators, but we must explain why it extends to their whole sigma-algebra.

Let $\mathcal{D} = \{B \subset S_1 \times S_2 : X^{-1}(B) \in \mathcal{F}\}$. The full product space is in $\mathcal{D}$ because its inverse image is Ω. If $B \in \mathcal{D}$, then $X^{-1}(B^c) = \Omega \setminus X^{-1}(B) \in \mathcal{F}$, so $B^c \in \mathcal{D}$. For $B_n \in \mathcal{D}$, inverse images distribute over countable unions, and therefore

$$X^{-1}(\textstyle\bigcup_n B_n) = \bigcup_n X^{-1}(B_n) \in \mathcal{F}.$$

Thus $\mathcal{D}$ is a sigma-algebra containing every measurable rectangle.

By the defining minimality of $\Sigma_1 \otimes \Sigma_2$, this product sigma-algebra is contained in $\mathcal{D}$. Accordingly every set measurable in the target product has a measurable inverse image, proving the assertion. Neither probabilities nor independence of X_1, X_2 are involved; this is purely a measurability statement.

The homework above shows that $X(\omega) := (X_1(\omega), X_2(\omega))$ is a random variable on $(\Omega, \mathcal{F})$ and it makes sense to define its law $\mathcal{L}_X = \mathsf{P} \circ X^{-1}$ on $(\mathcal{S}, \Sigma)$.

Proofs — Joint Laws, Marginals and Independence

Proof. HW 66 proves that (X, Y) is measurable for the product Borel sigma-algebra. Its law is a probability measure by the pre-image argument used for a single random variable: every Borel pre-image is measurable, the whole-space pre-image is Ω, and countable disjoint unions have disjoint pre-images, so probability additivity verifies the measure axioms on the joint codomain as well.

The lower-left rectangles form a pi-system, since intersecting two takes the smaller endpoint in each coordinate. They generate the product Borel sigma-algebra. For example, $\mathbb{R} \times (-\infty, b] = \bigcup_n (-\infty, n] \times (-\infty, b]$; both inclusions follow because a real first coordinate is bounded above by some integer. The analogous first-coordinate rays also belong to the generated class. Proposition 1.1 then makes each coordinate projection measurable for that class: the pre-images of its generating lower rays have just been included, and Lemma 3.2 applies. Hence all rectangles $A \times B$ with Borel A, B belong, as intersections of these two coordinate pre-images. This gives one inclusion of the generated sigma-algebras; the reverse holds because every lower-left rectangle is itself a Borel rectangle. Lemma 1.2, with equal total masses one, therefore proves that the joint CDF determines the joint law.

When a joint density exists, for every Borel $A \subseteq \mathbb{R}$,

$$\mathcal{L}_X(A) = \mathcal{L}_{X,Y}(A \times \mathbb{R}) = \int_{A \times \mathbb{R}} f_{X,Y}(x,y)\, dx\, dy = \int_A \left(\int_{\mathbb{R}} f_{X,Y}(x,y)\, dy \right) dx.$$

The first equality identifies the same event $\{X \in A\}$; the second is the joint-density definition; the third is the non-negative Fubini identity already proved. Thus the inner integral is a marginal density, finite almost everywhere because its total integral is one. Interchanging the coordinates proves the formula for the other marginal. These are density versions; arbitrary representatives agree only almost everywhere by Theorem 5.8.

For (i) implies (ii), independence gives, for every Borel rectangle,

$$\mathcal{L}_{X,Y}(A \times B) = \mathsf{P}(X \in A, Y \in B) = \mathcal{L}_X(A)\mathcal{L}_Y(B) = (\mathcal{L}_X \times \mathcal{L}_Y)(A \times B).$$

The first equality is the joint law definition, the second is independence together with the marginal law definitions, and the third is the rectangle rule for the product measure. The rectangle pi-system generates the product sigma-algebra, and both measures are probabilities. Lemma 1.2 proves their agreement on all Borel sets. Conversely (ii) gives the same rectangle identity; every event in $\sigma(X)$ or $\sigma(Y)$ is a Borel pre-image by Lemma 3.1, so this is independence of their generated sigma-algebras.

Under (ii), evaluating lower-left rectangles gives (iii). Conversely (iii) says that the joint and product laws agree on the lower-left rectangle pi-system. Their equal finite total masses allow Lemma 1.2 to give (ii). This proves the equivalence of (i)–(iii).

Finally suppose a joint density is present. The non-negative function $g(x,y) = f_X(x) f_Y(y)$ has total integral $(\int f_X)(\int f_Y) = 1$ by Fubini and marginal normalization. Its density measure agrees with $\mathcal{L}_X \times \mathcal{L}_Y$ on rectangles, since the iterated rectangle integral factors into $\int_A f_X \int_B f_Y$. The density measure and the product measure are probabilities, so Lemma 1.2 extends this agreement to all Borel sets. Consequently (ii) implies $f_{X,Y} = g$ almost everywhere by density uniqueness in Theorem 5.8, which is the precise version of (iv). Conversely this almost-everywhere equality gives identical integrals over every Borel set, hence (ii) and independence. All four assertions have now been proved under their stated density hypothesis, rather than treating pointwise density values as unique. QED

8. Conditional expectation

Suppose X and Z are two random variables on $(\Omega, \mathcal{F}, \mathsf{P})$ with $X \in \{x_1, x_2, \ldots x_m\}$ and $Z \in \{z_1, z_2, \ldots z_n\}$. Then

$$\mathsf{P}(X = x_i | Z = z_j) = \frac{\mathsf{P}(X = x_i, Z = z_j)}{\mathsf{P}(Z = z_j)}$$

and

$$\mathsf{E}[X | Z = z_j] = \sum_{i=1}^{m} x_i \mathsf{P}(X = x_i | Z = z_j).$$

Without specifying the value of Z the conditional expectation $Y = \mathsf{E}[X|Z]$ is itself a random variable defined as

$$Y(\omega) = \mathsf{E}[X|Z = z_j] = y_j, \text{ if } Z(\omega) = z_j.$$

The events $\{\omega \in \Omega : Z(\omega) = z_j\}$ form a partition of Ω, i.e., they are pairwise disjoint $\{Z = z_j\} \cap \{Z = z_k\} = \varnothing$ if $j \neq k$ and $\Omega = \cup_{j=1}^{n}\{Z = z_j\}$. In this case we can easily construct the σ-algebra $\sigma(Z)$ which contains at most 2^n sets.
From (source formula 8.1) we see that Y is constant on each event $\{Z = z_j\}$. This corresponds to the fact that Y is $\sigma(Z)$-measurable.
For a cell $\{Z = z_j\}$ of probability zero, the conditional value may be chosen arbitrarily; the displayed ratios are used only on positive-probability cells.

HW 67 — Measurability in a finite conditional model

Show that Y is $\sigma(Z)$-measurable.

Solution to HW 67. By definition, $\sigma(Z)$ contains each atom $G_j = Z^{-1}(\{z_j\})$, since singletons are Borel. The piecewise-constant variable can be written as $Y = \sum_{j=1}^{n} y_j \mathbf{1}_{G_j}$. Each indicator is measurable for $\sigma(Z)$, and their finite linear combination is therefore measurable. More directly, for an arbitrary Borel set $B \subset \mathbb{R}$, $\{Y \in B\} = \bigcup_{j:y_j \in B} G_j$. For the first inclusion in the pre-image identity, take an outcome with $Y(\omega_0) \in B$. It belongs to exactly one partition atom G_j, on which $Y = y_j$; hence that atom is selected in the union. Conversely, membership in a selected atom gives $Y(\omega_0) = y_j \in B$. These two inclusions prove the equality. Each G_j belongs to $\sigma(Z)$ by its Borel-singleton pre-image definition, so finite-union closure gives the final membership.
This is a finite union of events known from Z, so belongs to $\sigma(Z)$. This latter calculation verifies all Borel inverse images and explains the informational meaning: once the value of Z is known, the corresponding value y_j of Y is determined.
If an atom has probability zero, the ratio formula defining a conditional mean on that atom is not used, because its denominator is zero. Assigning any finite constant there still makes Y measurable. It changes Y only on a null event, so it has no effect on the test-integral property or on its almost-sure conditional-expectation class.

We also observe that

$$\int_{\{Z=z_j\}} Y(\omega)\mathsf{P}(\mathrm{d}\omega) = y_j\mathsf{P}(Z = z_j) = \mathsf{E}[X|Z = z_j]\mathsf{P}(Z = z_j)$$
$$= \sum_{i=1}^{m} x_i\mathsf{P}(X = x_i|Z = z_j)\mathsf{P}(Z = z_j) = \sum_{i=1}^{m} x_i\mathsf{P}(X = x_i, Z = z_j) = \int_{\{Z=z_j\}} X(\omega)\mathsf{P}(\mathrm{d}\omega).$$

The first equality integrates the constant value y_j on $\{Z = z_j\}$. The second substitutes its definition as the conditional mean. The third expands that finite conditional mean. The fourth uses the defining conditional-probability quotient, cancelling the positive probability $\mathsf{P}(Z = z_j)$. The fifth sums the values of X over that event, which is the displayed restricted integral. Cells of probability zero may be assigned arbitrary conditional values and contribute zero to both integrals.
The formula above holds for any $G_j = \{Z = z_j\}$ and reads

$$\mathsf{E}[Y1_{G_j}] = \mathsf{E}[X1_{G_j}].$$

Since $\sigma(Z)$ is finite and $(G_j)_{j=1}^{n}$ is a partition, then for any $G \in \sigma(Z)$ the function $1_G(\omega)$ is a finite linear combination of $1_{G_j}(\omega)$. Thus, by linearity of expectation, for any $G \in \sigma(Z)$ we have

$$\mathsf{E}[Y1_G] = \mathsf{E}\big[Y\sum_{k=1}^{N}\alpha_k 1_{G_k}\big] = \sum_{k=1}^{N}\alpha_k\mathsf{E}\big[Y1_{G_k}\big] = \sum_{k=1}^{N}\alpha_k\mathsf{E}\big[X1_{G_k}\big] = \mathsf{E}[X1_G].$$

The first equality substitutes the finite indicator representation of G. The second is finite linearity of expectation. The third applies the identity on each partition cell just proved. The fourth reverses finite linearity and reconstructs $\mathbf{1}_G$.

Theorem 8.1 — Conditional Expectation (Kolmogorov, 1933)
Let $(\Omega, \mathcal{F}, \mathsf{P})$ and $X \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$. Let $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$. Then there exists a random variable Y such that

(a) $Y \in m\mathcal{G}$
(b) $\mathsf{E}[|Y|] < \infty$
(c) for every $G \in \mathcal{G}$ we have

$$\mathsf{E}[Y 1_G] = \mathsf{E}[X 1_G].$$

Moreover, if there is another r.v. $\tilde{Y}$ that satisfies (a)–(c), then $\mathsf{P}(\tilde{Y} = Y) = 1$. A random variable with properties (a)–(c) is called conditional expectation of X given $\mathcal{G}$ *and it is denoted*

$$Y(\omega) = \mathsf{E}[X|\mathcal{G}](\omega).$$

Intuition. In the special case $\mathcal{G} = \sigma(Z)$ we have that $\mathsf{E}[X|\sigma(Z)]$ is the best estimate of the value of X given the knowledge about the value of Z.
Least squares predictor. If $\mathsf{E}[|X|^2] < \infty$, then we will see in the proof of the theorem that $Y = \mathsf{E}[X|\mathcal{G}]$ is a version of the orthogonal projection of X onto $\mathcal{L}^2(\Omega, \mathcal{G}, \mathsf{P})$. So Y is the random variable that minimises

$$T \mapsto \mathsf{E}[(X - T)^2]$$

among all random variables $T \in \mathcal{L}^2(\Omega, \mathcal{G}, \mathsf{P})$.

HW 68 — Conditioning on the trivial sigma-algebra
Prove that if $\mathcal{G} = \{\Omega, \varnothing\}$ then $\mathsf{E}[X|\mathcal{G}] = \mathsf{E}[X]$ (i.e., the conditional expectation is deterministic and equal to unconditional expectation).

Solution to HW 68. Conditional expectation is characterised by three requirements: a candidate Y must be $\mathcal{G}$-measurable, must be integrable, and must satisfy $\mathsf{E}[Y\mathbf{1}_G] = \mathsf{E}[X\mathbf{1}_G]$ for every $G \in \mathcal{G}$. Rather than manipulate a conditional-expectation symbol formally, we verify these requirements for the constant $Y = \mathsf{E}X$.
Integrability of X makes $\mathsf{E}X$ a finite real number. A constant variable is measurable for every sigma-algebra, including the trivial one, and has $\mathsf{E}|Y| = |\mathsf{E}X| < \infty$. There are exactly two measurable test events. For $G = \varnothing$, both test integrals are zero. For $G = \Omega$, the probability total is one, so

$$\mathsf{E}[Y\mathbf{1}_\Omega] = \mathsf{E}Y = \mathsf{E}X = \mathsf{E}[X\mathbf{1}_\Omega].$$

The first equality substitutes $\mathbf{1}_\Omega = 1$. The second uses that Y is the constant $\mathsf{E}X$ on a probability space of total mass one. The third inserts $\mathbf{1}_\Omega = 1$ in the expectation of X.
Thus the candidate satisfies the full defining test identity.
To see directly that every candidate has this value, let Z be measurable for the trivial sigma-algebra. If $Z(\omega_1) < Z(\omega_2)$ for two outcomes, choose a rational q strictly between the two real values. Then $\{Z \le q\}$ contains ω_1 and excludes ω_2; it is neither empty nor the whole space, contradicting measurability. Thus Z is constant. Its defining test on Ω forces that constant to equal $\mathbb{E}X$, because $\mathbb{P}(\Omega) = 1$. This proves the claim using the present definition, without assuming the general uniqueness result proved later.

Expanded proof. Let $Y, \widetilde{Y}$ be integrable, $\mathcal{G}$-measurable candidates. Their defining integral identities imply $\mathsf{E}[(Y - \widetilde{Y})\mathbf{1}_G] = 0$ for every $G \in \mathcal{G}$. Suppose $\mathsf{P}(Y > \widetilde{Y}) > 0$. We use

$$\{Y > \widetilde{Y}\} = \bigcup_{m \ge 1} \{Y > \widetilde{Y} + 1/m\},\ \{Y > \widetilde{Y} + 1/m\} \in \mathcal{G},$$

as we will see in HW 69. Some level $G_m = \{Y > \widetilde{Y} + 1/m\}$ has positive probability, since otherwise its countable union would be null. Use this level as the test event:

$$0 = \mathsf{E}[(Y - \widetilde{Y})\mathbf{1}_{G_m}] \ge m^{-1}\mathsf{P}(G_m) > 0,$$

The equality is the difference of the two defining test identities on G_m. The first inequality integrates $Y - \widetilde{Y} > 1/m$ there, and the final strict inequality uses $\mathbb{P}(G_m) > 0$. All subtractions are of finite integrals because both candidates are integrable.
a contradiction. Hence $\mathsf{P}(Y > \widetilde{Y}) = 0$. Interchanging the candidates gives the reverse inequality almost surely. Thus $Y = \widetilde{Y}$ almost surely, not necessarily at every outcome. QED

HW 69 — Measurable level sets in conditional uniqueness
Prove that $\{Y > \tilde{Y}\} = \cup_{n\geq 1}\{Y > \tilde{Y} + \frac{1}{n}\}$ *and* $\{Y > \tilde{Y} + n^{-1}\} \in \mathcal{G}$.

Solution to HW 69. Set $D = Y - \widetilde{Y}$. Subtraction is a continuous real operation, so D is $\mathcal{G}$-measurable. The set $\{D > 1/n\}$ is the inverse image of the open ray $(1/n, \infty)$ under D, hence belongs to $\mathcal{G}$. This proves measurability of every event in the union.
Each such event is contained in $\{D > 0\}$, giving one inclusion immediately. For the reverse inclusion, fix a point with $D(\omega) > 0$. The Archimedean property supplies an integer $n > 1/D(\omega)$, so $1/n < D(\omega)$; the point belongs to the nth event. Therefore every point with a strictly positive difference is captured by the countable union, establishing the equality.
This identity supplies the strict numerical margin needed in a uniqueness proof. If $\mathsf{P}(Y > \widetilde{Y}) > 0$, then at least one $G_n = \{D > 1/n\}$ must have positive probability; otherwise their countable union would be null. On that event, if D is integrable, $\mathsf{E}[D\mathbf{1}_{G_n}] \geq \frac{1}{n}\mathsf{P}(G_n) > 0$. Thus a difference whose integral is zero on every $\mathcal{G}$-event cannot be positive with positive probability. Applying the same argument to $-D$ rules out a negative difference as well. The countable union is important: no selection of a different uncontrolled event for every point is needed.

Orthogonal Projection and Construction
In order to prove *existence* of conditional expectation we need a result about *projections* in $\mathcal{L}^2$-spaces.

Theorem 8.2 — Orthogonal Projection
Let $\mathcal{K}$ *be a* complete *vector subspace of* $\mathcal{L}^2(\Omega, \mathcal{F}, \mathsf{P})$*, i.e., for any Cauchy sequence* $(v_n)_{n\in\mathbb{N}} \subset \mathcal{K}$ *the limit exists in* $\mathcal{K}$ *(*$\exists v \in \mathcal{K} : \|v_n - v\|_2 \to 0$ *as* $n \to \infty$*).*
Then, given $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathsf{P})$ *there exists* $Y \in \mathcal{K}$ *such that*
(i)

$$\|X - Y\|_2 = \Delta := \inf\{\|X - W\|_2 : W \in \mathcal{K}\};$$

(ii) $\langle X - Y, Z\rangle_2 = 0$ *for all* $Z \in \mathcal{K}$.

Proof of Theorem 8.2
Expanded proof. Put $\Delta = \inf_{W\in\mathcal{K}} \|X - W\|_2$. Since $0 \in \mathcal{K}$, this infimum is finite. Choose $Y_n \in \mathcal{K}$ with $\|X - Y_n\|_2 \to \Delta$. Their midpoints remain in the vector space $\mathcal{K}$. The parallelogram identity gives

$$\begin{aligned}\|Y_n - Y_m\|_2^2 &= 2\|X - Y_n\|_2^2 + 2\|X - Y_m\|_2^2 - 4\|X - (Y_n + Y_m)/2\|_2^2\\ &\leq 2\|X - Y_n\|_2^2 + 2\|X - Y_m\|_2^2 - 4\Delta^2 \longrightarrow 0.\end{aligned}$$

The equality is the parallelogram identity applied to $X - Y_n$ and $X - Y_m$. The inequality uses $\|X - (Y_n + Y_m)/2\|_2 \geq \Delta$, since the midpoint belongs to $\mathcal{K}$. The last limit substitutes the two chosen distances converging to Δ, giving $2\Delta^2 + 2\Delta^2 - 4\Delta^2 = 0$.
Completeness of $\mathcal{K}$ supplies $Y \in \mathcal{K}$ with $Y_n \to Y$ in norm. The reverse triangle inequality gives

$$|\|X - Y_n\|_2 - \|X - Y\|_2| \leq \|Y_n - Y\|_2,$$

so $\|X - Y\|_2 = \Delta$. For $Z \in \mathcal{K}$, $t \in \mathbb{R}$, minimality and $Y + tZ \in \mathcal{K}$ give

$$0 \leq \|X - Y - tZ\|_2^2 - \|X - Y\|_2^2 = t^2\|Z\|_2^2 - 2t\langle X - Y, Z\rangle.$$

If the inner product were nonzero, sufficiently small t of its sign would make the right-hand side negative. Thus it is zero. If Y' were another minimizer, the same argument would make $X - Y'$

orthogonal to $\mathcal{K}$. Subtracting the two orthogonality identities and testing with $Y - Y' \in \mathcal{K}$ gives $\|Y - Y'\|_2^2 = 0$. The projection is unique as an L^2 class, that is, almost surely. QED

Existence in Theorem 8.1

Expanded proof. Square-integrable case. The space of $\mathcal{G}$-measurable L^2 classes is closed: from $Y_n \to Y$ in L^2, choose a subsequence with $\|Y_{n_j} - Y\|_2 \le 2^{-j}$. Markov and Borel–Cantelli give almost sure convergence of this subsequence. Its pointwise limsup, set to zero wherever it is not finite, is $\mathcal{G}$-measurable and agrees with Y almost surely. Completeness of L^2 makes the closed subspace complete. The projection theorem 8.2 just proved supplies $Y \in L^2(\mathcal{G})$ with $\langle X - Y, Z\rangle = 0$ for all $Z \in L^2(\mathcal{G})$. Take $Z = \mathbf{1}_G$, $G \in \mathcal{G}$, which belongs to L^2 on a probability space. Then $\mathsf{E}[Y\mathbf{1}_G] = \mathsf{E}[X\mathbf{1}_G]$. Also $L^2 \subseteq L^1$, so all requirements of conditional expectation hold.

Non-negative integrable case. Let $X_n = X \wedge n$ and construct $Y_n = \mathsf{E}[X_n \mid \mathcal{G}]$ by the square-integrable case. Each $Y_n \ge 0$ almost surely: if a measurable negative level had positive probability, its integral would be negative, contradicting the non-negative integral of X_n over that same test event. We use the monotonicity $Y_n \le Y_{n+1}$, as we will see in HW 70. Removing one countable union of null events, choose versions non-negative and increasing simultaneously. Then $Y = \sup_n Y_n$ is $\mathcal{G}$-measurable, and MON gives

$$\mathsf{E}[Y\mathbf{1}_G] = \lim_n \mathsf{E}[Y_n\mathbf{1}_G] = \lim_n \mathsf{E}[X_n\mathbf{1}_G] = \mathsf{E}[X\mathbf{1}_G].$$

In the test-integral chain, the first equality is MON for $Y_n\mathbf{1}_G$, the second is the defining identity for each bounded approximation, and the last is MON for $X_n\mathbf{1}_G$. The same fixed event $G \in \mathcal{G}$ is used throughout.

Taking $G = \Omega$ yields $\mathsf{E}Y = \mathsf{E}X < \infty$. Thus Y is finite almost surely and integrable.

General integrable case. Apply the non-negative construction to X^+ and X^-. Their conditional expectations have finite expectations. Subtract them and use linearity of the test integrals to verify the definition for X; no undefined difference occurs. The previously proved uniqueness completes the construction. QED

HW 70 — Monotonicity of conditional expectation

For $Y_n = \mathsf{E}[X_n|\mathcal{G}]$, prove that $X_{n+1} \ge X_n \implies Y_{n+1} \ge Y_n$.

Solution to HW 70. Let $D = Y_{n+1} - Y_n$. It is integrable and $\mathcal{G}$-measurable, since each conditional expectation has these properties. For any $G \in \mathcal{G}$, subtracting their defining test identities yields

$$\mathsf{E}[D\mathbf{1}_G] = \mathsf{E}[(X_{n+1} - X_n)\mathbf{1}_G] \ge 0.$$

The inequality holds because the variable inside the last expectation is nonnegative almost surely. A nonnegative unconditional mean alone would not prove $D \ge 0$; what matters is that the inequality holds for *every* measurable test event.

Suppose, towards a contradiction, that $\mathsf{P}(D < 0) > 0$. Since $\{D < 0\} = \bigcup_{k\ge1}\{D < -1/k\}$, one of the events $G_k = \{D < -1/k\}$ has positive probability. It is a valid test event because D is $\mathcal{G}$-measurable. On that event $D \le -1/k$, and hence

$$0 \le \mathsf{E}[D\mathbf{1}_{G_k}] \le -\frac{1}{k}\mathsf{P}(G_k) < 0,$$

a contradiction. Therefore $D \ge 0$ almost surely, proving conditional monotonicity from its definition.

For an increasing sequence (X_n) there are countably many adjacent inequalities. For each pair the proof may give its own null exceptional set, but the union of those null sets is still null. Outside that common set, all the inequalities $Y_n \le Y_{n+1}$ hold simultaneously, allowing one to form a genuine pointwise increasing sequence of versions when proving conditional monotone convergence.

Conditional Densities and Operational Properties

How do we connect the abstract definition of conditional expectation and the classical one? Consider two r.v.s (X, Z) with joint p.d.f. $f_{X,Z}(x, z)$. Assume that h is Borel measurable and $\mathsf{E}|h(X)| < \infty$. Choose the marginal version $f_Z(z) = \int_{\mathbb{R}} f_{X,Z}(x, z)\, dx$. The traditional construction of conditional probability reads

$$\mathsf{E}[h(X)|Z = z] = \int_{\mathbb{R}} h(x) f_{X|Z}(x|z) \mathrm{d}x$$

with

$$f_{X|Z}(x|z) = \begin{cases} \frac{f_{X,Z}(x,z)}{f_Z(z)}, & \text{if } f_Z(z) \neq 0, \\ 0, & \text{otherwise.} \end{cases}$$

The connection to Kolmogorov's theorem is as follows: set

$$g(z) = \int_{\mathbb{R}} h(x) f_{X|Z}(x|z) \mathrm{d}x$$

and define $Y(\omega) = g(Z(\omega))$. We expect that $Y(\omega) = \mathsf{E}[h(X)|\sigma(Z)]$. Let's check it. The section integral defining g is measurable by Fubini. The estimate

$$\int_{\mathbb{R}} |g(z)| f_Z(z)\, dz \leq \int_{\mathbb{R}} \int_{\mathbb{R}} |h(x)| f_{X,Z}(x, z)\, dx\, dz = \mathsf{E}|h(X)| < \infty$$

uses the integral triangle inequality in each conditional section, then the joint-density formula and Fubini. Thus $g(Z)$ is measurable and integrable by composition and the transfer formula. On any Borel $f_Z(z)\, dz$-null set where a section is undefined or infinite, define $g(z) = 0$. This changes no probability value for Z. Where $f_Z = 0$, the joint section is zero almost everywhere by non-negativity, so it also contributes zero to all tested integrals.
Since $\sigma(Z) = (Z^{-1}(B), B \in \mathcal{B}(\mathbb{R}))$ then we can consider $G = \{Z \in B\}$ and notice that

$$\mathsf{E}[Y 1_G] = \mathsf{E}[g(Z) 1_{\{Z \in B\}}] = \int_B g(z) f_Z(z) \mathrm{d}z = \int_B \int_R h(x) f_{X|Z}(x|z) \mathrm{d}x f_Z(z) \mathrm{d}z$$
$$= \int_B \int_R h(x) f_{X,Z}(x, z) \mathrm{d}x \mathrm{d}z = \mathsf{E}[h(X) 1_{\{Z \in B\}}] = \mathsf{E}[h(X) 1_G],$$

The first equality substitutes $Y = g(Z)$ and $G = \{Z \in B\}$. The second is the transfer formula for Z. The third substitutes the defining integral for g. The fourth uses $f_{X|Z}(x \mid z) f_Z(z) = f_{X,Z}(x, z)$ almost everywhere. The fifth is the joint-density transfer formula and Fubini, under the stated integrability assumption. The sixth substitutes the description of G again.
as needed.

We now provide a long list of properties of conditional expectation. Many of those are straightforward to prove using arguments we know already for analogous properties of Lebesgue integrals. You are encouraged to attempt a proof of each property below. In the list that follows $(X_n)_{n \in \mathbb{N}}, X \subset \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$,$\mathcal{G}, \mathcal{H} \subset \mathcal{F}$ are σ-algebras.

(a) $Y = \mathsf{E}[X|\mathcal{G}] \implies \mathsf{E}[X] = \mathsf{E}[Y]$;
(b) $X \in m\mathcal{G} \implies \mathsf{E}[X|\mathcal{G}] = X$;
(c) $\mathsf{E}[\alpha X_1 + \beta X_2|\mathcal{G}] = \alpha \mathsf{E}[X_1|\mathcal{G}] + \beta \mathsf{E}[X_2|\mathcal{G}]$, for $\alpha, \beta \in \mathbb{R}$;
(d) $X \geq 0 \implies \mathsf{E}[X|\mathcal{G}] \geq 0$
(e) $0 \leq X_n \uparrow X \implies \mathsf{E}[X_n|\mathcal{G}] \uparrow \mathsf{E}[X|\mathcal{G}]$;
(f) $X_n \geq 0 \implies \liminf_{n\to\infty} \mathsf{E}[X_n|\mathcal{G}] \geq \mathsf{E}[\liminf_{n\to\infty} X_n|\mathcal{G}]$;
(g) $|X_n| \leq V \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ and $\mathsf{P}(X_n \to X) = 1$ imply $\lim_{n\to\infty} \mathsf{E}[X_n|\mathcal{G}] = \mathsf{E}[X|\mathcal{G}]$;
(h) If $c : \mathbb{R} \to \mathbb{R}$ is convex and $\mathsf{E}[|c(X)|] < \infty$, then $\mathsf{E}[c(X)|\mathcal{G}] \geq c(\mathsf{E}[X|\mathcal{G}])$;
(i) (**Tower property**) If $\mathcal{H} \subset \mathcal{G}$ then $\mathsf{E}[\mathsf{E}[X|\mathcal{G}]|\mathcal{H}] = \mathsf{E}[X|\mathcal{H}]$;
(j) $Z \in m\mathcal{G} \implies \mathsf{E}[ZX|\mathcal{G}] = Z\mathsf{E}[X|\mathcal{G}]$;
(k) If $\mathcal{H}$ is independent of $\sigma(\sigma(X), \mathcal{G}) = \sigma(\sigma(X) \cup \mathcal{G})$, then $\mathsf{E}[X|\sigma(\mathcal{H}, \mathcal{G})] = \mathsf{E}[X|\mathcal{G}]$ (special case: if $\mathsf{E}[X|\mathcal{H}] = \mathsf{E}[X]$).

Qualifications for the complete statement list. In (j), an unbounded measurable multiplier Z additionally requires $ZX \in L^1$. In (k), the special case $\mathsf{E}[X \mid \mathcal{H}] = \mathsf{E}X$ follows when X is independent of $\mathcal{H}$, for example by taking the other conditioning sigma-algebra trivial. That equality alone is not a characterization of independence.

Proofs of the Conditional Expectation Properties

All conditional identities below hold almost surely. The complete statement list (a)–(k) precedes these proofs.

Proof of (a). Use the defining test event $G = \Omega$: $\mathsf{E}[\mathsf{E}(X \mid \mathcal{G})] = \mathsf{E}X$.

Proof of (b). If X is already $\mathcal{G}$-measurable, it satisfies every condition in the definition. Uniqueness gives the identity.

Proof of (c). Linear combinations of the two conditional expectations are measurable and integrable; linearity of integrals verifies the test identity for the same linear combination of X, Y. Uniqueness identifies it.

Proof of (d). For $X \geq 0$, a negative measurable level of $Y = \mathsf{E}[X \mid \mathcal{G}]$ of positive probability would give

$$\int_G Y < 0 \leq \int_G X,$$

contradicting the defining test identity. Thus $Y \geq 0$. Applying this to a non-negative difference proves monotonicity, already established in HW 70.

Proof of (e). For $0 \leq X_n \uparrow X$, choose conditional versions Y_n simultaneously non-negative and increasing by removing one countable union of null events. MON on each test event gives $\int_G \sup_n Y_n = \int_G X$. If X is integrable, the test $G = \Omega$ gives integrability of the limit and uniqueness identifies it. For a non-integrable non-negative limit the same construction defines the extended conditional expectation.

Proof of (f). Put $V_n = \inf_{m \geq n} X_m$. Then $V_n \uparrow \liminf X_n$ and $V_n \leq X_m$ for $m \geq n$. Apply monotonicity and (e):

$$\mathsf{E}[\liminf X_n \mid \mathcal{G}] = \lim_n \mathsf{E}[V_n \mid \mathcal{G}] \leq \lim_n \inf_{m \geq n} \mathsf{E}[X_m \mid \mathcal{G}].$$

Proof of (g). Apply (f) to the non-negative sequences $V + X_n$ and $V - X_n$, where $|X_n| \leq V \in L^1$. Using linearity, the first gives $\mathsf{E}[X \mid \mathcal{G}] \leq \liminf_n \mathsf{E}[X_n \mid \mathcal{G}]$; the second gives the reverse upper bound for the limsup. The two inequalities imply the claimed limit.

Proof of (h). The elementary convexity argument proved with Theorem 6.2 represents a finite convex function as the supremum of its supporting affine lines chosen at rational points. That earlier proof established both continuity and the locally bounded slopes needed for this representation. For each line $a_j x + b_j \leq c(x)$, positivity and linearity give $a_j \mathsf{E}[X \mid \mathcal{G}] + b_j \leq \mathsf{E}[c(X) \mid \mathcal{G}]$. Outside a single countable union of null events, take the supremum in j to obtain conditional Jensen.

Proof of (i). If $\mathcal{H} \subseteq \mathcal{G}$, then $\mathsf{E}[\mathsf{E}[X \mid \mathcal{G}] \mid \mathcal{H}]$ is integrable and $\mathcal{H}$-measurable. For $H \in \mathcal{H}$, two test identities give its integral over H equal to $\int_H X$. Uniqueness gives $\mathsf{E}[X \mid \mathcal{H}]$. The other nesting order follows because an $\mathcal{H}$-measurable variable is already $\mathcal{G}$-measurable.

Proof of (j). For $Z = \mathbf{1}_A$, $A \in \mathcal{G}$, test on $G \cap A$ to obtain the product identity. Finite linearity proves it for simple Z. Bounded measurable Z follow by uniformly bounded simple approximation and DOM in the test integrals. For unbounded Z, additionally assume $ZX \in L^1$. Apply the bounded case first to $|Z| \wedge n$ and $|X|$. MON gives $\mathsf{E}[|Z|\mathsf{E}(|X| \mid \mathcal{G})] = \mathsf{E}|ZX| < \infty$. Since $|\mathsf{E}(X \mid \mathcal{G})| \leq \mathsf{E}(|X| \mid \mathcal{G})$, the proposed product is integrable. Truncate Z and use DOM on both sides of each test identity. Uniqueness completes the argument. Measurability alone would not guarantee these signed integrals exist.

Proof of (k). Let $Y = \mathsf{E}[X \mid \mathcal{G}]$. For $A \in \mathcal{H}, B \in \mathcal{G}$, independence of $\mathcal{H}$ from $\sigma(X, \mathcal{G})$ gives

$$\mathsf{E}[X\mathbf{1}_A\mathbf{1}_B] = \mathsf{P}(A)\mathsf{E}[X\mathbf{1}_B] = \mathsf{P}(A)\mathsf{E}[Y\mathbf{1}_B] = \mathsf{E}[Y\mathbf{1}_A\mathbf{1}_B].$$

The first equality uses independence of $\mathcal{H}$ from $\sigma(X, \mathcal{G})$, since $X\mathbf{1}_B$ is integrable and measurable for that sigma-algebra. The second is the defining test identity for $Y = \mathsf{E}[X \mid \mathcal{G}]$ on $B \in \mathcal{G}$. The third uses the same independence with the $\mathcal{G}$-measurable variable $Y\mathbf{1}_B$.

The rectangles $A \cap B$ form a pi-system generating $\sigma(\mathcal{H}, \mathcal{G})$. To extend the integral identity, use the two finite positive measures $\nu_+(D) = \mathsf{E}[(X-Y)^+ \mathbf{1}_D]$ and $\nu_-(D) = \mathsf{E}[(X-Y)^- \mathbf{1}_D]$. They agree on the rectangles by the preceding calculation and have equal total masses because $\mathsf{E}X = \mathsf{E}Y$. The previously established uniqueness Lemma 1.2 extends their agreement to the generated sigma-algebra, which is the required integral identity. Thus Y is also the conditional expectation given $\sigma(\mathcal{H}, \mathcal{G})$. In particular, if X is independent of $\mathcal{H}$, $\mathsf{E}[X \mid \mathcal{H}] = \mathsf{E}X$; that equality by itself does not imply independence.

HW 71 — Statement

Show that $\|\mathsf{E}[X|\mathcal{G}]\|_p \leq \|X\|_p$.

Solution to HW 71. First the conditional expectation exists as an ordinary integrable variable. Indeed $|x| \leq 1 + |x|^p$ for all real x, so $\mathsf{E}|X| \leq 1 + \mathsf{E}|X|^p < \infty$. The convex function $\phi(x) = |x|^p$ has integrable value $\phi(X)$ by hypothesis. Conditional Jensen's inequality says $\phi(\mathsf{E}[X \mid \mathcal{G}]) \leq \mathsf{E}[\phi(X) \mid \mathcal{G}]$ almost surely, provided X and $\phi(X)$ are integrable. Both required assumptions have just been checked. Substitution gives $|Y|^p \leq \mathsf{E}[|X|^p \mid \mathcal{G}]$. The right side is a nonnegative integrable variable with mean $\mathsf{E}|X|^p$: this follows by taking the whole sample space as the test event in the conditional-expectation definition. Integrating the inequality therefore proves at once that $Y \in L^p$ and that

$$\mathsf{E}|Y|^p \leq \mathsf{E}\big[\mathsf{E}(|X|^p \mid \mathcal{G})\big] = \mathsf{E}|X|^p.$$

Taking pth roots preserves order on nonnegative numbers and yields the desired norm contraction. For $p = 1$, the same argument is the familiar bound $|\mathsf{E}[X \mid \mathcal{G}]| \leq \mathsf{E}[|X| \mid \mathcal{G}]$ followed by integration. If an L^∞ version is desired, no power limit is needed: an almost-sure bound $-M \leq X \leq M$ is preserved by conditional monotonicity, so $|Y| \leq M$, and taking the infimum over such essential bounds gives $\|Y\|_\infty \leq \|X\|_\infty$. The contraction expresses loss, rather than amplification, of moment size when information is averaged out.

Lemma 8.3 — Conditioning under Independence

Let $X_1, \ldots, X_r$ be independent with laws $\Lambda_1, \ldots, \Lambda_r$, and let $h : \mathbb{R}^r \to \mathbb{R}$ be bounded and Borel measurable. Put $\gamma^h(x_1) = \mathsf{E}[h(x_1, X_2, \ldots, X_r)]$. Then $\gamma^h(X_1) = \mathsf{E}[h(X_1, \ldots, X_r) \mid \sigma(X_1)]$ almost surely.

Expanded proof. Independence identifies the joint law with $\Lambda_1 \times \cdots \times \Lambda_r$. The function $\gamma^h(x_1) = \int h(x_1, x_2, \ldots, x_r)\Lambda_2(\mathrm{d}x_2) \cdots \Lambda_r(\mathrm{d}x_r)$ is Borel measurable by product integration and bounded by $\|h\|_\infty$. Thus $\gamma^h(X_1)$ is $\sigma(X_1)$-measurable and integrable. For any $G = \{X_1 \in B\}$, B Borel, Fubini gives

$$\mathsf{E}[\mathbf{1}_G \gamma^h(X_1)] = \int \mathbf{1}_B(x_1) h(x_1, \ldots, x_r) \Lambda_1(\mathrm{d}x_1) \cdots \Lambda_r(\mathrm{d}x_r) = \mathsf{E}[\mathbf{1}_G h(X_1, \ldots, X_r)].$$

The first equality combines integration with respect to the law of X_1, the definition of γ^h, and Fubini. The second integrates the same function against the joint law of the independent coordinates. Thus the two equalities concern the same product integral.
Fubini is justified by boundedness and the probability masses. Every event in $\sigma(X_1)$ has this preimage form. The defining conditions and uniqueness identify the conditional expectation. QED

HW 72 — A coordinate conditioned on the sample mean

Let $(X_n)_{n \in \mathbb{N}}$ *be i.i.d. with* $\mathcal{L}_{X_n} = \Lambda$. *Let* $S_n = \frac{1}{n}\sum_{j=1}^n X_j$. *Calculate* $\mathsf{E}[X_1|\sigma(S_n)]$.

Solution to HW 72. The candidate is the realised sample mean S_n, not the fixed population mean. All coordinates are exchangeable, and observing their average cannot favour one coordinate over another. We turn this symmetry into the exact conditional test-integral identity.
The candidate is $\sigma(S_n)$-measurable by its definition. It is integrable because

$$\mathsf{E}|S_n| \leq \frac{1}{n}\sum_{j=1}^n \mathsf{E}|X_j| = \mathsf{E}|X_1| < \infty.$$

An arbitrary event in $\sigma(S_n)$ has the form $G = \{S_n \in B\}$ for a Borel set B: inverse images of all Borel sets form a sigma-algebra and are exactly the sigma-algebra generated by this one real variable.
Fix such a B. The joint law of $(X_1, \ldots, X_n)$ is invariant under any permutation, because it is the product of the same marginal law in each coordinate. A permutation swapping coordinates one and i leaves their sum and $\mathbf{1}_{\{S_n \in B\}}$ unchanged. Integrating under this permutation-invariant law gives

$$\mathsf{E}[X_i \mathbf{1}_G] = \mathsf{E}[X_1 \mathbf{1}_G], \ i = 1, \ldots, n.$$

All these signed expectations are finite by the common L^1 assumption. Add the n equalities and use the definition of the average:

$$n\mathsf{E}[X_1 \mathbf{1}_G] = \mathsf{E}\left[\left(\sum_{i=1}^{n} X_i\right) \mathbf{1}_G\right] = n\mathsf{E}[S_n \mathbf{1}_G].$$

The first equality adds the n preceding equal test expectations and uses finite linearity. The second substitutes $\sum_{i=1}^{n} X_i = nS_n$ and factors out the deterministic n.
Dividing by the positive integer n proves $\mathsf{E}[X_1 \mathbf{1}_G] = \mathsf{E}[S_n \mathbf{1}_G]$ for every event $G \in \sigma(S_n)$. Together with the candidate's measurability and integrability, this is the full definition of conditional expectation. Uniqueness therefore yields $\mathsf{E}[X_1 \mid \sigma(S_n)] = S_n$ almost surely. The argument only needs exchangeability of the finite vector and equal integrability, so independence is stronger than necessary. Without integrability, a signed conditional expectation may not exist at all; one would need a separate extended nonnegative formulation rather than reuse this proof unqualified.

9. Martingales

Definitions — Filtration and Filtered Probability Space
An increasing sequence of σ-algebras $(\mathcal{F}_n)_{n\in\mathbb{N}}$ with $\mathcal{F}_n \subset \mathcal{F}_{n+1}$ is called a *filtration*. A collection

$$(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\in\mathbb{N}}, \mathsf{P})$$

with $(\mathcal{F}_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ a filtration, is called a *filtered probability space*. We also define

$$\mathcal{F}_\infty = \sigma\Big(\cup_{n\in\mathbb{N}} \mathcal{F}_n\Big) \subseteq \mathcal{F}.$$

Example 5 — Natural Filtration
Very often a filtration is constructed from the σ-algebras generated by a given stochastic process: let $(W_n)_{n\in\mathbb{N}}$ be a stochastic process on $(\Omega, \mathcal{F}, \mathsf{P})$ and let $\mathcal{F}_k = \sigma(W_0, W_1, \ldots W_k)$, $k \geq 0$. Then, it is not hard to verify that $(\mathcal{F}_k)_{k\in\mathbb{N}}$ forms a filtration.
Notation: $\mathbb{Z}^+ = \mathbb{N} \cup \{0\}$.

Verification of the natural filtration. Every generator of $\mathcal{F}_k = \sigma(W_0, \ldots, W_k)$ is also a generator for $\mathcal{F}_{k+1}$. Minimality of the generated sigma-algebra gives $\mathcal{F}_k \subseteq \mathcal{F}_{k+1}$. Each is a sub-sigma-algebra of the ambient $\mathcal{F}$, because every W_j is a random variable.

Definition 9.1 — Adapted Process
A process $(X_n)_{n\in\mathbb{N}}$ is said to be adapted to a filtration $(\mathcal{F}_n)_{n\in\mathbb{Z}^+}$ if $X_n \in m\mathcal{F}_n$ for each $n \in \mathbb{Z}^+$.
Notice that if $(X_n)_{n\in\mathbb{N}}$ is adapted, then $X_n \in m\mathcal{F}_k$ for all $k \geq n$.

Definition 9.2 — Martingale
A process $X = (X_n)_{n\in\mathbb{Z}^+}$ is called a martingale relative to $(\mathcal{F}_n, \mathsf{P})$ if
(i) X is adapted;
(ii) $\mathsf{E}[|X_n|] < \infty$ for all $n \in \mathbb{Z}^+$;
(iii) $\mathsf{E}[X_{n+1}|\mathcal{F}_n] = X_n$, P-a.s., for all $n \in \mathbb{Z}^+$.
A super-martingale *is a process with properties (i), (ii) and*

(iv) $\mathsf{E}[X_{n+1}|\mathcal{F}_n] \le X_n$, P *-a.s., for all* $n \in \mathbb{Z}^+$.

A sub-martingale *is a process with properties (i), (ii) and*

(v) $\mathsf{E}[X_{n+1}|\mathcal{F}_n] \ge X_n$, P *-a.s., for all* $n \in \mathbb{Z}^+$.

Notice that if X is a super-martingale, then $-X$ is a sub-martingale.

HW 73 — One-step and multi-step martingale conditions

Show that $(X_n)_{n\in\mathbb{Z}^+}$ *is a martingale if and only if* $\mathsf{E}[X_m|\mathcal{F}_n] = X_n$ *for any* $m \ge n$. *(Analogue properties hold for super/sub-martingales.)*

Solution to HW 73. The implication from the multi-step condition to the martingale definition is immediate by taking $m = n+1$; adaptation and integrability are already part of the assumptions. The converse uses the tower property: if $\mathcal{H} \subset \mathcal{G}$, then $\mathsf{E}[\mathsf{E}[W \mid \mathcal{G}] \mid \mathcal{H}] = \mathsf{E}[W \mid \mathcal{H}]$ for integrable W.

Fix n. For $m = n$ the equality is $\mathsf{E}[X_n \mid \mathcal{F}_n] = X_n$, a consequence of its measurability. The case $m = n + 1$ is the one-step martingale condition. Suppose the equality is known with horizon $m - 1 \ge n$. Since $\mathcal{F}_n \subset \mathcal{F}_{m-1}$,

$$\mathsf{E}[X_m \mid \mathcal{F}_n] = \mathsf{E}\big[\mathsf{E}[X_m \mid \mathcal{F}_{m-1}] \mid \mathcal{F}_n\big] = \mathsf{E}[X_{m-1} \mid \mathcal{F}_n] = X_n.$$

The first step is the tower property, the second uses one-step fairness at date $m - 1$, and the last is the induction hypothesis. Induction proves every finite future horizon.

For a supermartingale, $\mathsf{E}[X_m \mid \mathcal{F}_{m-1}] \le X_{m-1}$. Conditional monotonicity preserves this inequality when conditioning further on $\mathcal{F}_n$, so the same induction yields $\mathsf{E}[X_m \mid \mathcal{F}_n] \le X_n$. A submartingale gives the reversed inequality. Conversely each multi-step inequality includes the corresponding one-step inequality. These conditional statements are meaningful because all variables being conditioned are integrable; they are not replacements for the adaptation and moment requirements.

Example 6 — Examples of Martingales

(a) Let $(X_n)_{n\in\mathbb{N}} \subset \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ *be a sequence of independent random variables with* $\mathsf{E}[X_k] = 0$. *Set* $S_0 = 0$ *and* $S_n = \sum_{j=1}^n X_j$ *and let* $\mathcal{F}_n = \sigma(X_1, \dots X_n)$ *with* $\mathcal{F}_0 = \{\Omega, \varnothing\}$. *The process* $(S_n)_{n\in\mathbb{Z}^+}$ *is a* $(\mathcal{F}_n, \mathsf{P})$*-martingale.*

(b) Let $(X_n)_{n\in\mathbb{N}}$ *be a sequence of positive independent random variables with* $\mathsf{E}[X_k] = 1$. *Let* $\mathcal{F}_n = \sigma(X_1, \dots X_n)$ *with* $\mathcal{F}_0 = \{\Omega, \varnothing\}$ *and define* $M_0 = 1$, $M_n = X_1 \cdot X_2 \cdots X_n$. *Then* $(M_n)_{n\in\mathbb{Z}^+}$ *is a* $(\mathcal{F}_n, \mathsf{P})$*-martingale.*

(c) Let $(\mathcal{F}_n)_{n\in\mathbb{Z}^+}$ *be a filtration and* $\xi \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$. *Set* $M_n = \mathsf{E}[\xi|\mathcal{F}_n]$. *Then* $(M_n)_{n\in\mathbb{Z}^+}$ *is a* $(\mathcal{F}_n, \mathsf{P})$*-martingale.*

HW 74 — Three basic martingales

Verify that all the processes in the example above are indeed martingales

Solution to HW 74. A martingale must be adapted, integrable at each date, and satisfy the one-step conditional equality. We check all three requirements in each construction.

In (a), the finite sum S_n is a measurable function of the first n recorded variables, hence is $\mathcal{F}_n$-measurable. The bound $|S_n| \le \sum_{j=1}^n |X_j|$ gives $\mathsf{E}|S_n| \le \sum_{j=1}^n \mathsf{E}|X_j| < \infty$. Independence makes X_{n+1} independent of the sigma-algebra generated by the previous variables, so

$$\mathsf{E}[X_{n+1} \mid \mathcal{F}_n] = \mathsf{E}X_{n+1} = 0.$$

Therefore

$$\mathsf{E}[S_{n+1} \mid \mathcal{F}_n] = \mathsf{E}[S_n + X_{n+1} \mid \mathcal{F}_n] = S_n + 0 = S_n.$$

The first equality substitutes the sum recurrence. The second uses conditional linearity, takes out the known S_n, and inserts $\mathsf{E}[X_{n+1} \mid \mathcal{F}_n] = 0$. The third adds zero.

In (b), the finite product is again adapted. Its nonnegativity and independence give

$$\mathsf{E}|M_n| = \mathsf{E}M_n = \prod_{j=1}^n \mathsf{E}X_j = 1,$$

so it is integrable. To condition the next product, M_n is known at date n and X_{n+1} is independent of that information. Consequently

$$\mathsf{E}[M_{n+1} \mid \mathcal{F}_n] = \mathsf{E}[M_n X_{n+1} \mid \mathcal{F}_n] = M_n \mathsf{E}[X_{n+1} \mid \mathcal{F}_n] = M_n.$$

The first equality substitutes the product recurrence. The second takes out the $\mathcal{F}_n$-measurable factor M_n. The third uses independence of the next factor and its mean one.
The taking-out step is legitimate for nonnegative products, or by bounded truncation of the known factor followed by monotone convergence. Its resulting conditional expectation is integrable because its unconditional mean is one.
In (c), the definition of conditional expectation already provides $\mathcal{F}_n$-measurability and integrability. More explicitly, $|M_n| \le \mathsf{E}[|\xi| \mid \mathcal{F}_n]$ almost surely, so $\mathsf{E}|M_n| \le \mathsf{E}|\xi| < \infty$. Since $\mathcal{F}_n \subset \mathcal{F}_{n+1}$, the tower property yields

$$\mathsf{E}[M_{n+1} \mid \mathcal{F}_n] = \mathsf{E}\big[\mathsf{E}[\xi \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\big] = \mathsf{E}[\xi \mid \mathcal{F}_n] = M_n.$$

The first equality substitutes the definition of M_{n+1}. The second is the tower property because $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$. The third is the definition of M_n.
No independence assumptions on ξ or the filtration are required in this construction; its fairness is a consequence of consistent conditioning as information increases.

Previsible Transforms and Fair Games

Fair and unfair games. Let $X_n - X_{n-1}$ be your net winnings per unit stake in a series of games played at times $n = 1, 2, \ldots$.

- If $(X_n)_{n\in\mathbb{Z}^+}$ is a martingale, then the game is *fair*: $\mathsf{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] = 0$.
- If $(X_n)_{n\in\mathbb{Z}^+}$ is a super-martingale, then the game is *unfair*: $\mathsf{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] \le 0$.
- If $(X_n)_{n\in\mathbb{Z}^+}$ is a sub-martingale, then the game is *unfair* (for the other player): $\mathsf{E}[X_n - X_{n-1}|\mathcal{F}_{n-1}] \ge 0$.

A natural question is whether it is possible to gamble in such a way so as to "beat" the casino. In order to answer this question we need to define a process that describes your stakes (what you gamble).

Definition 9.3 — Previsible Process

We say that a process $(C_n)_{n\in\mathbb{N}}$ is previsible if $C_n \in m\mathcal{F}_{n-1}$ for all $n \in \mathbb{N}$.
The net winnings obtained by using C_n in the n-th round of the game are

$$C_n(\omega)(X_n(\omega) - X_{n-1}(\omega)).$$

Given a previsible strategy $(C_n)_{n\in\mathbb{N}}$ the total net winnings until the end of the n-th round of the game read

$$Y_n(\omega) = \sum_{k=1}^{n} C_k(\omega)(X_k(\omega) - X_{k-1}(\omega)) =: (C \bullet X)_n.$$

We take $Y_0 = 0$ by convention and notice that

$$Y_n - Y_{n-1} = C_n(\omega)(X_n(\omega) - X_{n-1}(\omega)).$$

Theorem 9.1 — You Cannot Beat the Casino

(i) Let $(C_n)_{n\in\mathbb{N}}$ be bounded, non-negative, previsible so that there is $K > 0$ such that

$$0 \le C_n(\omega) \le K \text{ for all } \ n \in \mathbb{N}, \omega \in \Omega.$$

If X is a martingale (super-martingale), then $(C \bullet X)$ is a martingale (super-martingale).

(ii) Let $(C_n)_{n\in\mathbb{N}}$ be bounded and previsible and let X be a martingale. Then $(C \bullet X)$ is a martingale.

(iii) In (i) and (ii) we can replace the boundedness requirement with $(C_n)_{n\in\mathbb{N}}, (X_n)_{n\in\mathbb{Z}^+} \subset \mathcal{L}^2(\Omega, \mathcal{F}, \mathsf{P})$.

In all cases we take $(C \bullet X)_0 = 0$.

Expanded proof. We prove part (i), following the lecturer's proof. Let $Y_n = \sum_{k=1}^{n} C_k(X_k - X_{k-1})$. Each term is $\mathcal{F}_n$-measurable, so Y is adapted. If $0 \le C_k \le K$, then $\mathsf{E}|C_k(X_k - X_{k-1})| \le K(\mathsf{E}|X_k| + \mathsf{E}|X_{k-1}|) < \infty$. Thus the finite sum is integrable. Since C_n is $\mathcal{F}_{n-1}$-measurable and bounded,

$$\mathsf{E}[Y_n - Y_{n-1} \mid \mathcal{F}_{n-1}] = C_n \mathsf{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}].$$

The equality uses $Y_n - Y_{n-1} = C_n(X_n - X_{n-1})$ and the bounded $\mathcal{F}_{n-1}$-measurable pull-out property. Conditional linearity and $\mathbb{E}[X_{n-1} \mid \mathcal{F}_{n-1}] = X_{n-1}$ then identify the conditional drift.
For a martingale the right-hand side is zero, giving the martingale property for Y. For a supermartingale it is non-positive: the non-negativity of C_n preserves the inequality. This proves precisely part (i). QED

HW 75 — Martingale transforms beyond nonnegative bounded stakes
Prove statements (ii) and (iii) in the theorem above.

Solution to HW 75. Write $\Delta X_k = X_k - X_{k-1}$. Previsibility makes C_k measurable for $\mathcal{F}_{k-1} \subset \mathcal{F}_n$ when $k \le n$, and adaptation makes ΔX_k measurable for $\mathcal{F}_k \subset \mathcal{F}_n$. Every finite sum defining Y_n is therefore $\mathcal{F}_n$-measurable. Adaptation does not ensure integrability; we verify that before manipulating conditional expectations.
For (ii), let $|C_k| \le K$ for every date and path. Then

$$\mathsf{E}|Y_n| \le \sum_{k=1}^{n} \mathsf{E}|C_k \Delta X_k| \le K \sum_{k=1}^{n} (\mathsf{E}|X_k| + \mathsf{E}|X_{k-1}|) < \infty.$$

The increment of the transformed process is just $\Delta Y_n = C_n \Delta X_n$. Bounded $\mathcal{F}_{n-1}$-measurable factors can be taken outside conditional expectations; this identity follows first for indicator factors by the test-integral definition, then for simple factors and finally bounded limits. Thus

$$\mathsf{E}[\Delta Y_n \mid \mathcal{F}_{n-1}] = C_n \mathsf{E}[\Delta X_n \mid \mathcal{F}_{n-1}] = 0.$$

The last equality is the martingale property of X. Adding back the known Y_{n-1} proves $\mathsf{E}[Y_n \mid \mathcal{F}_{n-1}] = Y_{n-1}$. A sign restriction on C_n was unnecessary because it multiplies zero, rather than an inequality.
For (iii), $X_k, X_{k-1} \in L^2$ implies $\Delta X_k \in L^2$. Cauchy–Schwarz gives $\mathsf{E}|C_k \Delta X_k| \le \|C_k\|_2 \|\Delta X_k\|_2 < \infty$, so every finite Y_n is again integrable. The factor C_n can now be unbounded, so the taking-out step needs justification. Put $U = \Delta X_n$, $\mathcal{H} = \mathcal{F}_{n-1}$ and truncate $C = C_n$ by $C^{(m)} = \max(-m, \min(C, m))$. Bounded pull-out gives $\mathsf{E}[C^{(m)}U \mid \mathcal{H}] = C^{(m)}\mathsf{E}[U \mid \mathcal{H}]$. Since $C \in L^2$, dominated convergence applied to $|C - C^{(m)}|^2 \le |C|^2$ gives $\|C - C^{(m)}\|_2 \to 0$. Hence $\mathsf{E}|(C - C^{(m)})U| \le \|C - C^{(m)}\|_2 \|U\|_2 \longrightarrow 0$. Conditional expectation contracts L^1, because $|\mathsf{E}[V \mid \mathcal{H}]| \le \mathsf{E}[|V| \mid \mathcal{H}]$ and then one integrates. The left side of the truncated conditional identity therefore converges in L^1 to $\mathsf{E}[CU \mid \mathcal{H}]$.
The right side converges in L^1 to $C\mathsf{E}[U \mid \mathcal{H}]$ as well. Indeed conditional Jensen gives $\|\mathsf{E}[U \mid \mathcal{H}]\|_2 \le \|U\|_2$, and Cauchy–Schwarz bounds the product of that variable with $C - C^{(m)}$. Passing to the limit proves the unbounded pull-out identity with all products integrable. When X is a martingale, $\mathsf{E}[U \mid \mathcal{H}] = 0$; thus the transformed conditional increment is zero and Y is a martingale.
When X is a supermartingale, the same calculation gives $\mathsf{E}[U \mid \mathcal{H}] \le 0$. If $C_n \ge 0$, multiplication preserves that inequality and yields $\mathsf{E}[\Delta Y_n \mid \mathcal{F}_{n-1}] \le 0$. The adaptation and integrability checks already made then prove the supermartingale property. Negative stakes cannot generally be allowed in this last case, because multiplying a negative drift by a negative factor reverses its sign.

Stopping Times and Stopped Processes
One may think that there is a way of abandoning the game so that it is still possible to beat the casino. To formulate and answer such question we need the nest definition.

Definition 9.4 — Stopping Time
A map $\tau : \Omega \to \mathbb{Z}^+$ is a stopping time for a filtration $(\mathcal{F}_n)_{n\in\mathbb{N}}$ if for all $n \in \mathbb{Z}^+$

$$\{\omega \in \Omega : \tau(\omega) \le n\} \in \mathcal{F}_n.$$

HW 76 — Exact-time events and stopping times

Prove that τ is a stopping time if and only if for all $n \in \mathbb{Z}^+$

$$\{\omega \in \Omega : \tau(\omega) = n\} \in \mathcal{F}_n.$$

Solution to HW 76. By definition, a stopping time satisfies $\{\tau \leq n\} \in \mathcal{F}_n$ for every date n. If this holds, the date-zero event is $\{\tau = 0\} = \{\tau \leq 0\} \in \mathcal{F}_0$. For $n \geq 1$, an exact hit at date n means a hit by n but not by $n-1$:

$$\{\tau = n\} = \{\tau \leq n\} \setminus \{\tau \leq n-1\}.$$

The first event belongs to $\mathcal{F}_n$. The second belongs to $\mathcal{F}_{n-1} \subset \mathcal{F}_n$, so their difference is also in $\mathcal{F}_n$. This proves one direction, including the boundary date separately.

Conversely, suppose each exact-time event is in its corresponding sigma-algebra. A nonnegative integer-valued time has occurred by date n precisely when one of the finitely many exact times $0, \ldots, n$ has occurred. Hence $\{\tau \leq n\} = \bigcup_{k=0}^{n} \{\tau = k\}$. For $k \leq n$, the event in this union belongs to $\mathcal{F}_k \subset \mathcal{F}_n$, and the finite union remains in $\mathcal{F}_n$. This verifies the stopping-time definition. Paths with $\tau = \infty$ are outside every one of these finite unions, so allowing infinity does not change either implication.

Example 7 — A First Hitting Time

Suppose that $(X_n)_{n\in\mathbb{Z}^+}$ is a process taking values in $\mathbb{Z} = \{-\infty, \ldots -1, 0, 1, 2, \ldots \infty\}$ with $X_0 = 0$. Let $(\mathcal{F}_n)_{n\in\mathbb{Z}^+}$ be the filtration generated by the process. The random time

$$\tau_k = \inf\{n \in \mathbb{Z}^+ : X_n \geq k\}$$

is a stopping time for any $k \in \mathbb{Z}$. Let's see why

$$\{\tau_k \leq n\} = \cup_{j=0}^{n} \{X_j \geq k\} \in \mathcal{F}_n$$

because $\{X_j \geq k\} \in \mathcal{F}_j \subset \mathcal{F}_n$.

HW 77 — A last visit is not generally a stopping time

In the same framework of the example above, determine whether or note the random time

$$\sigma_k = \sup\{n \in \mathbb{Z}^+ : X_n \geq k\}$$

is a $(\mathcal{F}_n)_{n\in\mathbb{Z}^+}$-stopping time.

Solution to HW 77. A first visit can be identified from the observations already made, but deciding that a visit was the *last* one may require knowing all future observations. To disprove a universal stopping-time claim it is enough to give one process whose two possible paths have the same observed past but different last-visit outcomes.

Let $\Omega = \{a, b\}$, with probability one half at each point. Define $X_0 = 0$, $X_1 = 1$ on both paths, $X_2(a) = 0$, $X_2(b) = 1$, and $X_n = 0$ on both paths for every $n \geq 3$. This is an adapted integer-valued process in its natural filtration. Its observations through date one are deterministic, so $\mathcal{F}_1 = \{\varnothing, \Omega\}$. For the threshold $k = 1$, path a visits the upper set for the last time at date one, whereas path b does so at date two. Thus

$$\sigma_1(a) = 1,\ \sigma_1(b) = 2,\ \{\sigma_1 \leq 1\} = \{a\} \notin \mathcal{F}_1.$$

The stopping-time definition already fails at this finite date. Notice that both last-visit times in this counterexample are finite, so no issue about the convention for infinity is responsible for the failure.

In general one can set the supremum to ∞ if there are infinitely many visit dates. If the set is empty, a convention such as $-\infty$ must be specified, or the definition must be modified to take nonnegative extended values; this domain issue is separate from adaptation. For particular processes a last visit may be deterministic and thus a stopping time. The correct conclusion is that it is *not necessarily* a stopping time, not that it can never be one.

Given a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\in\mathbb{Z}^+}, \mathsf{P})$, an adapted stochastic process $(X_n)_{n\in\mathbb{Z}^+}$ and a stopping time τ, we can define the *stopped process* $X^\tau = (X_{n\wedge\tau})_{n\in\mathbb{Z}^+}$.

HW 78 — Adaptedness of a stopped process

Show that if X is adapted then X^τ is also adapted.

Solution to HW 78. Fix a deterministic date n. The stopped value equals X_k on $\{\tau = k\}$ for $k < n$, and equals X_n if stopping has not occurred strictly before n. These mutually disjoint cases yield the exact finite representation $X_{n\wedge\tau} = \sum_{k=0}^{n-1} X_k \mathbf{1}_{\{\tau=k\}} + X_n \mathbf{1}_{\{\tau\geq n\}}$. For each $k < n$, X_k and $\{\tau = k\}$ are measurable for $\mathcal{F}_k$, hence for $\mathcal{F}_n$. For the last term, X_n is $\mathcal{F}_n$-measurable and, when $n \geq 1$,

$$\{\tau \geq n\} = \{\tau \leq n-1\}^c \in \mathcal{F}_{n-1} \subset \mathcal{F}_n.$$

At $n = 0$ this event is the whole space and $X_{0\wedge\tau} = X_0$. Every term in the representation is therefore $\mathcal{F}_n$-measurable, as is the finite sum. This proves adaptation for all dates.
If $X_k \in L^1$ for every fixed k, the same partition gives a simple deterministic-date bound:

$$|X_{n\wedge\tau}| \leq \sum_{k=0}^{n} |X_k|,\ \mathsf{E}|X_{n\wedge\tau}| \leq \sum_{k=0}^{n} \mathsf{E}|X_k| < \infty.$$

Thus fixed-date stopping preserves integrability. This does not by itself show integrability of X_τ for an unbounded time, since the number of potentially used dates would then be unbounded. That later question needs separate optional-sampling hypotheses or a nonnegativity argument.

Theorem 9.2 — Stopped (Super)martingales

(i) If X is a super-martingale and τ is a stopping time, then the stopped process X^τ us a super-martingale and

$$\mathsf{E}[X_n^\tau] = \mathsf{E}[X_{n\wedge\tau}] \leq \mathsf{E}[X_0],\ n \in \mathbb{Z}^+.$$

(ii) If X is a martingale and τ is a stopping time, then the stopped process X^τ us a martingale and

$$\mathsf{E}[X_n^\tau] = \mathsf{E}[X_{n\wedge\tau}] = \mathsf{E}[X_0],\ n \in \mathbb{Z}^+.$$

Expanded proof. Set $C_k = \mathbf{1}_{\{k\leq\tau\}}$. This process is bounded and non-negative. Since $\{k \leq \tau\} = \{\tau \leq k-1\}^c \in \mathcal{F}_{k-1}$, it is previsible. Path by path, $X_{n\wedge\tau} = X_0 + \sum_{k=1}^{n} \mathbf{1}_{\{k\leq\tau\}}(X_k - X_{k-1})$. If $\tau = m \leq n$, the sum contains exactly the first m increments and telescopes to $X_m - X_0$. If $\tau > n$, it contains the first n increments and telescopes to $X_n - X_0$. Theorem 9.1(i) makes the stochastic-integral part a (super)martingale. Adding the integrable $\mathcal{F}_0$-measurable constant-in-time process X_0 preserves that property. Adaptedness was also checked independently in the earlier HW 78. This is a statement at deterministic n; an unbounded random terminal time requires additional hypotheses. QED

WARNING: we may be tempted to conclude that if X is a a martingale, then $\mathsf{E}[X_\tau] = \mathsf{E}[X_0]$ for any stopping time. That is WRONG. Here is a counterexample: let $(Z_n)_{n\in\mathbb{N}}$ be i.i.d. with $\mathsf{P}(Z_k = +1) = \mathsf{P}(Z_k = -1) = \frac{1}{2}$, let $(\mathcal{F}_n)_{n\in\mathbb{Z}^+}$ be the filtration generated by $(Z_n)_{n\in\mathbb{N}}$ with $\mathcal{F}_0 = \{\Omega, \varnothing\}$. The process

$$X_0 = 0,\ X_n = \sum_{k=1}^{n} Z_k.$$

is a $(\mathcal{F}_n, \mathsf{P})$-martingale (called *random walk*). Setting

$$\tau = \inf\{k \in \mathbb{N} : X_k = 1\}$$

it is easy to check that τ is $(\mathcal{F}_n)$-stopping time and it can also be shown that $\mathsf{P}(\tau < \infty) = 1$. So, if our (wrong) conjecture were correct we would have

$$0 = \mathsf{E}[X_0] = \mathsf{E}[X_\tau] = 1,$$

which is clearly impossible.

Optional Sampling and Its Hypotheses

Verification — The Random Walk Hits One Almost Surely

Proof. The event $\{\tau \le n\} = \bigcup_{j=1}^{n}\{X_j = 1\}$ belongs to $\mathcal{F}_n$: each level event is measurable by adaptedness, and finite-union closure applies. Both directions of the equality express the existence of a first hit among those dates. Thus τ is a stopping time. To prove its almost sure finiteness without an optional-sampling assumption, fix a positive integer m, and let $\sigma_m = \inf\{n \ge 0 : X_n \in \{-m, 1\}\}$. This is a stopping time by the same finite-union argument. From any interior integer state, $m+1$ consecutive upward steps force an exit by the upper boundary. Those steps have probability $2^{-(m+1)}$, independently of the past. Conditioning successively at block endpoints therefore gives

$$\mathbb{P}(\sigma_m > k(m+1)) \le (1 - 2^{-(m+1)})^k \longrightarrow 0.$$

Hence $\sigma_m < \infty$ almost surely. The stopped-process theorem already proved makes $X_{n\wedge\sigma_m}$ a martingale of mean zero. Its absolute value is bounded by the integrable constant m, because the walk cannot jump over either boundary. DOM gives $\mathbb{E}X_{\sigma_m} = 0$. Writing $q_m = \mathbb{P}(X_{\sigma_m} = 1)$, we obtain

$$0 = \mathbb{E}X_{\sigma_m} = q_m - m(1 - q_m),\ q_m = \frac{m}{m+1}.$$

The first equality follows from DOM and the stopped means. The second averages the two possible boundary values. Solving this finite linear equation gives the last equality. An upper-boundary exit is a hit of one, so $\mathbb{P}(\tau < \infty) \ge q_m$. Letting $m \to \infty$, and using the upper bound one for a probability, proves $\mathbb{P}(\tau < \infty) = 1$. Then $X_\tau = 1$ almost surely, whereas $\mathbb{E}X_0 = 0$. This verifies every assertion of the counterexample without using the optional-sampling theorem that follows. QED

Theorem 9.3 — Doob Optional Sampling

Let X be a super-martingale and τ be a stopping time. Then $\mathsf{E}[X_\tau] \le \mathsf{E}[X_0]$ if one of the following assumptions holds:

(i) τ is bounded (i.e., $\tau(\omega) \le N$ for all $\omega \in \Omega$ and a fixed $N \in \mathbb{N}$);

(ii) X is bounded and $\mathsf{P}(\tau < \infty) = 1$ (i.e., $|X_n(\omega)| \le K$ for all $(n, \omega) \in \mathbb{Z}^+ \times \Omega$ and a fixed $K > 0$);

(iii) $\mathsf{E}[\tau] < \infty$ and there is $K > 0$ such that

$$|X_n(\omega) - X_{n-1}(\omega)| \le K \text{ for all } (n, \omega) \in \mathbb{Z}^+ \times \Omega.$$

If X is a martingale and one of the conditions above holds, then $\mathsf{E}[X_\tau] = \mathsf{E}[X_0]$.

Expanded proof. For every deterministic n, the stopped-process result gives $\mathsf{E}[X_{n\wedge\tau} - X_0] \le 0$. The three alternative assumptions justify passing to the terminal time.

Case (i). If $\tau \le N$, then $X_{n\wedge\tau} = X_\tau$ for $n \ge N$; no limiting interchange is needed.

Case (ii). If $\tau < \infty$ almost surely and $|X_n| \le K$, then $X_{n\wedge\tau} \to X_\tau$ almost surely, dominated by the integrable constant K. DOM passes the limit through expectation.

Case (iii). If $|X_k - X_{k-1}| \le K$, then $|X_{n\wedge\tau} - X_0| \le \sum_{k=1}^{n\wedge\tau} |X_k - X_{k-1}| \le K\tau$. The bound is integrable because $\mathsf{E}\tau < \infty$. This also implies $\tau < \infty$ almost surely by HW 54. DOM applies to these differences, with X_0 already integrable.

Each case yields $\mathsf{E}X_\tau \le \mathsf{E}X_0$. For a martingale, apply the same result to $-X$ to obtain equality. Almost sure finiteness of τ alone does not provide the domination needed in either limit argument. QED

HW 79 — Sampling a nonnegative supermartingale

Prove that if X is a non-negative martingale or super-martingale, then $\mathsf{E}[X_\tau] \le \mathsf{E}[X_0]$ for any stopping time τ such that $\mathsf{P}(\tau < \infty) = 1$.

Solution to HW 79. The difficulty is the unbounded time τ: fixed-date martingale equalities cannot simply be evaluated at an arbitrary random horizon. Truncate it at n. The bounded stopping result gives $\mathsf{E}X_{n\wedge\tau} \le \mathsf{E}X_0$ $(n \ge 0)$, with equality for each n if X is a martingale. This bounded-time fact can also be seen from the gain identity

$$X_{n\wedge\tau} - X_0 = \sum_{k=1}^{n} \mathbf{1}_{\{\tau \ge k\}}(X_k - X_{k-1}).$$

The indicator is $\mathcal{F}_{k-1}$-measurable because its complement is $\{\tau \le k-1\}$; it is bounded and nonnegative. Conditioning each integrable summand therefore gives nonpositive mean for a supermartingale, and zero for a martingale, proving the bound directly.

On every path where $\tau < \infty$, the values $X_{n\wedge\tau}$ eventually equal X_τ, so converge to it. All these variables are nonnegative, and therefore Fatou's lemma applies without any common integrable dominating variable:

$$\mathsf{E}X_\tau = \mathsf{E}\big[\liminf_n X_{n\wedge\tau}\big] \le \liminf_n \mathsf{E}X_{n\wedge\tau} \le \mathsf{E}X_0.$$

The first equality in the final chain uses the almost-sure eventual constancy at the finite stopping time. The first inequality is Fatou for the non-negative stopped variables, and the second is the bound just proved for every truncation. The first step is an almost-sure replacement, which preserves the integral.

The finite upper bound proves integrability of the sampled value at the same time as the inequality. On the null event where $\tau = \infty$, X_τ can be assigned any finite value without changing this conclusion.

Even when each truncated expectation equals $\mathsf{E}X_0$, Fatou yields only a limiting inequality. Equality would require additional control such as an integrable bound or uniform integrability. Nonnegativity supplies the lower bound needed for Fatou, not the two-sided domination needed to exchange expectation and limit as an equality.

A beautiful result from the general theory is about a decomposition of adapted processes.

Theorem 9.4 — Doob Decomposition

Let X be an adapted process on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\in\mathbb{Z}^+}, \mathsf{P})$ with $X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathsf{P})$ for all $n \in \mathbb{Z}^+$.

(a) X admits the decomposition

$$X_n = X_0 + M_n + A_n,\ M_0 = A_0 = 0,$$

where $(M_n)_{n\in\mathbb{Z}^+}$ is a a martingale and $(A_n)_{n\in\mathbb{Z}^+}$ is a previsible process. Such decomposition is unique, in the sense that if $X_n = X_0 + \tilde{M}_n + \tilde{A}_n$ is another decomposition, then

$$\mathsf{P}(M_n = \tilde{M}_n, A_n = \tilde{A}_n, \forall n \in \mathbb{Z}^+) = 1.$$

(b) X is a sub-martingale (or super-martingale) if and only if $\mathsf{P}(A_n \le A_{n+1}, \forall n \in \mathbb{Z}^+) = 1$ (or $\mathsf{P}(A_n \ge A_{n+1}, \forall n \in \mathbb{Z}^+) = 1$).

Expanded proof. Uniqueness. Suppose $X_n = X_0 + M_n + A_n$, with $M_0 = A_0 = 0$. The martingale and previsibility properties give

$$\mathsf{E}[\Delta X_n \mid \mathcal{F}_{n-1}] = \mathsf{E}[\Delta M_n \mid \mathcal{F}_{n-1}] + \Delta A_n = \Delta A_n.$$

The first equality uses conditional linearity and pulls out the integrable $\mathcal{F}_{n-1}$-measurable increment ΔA_n. The second uses the zero conditional mean of a martingale increment.

Thus both components are forced by X:

$$A_n = \sum_{k=1}^{n} \mathsf{E}[\Delta X_k \mid \mathcal{F}_{k-1}],\ M_n = X_n - X_0 - A_n.$$

Two decompositions therefore agree almost surely at each date. A countable intersection makes the agreement simultaneous at all dates.

Existence. Define A, M by these formulas. We take previsibility of A as given, as we will see in HW 80. Conditional Jensen gives $\mathsf{E}|\mathsf{E}[\Delta X_k \mid \mathcal{F}_{k-1}]| \le \mathsf{E}|\Delta X_k| < \infty$. Hence the finite sums A_n, and therefore M_n, are integrable; they are also adapted. Their increments satisfy

$$\mathsf{E}[\Delta M_n \mid \mathcal{F}_{n-1}] = \mathsf{E}[\Delta X_n \mid \mathcal{F}_{n-1}] - \Delta A_n = 0.$$

The first equality substitutes $\Delta M_n = \Delta X_n - \Delta A_n$, uses conditional linearity, and takes out the $\mathcal{F}_{n-1}$-measurable increment ΔA_n. The second substitutes its definition as $\mathsf{E}[\Delta X_n \mid \mathcal{F}_{n-1}]$, so the two finite terms cancel.

Thus M is a martingale.

Part (b). The same identity $\Delta A_n = \mathsf{E}[\Delta X_n \mid \mathcal{F}_{n-1}]$ shows that X is a submartingale exactly when every increment of A is non-negative almost surely. For a supermartingale use non-positive increments. Countability again makes the inequalities simultaneous. QED

HW 80 — Predictability of the Doob compensator

Check that $A_n := \sum_{k=1}^{n} \mathsf{E}[X_k - X_{k-1}|\mathcal{F}_{k-1}]$ introduced in the proof is indeed previsible.

Solution to HW 80. Conditional expectation of the integrable increment $X_k - X_{k-1}$ exists and is measurable for its conditioning sigma-algebra $\mathcal{F}_{k-1}$. When $k \le n$, this sigma-algebra is contained in $\mathcal{F}_{n-1}$ because the filtration increases. Thus every term of the finite sum defining A_n is $\mathcal{F}_{n-1}$-measurable, and so is A_n. This is exactly previsibility, one time step earlier than ordinary adaptedness of the same value.

Integrability of the increments also gives a bound on the compensator. The conditional absolute-value inequality yields

$$|\mathsf{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}]| \le \mathsf{E}[|X_k - X_{k-1}| \mid \mathcal{F}_{k-1}].$$

Taking expectations and summing finitely many terms,

$$\mathsf{E}|A_n| \le \sum_{k=1}^{n} \mathsf{E}|X_k - X_{k-1}| \le \sum_{k=1}^{n} (\mathsf{E}|X_k| + \mathsf{E}|X_{k-1}|) < \infty.$$

Hence the proposed process has both required properties. Its increment $A_n - A_{n-1} = \mathsf{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}]$ is the expected next drift known before that observation arrives. Subtracting it is what removes the predictable part of X in Doob's decomposition.

Qualification — Square Integrability

For this exercise assume $M_n \in L^2$ at each date. Integrability of a martingale alone does not imply integrability of its squares.

HW 81 — Squaring a martingale and its Doob decomposition

Show that if M is a martingale, then $Y = (M_n^2)_{n\in\mathbb{Z}^+}$ is a sub-martingale. Find the Doob's decomposition of Y.

Solution to HW 81. Assume square integrability henceforth. Squaring preserves adaptation, and $\mathsf{E}Y_n = \mathsf{E}M_n^2 < \infty$ ensures the required integrability. Conditional Jensen with the convex function x^2 gives

$$\mathsf{E}[M_n^2 \mid \mathcal{F}_{n-1}] \ge \big(\mathsf{E}[M_n \mid \mathcal{F}_{n-1}]\big)^2 = M_{n-1}^2.$$

Thus the conditional value of the square is at least its previous square, which proves the sub-martingale inequality with all defining conditions checked.

To compute its predictable drift, put $\Delta M_k = M_k - M_{k-1}$. The square expands as $M_k^2 - M_{k-1}^2 = 2M_{k-1}\Delta M_k + (\Delta M_k)^2$. The increment is square-integrable, and its product with M_{k-1} is integrable by Cauchy–Schwarz. Because M_{k-1} is known at date $k-1$ and the martingale increment has conditional mean zero, the mixed term has conditional expectation zero. Taking out this possibly unbounded known factor is justified by bounded truncation and L^2 product control. Therefore

$$\mathsf{E}[Y_k - Y_{k-1} \mid \mathcal{F}_{k-1}] = \mathsf{E}[(\Delta M_k)^2 \mid \mathcal{F}_{k-1}].$$

Define

$$A_0 = N_0 = 0,\ A_n = \sum_{k=1}^{n} \mathsf{E}[(\Delta M_k)^2 \mid \mathcal{F}_{k-1}],\ N_n = M_n^2 - M_0^2 - A_n.$$

Every summand in A_n is $\mathcal{F}_{k-1}$-measurable and integrable, so A_n is $\mathcal{F}_{n-1}$-measurable and integrable. The summands are nonnegative, so A increases. The variable N_n is adapted and integrable, and its conditional increment is

$$\mathsf{E}[N_n - N_{n-1} \mid \mathcal{F}_{n-1}] = \mathsf{E}[M_n^2 - M_{n-1}^2 \mid \mathcal{F}_{n-1}] - \mathsf{E}[(\Delta M_n)^2 \mid \mathcal{F}_{n-1}] = 0.$$

The first equality substitutes the increment of N, uses conditional linearity, and takes out the predictable increment of A. The second is the conditional second-moment identity obtained above, with the identical finite conditional expectations cancelling.

Thus N is a martingale and $M_n^2 = M_0^2 + N_n + A_n$ is the desired Doob decomposition. The increasing process accumulates the conditional second moments of the martingale increments, rather than the realised squared increments themselves.

For uniqueness, any decomposition $Y_n = Y_0 + \widetilde{N}_n + \widetilde{A}_n$ with zero initial martingale and previsible drift must have $\Delta\widetilde{A}_n = \mathsf{E}[Y_n - Y_{n-1} \mid \mathcal{F}_{n-1}]$ by conditioning. This is the same increment as A, so the initial normalization makes both drift and martingale coincide at each time, and countability makes this simultaneous almost surely.

The missing L^2 hypothesis is substantive: take a variable $Z \in L^1 \setminus L^2$ and the constant-in-time martingale $M_n = Z$ with filtration $\sigma(Z)$. Then M_n^2 has infinite expectation, so it is not an integrable submartingale. The corrected assertion explicitly rules out this counterexample.

Versions and Indistinguishability

HW 82 — One common exceptional set in discrete time

Given two processes X and Y, prove that the two conditions below are equivalent:

(a) $\mathsf{P}(X_n = Y_n) = 1$, *for all* $n \in \mathbb{Z}^+$;

(b) $\mathsf{P}(X_n = Y_n, forall\ n \in \mathbb{Z}^+) = 1$.

Solution to HW 82. Suppose (a) holds. For each fixed n, let $N_n = \{X_n \neq Y_n\}$. This is a measurable null event. Their union $N = \bigcup_{n\geq 0} N_n$ is countable, and countable subadditivity gives $\mathsf{P}(N) \leq \sum_{n\geq 0} \mathsf{P}(N_n) = 0$. On the complement N^c, none of the individual inequalities occurs; thus every equality $X_n = Y_n$ holds on that same event. Since $\mathsf{P}(N^c) = 1$, this proves (b).

Conversely, the event of simultaneous equality is contained in $\{X_n = Y_n\}$ for each chosen n. If the smaller event has probability one, monotonicity forces the larger event to have probability one as well, proving (a).

The key point is not the ability to choose one version at a particular date, but the countability of all dates under consideration. Each date can initially have a different exceptional null set; a countable union combines them into one null set. No independence of the events or continuity of trajectories is needed in discrete time.

HW 83 — Equality at each time without equal trajectories

Let $(X_t)_{t\geq 0}$ and $(Y_t)_{t\geq 0}$ be stochastic processes in continuous time. Find an example of such X and Y for which

$$\mathsf{P}(X_t = Y_t) = 1, \text{ for all } \ t \geq 0,$$

but $\mathsf{P}(X_t = Y_t, \text{for all } t \geq 0) = 0$. Hint: do not try continuous trajectories.

Solution to HW 83. Take $\Omega = [0, 1]$ with its Borel sigma-algebra and Lebesgue probability, and let $U(\omega) = \omega$. Define

$$X_t(\omega) = \mathbf{1}_{\{U(\omega)=t\}},\ Y_t(\omega) = 0,\ t \geq 0.$$

For every fixed $t \in [0, 1]$, the exceptional event is the singleton $\{U = t\} = \{t\}$, which is Borel and has probability zero. For $t > 1$ it is empty. Thus each X_t is a random variable and $\mathsf{P}(X_t = Y_t) = 1$ for every deterministic time.

On the other hand, fix any path $\omega \in [0,1]$ and choose its own time $t = U(\omega)$. At that time $X_t(\omega) = 1$ whereas $Y_t(\omega) = 0$. Every single path therefore fails equality at some time. The event of equality at all times is exactly the empty set, which is measurable and has probability zero.
The difference from discrete time is explicit: the union of the individual null exceptional sets is $\bigcup_{t\in[0,1]}\{t\} = [0,1]$, an uncountable union of full total mass. No countable-null-union theorem applies to it. Each path of X has an isolated spike at its random time, so the example intentionally lacks continuity. If both processes had continuous paths, equality on all nonnegative rational times outside one null set would extend to all real times by density and continuity, excluding this phenomenon.

10. Convergence theorem for martingales

Upcrossings and a Previsible Strategy

Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_n), \mathsf{P})$ and an adapted stochastic process $(X_n)_{n\in\mathbb{Z}^+}$. Given $a < b$ we say that X performs an *upcrossing* of $[a,b]$ by time n if, starting from $X_0 < a$ we have $X_n > b$ and $X_k \le b$ for all $k < n$. The symmetric situation is called a *downcrossing* of $[a,b]$. The terminal date is excluded from the preceding $X_k \le b$ condition: including it would contradict $X_n > b$. This corrects the boundary index in the source description.
We denote by $U_N[a,b](\omega)$ the number of upcrossing of $[a,b]$ made by time N. That is, for fixed $\omega \in \Omega$, $U_N[a,b](\omega)$ is the largest $k \in \mathbb{Z}^+$ for which we can find a sequence

$$0 \le s_1 < t_1 < s_2 < t_2 < \ldots < s_k < t_k \le N$$

such that $X_{s_i}(\omega) < a$ and $X_{t_i}(\omega) > b$ for all $1 \le i \le k$. Our next task is to estimate $\mathsf{E}[U_N[a,b]]$. Let $(C_n)_{n\in\mathbb{N}}$ be a previsible process that is equal to one for as long as X is performing an upcrossing and it is equal to zero during a downcrossing of an interval $[a,b]$. Set $C_0 = 0$. Formally:

$$C_n = 1_{\{C_{n-1}=1\}\cap\{X_{n-1}\le b\}} + 1_{\{C_{n-1}=0\}\cap\{X_{n-1}<a\}}.$$

Define $Y_0 = 0$ and

$$Y_N = \sum_{n=1}^{N} C_n(X_n - X_{n-1}).$$

HW 84 — Upcrossing gains and the terminal-loss correction

Show that $Y_N(\omega) \ge (b-a)U_N[a,b](\omega) - \big(X_N(\omega) - a\big)^-$.

Solution to HW 84. The two cases in the recurrence are disjoint, so C_n always equals zero or one. Each condition uses only C_{n-1} and X_{n-1}, known by time $n-1$. Inductively the holding process is therefore previsible and bounded. The convention $C_0 = 0$ gives $C_1 = \mathbf{1}_{\{X_0<a\}}$, so an initial value below a starts the first holding interval correctly.
Fix a path. On a completed purchase–sale interval, buying occurs at a date s with $X_s < a$ and selling at t with $X_t > b$. The gain accumulated while holding telescopes: $\sum_{n=s+1}^{t}(X_n - X_{n-1}) = X_t - X_s > b - a$. The greedy rule of taking the first available value below a and the first following value above b completes as many disjoint crossings as possible. To see this, compare with any admissible sequence of crossing pairs: the greedy first purchase and sale occur no later than the first pair of that sequence; after its sale the same argument applies inductively to the remaining pairs. Thus the strategy completes $U_N[a,b]$ crossings by N.
There can be at most one unfinished holding interval after those completed crossings. If its purchase date is s, its terminal gain is $X_N - X_s \ge X_N - a$. This lower bound can be negative, so the bound uniformly valid in both signs is $X_N - X_s \ge X_N - a \ge -(X_N - a)^-$. If no interval remains open, its contribution is zero and the same lower bound still holds. Summing the completed gains and this possible unfinished gain proves $Y_N \ge (b-a)U_N[a,b] - (X_N - a)^-$. It is the downside below the purchase threshold that controls an open loss; the positive part above that threshold cannot do so.

For a direct counterexample to the printed sign, let the deterministic path have $X_0 = -1$, $X_1 = -2$ and choose $a = 0, b = 1$. The strategy holds one unit, so $Y_1 = -1$, and no upcrossing is completed, so $U_1 = 0$. The printed right side with $(X_1 - a)^+$ equals zero and would assert $-1 \geq 0$, which is false. This deterministic decreasing path is even a supermartingale, so the error cannot be removed by the later martingale hypothesis.
This completes the pathwise assertion required here. Its expectation consequence is proved separately in the following Lemma 10.1, in the lecturer's order.

Lemma 10.1 — Doob Upcrossing Lemma
Let X be a supermartingale. Then $(b-a)\mathsf{E}\big[U_N[a,b]\big] \leq \mathsf{E}\big[\big(X_N - a\big)^-\big]$

Expanded proof. The process C defined above is bounded, non-negative and previsible. Thus Theorem 9.1(i) makes $Y = (C \bullet X)$ a supermartingale with $Y_0 = 0$, and $\mathsf{E}Y_N \leq 0$. The pathwise inequality already proved in HW 84 gives

$$Y_N \geq (b-a)U_N[a,b] - (X_N - a)^-.$$

The negative part is integrable because $(X_N - a)^- \leq |X_N| + |a|$. The pathwise bound also gives

$$0 \leq (b-a)U_N[a,b] \leq Y_N + (X_N - a)^-.$$

The upper bound is integrable, so $U_N[a,b]$ is integrable before expectations are rearranged.
Taking expectations and rearranging yields

$$(b-a)\mathsf{E}U_N[a,b] \leq \mathsf{E}Y_N + \mathsf{E}[(X_N - a)^-] \leq \mathsf{E}[(X_N - a)^-].$$

Here the negative part is the rigorously corrected source convention: the positive part printed in the source does not control a loss when $X_N < a$. The same correction is used in the preceding HW 84. QED

There is a useful corollary, concerning the total number of upcrossings. First notice that $U_N[a,b](\omega) \leq U_{N+1}[a,b](\omega)$ and therefore the limit below is well-defined:

$$U_\infty[a,b](\omega) := \lim_{N\to\infty} U_N[a,b].$$

Corollary 10.2 — Finitely Many Upcrossings
Let X be a supermartingale bounded in $\mathcal{L}^1$, i.e.,

$$\sup_n \mathsf{E}[|X_n|] < \infty.$$

Then

$$(b-a)\mathsf{E}\big[U_\infty[a,b]\big] \leq |a| + \sup_n \mathsf{E}|X_n| < \infty,$$

so that

$$\mathsf{P}\big(U_\infty[a,b] = \infty\big) = 0.$$

Expanded proof. Completed upcrossings cannot decrease with the horizon: $U_N[a,b] \uparrow U_\infty[a,b]$. Since $(X_N - a)^- \leq |X_N| + |a|$, the preceding lemma gives

$$(b-a)\mathsf{E}U_N[a,b] \leq |a| + \sup_n \mathsf{E}|X_n| =: C < \infty.$$

MON passes to the limit: $(b-a)\mathsf{E}U_\infty[a,b] \leq C$. Because $b - a > 0$, the non-negative random variable $U_\infty[a,b]$ has finite expectation and is therefore finite almost surely by HW 54. Thus infinitely many upcrossings of this fixed interval form a null event. QED

From Oscillation Bounds to Almost Sure Limits
We can use the corollary to prove a martingale convergence result. For each $\omega \in \Omega$ we define the $\mathcal{F}_\infty$-measurable random variable

$$X_\infty(\omega) := \limsup_{n\to\infty} X_n(\omega).$$

The next theorem states conditions under which supermartingales admit limit as $n \to \infty$.

Theorem 10.3 — Almost Sure Martingale Convergence
If X is a supermartingale bounded in $\mathcal{L}^1$ (i.e., $\sup_n \mathsf{E}|X_n| < \infty$), then almost surely

$$\lim_{n\to\infty} X_n = X_\infty$$

and it is almost surely finite.

Expanded proof. If a path has distinct lower and upper limits, density of the rationals supplies $a < b$, both rational, strictly between them. Infinitely many values lie below a and infinitely many lie above b. After each such lower value choose a later upper value, then a later lower one, and continue. This constructs infinitely many upcrossings. Consequently

$$\{\liminf_n X_n < \limsup_n X_n\} \subseteq \bigcup_{\substack{a,b\in\mathbb{Q}\\ a<b}} \{U_\infty[a,b] = \infty\}.$$

Each event on the right is null by the preceding corollary. There are only countably many rational pairs, so their union is null. Outside it an extended limit X_∞ exists. Fatou excludes infinite values:

$$\mathsf{E}|X_\infty| \le \liminf_n \mathsf{E}|X_n| \le \sup_n \mathsf{E}|X_n| < \infty.$$

In the last chain, the first inequality is Fatou for $|X_n|$, whose pointwise limit is the absolute value of the extended limit on the convergence event. The second bounds a liminf by the supremum of the same numerical sequence. Finite expectation of a non-negative extended random variable excludes an infinite value on a set of positive probability.
Hence the limit is finite almost surely. This establishes almost sure convergence; bounded L^1 norms alone do not imply convergence in L^1. QED

HW 85 — Convergence of a nonnegative supermartingale
Show that any non-negative supermartingale is bounded in $\mathcal{L}^1$. Deduce that if X is a non-negative supermartingale, then $X_\infty = \lim_{n\to\infty} X_n$.
Solution to HW 85. Each X_n is integrable by the supermartingale definition and nonnegative by hypothesis. Therefore $\mathsf{E}|X_n| = \mathsf{E}X_n$. Taking ordinary expectations in $\mathsf{E}[X_{n+1} \mid \mathcal{F}_n] \le X_n$ and using the defining whole-space test identity gives $\mathsf{E}X_{n+1} \le \mathsf{E}X_n$. Induction consequently yields $\mathsf{E}|X_n| \le \mathsf{E}X_0$ for every n, so

$$\sup_n \mathsf{E}|X_n| \le \mathsf{E}X_0 < \infty.$$

This is exactly the L^1 boundedness condition, even though no deterministic pointwise bound on all values X_n has been assumed.
The supermartingale convergence theorem states that an L^1-bounded discrete-time supermartingale converges almost surely to a finite random variable. Its mechanism is the upcrossing estimate: for any rational $a < b$, the expected number of completed upcrossings is uniformly bounded by a constant proportional to $\sup_n \mathsf{E}|X_n| + |a|$. Monotone convergence in the crossing count then makes its total number finite almost surely. If lower and upper limits differed, some rational interval between them would have infinitely many upcrossings. There are countably many rational intervals, so that oscillation event is null. Thus an extended-valued limit exists almost surely.
Finiteness follows from Fatou rather than from an unproved pointwise bound. On the convergence event, nonnegativity and Fatou give $\mathsf{E}X_\infty \le \liminf_n \mathsf{E}X_n \le \mathsf{E}X_0 < \infty$. A nonnegative variable with finite expectation is finite almost surely, so $X_\infty < \infty$ almost surely. This also shows $X_\infty \in L^1$. The conclusion is almost-sure convergence; it does not by itself assert convergence in L^1 or equality of the limiting and initial expectations.

11. Black and Scholes model in discrete time

Let $\mathcal{S} = \{-1, 1\}$ and $\Sigma = \mathcal{P}(\mathcal{S})$. Take $(\mathcal{S}, \Sigma, \mu)$ with $\mu(\{1\}) = p \in [0, 1]$. In order to properly define a *ramdom walk* we need the product space

$$(\Omega, \mathcal{F}, \mathsf{P}) = (\mathcal{S}, \Sigma, \mu)^{\mathbb{N}}.$$

An element from Ω is:

$$\omega = (\omega_1, \omega_2, \dots, \omega_k, \dots)$$

with each ω_j equal to either -1 or 1. We define the *canonical process* $(\varepsilon_k)_{k\in\mathbb{Z}^+}$ with $\varepsilon_0(\omega) = 0$ and $\varepsilon_k(\omega) = \omega_k$, $k \in \mathbb{N}$. Since ω is defined on a product space, then ε_j and ε_k are independent. Actually $(\varepsilon_n)_{n\in\mathbb{N}}$ us an i.i.d. sequence on $(\Omega, \mathcal{F}, \mathsf{P})$ with $\mathsf{P}(\varepsilon_k = 1) = p$ and $\mathsf{P}(\varepsilon_k = -1) = 1 - p$.
A random walk is a process Z defined as

$$Z_0 = 0,\ Z_n = \sum_{k=1}^{n} (\varepsilon_k - 2p + 1).$$

Its natural filtration is

$$\mathcal{F}_n = \sigma(Z_0, Z_1, \dots Z_n) = \sigma(\varepsilon_0, \varepsilon_1, \dots \varepsilon_n).$$

Notice that $\mathcal{F}_0 = \{\Omega, \varnothing\}$, so any $X \in m\mathcal{F}_0$ must be deterministic.

HW 86 — Centring the canonical sign walk

Show that Z is a $(\mathcal{F}_n, \mathsf{P})$-martingale.

Solution to HW 86. The mean of one sign is $p - (1 - p) = 2p - 1$. Thus the increment $\Delta Z_n = \varepsilon_n - (2p - 1)$ is precisely the sign with its mean subtracted. Its two values are $2(1 - p)$ on an up sign and $-2p$ on a down sign; both have absolute value at most two.
The partial sum is measurable with respect to the first n signs, so it is adapted. Also $|Z_n| \le 2n$, yielding $\mathsf{E}|Z_n| < \infty$ at each fixed date. For the conditional identity, Z_{n-1} is already known at $\mathcal{F}_{n-1}$, while the next independent sign has its original distribution under that conditioning. Hence

$$\mathsf{E}[Z_n \mid \mathcal{F}_{n-1}] = Z_{n-1} + \mathsf{E}[\varepsilon_n - 2p + 1 \mid \mathcal{F}_{n-1}] = Z_{n-1} + (2p - 1) - 2p + 1 = Z_{n-1}.$$

The first equality uses the increment formula, conditional linearity, and measurability of Z_{n-1}. The second uses independence and $\mathsf{E}\varepsilon_n = 2p - 1$, leaving deterministic constants unchanged. The third cancels $(2p - 1) - 2p + 1 = 0$.
This proves all martingale conditions. At the degenerate endpoints $p = 0$ or $p = 1$ the centred increments vanish almost surely, so the conclusion still holds. Later representation formulas involving division by p or $1 - p$ require $0 < p < 1$ separately; they do not follow at those endpoints from this martingale calculation alone.

Representing Binary Risks

Qualification — Non-degeneracy

Uniqueness of the binary representation requires $0 < p < 1$. At the endpoints the random branch degenerates and its coefficient is not identifiable.

Lemma 11.1 — Martingale Representation

If $M = (M_n)_{n\in\mathbb{Z}^+}$ is a $(\mathcal{F}_n, \mathsf{P})$-martingale, then there exists a unique previsible process $H = (H_n)_{n\in\mathbb{N}}$ such that $M = M_0 + (H \bullet Z)$, i.e.,

$$M_n = M_0 + \sum_{k=1}^{n} H_k(Z_k - Z_{k-1}).$$

Expanded proof. Assume $0 < p < 1$. At a fixed node with history through date $n-1$, the next value of M_n has two possibilities m^+, m^-. The martingale property gives $M_{n-1} = pm^+ + (1 - p)m^-$. Thus its two increments are $(1 - p)(m^+ - m^-)$ and $-p(m^+ - m^-)$. The corresponding increments of Z are $2(1 - p)$ and $-2p$. Set $H_n = (m^+ - m^-)/2$ at that node. Both branches then satisfy $\Delta M_n = H_n \Delta Z_n$. The coefficient depends only on the preceding node, so it is $\mathcal{F}_{n-1}$-measurable. Summing gives $M_n = M_0 + \sum_{k=1}^{n} H_k \Delta Z_k$. For uniqueness, the difference of the two branch increments equals $2H_n$, determining H_n at every positive-probability node. Thus the coefficients at dates $n \ge 1$ are unique up to almost sure equality. A coefficient at date zero is unused and needs a convention if one chooses to define it. At $p = 0$ or $p = 1$ the reference increments vanish, so uniqueness would fail. The non-degeneracy assumption is necessary. QED

The Black and Scholes model.
We consider a financial market with two traded assets:

- A riskless asset (bond) growing at a constant interest rate $r > 0$. The bond price follows the dynamics:

$$B_{n+1} = (1 + r)B_n$$

  so that $B_n = B_0(1 + r)^n$ (deterministic).

- A risky asset whose interest rate is a stochastic process $(R_n)_{n\in\mathbb{N}}$. The dynamics of the risky asset reads

$$S_n = (1 + R_n)S_{n-1}$$

  so that

$$S_n = S_0 \prod_{k=1}^{n} (1 + R_k).$$

  Notice that $R_k = R_k(\omega)$ and therefore $S_k = S_k(\omega)$.

An investor with initial wealth $x \geq 0$ builds a portfolio containing shares of the two assets. The dynamics of the portfolio is described below:

- *At time 0*: she buys A_0 units of the stock and invests the rest in bonds. The number of units of bonds reads

$$V_0 = \frac{x - A_0S_0}{B_0}$$

  so that $x = A_0S_0 + V_0B_0$.

- *Between time 0 and time 1*: the investor can rebalance her portfolio but without changing its value (*self-financing*). The new holdings (A_1, V_1) are such that

$$x = A_1S_0 + V_1B_0.$$

- *At time 1*: the investor observes the new market prices of stock and bond (A_1, V_1). Her portfolio value becomes

$$X_1 = A_1S_1 + V_1B_1.$$

- *Between time 1 and time 2*: the investor can rebalance her portfolio but without changing its value (*self-financing*). The new holdings (A_2, V_2) are such that

$$X_1 = A_2S_1 + V_2B_1.$$

- *At time 2*: the investor observes the new market prices of stock and bond (A_2, V_2). Her portfolio value becomes

$$X_2 = A_2S_2 + V_2B_2.$$

- *etc. etc.*

Iterating the procedure described above we arrive at the dynamics of the portfolio value described by two equations:

$$X_{n-1} = A_nS_{n-1} + V_nB_{n-1}$$

$$X_n = A_nS_n + V_nB_n.$$

Subtracting the two expressions we arrive at

$$X_n - X_{n-1} = A_n(S_n - S_{n-1}) + V_n(B_n - B_{n-1}).$$

Substituting the dynamics of the two assets $B_n - B_{n-1} = rB_{n-1}$ and $S_n - S_{n-1} = R_nS_{n-1}$ gives

$$X_n - X_{n-1} = A_nR_nS_{n-1} + rV_nB_{n-1}.$$

Expressing $V_n B_{n-1} = X_{n-1} - A_n S_{n-1}$ we have

$$X_n - (1+r)X_{n-1} = A_n(R_n - r)S_{n-1}.$$

It is financially natural and mathematically preferable to work with discounted quantities $Y_n = (1+r)^{-n}X_n$:

$$Y_n - Y_{n-1} = (1+r)^{-n}A_n(R_n - r)S_{n-1}.$$

Given the random walk $(Z_n)_{n\in\mathbb{Z}^+}$ we assume that the dynamics of the interest rate $(R_n)_{n\in\mathbb{Z}^+}$ be adapted to the filtration $(\mathcal{F}_n)_{n\in\mathbb{Z}^+}$ generated by Z (or equivalently by $(\varepsilon_n)_{n\in\mathbb{N}}$). In an arbitrage-free market we must have that $(Y_n)_{n\in\mathbb{Z}^+}$ be a martingale. Thus we must choose a suitable model for the interest rate. Assuming for example that there are $-1 < a < r < b$ such that for all $n \in \mathbb{N}$

$$R_n(\omega) = \frac{a+b}{2} + \frac{b-a}{2}\varepsilon_n(\omega)$$

we notice that R is $\mathcal{F}_n$-adapted and

$$R_n \in \{a, b\},\ \mathsf{P}(R_n = b) = p,\ \mathsf{P}(R_n = a) = 1 - p.$$

Source Qualification — The Up-Branch Probability

Under the convention $\mathsf{P}(\varepsilon_n = 1) = p$, the return formula gives $\mathsf{P}(R_n = b) = p$ and

$$\mathsf{P}(R_n = a) = 1 - p.$$

The source reverses these two probabilities in its description; the consistent convention is used here.

Discounting and Martingale Probabilities

HW 87 — Discounted portfolios and the risk-neutral probability

Show that Y is a martingale if and only if $p = \frac{r-a}{b-a}$.

Solution to HW 87. The return formula fixes which probability belongs to which branch. Substituting $\varepsilon_n = 1$ gives $R_n = b$, while substituting -1 gives $R_n = a$. Thus the probabilities consistent with the formula are

$$\mathsf{P}(R_n = b) = p,\ \mathsf{P}(R_n = a) = 1 - p.$$

The sentence assigning these probabilities in the opposite order is inconsistent with this displayed model. We keep the original definition of p as the positive-sign probability throughout.

At date $n-1$, the previsible A_n and the stock value S_{n-1} are known. The future return is independent of this past and has conditional mean

$$\mathsf{E}[R_n \mid \mathcal{F}_{n-1}] = pb + (1-p)a = a + p(b-a).$$

The first equality uses independence from the past and the two return values with their probabilities. The second expands $(1-p)a = a - pa$ and collects the coefficient of p.

Taking the conditional mean of the discounted gain gives

$$\mathsf{E}[Y_n - Y_{n-1} \mid \mathcal{F}_{n-1}] = (1+r)^{-n}A_n S_{n-1}\,[a + p(b-a) - r].$$

The multiplier before the bracket is known at the conditioning date; for fixed finite dates all values are integrable in this finite-branch model with finite-valued holdings. Setting the deterministic bracket to zero therefore makes every strategy's discounted conditional gain zero. Solving the scalar equation gives $p(b-a) = r-a,\ p = \frac{r-a}{b-a}$. The inequalities $a < r < b$ imply $0 < r-a < b-a$, so this is a valid strictly interior probability. Adaptation comes from previsible holdings and adapted prices, and finite-date integrability from the finite tree. Thus zero conditional gain establishes the full martingale property, not only an equality of unconditional means.

Conversely, suppose *every* discounted self-financing portfolio is a martingale. Choose the buy-and-hold stock portfolio with a nonzero stock position. Already at its first step the known stock value is positive, so the preceding conditional equality can vanish only if the bracket is zero. This forces the same probability, proving necessity for the statement about all portfolios.

For one particular strategy the bare "if and only if" is false without a nonzero-exposure condition. An all-bond portfolio has $A_n = 0$ and constant discounted value for every p, so cannot identify the probability. More generally, the displayed drift only imposes the scalar equation at dates and states with nonzero stock exposure. The corrected assertion is that the computed p makes all discounted portfolios martingales, and is necessary for that universal claim or for a suitable nonzero-stock portfolio.

HW 88 — The discounted stock martingale condition

Show that $\{(1+r)^{-n}S_n, n \in \mathbb{Z}^+\}$ *is a martingale if and only if* $p = \frac{r-a}{b-a}$.

Solution to HW 88. Both stock multipliers $1+a, 1+b$ and the bank multiplier $1+r$ are positive. Thus S_n and D_n remain positive on every finite path. They are adapted functions of the observed return history. At fixed date n there are only finitely many values, all finite, so they are integrable. From the definition of the discount, $D_n = D_{n-1}\frac{1+R_n}{1+r}$. The previous value is $\mathcal{F}_{n-1}$-measurable. Independence of R_n from that past gives

$$\mathsf{E}[D_n \mid \mathcal{F}_{n-1}] = D_{n-1}\frac{p(1+b)+(1-p)(1+a)}{1+r} = D_{n-1}\frac{1+a+p(b-a)}{1+r}.$$

The first equality uses the discounted-stock recurrence, takes out the known D_{n-1}, and averages the two independent return factors. The second expands and collects the numerator:

$$p + pb + 1 - p + a - pa = 1 + a + p(b-a).$$

This equals D_{n-1} exactly when the conditional gain is zero: $D_{n-1}\frac{a+p(b-a)-r}{1+r} = 0$. Since $D_{n-1} > 0$ and $1+r > 0$, division is legitimate and yields $a + p(b-a) = r$. Solving and using $b - a > 0$ gives $p = (r-a)/(b-a)$. Conversely this probability makes the factor in the conditional expectation equal one, and the previously checked adaptation and integrability complete sufficiency. Its value lies in $(0,1)$ because $a < r < b$.

The positive initial stock value is needed for the necessity argument. If $S_0 = 0$, the stock remains identically zero and is a discounted martingale for any probability. Also, using the opposite branch convention would produce the down probability $1-p = (b-r)/(b-a)$; this is the same risk-neutral law with different notation, not a competing formula for the same p.

From now on we assume $p = \frac{r-a}{b-a}$. In particular, R can be expressed in terms of Z as

$$R_n(\omega) - r = \frac{1}{2}(b-a)\big(Z_n(\omega) - Z_{n-1}(\omega)\big).$$

Bear in mind that since Y is a martingale, then there is a previsible process H such that

$$Y_n = Y_0 + \sum_{k=1}^{n} H_k(Z_k - Z_{k-1}).$$

Pricing a European option on the stock

A European option gives the holder the right to buy or sell (call or put option) the underlying asset at a fixed future date N (maturity) for a fixed price K (strike price) which is set at time zero. In particular, the call option payoff at maturity reads:

$$\text{payoff} = (S_N(\omega) - K)^+$$

which means payoff $= S_N(\omega) - K$ if $S_N(\omega) > K$ and zero otherwise. The option price corresponds to the minimal amount of money needed at time zero for the seller to set-up a so-called replicating portfolio (hedging portfolio). That is a bond-stock portfolio that at time N is worth at least as much as the option payoff. This is made rigorous in the next definition.

Definition 11.1 — Hedging Strategy

A pair $(A_n, V_n)_{n=1}^N$ *of* $\mathcal{F}_n$*-adapted processes, where* $(A_n)_{n=1}^N$ *is previsible, is a hedging strategy for the European call option if the associated portfolio* $(X_n)_{n=0}^N$ *has the following properties:*

(h1) $X_0(\omega) = x$ *for all* $\omega \in \Omega$,

(h2) $X_n(\omega) \geq 0$ *for* $n = 1, \ldots N$ *and all* $\omega \in \Omega$,

(h3) $X_N(\omega) = (S_N(\omega) - K)^+$ *for all* $\omega \in \Omega$.

Theorem 11.2 — Black and Scholes Pricing Formula

A hedging strategy for the European call option described above, with initial value x exists if and only if

$$x = x_0 := \mathsf{E}\big[(1+r)^{-N}\big(S_N - K\big)^+\big],$$

where $(R_n)_{n\in\mathbb{N}}$ is given by (source formula 11.1) *and $p = \frac{r-a}{b-a}$ under the measure P.*
The value x_0 is the so-called rational price (fair price) of the option.

Expanded proof. Let $B_n = (1+r)^n$ and $\widetilde{S}_n = S_n/B_n$. For $p = (r-a)/(b-a)$, the discounted price is a martingale. The assumption $a < r < b$ ensures $0 < p < 1$. A self-financing portfolio satisfies $\Delta(X_n/B_n) = A_n \Delta\widetilde{S}_n$. On a finite binary tree there are finitely many nodes at each date, so finite real-valued positions and values are integrable. Therefore a hedge with $X_N = \Phi(S_N)$ necessarily has $x = X_0 = \mathsf{E}[B_N^{-1}\Phi(S_N)]$.
Conversely, define $M_n = \mathsf{E}[B_N^{-1}\Phi(S_N) \mid \mathcal{F}_n]$. This is a martingale by the tower property. The binary representation gives $\Delta M_n = H_n \Delta Z_n$. Directly from the return formula, $\Delta\widetilde{S}_n = \frac{1}{2}(b-a)(1+r)^{-n}S_{n-1}\Delta Z_n$. Choose

$$A_n = \frac{2}{b-a}(1+r)^n \frac{H_n}{S_{n-1}},\ X_n = (1+r)^n M_n.$$

The factor multiplying ΔZ_n in the discounted stock increment is $(b-a)(1+r)^{-n}S_{n-1}/2$. Its reciprocal gives the displayed holding A_n. Thus

$$A_n \Delta\widetilde{S}_n = H_n \Delta Z_n = \Delta M_n,$$

where the first equality is cancellation of that strictly positive factor and the second is martingale representation.
Prices are strictly positive, so division is valid, and the coefficient is previsible. The discounted-increment identity proves self-financing. Choose the bank holding at the preceding date as $V_n = \frac{X_{n-1} - A_n S_{n-1}}{B_{n-1}}$. The same increment identity then gives

$$X_n = A_n S_n + V_n B_n.$$

Finally, $M_N = B_N^{-1}\Phi(S_N)$, so $X_N = \Phi(S_N)$. The exponent $+n$ in the share holding comes from inverting the discount factor and corrects the sign in the source formula. QED

It is worth notice that we do not quite know explicitly the hedging portfolio because H in the proof above is not explicit...

12. Technical appendix

Definition 12.1 — Topological Space

A topological space is a pair $(X, \mathcal{X})$ where X is a set and $\mathcal{X}$ is a collection of subsets of X with the following properties

(i) $X \in \mathcal{X}$ *and* $\varnothing \in \mathcal{X}$;
(ii) $A, B \in \mathcal{X}$ *implies* $A \cap B \in \mathcal{X}$;
(iii) given an arbitrary collection $(A_\alpha)_\alpha \subset \mathcal{X}$ *it holds* $\bigcup_\alpha A_\alpha \in \mathcal{X}$.

The sets in the class $\mathcal{X}$ are called open *sets. The complement $A^c = X \setminus A$ of an open set A is a* closed *set.*
The main example of topological space that we have in mind is the real line $\mathbb{R}$ equipped with the topology generated by open intervals (a, b).

Definition 12.2 — d-System

Given a set $\mathcal{S}$ a class $\mathcal{D}$ of subsets of $\mathcal{S}$ is a d-system if

(i) $\mathcal{S} \in \mathcal{D}$;
(ii) $A, B \in \mathcal{D}$ *and* $A \subset B$ *imply* $B \setminus A \in \mathcal{D}$;
(iii) $(A_n)_{n\in\mathbb{N}}$ *and* $A_n \uparrow A$ *imply* $A \in \mathcal{D}$.

It is worth noticing that conditions (i)–(ii) above also yield $A \in \mathcal{D} \implies A^c = \mathcal{S} \setminus A \in \mathcal{D}$. Given an arbitrary collection $\mathcal{C}$ of subsets of $\mathcal{S}$, we denote by $d(\mathcal{C})$ the smallest d-system that contains $\mathcal{C}$ — we say that $d(\mathcal{C})$ is the d-system generated by $\mathcal{C}$. There is a link between π-systems, d-systems and σ-algebras, which is explored in the next proposition.

Proposition 12.1 — Pi-Systems and d-Systems
A class Σ is a σ-algebra if and only if it is both a π-system and d-system.

Expanded proof. A sigma-algebra is a pi-system because it is closed under finite intersections. It is a d-system because it contains the whole space, is closed under differences of nested members, and under increasing countable unions. Conversely, suppose the class is both a pi-system and a d-system. The whole space belongs to it, so the nested-difference property gives $A^c = \mathcal{S} \setminus A$ for every member A. Intersections and De Morgan give finite unions: $A \cup B = (A^c \cap B^c)^c$. For a sequence (A_n) of members, the finite unions $G_k = \bigcup_{n=1}^{k} A_n$ are members and increase to $\bigcup_{n\geq 1} A_n$. The increasing-union property of the d-system includes that limit. All sigma-algebra axioms have now been checked. QED

Lemma 12.2 — The Pi–d Lemma
If $\mathcal{I}$ is a π-system, then $d(\mathcal{I}) = \sigma(\mathcal{I})$.

Proof. A d-system is equivalently a Dynkin class in the whole-space, complement and disjoint-union formulation of the pi–d theorem in the lecturer's technical appendix. To check this, nested differences give complements; finite disjoint unions follow by complementing a nested difference, and increasing-union closure then gives countable disjoint unions. Conversely a Dynkin class has nested-difference closure by the nested-difference property; for an increasing sequence, its successive differences are disjoint and have the original union, giving increasing-union closure. Thus the two formulations define the same smallest generated class. The pi–d theorem in that formulation therefore gives $d(\mathcal{I}) = \sigma(\mathcal{I})$, as asserted. QED

Source Correction — Order of the Set Difference
The source reverses the two terms on the right. The correctly ordered identity is stated in the exercise below.

HW 89 — Intersection distributes over set difference
Prove $\big(B \setminus A\big) \cap C = (B \cap C) \setminus (A \cap C)$.

Solution to HW 89. *We prove the correctly ordered identity* $(B \setminus A) \cap C = (B \cap C) \setminus (A \cap C)$. For the first inclusion, let $s_0 \in (B \setminus A) \cap C$. Then $s_0 \in B$, $s_0 \notin A$, and $s_0 \in C$. The first and third facts give $s_0 \in B \cap C$. The second excludes membership in $A \cap C$. Thus s_0 belongs to the right-hand difference. For the reverse inclusion, let $s_0 \in (B \cap C) \setminus (A \cap C)$. Then $s_0 \in B \cap C$, but $s_0 \notin A \cap C$. Since $s_0 \in C$ is already known, $s_0 \in A$ would contradict the latter fact. Hence $s_0 \notin A$. It follows that $s_0 \in B \setminus A$ and $s_0 \in C$, giving the desired membership on the left. Combining the two inclusions proves the identity.
Equivalently, the set-algebra calculation is

$$\begin{aligned}(B \cap C) \setminus (A \cap C) &= (B \cap C) \cap (A \cap C)^c = (B \cap C) \cap (A^c \cup C^c) \\ &= (B \cap C \cap A^c) \cup (B \cap C \cap C^c) = B \cap C \cap A^c = (B \setminus A) \cap C.\end{aligned}$$

The first equality defines set difference. The second is De Morgan's law; the third is distributivity. The fourth uses $C \cap C^c = \varnothing$, and the last rearranges intersections and uses the definition of $B \setminus A$. For the reversed order printed in the source, take a nonempty space S, $A = \varnothing$, and $B = C = S$. Then the left side is S, whereas $(A \cap C) \setminus (B \cap C) = \varnothing$. Thus the printed reversal is false. The corrected order above is the one needed for nested intersections in the d-system argument.

Uniqueness of Measures

Using the above lemma we can prove the uniqueness lemma for measures (Lemma 1.2).

Proof. Lemma 1.2 is stated at its original position as the lecturer's finite-measure uniqueness result. Its finite equal total masses and generator agreement are precisely the hypotheses recalled here; applying it proves uniqueness on the whole generated sigma-algebra. QED

The use of d-systems also gives a functional monotone-class theorem, making precise a general principle behind the generating-class arguments used in integration.

Theorem 12.3 — Functional Monotone Class Theorem

Let $\mathcal{H}$ be a class of bounded functions $f : \mathcal{S} \to \mathbb{R}$ such that

(i) $\mathcal{H}$ is a real vector space;

(ii) $\mathcal{H}$ contains the function $f(s) = 1$, $s \in \mathcal{S}$;

(iii) if $(f_n)_{n\in\mathbb{N}} \subset \mathcal{H}$ is such that $0 \le f_n \uparrow f$ and f is bounded, then $f \in \mathcal{H}$.

If in addition $\mathcal{H}$ contains $f(s) = 1_I(s)$ for all $I \in \mathcal{I}$, $\mathcal{I}$ a π-system, then $\mathcal{H}$ contains all bounded $\sigma(\mathcal{I})$-measurable functions on $\mathcal{S}$.

Expanded proof. Define $\mathcal{D} = \{A \subseteq \mathcal{S} : \mathbf{1}_A \in \mathcal{H}\}$. This class is a d-system, as we will see in HW 90. It contains $\mathcal{I}$, so Lemma 12.2 gives $\sigma(\mathcal{I}) = d(\mathcal{I}) \subseteq \mathcal{D}$. Thus $\mathcal{H}$ contains the indicator of every set in the generated sigma-algebra. Its vector-space property then includes every measurable simple function. For $0 \le f \le K$, use the dyadic simple functions $f_n = 2^{-n}\lfloor 2^n f \rfloor$. Their ranges are finite and $f_n \uparrow f$. They belong to $\mathcal{H}$, so assumption (iii) gives $f \in \mathcal{H}$. For bounded signed f, both f^+ and f^- are bounded non-negative measurable functions. They belong to $\mathcal{H}$; subtracting them in this vector space gives $f \in \mathcal{H}$. QED

HW 90 — The indicator class is a d-system

Show that the class $\mathcal{D}$ in the proof above is a d-system on $\mathcal{S}$.

Solution to HW 90. Each d-system axiom corresponds to one of the function-space assumptions. First $\mathbf{1}_S = 1 \in H$ by hypothesis, so $S \in \mathcal{D}$.

Next take $A, B \in \mathcal{D}$ with $A \subset B$. Their indicators lie in H, and a real vector space is closed under their difference. The nesting is important: pointwise, $\mathbf{1}_B - \mathbf{1}_A$ is zero on A, one on $B \setminus A$, and zero outside B, so $\mathbf{1}_{B\setminus A} = \mathbf{1}_B - \mathbf{1}_A \in H$. Therefore $B \setminus A \in \mathcal{D}$. Without nesting, an indicator difference could take the value -1 and would not be the indicator of that set difference.

Finally let $A_n \in \mathcal{D}$ increase to $A = \bigcup_n A_n$. The indicator functions satisfy $0 \le \mathbf{1}_{A_n} \uparrow \mathbf{1}_A$ pointwise: once a point enters the increasing sets, it remains in them. The limit is bounded by one. All the hypotheses of the assumed monotone-limit closure are therefore satisfied, giving $\mathbf{1}_A \in H$ and hence $A \in \mathcal{D}$.

These are exactly the three stated d-system properties. In particular complements also follow by taking $B = S$ in the nested-difference property. We have not presumed that the class is closed under arbitrary intersections or unions; those require the additional pi-system argument in the monotone class theorem, rather than being part of this verification.

Definition — A Set That Splits Every Test Set

We now prepare the ground for the proof of Caratheodory's theorem (Theorem 1.3). Given an algebra $\mathcal{G}_0$ on $\mathcal{S}$ and a set-function $\lambda : \mathcal{G}_0 \to [0, \infty]$, we say that a set $L \in \mathcal{G}_0$ is λ-measurable if for any $G \in \mathcal{G}_0$

$$\lambda(G) = \lambda\big(G \cap L\big) + \lambda\big(G \cap L^c\big).$$

We denote $\mathcal{L}_0 \subset \mathcal{G}_0$ the class of all λ-measurable sets.

Qualification — The Empty-Set Convention

The next lemma requires $\lambda(\varnothing) = 0$, as for the pre-measure to which it is applied.

Lemma 12.4 — The Algebra of Splitting Sets

The class $\mathcal{L}_0$ forms an algebra on $\mathcal{S}$ and λ is finitely additive on $\mathcal{L}_0$. Moreover, given pairwise disjoint sets $(L_i)_{i=1}^n \subset \mathcal{L}_0$ and $G \in \mathcal{G}_0$, we have

$$\lambda\Big(\bigcup_{i=1}^{n}\big(L_i \cap G\big)\Big) = \sum_{i=1}^{n}\lambda\big(G \cap L_i\big).$$

Expanded proof. We use the necessary convention $\lambda(\varnothing) = 0$. The whole space splits every test set, and the splitting identity is symmetric in L, L^c; hence $\mathcal{S}$ belongs to the class and the class is closed under complements. Let L_1, L_2 belong to it and put $L = L_1 \cap L_2$. First split G by L_2, then $G \cap L_2$ by L_1: $\lambda(G) = \lambda(G \cap L) + \lambda(G \cap L_2 \cap L_1^c) + \lambda(G \cap L_2^c)$. We use

$$L^c \cap L_2 = L_1^c \cap L_2,\ L^c \cap L_2^c = L_2^c,$$

as we will see in HW 91. Splitting $G \cap L^c$ by L_2 identifies the last two terms with $\lambda(G \cap L^c)$. Thus L splits G, without subtracting possibly infinite quantities. Complement and finite-intersection closure make the class an algebra.

If L_1, L_2 are disjoint, split $G \cap (L_1 \cup L_2)$ by L_1. The pieces are $G \cap L_1$ and $G \cap L_2$, yielding the two-set additivity formula. To pass from k to $k+1$ disjoint pieces, split $G \cap \bigcup_{i=1}^{k+1} L_i$ by L_{k+1}. The two terms are $\lambda(G \cap L_{k+1})$ and the expression for the first k pieces. Induction gives the asserted finite-sum identity. Taking $G = \mathcal{S}$ proves finite additivity of the restriction. QED

HW 91 — Two identities for intersecting complements

For $L = L_1 \cap L_2$, prove $L^c \cap L_2 = L_1^c \cap L_2$ and $L^c \cap L_2^c = L_2^c$.

Solution to HW 91. Let $L = L_1 \cap L_2$. We prove each identity by two inclusions.

First identity: $L^c \cap L_2 = L_1^c \cap L_2$. Let $s_0 \in L^c \cap L_2$. Then $s_0 \notin L_1 \cap L_2$ and $s_0 \in L_2$. If s_0 belonged to L_1, these facts would put it in L, a contradiction. Hence $s_0 \in L_1^c \cap L_2$, proving the first inclusion. Conversely, let $s_0 \in L_1^c \cap L_2$. Its failure to belong to L_1 excludes membership in L. Thus $s_0 \in L^c$, and its given membership in L_2 places it in $L^c \cap L_2$. The two inclusions prove the identity.

Second identity: $L^c \cap L_2^c = L_2^c$. If $s_0 \in L^c \cap L_2^c$, intersection membership immediately gives $s_0 \in L_2^c$, proving the first inclusion. Conversely, let $s_0 \in L_2^c$. Then $s_0 \notin L_2$, so $s_0 \notin L_1 \cap L_2 = L$. Consequently it belongs to L^c as well as to L_2^c, which proves the reverse inclusion and the equality. These identities require neither measurability nor finite measure. When used in a measure argument, their disjoint splittings justify additive equalities; they do not justify an undefined subtraction of two infinite masses.

The construction of measures starts with the definition of so-called outer measures.

Definition 12.3 — Outer Measure

Let $\mathcal{G}$ be a σ-algebra of subsets of $\mathcal{S}$. A set function $\lambda : \mathcal{G} \to [0, \infty]$ is an outer measure if

(i) $\lambda(\varnothing) = 0$;

(ii) (Monotonicity) *for $G_1, G_2 \in \mathcal{G}$, $G_1 \subset G_2$, it holds $\lambda(G_1) \leq \lambda(G_2)$;*

(iii) (Sub-additivity) *for any sequence $(G_n)_{n\in\mathbb{N}} \subset \mathcal{G}$ we have*

$$\lambda\Big(\bigcup_{n\in\mathbb{N}} G_n\Big) \leq \sum_{n\in\mathbb{N}} \lambda(G_n).$$

The next lemma shows that we can construct a measure starting from an outer measure.

Lemma 12.5 — Measurability for an Outer Measure

Let λ be an outer measure on $(\mathcal{S}, \mathcal{G})$. Then the class $\mathcal{L}$ of λ-measurable sets forms a σ-algebra on $\mathcal{S}$. Moreover, λ is countably additive on $\mathcal{L}$ and therefore $(\mathcal{S}, \mathcal{L}, \lambda)$ forms a measure space.

Proof. The splitting-class argument in the authorised Lebesgue–Stieltjes appendix uses only outer-measure properties and sigma-algebra operations, so it also applies with all test sets restricted to a

sigma-algebra. Apply it with test sets in $\mathcal{G}$: the class of splitting members of $\mathcal{G}$ is a sigma-algebra, and the same finite-splitting estimates and outer subadditivity prove countable additivity on it. These are exactly the class $\mathcal{L}$ and conclusions in this statement. QED

The good news about the previous lemma is that it is possible to construct measures! The annoying fact is that the σ -algebra over which such measure is constructed is determined endogenously (and $\mathcal{L} \subset \mathcal{G}$). We would like to be able to construct a measure on an arbitrary σ -algebra and this is the object of interest in the proof of Theorem 1.3, which follows. We recall also the statement for convenience.

Theorem 12.6 — Extension of a Pre-measure

Let Σ_0 be an algebra of subsets of $\mathcal{S}$ and $\Sigma = \sigma(\Sigma_0)$. If $\mu_0 : \Sigma_0 \to [0, \infty]$ is countably additive then there is a measure $\mu : \Sigma \to [0, \infty]$ such that $\mu = \mu_0$ on Σ_0. If $\mu_0(\mathcal{S}) < \infty$, the choice of μ is unique.

Proof. Theorem 1.3 has the same algebra, premeasure and generated sigma-algebra hypotheses. The outer-measure construction is exhibited in the authorised Lebesgue–Stieltjes appendix, and the finite-total-mass uniqueness assertion is the named result Lemma 1.2. Applying those extension and uniqueness results gives every conclusion; no new extension assumption is introduced. QED

Remark 12.7 — Completion and Additional Measurable Sets

The σ-algebra $\mathcal{L}$ may contain many more sets than $\sigma(\Sigma_0)$ and therefore $\mu = \lambda$ is a measure on a much larger σ-algebra than Σ. In particular, notice that if $A \in \mathcal{G}$ is such that $\lambda(A) = 0$, then for any $G \in \mathcal{G}$

$$\lambda(G) \leq \lambda(G \cap A) + \lambda(G \cap A^c) = \lambda(G \cap A^c) \leq \lambda(G),$$

where the first inequality holds by sub-additivity and the second one by monotonicity of the outer measure. Hence, $\mathcal{L}$ is λ-complete (equiv. μ-complete) because all λ-null sets split well.
The above fact, implies that in the case of the Lebesgue measure, any subset of the Cantor set on $\mathbb{R}$ is measurable. Since the Cantor set has the cardinality of continuum, its power set has higher cardinality. This shows that the cardinality of the class $\mathcal{L}$ of Lebesgue measurable sets is higher than the cardinality of continuum (hence, higher cardinality of the class of Borel sets).

Verification — Every Subset of an Outer-Null Set Is Measurable

Proof. Assume here that the outer measure is defined on all subsets of $\mathcal{S}$. Let N have outer measure zero and let $A \subseteq N$. Monotonicity gives

$$0 \leq \lambda(A) \leq \lambda(N) = 0.$$

For an arbitrary test set $G \subseteq \mathcal{S}$,

$$\lambda(G) \leq \lambda(G \cap A) + \lambda(G \cap A^c) = \lambda(G \cap A^c) \leq \lambda(G).$$

The first inequality is subadditivity on the disjoint union $G = (G \cap A) \cup (G \cap A^c)$. The equality uses $G \cap A \subseteq A$ and the zero outer measure of A. The last inequality uses $G \cap A^c \subseteq G$ and monotonicity. The two bounding inequalities force equality, even if the common value is infinite. Therefore A splits every test set and belongs to the Caratheodory sigma-algebra, with measure zero. In particular every subset of a measurable null set is measurable: the extended measure is complete. The Cantor-set example uses that the Cantor set is Lebesgue-null. Each of its subsets is thus measurable by this argument. Its cardinality is the continuum, while its power set has strictly larger cardinality by Cantor's theorem. The additional elementary set-theoretic fact that there are only continuum many Borel subsets of $\mathbb{R}$ follows by coding each one with a countable tree of countable-union and complement operations on rational open intervals. Thus the measurable class is strictly larger than the Borel class, as asserted in the source. QED

Scope of completion. The assertion about all subsets of null sets applies when the outer measure is defined on $\mathcal{P}(\mathcal{S})$, as in the extension construction above. If its domain is only a smaller sigma-algebra $\mathcal{G}$, the splitting argument includes only subsets that already belong to $\mathcal{G}$; arbitrary subsets outside its domain cannot be tested.
Next we prove Theorem 8.2, whose statement we recall for convenience.

Theorem 12.8 — Existence and Uniqueness of Projection
Let $\mathcal{K}$ be a complete *vector subspace of $\mathcal{L}^2(\Omega, \mathcal{F}, \mathsf{P})$, i.e., for any Cauchy sequence $(v_n)_{n\in\mathbb{N}} \subset \mathcal{K}$ the limit exists in $\mathcal{K}$ ($\exists v \in \mathcal{K} : \|v_n - v\|_2 \to 0$ as $n \to \infty$).*
Then, given $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathsf{P})$ there exists $Y \in \mathcal{K}$ such that
(i)

$$\|X - Y\|_2 = \Delta := \inf\{\|X - W\|_2 : W \in \mathcal{K}\};$$

(ii) $\langle X - Y, Z\rangle_2 = 0$ for all $Z \in \mathcal{K}$.

Proof. This is Theorem 8.2, whose full existence, orthogonality and uniqueness proof was given before constructing conditional expectation. The hypotheses here are identical, so that established proof applies directly. QED

Appendix — Lebesgue Measures and Borel Approximation

Construction — Lebesgue Measure on the Whole Line

The interval-length assignment and the extension result stated as Theorem 1.3 give the probability measure ℓ on $(0,1]$ with its prescribed interval lengths. For $A \in \mathcal{B}(\mathbb{R})$, define $\mathcal{L}eb(A) = \sum_{k\in\mathbb{Z}} \ell\big((A-k)\cap(0,1]\big)$. Each set in the sum is Borel in $(0,1]$. Indeed, the class of A with this property is closed under complements and countable unions, by the corresponding pre-image identities for translation; it contains open intervals, whose translated traces are relatively open. Thus it contains the Borel sigma-algebra. The sum of these translated measures is a measure: the empty set has mass zero, and for disjoint A_n,

$$\mathcal{L}eb\left(\bigcup_n A_n\right) = \sum_k \sum_n \ell\big((A_n - k)\cap(0,1]\big) = \sum_n \sum_k \ell\big((A_n - k)\cap(0,1]\big) = \sum_n \mathcal{L}eb(A_n).$$

The first equality applies countable additivity to each translated trace; the second rearranges a non-negative double series, justified in the non-negative-series argument in the Lebesgue–Stieltjes appendix; the third uses the definition of $\mathcal{L}eb$. A bounded interval $(a,b]$ meets finitely many cells $(k,k+1]$. Its trace lengths add to $b-a$ by telescoping at the intervening integer endpoints. Hence $\mathcal{L}eb((a,b]) = b-a$ on the whole line. The integer cells also provide a countable finite-mass cover.

Construction — Lebesgue Measure in Finite Dimensions

For completeness, the corresponding finite-dimensional construction uses boxes $\prod_{i=1}^d (a_i, b_i]$, allowing infinite endpoints with the same whole-line conventions. Finite unions of these boxes form an algebra: collect the finitely many endpoints separately in each coordinate; their grid cells partition the whole space, each original box and each complement into finitely many boxes. Assign a nonempty box the volume $\prod_{i=1}^d (b_i - a_i)$ and an empty box volume zero; an unbounded nonempty box has infinite volume. Assign disjoint finite unions the sum. Common grid refinement proves well-definedness and finite additivity: in each coordinate, lengths add across a finite partition, and expanding the product of these sums gives precisely the sum of all cell volumes. The same grid argument proves that a finite cover has total covering volume at least the volume of the covered finite union.

The notation $v(K)$ for a closed box or a finite disjoint union of closed boxes denotes its geometric product-of-side-lengths value; it does not apply the still-unextended premeasure to a set outside its algebra. The finite grid-cover inequality also bounds this geometric value, by subdividing at the finitely many covering endpoints.

Here are the details of the countable step, which cannot be inferred from finite additivity alone. A cover of infinite total volume gives the required covering inequality immediately; otherwise every nonempty covering box is bounded, so the following enlargements have finite values. For a bounded finite union E of disjoint boxes, trim each finite left endpoint slightly to obtain a compact finite union of closed boxes $K \subseteq E$ of volume greater than $v(E) - \varepsilon$. This is possible by continuity of finite products of positive lengths. Enlarge every box in a countable algebra cover E_n slightly to an open box, with total added volume below $\varepsilon 2^{-n}$ for its nth finite union; the same product continuity permits this. Closed bounded boxes are compact: the bisection proof for intervals works by bisecting every coordinate, obtaining nested boxes whose coordinate lengths tend to zero and hence a unique common point by completeness of $\mathbb{R}$. An open set containing that point contains all of a sufficiently small nested box, contradicting the absence of a finite subcover. Finite unions are compact as before. A closed subset of a closed bounded box is compact by adding its open complement to a cover of that box, so every closed bounded set needed in the subsequent approximation lemma is compact. Thus the open boxes covering K have a finite subcover, and the finite grid-cover inequality gives

$$v(E) - \varepsilon < v(K) \le \sum_{n\ge 1} v(E_n) + \varepsilon.$$

The first inequality is the inner trimming choice. The second uses the finite subcover, bounds its

total volume by that of all enlarged boxes, and adds the chosen total enlargement budget. Letting $\varepsilon \downarrow 0$ proves the covering bound. For a disjoint sequence with bounded algebra union E, finite additivity gives the opposite bound on every partial sum; the covering bound therefore proves countable additivity. For a possibly unbounded algebra union, intersect it and all its disjoint pieces with $Q_m = (-m, m]^d$. The bounded case gives $v(E \cap Q_m) \leq \sum_n v(E_n)$. The left side increases to $v(E)$: check this on each of the finitely many disjoint boxes by the ordinary limits of their truncated coordinate lengths, then add the finite list. This proves the remaining bound also when $v(E) = \infty$.

Theorem 1.3 now extends volume to the generated sigma-algebra. Every box is Borel; conversely rational open boxes form a countable base for Euclidean open sets, and each is a countable union of the half-open grid boxes. The generated sigma-algebra is therefore $\mathcal{B}(\mathbb{R}^d)$. Uniqueness on each finite-mass box follows from Lemma 1.2 applied to its trace algebra. Partition the space into $Q_1, Q_2 \setminus Q_1, \ldots$ and add the equal restricted measures on this disjoint cover to obtain global uniqueness. This constructs the Lebesgue measure and its rectangle-volume rule in every finite dimension, without assuming a product-measure theorem not yet proved.

Auxiliary Lemma — Approximating Borel Sets by Closed and Open Sets

For a finite Borel measure ν on $\mathbb{R}^d$, each Borel set A and $\varepsilon > 0$ admit a closed F and an open G with $F \subseteq A \subseteq G$ and $\nu(G \setminus F) < \varepsilon$. The inner set can be chosen compact. The same conclusion holds for Lebesgue measure when A has finite measure.

Proof. For a closed set F, the sets $G_n = \{x : \operatorname{dist}(x, F) < 1/n\}$ are open and decrease to F. The distance is continuous because the triangle inequality gives $|\operatorname{dist}(x, F) - \operatorname{dist}(y, F)| \leq |x - y|$; distance zero means membership in the closed set. Thus the intersection identity holds in both directions. Lemma 1.6 gives $\nu(G_n \setminus F) \to 0$ for finite ν. The empty closed set is handled with the empty open set.

Let $\mathcal{C}$ be the class of sets with the stated closed/open approximation for every error tolerance. It contains all closed sets. It is closed under complements, since $G^c \subseteq A^c \subseteq F^c$ and the new error region is the same $G \setminus F$. For $A = \bigcup_j A_j$ with $A_j \in \mathcal{C}$, choose $F_j \subseteq A_j \subseteq G_j$ of error below $\varepsilon 2^{-j-2}$. The open union $G = \bigcup_j G_j$ satisfies

$$\nu(G \setminus A) \leq \sum_j \nu(G_j \setminus A_j) < \varepsilon/4.$$

The inequality is countable subadditivity and the inclusion that a point of G outside A lies in such an error piece. Continuity from below and finite total mass give an N with $\nu(A \setminus \bigcup_{j \leq N} A_j) < \varepsilon/2$. For the closed $F = \bigcup_{j \leq N} F_j$, finite subadditivity gives $\nu(A \setminus F) < 3\varepsilon/4$. Adding the two disjoint error regions yields $\nu(G \setminus F) < \varepsilon$. Thus $\mathcal{C}$ is a sigma-algebra containing the closed sets and hence all Borel sets. Since $[-m, m]^d$ increases to the whole space, Lemma 1.5 makes $\nu(F \setminus [-m, m]^d) \to 0$. Intersecting F with a sufficiently large box makes it compact, by the proved compactness of closed bounded boxes, and adds an arbitrarily small error.

For finite-Lebesgue-measure A, a countable open-box cover with total volume below $\mathcal{L}eb(A) + \delta$ exists from the outer-measure construction: start with an algebra cover within that error and enlarge each of its bounded boxes with a summable additional error. It yields an open $G \supseteq A$ with $\mathcal{L}eb(G \setminus A) < 2\delta$. Choose a large m with $\mathcal{L}eb(A \setminus [-m, m]^d) < \delta$ by continuity from below. Apply the finite-measure result to the restriction of Lebesgue measure to $[-m - 1, m + 1]^d$ and the set $A \cap [-m, m]^d$ to obtain a closed inner approximation there. It can be intersected with $[-m, m]^d$, retaining the error bound and making it compact in $\mathbb{R}^d$. Combining these error budgets and taking δ sufficiently small proves the assertion. QED

Appendix — Consistent Finite-Dimensional Laws

Let I be an index set; the empty-index case is the unit mass on its one-point product and needs no construction. For each finite $J \subseteq I$, let ν_J be a probability measure on $\mathcal{B}(\mathbb{R}^J)$. Assume consistency:

if $J \subseteq K$, projecting the law ν_K onto the coordinates in J gives ν_J. Then there is a probability measure on $\mathbb{R}^I$, with its coordinate-generated sigma-algebra, whose finite coordinate laws are the ν_J.

Supporting Approximation Fact

The finite-Borel-measure approximation result proved in the preceding appendix provides compact inner approximations in each finite-dimensional space. The proof uses only continuity of finite measures and Euclidean compactness.

Proof. For finite J and Borel $A \subseteq \mathbb{R}^J$, write $C(J, A) = \{x \in \mathbb{R}^I : (x_t)_{t\in J} \in A\}$. Such cylinders form an algebra: complements and finite unions can be expressed over the union of their finitely many coordinates. Define $\mu_0(C(J, A)) = \nu_J(A)$. If two cylinders are the same set, lift both to their common finite coordinate union. Their lifted bases are equal because every finite coordinate vector can be extended to an element of $\mathbb{R}^I$ by assigning zero to the remaining coordinates. Consistency then gives the same value to both representations. Thus μ_0 is well defined. The same common-coordinate representation and finite additivity of ν_J prove finite additivity of μ_0, with total mass one.

We show continuity at the empty set for decreasing cylinders C_n. Suppose $\bigcap_n C_n = \varnothing$ but $\mu_0(C_n) \geq \varepsilon > 0$ for all n. Only countably many coordinates occur in these cylinders; list them as $t_1, t_2, \ldots$. A finite list is handled by the same argument without the diagonal extraction across additional coordinates. Write $C_n = C(J_n, A_n)$. Choose compact $K_n \subseteq A_n$ with $\nu_{J_n}(A_n \setminus K_n) < \varepsilon 2^{-n-3}$, and put $H_n = C(J_n, K_n) \subseteq C_n$. For each listed coordinate choose $M_j \geq 1$ with $\nu_{\{t_j\}}(\{|x| > M_j\}) < \varepsilon 2^{-j-3}$. These bounds exist because the boxes increase to the line and the marginal measure is a probability measure.

For fixed N, choose m large enough that $J_1, \ldots, J_N$ are all among the first m coordinates. Finite subadditivity in their common finite-coordinate space gives

$$\mu_0\left(C_N \cap \bigcap_{n\leq N} H_n \cap \bigcap_{j\leq m} \{|x_{t_j}| \leq M_j\}\right) \geq \varepsilon - \sum_{n\leq N} \varepsilon 2^{-n-3} - \sum_{j\leq m} \varepsilon 2^{-j-3} > 0.$$

For the first inequality, remove from C_N the sets $C_n \setminus H_n$ and the marginal tail events. The nesting $C_N \subseteq C_n$ bounds the cost of each removal by its displayed error. The strict inequality uses that both full geometric error sums are $\varepsilon/8$, so their combined cost is below ε. Choose one point satisfying these finite conditions and set all its other coordinates to zero. This produces a vector $x^{(N)}$ in the countable box $\prod_j [-M_j, M_j]$ that belongs to $H_1, \ldots, H_N$.

This sequence has a subsequence converging in every listed coordinate. For the first coordinate select a convergent subsequence in its compact interval, then a subsequence of it converging in the second coordinate, and continue. The diagonal subsequence, using its kth entry from the kth nested subsequence, eventually lies in each previously selected subsequence and therefore converges in each fixed coordinate. For completeness, a sequence in a compact interval has a convergent subsequence by successively choosing a closed half-interval containing infinitely many terms. Nested endpoints converge to a common point, and selecting later terms in the successive intervals gives convergence. Let x be the coordinatewise limit, extended by zero to coordinates outside the list. For fixed n, all sufficiently late selected points are in H_n. Its base K_n is closed and depends on only finitely many coordinates; hence their limit is in K_n, and $x \in H_n \subseteq C_n$. This holds for every n, contradicting the empty intersection. Thus $\mu_0(C_n) \to 0$.

For disjoint cylinders E_n whose union E is a cylinder, the remainders $R_N = E \setminus \bigcup_{n\leq N} E_n$ are decreasing cylinders with empty intersection. Finite additivity gives

$$\mu_0(E) = \lim_{N\to\infty} \left(\sum_{n=1}^{N} \mu_0(E_n) + \mu_0(R_N)\right) = \sum_{n\geq 1} \mu_0(E_n).$$

The first equality uses the finite disjoint decomposition of E at every N, so the expression inside the limit is constant. The second uses the proved continuity at the empty set and the definition of

the non-negative series. Hence μ_0 is a premeasure on the cylinder algebra. The extension result stated as Theorem 1.3 extends it to a probability measure on the coordinate-generated sigma-algebra. Its agreement on every cylinder gives exactly the prescribed finite-dimensional laws. The coordinate maps are measurable because their Borel pre-images are cylinders. QED

Appendix — Sign Decomposition of a Finite Signed Measure

Auxiliary Lemma — Positive and Negative Sets of a Finite Signed Measure
Let ρ, ν be finite measures and $c \geq 0$. Write $\gamma(A) = \rho(A) - c\nu(A)$. There is a measurable partition N, P such that $\gamma(A) \leq 0$ for every measurable $A \subseteq N$, and $\gamma(A) \geq 0$ for every measurable $A \subseteq P$.

Proof. Put $\beta = \inf_{A \in \Sigma} \gamma(A)$, which is finite because

$$-c\nu(\mathcal{S}) \leq \gamma(A) \leq \rho(\mathcal{S}).$$

Choose A_j with $\gamma(A_j) < \beta + 2^{-j}$. Finite additivity and the partition of two sets into their common and separate parts give $\gamma(A \cap B) = \gamma(A) + \gamma(B) - \gamma(A \cup B)$. This equality follows by writing each side as the sum of its disjoint parts; all terms are finite. Since $\gamma(A \cup B) \geq \beta$, induction gives $\gamma(\bigcap_{j=n}^{m} A_j) \leq \beta + \sum_{j=n}^{m} 2^{-j}$. For the induction step, apply the displayed identity to the previous intersection and A_{m+1}, subtracting at least β.
Let $B_n = \bigcap_{j \geq n} A_j$. Each of the finite measures is continuous from above and below by Lemmas 1.5–1.6; subtraction of their finite limits therefore gives the same continuity for γ. Consequently $\gamma(B_n) \leq \beta + 2^{1-n}$, by taking the decreasing intersection limit in m and summing the geometric tail. The sets B_n increase. Their union N satisfies

$$\beta \leq \gamma(N) = \lim_n \gamma(B_n) \leq \lim_n (\beta + 2^{1-n}) = \beta.$$

The first inequality is the definition of the infimum, the equality is finite signed-measure continuity from below, the next inequality is the bound just obtained, and the last equality uses $2^{1-n} \to 0$. Thus $\gamma(N) = \beta$. If $A \subseteq N$ had $\gamma(A) > 0$, finite additivity would give $\gamma(N \setminus A) = \beta - \gamma(A) < \beta$, a contradiction. If $A \subseteq P = N^c$ had $\gamma(A) < 0$, disjoint additivity would give $\gamma(N \cup A) = \beta + \gamma(A) < \beta$, also a contradiction. This proves both asserted sign properties. QED

Past Examination Papers

17 December 2021

Two questions; show all workings.

Q.1 Let $(\varepsilon_k)_{k\in\mathbb{N}}$ be a sequence of i.i.d. random variables taking values $+1$ and -1, with $\mathsf{P}(\varepsilon_1 = 1) = p$. Consider a financial market in which the risk-free rate is $r > 0$. The stock price is defined by

$$S_k = S_0 \exp\left(\sum_{n=1}^{k} \varepsilon_n\right),\ S_0 = s \in (0,\infty).$$

Assume that P is the risk-neutral measure, so that the discounted asset is an $(\mathcal{F}_k, \mathsf{P})$-martingale.

(a) Explain why (S_k) is adapted.
(b) What is the main advantage of the model in the stock-price formula compared to a model with $S_k = S_0 + \sum_{n=1}^{k} \varepsilon_n$?
(c) Verify that $\mathsf{E}[|S_k|] < \infty$ for any $k \in \mathbb{N}$.
(d) Prove that, since P is risk-neutral, it must be $p = \frac{1+r-e^{-1}}{e-e^{-1}}$.
(e) Let $S_0 = 1$ and $r = 0$. Consider European put options with strike price $K = 1$ and maturities $T = 1$ and $T = 2$. Calculate the two option prices.

Q.2 Consider the measure space $\big((0,\infty), \mathcal{B}((0,\infty)), \lambda\big),\ \lambda = \mathcal{L}eb$.

(a) State Fatou's lemma.
(b) Take the sequence (f_n) of functions $f_n : (0,\infty) \to \mathbb{R}$ defined by $f_n(s) = \frac{1}{n}\mathbf{1}_{(0,n)}(s)$. Show that f_n is $\mathcal{B}((0,\infty))$-measurable for each n.
(c) Show that for this sequence Fatou's lemma holds with strict inequality.

Consider now

$$g_n(s) = -f_n(s) = -\frac{1}{n}\mathbf{1}_{(0,n)}(s).$$

(d) Deduce from (c) that Fatou's lemma does not hold for the sequence (g_n).
(e) Why does Fatou's lemma fail in (d)?

AI-Generated Worked Solution

Q.1: a positive stock model and risk-neutral probabilities

(a) Use the natural filtration $\mathcal{F}_k = \sigma(\varepsilon_1, \ldots, \varepsilon_k)$, with the constant initial value known at time zero. Each of the first k increments is measurable with respect to $\mathcal{F}_k$, so their finite sum is measurable. The map $x \mapsto se^x$ is continuous and therefore Borel-measurable; composing it with this sum makes S_k measurable with respect to $\mathcal{F}_k$. This is adaptedness: the price at time k can be determined from information already available then.

(b) For any finite sum of increments, $s\exp(\sum_{j=1}^{k}\varepsilon_j) > 0$ because $s > 0$ and exponentials are positive. In the additive model, on a path with k consecutive down moves the price would instead be $s - k$. For $k > s$ this is negative, whenever the path has positive probability. The multiplicative model preserves positivity, appropriate for an ordinary stock with limited liability, without needing a reflecting boundary or a special correction at zero.

(c) Since every increment is at most one, $0 < S_k \le se^k$. Thus $\mathsf{E}|S_k| \le se^k < \infty$ for each fixed date. For the exact expectation, independence permits factorization of $S_k = s\prod_{j=1}^{k} e^{\varepsilon_j}$, giving

$$\mathsf{E}S_k = s\prod_{j=1}^{k} \mathsf{E}e^{\varepsilon_j} = s[pe + (1-p)e^{-1}]^k.$$

Integrability is a separate martingale requirement; it is not enough just to verify a formal conditional-mean identity.

(d) Define the discounted stock $D_k = (1+r)^{-k}S_k$. The next-period equation is $D_{k+1} = D_k e^{\varepsilon_{k+1}}/(1+r)$. Since D_k is known at time k and the next increment is independent of $\mathcal{F}_k$,

$$\mathsf{E}[D_{k+1} \mid \mathcal{F}_k] = D_k \frac{pe + (1-p)e^{-1}}{1+r}.$$

The martingale condition is equality to D_k. Dividing by its strictly positive value and rearranging,

$$p(e - e^{-1}) + e^{-1} = 1 + r,\ p = \frac{1 + r - e^{-1}}{e - e^{-1}}.$$

Conversely this probability makes the conditional identity hold, and adaptedness and integrability were established above. Feasibility requires $e^{-1} \le 1 + r \le e$; strict inequalities ensure two positive branch probabilities. If r lies outside these bounds, no probability in this two-branch model can give the stipulated martingale property.

(e) Here the exercise resets the rate to zero. Substitution gives $p = 1/(e+1)$ and $q = 1 - p = e/(e+1)$. Under the stipulated risk-neutral law the price is the discounted expected payoff, with discount factor one. For $T = 1$ the prices are e, e^{-1}; the put payoffs are zero and $1 - e^{-1}$. Hence

$$P_0^{(1)} = q(1 - e^{-1}) = \frac{e-1}{e+1}.$$

For $T = 2$, the paths uu, ud, du, dd give prices $e^2, 1, 1, e^{-2}$ with probabilities p^2, pq, qp, q^2. Only dd has a positive put payoff. Therefore

$$P_0^{(2)} = q^2(1 - e^{-2}) = \frac{e^2 - 1}{(e+1)^2} = \frac{e-1}{e+1}.$$

The two maturities happen to have the same price in this particular tree; this is the result of payoff enumeration, not an assumption about maturity effects in general.

Q.2: Fatou's lemma and why its sign hypothesis matters

(a) If h_n are nonnegative measurable functions on a measure space, Fatou's lemma states

$$\int \liminf_n h_n \, d\lambda \le \liminf_n \int h_n \, d\lambda,$$

allowing infinite nonnegative integrals. To understand the direction, put $u_m = \inf_{n \ge m} h_n$. Then $u_m \uparrow \liminf h_n$ and $u_m \le h_n$ for every $n \ge m$. Monotone convergence and integral monotonicity give $\int \liminf h_n = \lim_m \int u_m \le \lim_m \inf_{n \ge m} \int h_n$, exactly the stated inequality.

(b) The interval $(0, n)$ is Borel in the domain $(0, \infty)$. An indicator of a measurable set is measurable, since every preimage is either that set, its complement, the whole space or the empty set. Multiplication by the constant $1/n$ preserves measurability. Thus f_n is measurable, even though it has a discontinuity at n.

(c) Fix $s > 0$ and choose an integer $N > s$. For every $n \ge N$ we have $s \in (0, n)$, so $f_n(s) = 1/n \to 0$. Thus the pointwise lower limit is identically zero and its integral is zero. In contrast, for every n, height times interval length gives

$$\int_0^\infty f_n(s)\, ds = \frac{1}{n}\mathcal{L}eb((0,n)) = \frac{1}{n}n = 1.$$

Fatou's inequality for this sequence is therefore $0 \le 1$, with strict inequality. Pointwise mass tends to zero while the support expands, so equality is not justified.

(d) Each $g_n = -f_n$ has pointwise limit zero and finite absolute integral one. Signed linearity gives $\int g_n = -1$. If one tried to apply the same Fatou inequality, it would assert

$$0 = \int \liminf_n g_n \, d\lambda \le \liminf_n \int g_n \, d\lambda = -1,$$

which is false. Every displayed integral is finite. The failed assertion comes from omitting Fatou's nonnegativity hypothesis, as addressed in (e).

(e) Nonnegativity fails because $g_n < 0$ on $(0, n)$. Fatou can be extended to signed functions if they share a single integrable lower bound, by subtracting that bound first; such a bound does not exist here. Indeed if an integrable h satisfied $h \le g_n$ for every n, then for $s \in (m-1, m)$ with $m \ge 2$, using $n = m$ gives $h(s) \le -1/m$. Therefore

$$\int_0^\infty |h(s)|\, ds \ge \sum_{m=2}^{\infty} \int_{m-1}^{m} \frac{1}{m}\, ds = \sum_{m=2}^{\infty} \frac{1}{m} = \infty,$$

a contradiction. This confirms that neither the nonnegative version nor the common-integrable-lower-bound version applies.

17 January 2022

Two questions; show all workings.

Q.1 Let (ε_k) be a sequence of i.i.d. random variables taking values such that

$$\mathsf{P}(\varepsilon_1 = -2) = p,\ \mathsf{P}(\varepsilon_1 = 1) = q.$$

Consider a financial market with zero risk-free interest rate ($r = 0$) and define

$$X_k = X_{k-1} + \varepsilon_k,\ X_0 = 0.$$

Let $(\mathcal{F}_k)$ denote the filtration generated by the sequence (ε_k).

(a) Write the expression of $\mathsf{P}(\varepsilon_k = 0)$ for every $k \in \mathbb{N}$.
(b) Show that (X_k) is an adapted process.
(c) Show that $X_k \in L^2(\Omega, \mathcal{F}, \mathsf{P})$ for every $k \in \mathbb{N}$.
Hint: use induction; $(a+b)^2 \le 2a^2 + 2b^2$ *may help you.*
(d) On this market define $Y_k = (X_k)^2 - k,\ Y_0 = 0$. Find p and q such that both (X_k) and (Y_k) are $(\mathcal{F}_k)$-martingales.
(e) Calculate the price at time zero of a European put option on Y with maturity $T = 1$ and strike price $K = 1$.

Q.2

(a) Define the Borel σ-algebra on $(0, 1)$.
(b) Prove that, for any $x \in (0, 1)$, the singleton $\{x\}$ is a Borel-measurable set.
(c) Let λ be the Lebesgue measure on $(0, 1)$. Prove that $\lambda(\{x\}) = 0$ for every $x \in (0, 1)$, and then deduce that $\lambda\big((0, 1) \cap \mathbb{Q}\big) = 0$.

AI-Generated Worked Solution

Q.1: imposing two martingale identities

(a) The intended support is $\{-2, 0, 1\}$. Under that support assumption, disjoint alternatives have probabilities summing to one, giving $\mathsf{P}(\varepsilon_k = 0) = 1 - p - q$. The parameters must satisfy $p \ge 0$, $q \ge 0$, $p + q \le 1$. If other values were possible, the two probabilities printed in the question alone would not determine the mass at zero, so the support qualification is necessary.
(b) Iterating $X_k = X_{k-1} + \varepsilon_k$ from zero gives $X_k = \sum_{j=1}^{k} \varepsilon_j$. Each summand is measurable with respect to $\mathcal{F}_k = \sigma(\varepsilon_1, \ldots, \varepsilon_k)$. A finite sum is measurable, so X_k is measurable at its own date for every k. Hence the process is adapted.
(c) *We prove square integrability by induction, following the hint.* The initial value $X_0 = 0$ is square-integrable. If $\mathsf{E}X_k^2 < \infty$, the inequality $(a+b)^2 \le 2a^2 + 2b^2$ gives

$$\mathsf{E}X_{k+1}^2 \le 2\mathsf{E}X_k^2 + 2\mathsf{E}\varepsilon_{k+1}^2 = 2\mathsf{E}X_k^2 + 2(4p + q) < \infty.$$

The equality uses the three-point support specified in (a). This completes the induction. Cauchy–Schwarz on the probability space gives $X_k \in L^1$. Moreover $\mathsf{E}|X_k^2 - k| \le \mathsf{E}X_k^2 + k < \infty$, so both processes in (d) meet the integrability requirement before their drifts are calculated.

(d) Let $m = \mathsf{E}\varepsilon_1 = -2p + q$ and $v = \mathsf{E}\varepsilon_1^2 = 4p + q$. Independence from $\mathcal{F}_k$ gives $\mathsf{E}[X_{k+1} \mid \mathcal{F}_k] = X_k + m$, so fairness of X requires $m = 0$. Expanding the square before conditioning,

$$Y_{k+1} - Y_k = 2X_k\varepsilon_{k+1} + \varepsilon_{k+1}^2 - 1,\ \mathsf{E}[Y_{k+1} - Y_k \mid \mathcal{F}_k] = 2mX_k + v - 1.$$

With $m = 0$, this vanishes exactly when $v = 1$. Thus $q = 2p$ and $4p + q = 1$, giving $p = 1/6$, $q = 1/3$, and zero mass $1/2$. Conversely these values make both drifts zero at every date. Together with adaptedness and the moment checks, this proves that both are martingales, and that these probabilities are necessary.

(e) At time one the full state calculation is

ε_1	-2	0	1
P	$1/6$	$1/2$	$1/3$
$Y_1 = \varepsilon_1^2 - 1$	3	-1	0
$(1 - Y_1)^+$	0	2	1

Zero interest means there is no discount factor, so the requested martingale-measure expectation is $P_0 = (1/6)0 + (1/2)2 + (1/3)1 = 4/3$. In particular the underlying Y is allowed to be negative in this mathematical payoff calculation; the put payoff is evaluated from its actual values, not from X_1.

Q.2: Borel sets of rational points

(a) The Borel sigma-algebra $\mathcal{B}((0,1))$ is the smallest sigma-algebra on $(0,1)$ containing its relatively open sets. A relatively open set has the form $U \cap (0,1)$ with U open in $\mathbb{R}$. Equivalently, $\mathcal{B}((0,1)) = \{B \cap (0,1) : B \in \mathcal{B}(\mathbb{R})\}$, the trace of the real Borel sigma-algebra on the interval.

(b) For a fixed $x \in (0,1)$,

$$\{x\} = \bigcap_{n \geq 1} \big((x - 1/n, x + 1/n) \cap (0,1)\big).$$

The point x belongs to every factor. Any $y \neq x$ is excluded once $1/n < |y - x|$, so the intersection contains no other point. Every factor is relatively open, hence Borel. A sigma-algebra is closed under countable intersections, proving the singleton is Borel.

(c) The same neighborhoods have length at most $2/n$. Monotonicity therefore gives $0 \leq \lambda(\{x\}) \leq 2/n$ for every n, forcing $\lambda(\{x\}) = 0$. Enumerate the distinct rational points in $(0,1)$ as $q_1, q_2, \ldots$. Their set is the countable union of the Borel singletons, so is Borel, and countable additivity on disjoint singletons gives

$$\lambda((0,1) \cap \mathbb{Q}) = \sum_{j \geq 1} \lambda(\{q_j\}) = 0.$$

The countability is essential: the interval itself is an uncountable union of zero-measure singletons and has positive measure.

7 February 2022

One question; show all workings.

Q.1 Let (ε_k) be an i.i.d. sequence of random variables with $\mathsf{E}[\varepsilon_1] = 0,\ \mathsf{E}[|\varepsilon_1|^2] < \infty$. Set

$$X_n = X_0 + \sum_{k=1}^{n} \varepsilon_k,\ n \geq 1.$$

(a) Assume $X_0 \in \mathcal{F}_0$ and finite. Show that (X_k) is an $(\mathcal{F}_k)$-martingale.
Hint:

$$Z \in L^2(\Omega, \mathcal{F}, \mathsf{P}) \Rightarrow Z \in L^1(\Omega, \mathcal{F}, \mathsf{P}).$$

(b) Let $c : \mathbb{R} \to [0, \infty)$ be a convex function with $|c(x)| \leq K(1 + |x|^2)$ for some $K > 0$. Set $Y_k = c(X_k)$. Prove that (Y_k) is an $(\mathcal{F}_k)$-submartingale.
Hint: it may be useful to know that $\left(\sum_{k=1}^n x_k\right)^2 \leq n \sum_{k=1}^n x_k^2$.

(c) Fix $a > 0$ and define $\tau_a := \inf\{k \in \mathbb{Z}_+ : Y_k \geq a\}$. Prove that τ_a is a stopping time for the filtration $(\mathcal{F}_k)$.

(d) Show that, for any $n \in \mathbb{Z}_+$, $\mathsf{P}(\tau_a \le n) = \mathsf{P}(\max_{0\le k\le n} Y_k \ge a)$. *Hint: you must prove the equivalence of the two events.*

(e) Explain in detail, showing all steps, how to use (d), Markov's inequality and Jensen's inequality to obtain $\mathsf{P}(\tau_a \le n) \le \frac{1}{a}\left(\mathsf{E}\left[\max_{0\le k\le n} Y_k^2\right]\right)^{1/2}$.

(f) Recall that for any non-negative submartingale (Y_k), Doob's inequality holds in the form $\mathsf{E}\left[\max_{0\le k\le n} Y_k^2\right] \le 4\mathsf{E}[Y_n^2]$. Assume $\varepsilon_1 \sim \mathcal{N}(0,1)$ and $X_0 = 0$. Taking $c(x) = |x|$ in (b), derive from the hitting-probability estimate the estimate $\mathsf{P}(\tau_a \le 2) \le \frac{2\sqrt{2}}{a}$.

AI-Generated Worked Solution

Q.1: convex transforms, hitting times and a maximal bound

The phrase "X_0 finite" means a finite value almost surely, which alone does not imply integrability. To justify both (a) and the quadratic-growth transform in (b), assume $X_0 \in L^2$ (a deterministic finite initial value is a special case). Also assume the increments are adapted and $\mathsf{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] = 0$, for example under the natural nonanticipating filtration when X_0 is independent of the iid increments. IID marginal increments do not provide this conditional equality for an arbitrary anticipative filtration.

(a) The sum representation makes X_n measurable with respect to $\mathcal{F}_n$. For its moments use the finite-sum Cauchy–Schwarz inequality:

$$|X_n|^2 \le (n+1)\left(|X_0|^2 + \sum_{j=1}^{n} |\varepsilon_j|^2\right).$$

Taking expectations gives a finite bound, hence $X_n \in L^2 \subseteq L^1$. Conditional linearity and centering of the next increment now give $\mathsf{E}[X_{n+1} \mid \mathcal{F}_n] = X_n + 0 = X_n$. This verifies all three requirements for a martingale: adaptedness, integrability, and the conditional identity. For (a) alone $X_0 \in L^1$ would suffice, but (b) needs the stronger moment control just used.

(b) A finite convex function on $\mathbb{R}$ is continuous, so $Y_n = c(X_n)$ is adapted by composition. The given growth bound ensures $0 \le Y_n \le K(1 + X_n^2)$; therefore Y_n is integrable. Apply conditional Jensen to the convex function c and the integrable variable X_{n+1}, with the just-verified integrability of its transform:

$$\mathsf{E}[Y_{n+1} \mid \mathcal{F}_n] \ge c(\mathsf{E}[X_{n+1} \mid \mathcal{F}_n]) = c(X_n) = Y_n.$$

This is the submartingale inequality. Convexity is what changes equality for X into a one-sided inequality for its transform.

(c) If the notation $\mathbb{Z}^+$ includes zero, a hit has occurred by date n exactly when at least one of dates $0, \dots, n$ has $Y_j \ge a$. Thus $\{\tau_a \le n\} = \bigcup_{j=0}^{n} \{Y_j \ge a\} \in \mathcal{F}_n$, since each member is in $\mathcal{F}_j \subseteq \mathcal{F}_n$. If $\mathbb{Z}^+$ instead means strictly positive integers, start the union at one; the date-zero event is then empty. Both conventions give a stopping time. The infimum of an empty set is understood to be infinity.

(d) A maximum over a finite set of dates is at least a exactly when one of the values at those dates is at least a. Hence with the zero-inclusive convention,

$$\{\tau_a \le n\} = \{\max_{0\le j\le n} Y_j \ge a\},$$

and equality of probabilities follows. With positive hitting indices, the maximum must instead start at one. It can be extended to date zero if $Y_0 < a$. Without this condition the printed equality can fail: take $X_0 = 0$, $c(x) = (x-1)^2$, $a = 1$ and symmetric ± 1 increments. Then $Y_0 = 1$, while on the first up step $Y_1 = 0$. At date one the zero-inclusive maximum has reached the barrier but the positive-time hitting event has not. This example respects convexity and the quadratic growth bound.

(e) Let $M_n = \max_{0\le j\le n} Y_j$. Nonnegativity and a finite number of integrable Y_j ensure $M_n \le \sum_{j=0}^{n} Y_j \in L^1$. In either hitting convention, a hit by n implies $M_n \ge a$, giving the pointwise bound $\mathbf{1}_{\{\tau_a \le n\}} \le M_n/a$. Taking expectations is Markov's argument. Next Jensen for the

square, or Cauchy–Schwarz against the constant one, gives $(\mathsf{E}M_n)^2 \le \mathsf{E}M_n^2$. Therefore

$$\mathsf{P}(\tau_a \le n) \le \frac{\mathsf{E}M_n}{a} \le \frac{(\mathsf{E}M_n^2)^{1/2}}{a} = \frac{1}{a}\left(\mathsf{E}\max_{0\le j\le n} Y_j^2\right)^{1/2}.$$

If the second moment is infinite this remains a true but uninformative bound. Its finiteness is supplied in the next special case, not by the general quadratic growth assumption alone.

(f) Now $Y_j = |X_j|$, $X_0 = 0$ and $X_2 = \varepsilon_1 + \varepsilon_2$. Independence, zero means and unit variances give $\mathsf{E}X_2^2 = 1 + 1 + 2(0) = 2$. The process $|X|$ is a nonnegative submartingale by (b). The stated Doob inequality yields

$$\mathsf{E}M_2^2 \le 4\mathsf{E}Y_2^2 = 4\mathsf{E}X_2^2 = 8.$$

Substitute into (e) to get $\mathsf{P}(\tau_a \le 2) \le \sqrt{8}/a = 2\sqrt{2}/a$. Since any probability is at most one, the bound can be written $\min(1, 2\sqrt{2}/a)$. Here $Y_0 = 0 < a$, so the date-zero ambiguity has no effect.

12 December 2022

Two questions; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space and let $(\varepsilon_k)_{k\in\mathbb{N}}$ be a sequence of i.i.d. random variables with

$$\mathsf{P}(\varepsilon_1 = 1) = p,\ \mathsf{P}(\varepsilon_1 = -1) = 1 - p$$

for some $p \in [0, 1]$. Let $(X_n)_{n\in\mathbb{Z}_+}$ be defined as follows:

$$X_0 = 0 \text{ and } X_n = \sum_{j=1}^{n} \varepsilon_j,\ \text{for } n \in \mathbb{N},$$

and let $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$ be the filtration generated by $(\varepsilon_k)_{k\in\mathbb{N}}$, i.e.,

$$\mathcal{F}_0 = \{\Omega, \emptyset\},\ \mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n).$$

(a) Show that in general the process $(X_n)_{n\in\mathbb{Z}_+}$ is not a $(\mathcal{F}_n, \mathsf{P})$-martingale and find the value of p for which it is indeed a martingale.

(b) Find Doob's decomposition of the process $(X_n)_{n\in\mathbb{Z}_+}$.

From now on take $p = \frac{1}{2}$. Let the process $(Y_n)_{n\in\mathbb{Z}_+}$ be defined as $Y_n = e^{X_n},\ n \in \mathbb{Z}_+$.

(c) Prove that the process $(Y_n)_{n\in\mathbb{Z}_+}$ is a $(\mathcal{F}_n, \mathsf{P})$-sub-martingale.
Hint: recall that $e \approx 2.72$.

(d) In a possible financial application, why is the process $(Y_n)_{n\in\mathbb{Z}_+}$ a better model for a stock price than the process $(X_n)_{n\in\mathbb{Z}_+}$?

(e) In an arbitrage-free market the discounted value of any asset must be a martingale. Let the discounted stock process $(\widetilde{Y}_n)_{n\in\mathbb{Z}_+}$ be defined as $\widetilde{Y}_n := (1 + r)^{-n} Y_n,\ n \in \mathbb{Z}_+$, for some $r > 0$. Show that $(\widetilde{Y}_n)_{n\in\mathbb{Z}_+}$ is a $(\mathcal{F}_n, \mathsf{P})$-martingale if and only if the discount rate $r > 0$ is given by $r = \frac{1}{2}\left(e^1 + e^{-1}\right) - 1$.

(f) Calculate the price at time zero of a European option with maturity at time 1 and payoff $\Phi(Y_1) = \max\{1, Y_1\}$. Use the discount rate from (e).

Q.2 On a probability space $(\Omega, \mathcal{F}, \mathsf{P})$ let $Z \in L^1(\Omega, \mathcal{F}, \mathsf{P})$ be such that $Z \ge 0$ and $\mathsf{E}[Z] = 1$.

(a) Show that the set function $\mathbb{Q} : \mathcal{F} \to [0, \infty)$ given by

$$\mathbb{Q}(A) := \mathsf{E}[Z\mathbf{1}_A],\ A \in \mathcal{F},$$

is a probability measure on $(\Omega, \mathcal{F})$.

Notice that $\mathsf{E}[\,\cdot\,]$ is the expectation under the measure P: for any $X \in m\mathcal{F}$ with $X \ge 0$ we have $\mathsf{E}[X] = \int_\Omega X(\omega)\, d\mathsf{P}(\omega)$. Let $\mathsf{E}^{\mathbb{Q}}$ denote the expectation under the probability measure $\mathbb{Q}$: for any $X \in m\mathcal{F}$ with $X \ge 0$ we have $\mathsf{E}^{\mathbb{Q}}[X] = \int_\Omega X(\omega)\, d\mathbb{Q}(\omega)$.

(b) Show that $\mathsf{E}^{\mathbb{Q}}[X] = \mathsf{E}[ZX]$ for any $X \in (m\mathcal{F})^+$.
Hint: start by proving the claim for indicator functions, then simple functions, etc.

(c) Assume now that $Z \in L^2(\Omega, \mathcal{F}, \mathsf{P})$. Given $X \in L^2(\Omega, \mathcal{F}, \mathsf{P})$, prove that the result in (b) still holds. Deduce from it that $X \in L^1(\Omega, \mathcal{F}, \mathbb{Q})$.

Let $\mathcal{G} \subset \mathcal{F}$ be a sub-σ-algebra. Assume $X \in m\mathcal{F}$ and bounded.

(d) Give the definition of $Y = \mathsf{E}^{\mathbb{Q}}[X \mid \mathcal{G}]$.

(e) For $Y = \mathsf{E}^{\mathbb{Q}}[X \mid \mathcal{G}]$, show that for any $A \in \mathcal{G}$, $\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_A Y] = \mathsf{E}[\mathbf{1}_A Y \mathsf{E}[Z \mid \mathcal{G}]]$ and $\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_A X] = \mathsf{E}[\mathbf{1}_A \mathsf{E}[ZX \mid \mathcal{G}]]$.

(f) Assume that $\mathsf{E}[Z \mid \mathcal{G}] > 0$ and deduce from (e) that $\mathsf{E}^{\mathbb{Q}}[X \mid \mathcal{G}] = \frac{\mathsf{E}[ZX|\mathcal{G}]}{\mathsf{E}[Z|\mathcal{G}]}$, P-a.s.

You have just obtained the formula that connects conditional expectations under the probability measure P to those under the measure $\mathbb{Q}$!

AI-Generated Worked Solution

Q.1: drift decomposition, exponential stock and discounting

(a) The finite sum $X_n = \sum_{j=1}^n \varepsilon_j$ is adapted and bounded by n in absolute value, hence integrable. Independence of the next increment from $\mathcal{F}_n$ gives

$$\mathsf{E}[X_{n+1} \mid \mathcal{F}_n] = X_n + \mathsf{E}\varepsilon_{n+1} = X_n + (2p - 1).$$

The equality to X_n required for a martingale holds exactly when $2p - 1 = 0$, or $p = 1/2$. Thus the general process has a deterministic drift unless the two directions are equiprobable.

(b) Write $m = 2p-1$. Its predictable accumulated conditional drift is $A_0 = 0$, $A_n = \sum_{j=1}^n \mathsf{E}[X_j - X_{j-1} \mid \mathcal{F}_{j-1}] = nm$. Put $M_n = X_n - nm$. Then $M_0 = 0$ and $\mathsf{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = \mathsf{E}\varepsilon_{n+1} - m = 0$. Hence M is an integrable martingale, A_n is deterministic and therefore known at date $n - 1$, and $X_n = X_0 + M_n + A_n$. The predictable part increases if $p > 1/2$ and decreases if $p < 1/2$; a general adapted integrable process need not have an increasing compensator.

(c) From this point $p = 1/2$. The transform $Y_n = e^{X_n}$ is adapted and satisfies $0 < Y_n \leq e^n$, ensuring integrability. Since $Y_{n+1} = Y_n e^{\varepsilon_{n+1}}$,

$$\mathsf{E}[Y_{n+1} \mid \mathcal{F}_n] = Y_n \frac{e + e^{-1}}{2} \geq Y_n.$$

The multiplier is greater than one because $(e^{1/2} - e^{-1/2})^2 > 0$ implies $e + e^{-1} > 2$. Thus Y is a submartingale. Equivalently conditional Jensen for the convex exponential gives $\mathsf{E}[e^{X_{n+1}} \mid \mathcal{F}_n] \geq e^{\mathsf{E}[X_{n+1}|\mathcal{F}_n]} = e^{X_n}$.

(d) The exponential preserves strict positivity for every real value of the random walk, while the additive walk itself can take negative values. For example $X_1 = -1$ on a down step, but $Y_1 = e^{-1} > 0$. This makes Y a suitable positive stock-price model without constraining or reflecting the underlying random-walk states.

(e) Put $D_n = (1 + r)^{-n} Y_n$. The known factor D_n can be taken outside the conditional expectation, giving

$$\mathsf{E}[D_{n+1} \mid \mathcal{F}_n] = D_n \frac{(e + e^{-1})/2}{1 + r}.$$

Since $D_n > 0$, this equals D_n if and only if $1 + r = (e + e^{-1})/2$. Consequently $r = (e + e^{-1})/2 - 1 > 0$. The discount rate exactly offsets the upward mean effect of exponentiating a symmetric increment. Adaptedness and fixed-date boundedness ensure the other martingale requirements, proving sufficiency as well as necessity.

(f) At maturity one, the two payoffs are $\max(1, e) = e$ on an up step and $\max(1, e^{-1}) = 1$ on a down step. Each has probability $1/2$. Under the rate just found the time-zero value is

$$V_0 = (1 + r)^{-1} \left(\frac{e}{2} + \frac{1}{2} \right) = \frac{(e + 1)/2}{(e + e^{-1})/2} = \frac{e + 1}{e + e^{-1}}.$$

Using an undiscounted expectation would give a different value: the martingale is the discounted exponential stock, not the exponential stock itself.

Q.2: a density changes both integration and conditioning

(a) The density variable satisfies $Z \geq 0$ and $\mathsf{E}Z = 1$. Thus $\mathbb{Q}(A) = \mathsf{E}[Z\mathbf{1}_A] \geq 0$, $\mathbb{Q}(\varnothing) = 0$, and $\mathbb{Q}(\Omega) = 1$. If A_j are disjoint, $Z\sum_{j=1}^{m} \mathbf{1}_{A_j} \uparrow Z\mathbf{1}_{\cup_j A_j}$. Monotone convergence and finite linearity imply $\mathbb{Q}(\cup_j A_j) = \lim_m \sum_{j=1}^{m} \mathbb{Q}(A_j) = \sum_j \mathbb{Q}(A_j)$. These are exactly the probability-measure axioms. Independence of Z and the events is neither required nor asserted.

(b) For $X = \mathbf{1}_A$, the identity $\mathsf{E}^{\mathbb{Q}}X = \mathsf{E}[ZX]$ is the definition. For a nonnegative simple $X = \sum_{j=1}^{m} a_j\mathbf{1}_{A_j}$ it follows by multiplying each indicator identity by $a_j \geq 0$ and summing. For arbitrary measurable $X \geq 0$, choose simple $X_m \uparrow X$. Then $ZX_m \uparrow ZX$, so monotone convergence under each measure gives

$$\mathsf{E}^{\mathbb{Q}}X = \lim_m \mathsf{E}^{\mathbb{Q}}X_m = \lim_m \mathsf{E}[ZX_m] = \mathsf{E}[ZX].$$

The equality remains meaningful if the common value is infinite. A product with zero density is interpreted as zero on that set.

(c) For $Z, X \in L^2(\mathsf{P})$, Cauchy–Schwarz gives $\mathsf{E}[Z|X|] \leq (\mathsf{E}Z^2)^{1/2}(\mathsf{E}X^2)^{1/2} < \infty$. Applying (b) to $|X|$ proves $\mathsf{E}^{\mathbb{Q}}|X| < \infty$, namely $X \in L^1(\mathbb{Q})$. Apply (b) separately to X^+ and X^-; their weighted integrals are finite by the same bound, so subtraction is legitimate and yields the signed identity $\mathsf{E}^{\mathbb{Q}}X = \mathsf{E}[ZX]$.

(d) A conditional expectation $Y = \mathsf{E}^{\mathbb{Q}}[X \mid \mathcal{G}]$ is a $\mathcal{G}$-measurable, $\mathbb{Q}$-integrable variable such that $\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_A Y] = \mathsf{E}^{\mathbb{Q}}[\mathbf{1}_A X]$ for every $A \in \mathcal{G}$. It is unique up to $\mathbb{Q}$-null sets. If $|X| \leq C$, conditional monotonicity gives a version with $|Y| \leq C$. Using this bounded version will justify all products with Z below, even if Z can vanish.

(e) The signed density identity applies to the bounded variables $\mathbf{1}_A Y$ and $\mathbf{1}_A X$. Moreover $\mathbf{1}_A Y$ is $\mathcal{G}$-measurable. The defining test identity for ordinary conditional expectation, or the tower and taking-out properties, gives

$$\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_A Y] = \mathsf{E}[Z\mathbf{1}_A Y] = \mathsf{E}[\mathbf{1}_A Y \mathsf{E}[Z \mid \mathcal{G}]],$$
$$\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_A X] = \mathsf{E}[Z\mathbf{1}_A X] = \mathsf{E}[\mathbf{1}_A \mathsf{E}[ZX \mid \mathcal{G}]].$$

For the first test identity, approximate the bounded $\mathcal{G}$-measurable factor by simple functions. The defining property applies to each indicator term, and dominated convergence passes to the bounded limit. All expectations are finite since the bounded factors multiply integrable Z.

(f) Write $D = \mathsf{E}[Z \mid \mathcal{G}]$ and $N = \mathsf{E}[ZX \mid \mathcal{G}]$. The conditional-expectation definition under $\mathbb{Q}$, combined with (e), says $\mathsf{E}[\mathbf{1}_A YD] = \mathsf{E}[\mathbf{1}_A N]$ for every $A \in \mathcal{G}$. The variables YD, N are integrable and $\mathcal{G}$-measurable. Their equality of integrals on all $\mathcal{G}$ events implies equality almost surely: if their difference were positive on a measurable event of positive probability, a positive-level subset would have strictly positive integral, contradicting the test identity; apply the same argument to a negative difference. Thus $YD = N$ P-a.s. Under the assumption $D > 0$ a.s., division gives

$$\mathsf{E}^{\mathbb{Q}}[X \mid \mathcal{G}] = \frac{\mathsf{E}[ZX \mid \mathcal{G}]}{\mathsf{E}[Z \mid \mathcal{G}]} \ \mathsf{P}\text{-a.s.}$$

The positivity also aligns null events on $\mathcal{G}$: $\mathbb{Q}(A) = \mathsf{E}[\mathbf{1}_A D] = 0$ iff $\mathsf{P}(A) = 0$ for $A \in \mathcal{G}$. This explains why the conditional version, initially unique only $\mathbb{Q}$-a.s., can be specified by a P-a.s. ratio here.

11 January 2023

Two questions; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space and let $(\varepsilon_k, \delta_k)$ be a sequence of i.i.d. pairs of random variables with

$$\mathsf{P}\big((\varepsilon_1, \delta_1) = (-1, -1)\big) = 0,\ \mathsf{P}\big((\varepsilon_1, \delta_1) = (1, 1)\big) = p,$$
$$\mathsf{P}\big((\varepsilon_1, \delta_1) = (-1, 1)\big) = q,\ \mathsf{P}\big((\varepsilon_1, \delta_1) = (1, -1)\big) = r.$$

Let $(\mathcal{F}_n)$ be the filtration generated by the pairs and define

$$S_n^1 = \sum_{j=1}^{n} \varepsilon_j,\ S_n^2 = \sum_{j=1}^{n} \delta_j.$$

(a) Prove that (S_n^1) and (S_n^2) are adapted processes.
(b) Prove that (S_n^1) and (S_n^2) are martingales if and only if

$$p = 0,\ q = r = \frac{1}{2}.$$

Now assume (S_n^1) and (S_n^2) are martingales. Define

$$\rho_n = \frac{\mathrm{Cov}(S_n^1, S_n^2 \mid \mathcal{F}_{n-1})}{\sqrt{\mathrm{Var}(S_n^1 \mid \mathcal{F}_{n-1})}\sqrt{\mathrm{Var}(S_n^2 \mid \mathcal{F}_{n-1})}},$$

where

$$\mathrm{Cov}(S_n^1, S_n^2 \mid \mathcal{F}_{n-1}) = \mathsf{E}\left[\left(S_n^1 - \mathsf{E}[S_n^1 \mid \mathcal{F}_{n-1}]\right)\left(S_n^2 - \mathsf{E}[S_n^2 \mid \mathcal{F}_{n-1}]\right) \,\middle|\, \mathcal{F}_{n-1}\right],$$

and, for $i = 1, 2$, $\mathrm{Var}(S_n^i \mid \mathcal{F}_{n-1}) = \mathsf{E}\left[\left(S_n^i - \mathsf{E}[S_n^i \mid \mathcal{F}_{n-1}]\right)^2 \,\middle|\, \mathcal{F}_{n-1}\right]$.

(c) Calculate ρ_n for every $n \in \mathbb{N}$.
(d) Find the predictable process (A_n) that makes $Y_n = S_n^1 S_n^2 + A_n$ a martingale.
(e) Show that, for $a > 1$ and $\varepsilon > 0$, $\lim_{n\to\infty} \mathsf{P}\left(\frac{1}{n^a}|S_n^1 S_n^2| > \varepsilon\right) = 0$. *Use Markov's inequality.*

Q.2 Let (X_n) be a sequence of random variables with

$$X_n \sim \mathrm{Exp}(\lambda_n),\ Y_0 = 1,\ Y_n = \prod_{j=1}^{n} X_j.$$

(a) Show that (Y_n) is adapted.
(b) Find the unique non-decreasing sequence of numbers (λ_n) for which (Y_n) is a martingale.
(c) Find the predictable process (A_n) that makes $Z_n := Y_n + A_n$ a martingale.
(d) Let

$$\widetilde{Y} := \liminf_{n\to\infty} Y_n,\ \widehat{Y} := \limsup_{n\to\infty} Y_n.$$

Show that $\mathsf{E}[\widetilde{Y}] = 0$ and deduce that $\widetilde{Y} = 0$, P-a.s.
Use Fatou's lemma.
(e) Set $Q := \sum_n Y_n$ and show that $\mathsf{E}[Q] < \infty$.
(f) Deduce from (e) that $\widehat{Y} = 0$, P-a.s.
(g) Explain why $Y_\infty := \liminf_{n\to\infty} Y_n$ is a well-defined random variable and determine its value.

AI-Generated Worked Solution

Q.1: two perfectly negatively correlated martingales

(a) At date n the joint natural filtration contains all observed pairs $(\varepsilon_j, \delta_j)$ for $j \le n$. Their two coordinate sums are therefore measurable with respect to it, proving adaptedness. Since both coordinates are signs, $|S_n^i| \le n$, so both sums have finite first and second moments at every fixed date. These bounds will also justify conditioning their product.

(b) Independence of the next pair from the past gives the conditional increment means

$$\mathsf{E}\varepsilon_1 = p - q + r,\ \mathsf{E}\delta_1 = p + q - r.$$

($\Rightarrow$) If both sums are martingales, these two means vanish. Adding their equations yields $2p = 0$, hence $p = 0$ and $q = r$; normalization then gives $q = r = 1/2$.
($\Leftarrow$) Conversely, these parameters center both next increments. The adaptedness and integrability established in (a), together with these zero conditional drifts, prove that both sums are martingales.

(c) Under those probabilities each pair is either $(-1, 1)$ or $(1, -1)$, so $\delta_n = -\varepsilon_n$ almost surely. Conditional on $\mathcal{F}_{n-1}$, each sum differs from its previous known value by its next sign. Thus

$\mathrm{Var}(S_n^1 \mid \mathcal{F}_{n-1}) = \mathrm{Var}(\varepsilon_n) = 1$ and similarly the other variance is one. The centered conditional product is $\varepsilon_n \delta_n = -1$; hence the conditional covariance is minus one. Dividing by the two unit standard deviations gives $\rho_n = -1$ for every $n \geq 1$.

(d) Expand before conditioning:

$$S_{n+1}^1 S_{n+1}^2 - S_n^1 S_n^2 = S_n^1 \delta_{n+1} + S_n^2 \varepsilon_{n+1} + \varepsilon_{n+1} \delta_{n+1}.$$

The first two terms have conditional mean zero and the last conditional mean is -1. Therefore the product has conditional drift -1 per period. To cancel it in $Y_n = S_n^1 S_n^2 + A_n$, choose $A_0 = 0$, $A_{n+1} - A_n = 1$, namely $A_n = n$. This is deterministic and predictable, and $\mathsf{E}[Y_{n+1} - Y_n \mid \mathcal{F}_n] = 0$. The correction is unique with $A_0 = 0$ among predictable integrable corrections: its increment is already known at time n, so the zero-drift condition forces that increment to be one.

(e) Summing $\delta_j = -\varepsilon_j$ yields $S_n^2 = -S_n^1$, so $|S_n^1 S_n^2| = (S_n^1)^2$. The first-coordinate increments are independent centered signs; expanding the square gives $\mathsf{E}(S_n^1)^2 = n$, since all cross expectations vanish. Markov's inequality applied to this nonnegative square gives

$$\mathsf{P}(n^{-a} |S_n^1 S_n^2| > \epsilon) \leq \frac{\mathsf{E}(S_n^1)^2}{\epsilon n^a} = \frac{n^{1-a}}{\epsilon} \longrightarrow 0 \ (a > 1).$$

This proves convergence in probability at the requested scale. It does not require that the two random walks themselves be independent; they are maximally dependent.

Q.2: products of exponentials and inconsistent later requirements

Only the marginal laws $X_n \sim \mathrm{Exp}(\lambda_n)$ are printed. To make the intended product calculations determined, assume the variables are independent, use rate parameters ($\mathsf{E}X_n = 1/\lambda_n$), and use the natural filtration $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$. Without independence or conditional mean assumptions, marginals do not determine the conditional product drift. Even with this natural interpretation, part (e) contradicts the martingale requirement in (b); the contradiction is shown explicitly rather than silently changing the rates.

(a) Each $Y_n = \prod_{j=1}^n X_j$ is a finite product of $\mathcal{F}_n$-measurable variables, so is adapted. The exponential variables are positive and have finite means. Independence gives $\mathsf{E}Y_n = \prod_{j=1}^n \lambda_j^{-1} < \infty$ for every fixed n. Thus the nonnegative product is integrable as needed for a martingale calculation.

(b) Since $Y_n = Y_{n-1} X_n$ and Y_{n-1} is already known,

$$\mathsf{E}[Y_n \mid \mathcal{F}_{n-1}] = Y_{n-1} \mathsf{E}X_n = Y_{n-1}/\lambda_n.$$

Strict positivity of Y_{n-1} makes equality to Y_{n-1} equivalent to $\lambda_n = 1$. This holds for each n, including $n = 1$ with $Y_0 = 1$. Hence the unique rate sequence is the constant sequence one, which is indeed nondecreasing. Conversely those rates make the conditional identity hold and all fixed-date moments from (a) are finite.

(c) For arbitrary positive independent rates, the conditional drift of $Y_n - Y_{n-1}$ is $Y_{n-1}(\lambda_n^{-1} - 1)$. Choose its negative as the predictable increment of A:

$$A_0 = 0, \ A_n = \sum_{j=1}^n Y_{j-1}(1 - \lambda_j^{-1}).$$

Every summand is known at date $j - 1$ and is integrable, so A is predictable and each A_n integrable. Then $Z_n - Z_{n-1} = Y_n - Y_{n-1} + Y_{n-1}(1 - \lambda_n^{-1})$ has conditional mean zero, making Z a martingale. With the rates required in (b), this correction is identically zero.

(d) Under $\lambda_n = 1$, Fatou applied directly to Y_n only gives $\mathsf{E} \liminf Y_n \leq 1$; it does not by itself give zero. To obtain the requested conclusion, use the smaller moment that decreases geometrically. For a rate-one exponential,

$$c = \mathsf{E}\sqrt{X_1} = \int_0^\infty x^{1/2} e^{-x} dx = \Gamma(3/2) = \frac{\sqrt{\pi}}{2} < 1.$$

Independence gives $\mathsf{E}\sqrt{Y_n} = c^n$. Fatou applied to $\sqrt{Y_n}$ gives $\mathsf{E}\liminf\sqrt{Y_n} \le \liminf c^n = 0$, so $\liminf\sqrt{Y_n} = 0$ a.s., hence $\widetilde{Y} = \liminf Y_n = 0$ a.s. The last implication uses nonnegativity and the continuous increasing square-root map, which preserves this lower limit. Therefore $\mathsf{E}\widetilde{Y} = 0$, as claimed, but the useful Fatou application is to the fractional power.

(e) For the nonnegative sum $Q = \sum_{n\ge1} Y_n$, monotone convergence allows interchange of expectation and sum. The martingale rates have $\mathsf{E}Y_n = 1$ for every n, so

$$\mathsf{E}Q = \sum_{n\ge1} \mathsf{E}Y_n = \sum_{n\ge1} 1 = \infty.$$

Thus the printed assertion $\mathsf{E}Q < \infty$ is false under (b). No proof can establish all the requested statements simultaneously under these rates. Finite expectation at each separate date does not imply finite expected total sum.

(f) Despite the failure of the proposed route through (e), the upper limit is also zero. Since $\sum_n \mathsf{E}\sqrt{Y_n} = \sum_n c^n < \infty$, monotone convergence gives a finite expectation for the nonnegative random series $\sum_n \sqrt{Y_n}$. Such a variable is finite almost surely; consequently its terms tend to zero on that probability-one event. Thus $\sqrt{Y_n} \to 0$, and hence $Y_n \to 0$ a.s., proving $\widehat{Y} = 0$. In fact eventually $Y_n \le \sqrt{Y_n}$, so Q itself is finite a.s.; this is compatible with the infinite expectation in (e).

(g) A countable liminf of measurable variables is measurable, because it is a countable supremum of countable infima. Therefore Y_∞ is a well-defined nonnegative extended random variable even before any convergence is proved. Part (f) now shows the whole sequence converges to zero a.s., so this lower limit is finite and equals zero a.s. Its upper limit agrees there as well.

A consistent alternative for the later estimates is to choose independent rate-n exponentials: then $\mathsf{E}Y_n = 1/n!$, so direct Fatou gives zero expected lower limit and $\mathsf{E}Q = e - 1 < \infty$. This is an explicit different model: the products are supermartingales, not the martingales required by (b).

3 February 2023

One question; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space. Let (X_n) be a positive $(\mathcal{F}_n)$-submartingale and define $Y_n(\omega) := \max_{0\le j\le n} X_j(\omega)$, $\lambda > 0$, and $\tau_\lambda(\omega) := \inf\{j \in \mathbb{Z}_+ : X_j(\omega) > \lambda\}$.

(a) Prove that for any $n \in \mathbb{Z}_+$ and for all $\ell \in \mathbb{R}^+$, $\{Y_n \le \ell\} = \bigcap_{j=0}^n \{X_j \le \ell\}$. Deduce from it that (Y_n) is $(\mathcal{F}_n)$-adapted.

(b) Prove that $Y_{n+1}(\omega) \ge Y_n(\omega)$ for a.e. $\omega \in \Omega$. Use it to show that (Y_n) is an $(\mathcal{F}_n)$-submartingale.

(c) Prove that $\{\tau_\lambda > n\} = \{Y_n \le \lambda\}$. Use it to show that τ_λ is an $(\mathcal{F}_n)$-stopping time.

(d) Prove that $\{Y_n > \lambda\} = \bigcup_{j=0}^n \{\tau_\lambda = j\}$. Deduce the equality $\mathbf{1}_{\{Y_n>\lambda\}} = \sum_{j=0}^n \mathbf{1}_{\{\tau_\lambda=j\}}$.

(e) Prove $Y_n^2 = 2\sum_{j=0}^n \int_0^\infty \mathbf{1}_{\{\tau_\lambda=j\}}\, \lambda\, d\lambda$. You should use the maximum-indicator identity and the fact that, for any $p > 1$,

$$y^p = p\int_0^y \lambda^{p-1}\, d\lambda = p\int_0^\infty \mathbf{1}_{\{\lambda<y\}}\lambda^{p-1}\, d\lambda,\ y > 0.$$

(f) Prove that $\{\tau_\lambda = j\} \subseteq \{X_j > \lambda\}$ and use it to deduce $\mathbf{1}_{\{\tau_\lambda=j\}}\lambda \le \mathbf{1}_{\{\tau_\lambda=j\}}X_j$.

(g) From the tail-integral identity and the level-set bound, deduce

$$\mathsf{E}[Y_n^2] \le 2\sum_{j=0}^n \int_0^\infty \mathsf{E}\big[X_j \mathbf{1}_{\{\tau_\lambda=j\}}\big]\, d\lambda.$$

(h) Use the submartingale property to prove $\mathsf{E}\big[X_j\mathbf{1}_{\{\tau_\lambda=j\}}\big] \le \mathsf{E}\big[X_n\mathbf{1}_{\{\tau_\lambda=j\}}\big]$.

(i) Substitute the submartingale comparison into the expectation bound and use once again the tail-integral identity to prove that $\mathsf{E}[Y_n^2] \le 2\mathsf{E}[X_nY_n]$.

(j) Use the Cauchy–Schwarz inequality to deduce from (i) that

$$\mathsf{E}\left[\left(\max_{0\le j\le n} X_j\right)^2\right] = \mathsf{E}[Y_n^2] \le 4\mathsf{E}[X_n^2].$$

(k) Prove that $\mathsf{E}[Z \mid \mathcal{G}] \geq \mathsf{E}[W \mid \mathcal{G}]$ a.s. whenever $Z \geq W$.

AI-Generated Worked Solution

Q.1: a proof of the maximal inequality

The process X is nonnegative and integrable at every date, as required for a positive submartingale. We take the hitting-time index set to include zero, consistently with the maximum $Y_n = \max_{0 \leq j \leq n} X_j$ and the sums starting at zero in the question. If positive integers alone are intended, these identities require matching the maximum's index set or excluding an initial crossing. An empty hitting set has infimum infinity.

(a) For a fixed outcome, the maximum of finitely many numbers is at most ℓ if and only if each of those numbers is at most ℓ. Therefore

$$\{Y_n \leq \ell\} = \bigcap_{j=0}^{n} \{X_j \leq \ell\}.$$

Each event on the right is in $\mathcal{F}_j$ by adaptedness of X, hence in $\mathcal{F}_n$ because the filtration increases. A finite intersection remains in $\mathcal{F}_n$. Lower-ray preimages characterize real-variable measurability, so this proves Y_n is $\mathcal{F}_n$-measurable. Checking positive ℓ suffices here: for $\ell < 0$ the event is empty since $Y_n \geq 0$, and the zero threshold can be obtained by decreasing positive thresholds.

(b) Adding one more candidate to a maximum cannot reduce its value, so $Y_{n+1} \geq Y_n$ pointwise. Also $0 \leq Y_n \leq \sum_{j=0}^{n} X_j$, whose expectation is finite. Thus Y is integrable and adapted. Conditional monotonicity now gives

$$\mathsf{E}[Y_{n+1} \mid \mathcal{F}_n] \geq \mathsf{E}[Y_n \mid \mathcal{F}_n] = Y_n.$$

This verifies the submartingale property. Its justification needs both the pathwise monotonicity and integrability; monotonicity of an arbitrary infinite-valued process alone would not meet the usual submartingale definition.

(c) The statement $\tau_\lambda > n$ means that no date $j \leq n$ has crossed strictly above λ. Equivalently every $X_j \leq \lambda$, which is $Y_n \leq \lambda$. Taking complements gives

$$\{\tau_\lambda > n\} = \{Y_n \leq \lambda\},\ \{\tau_\lambda \leq n\} = \{Y_n > \lambda\} \in \mathcal{F}_n.$$

The second condition is exactly the stopping-time definition. Notice the weak inequality on the first maximum event: a value exactly at the threshold has not crossed the strictly greater barrier.

(d) If $Y_n > \lambda$, at least one of the finitely many dates $0, \ldots, n$ crosses; the set of such dates has a unique smallest member j. Thus

$$\{Y_n > \lambda\} = \bigcup_{j=0}^{n} \{\tau_\lambda = j\}.$$

The union is disjoint because a time cannot equal two different indices. For a path with a crossing exactly one summand indicator is one, and for a path without a crossing they are all zero. Consequently

$$\mathbf{1}_{\{Y_n > \lambda\}} = \sum_{j=0}^{n} \mathbf{1}_{\{\tau_\lambda = j\}}.$$

Disjointness is what permits replacing the indicator by a sum rather than merely bounding it by one.

(e) For any nonnegative finite number y, elementary integration gives

$$2 \int_0^\infty \lambda \mathbf{1}_{\{y > \lambda\}}\, d\lambda = 2 \int_0^y \lambda\, d\lambda = y^2.$$

Use $y = Y_n(\omega)$ for each outcome and substitute the disjoint-indicator identity from (d).

Moving the finite sum outside this nonnegative integral gives the required pathwise identity

$$Y_n(\omega)^2 = 2\sum_{j=0}^{n}\int_0^\infty \lambda \mathbf{1}_{\{\tau_\lambda = j\}}(\omega)\, d\lambda.$$

The values of an integrand at an individual threshold do not affect the Lebesgue integral; the strict-crossing convention nevertheless stays consistent throughout.

(f) On $\{\tau_\lambda = j\}$, the definition of a crossing gives $X_j > \lambda$. Multiplying this by the event indicator yields $\lambda \mathbf{1}_{\{\tau_\lambda = j\}} \le X_j \mathbf{1}_{\{\tau_\lambda = j\}}$. Off the event both sides are zero, so the inequality holds everywhere. It is this local information at the first crossing that will allow a probability of a maximum to be bounded by the terminal variable.

(g) Taking expectations in the identity in (e), we need to exchange integration over λ with integration over outcomes. The integrands are jointly measurable: for fixed j,

$$\{(\omega,\lambda) : \tau_\lambda(\omega) = j\} = \{X_j(\omega) > \lambda\} \cap \bigcap_{i<j} \{X_i(\omega) \le \lambda\},$$

with an empty intersection if $j = 0$. Each comparison is measurable on the product space. The integrands are also nonnegative, so Tonelli's theorem applies without presupposing a finite second moment. Applying (f) after this interchange gives

$$\mathsf{E}Y_n^2 = 2\sum_{j=0}^{n}\int_0^\infty \lambda \mathsf{P}(\tau_\lambda = j)\, d\lambda \le 2\sum_{j=0}^{n}\int_0^\infty \mathsf{E}[X_j \mathbf{1}_{\{\tau_\lambda = j\}}]\, d\lambda.$$

Both sides may still be infinite at this stage; the later estimate addresses finiteness.

(h) For $j \le n$, repeated tower conditioning of the one-step submartingale inequalities yields $\mathsf{E}[X_n \mid \mathcal{F}_j] \ge X_j$. The event $A = \{\tau_\lambda = j\}$ belongs to $\mathcal{F}_j$; for example it equals $\{\tau_\lambda \le j\} \setminus \{\tau_\lambda \le j-1\}$ when $j > 0$. Multiplying the conditional inequality by its nonnegative indicator and integrating gives

$$\mathsf{E}[X_j \mathbf{1}_A] \le \mathsf{E}[\mathsf{E}[X_n \mid \mathcal{F}_j]\mathbf{1}_A] = \mathsf{E}[X_n \mathbf{1}_A].$$

The equality is the defining test-integral property of conditional expectation. This is the required eventwise inequality, not a claim that $X_n \ge X_j$ pointwise.

(i) Substitute the bound in (h) into (g), sum over the disjoint first-crossing events, and use Tonelli again:

$$\begin{aligned}
\mathsf{E}Y_n^2 &\le 2\int_0^\infty \mathsf{E}\left[X_n \sum_{j=0}^{n} \mathbf{1}_{\{\tau_\lambda = j\}}\right] d\lambda \\
&= 2\int_0^\infty \mathsf{E}[X_n \mathbf{1}_{\{Y_n > \lambda\}}] d\lambda \\
&= 2\mathsf{E}\left[X_n \int_0^\infty \mathbf{1}_{\{Y_n > \lambda\}} d\lambda\right] = 2\mathsf{E}[X_n Y_n].
\end{aligned}$$

The indicator identity is the one proved in (d). The last integral is the length of $(0, Y_n)$, namely Y_n, which gives the final equality.

(j) We must not apply Cauchy–Schwarz and divide by $\|Y_n\|_2$ before knowing this norm is finite. If $\mathsf{E}X_n^2 = \infty$ the desired bound is already true in the extended sense. Otherwise truncate the threshold integrals above at $L < \infty$ and put $Y_n^L = Y_n \wedge L$. The truncated layer-cake identity is $(Y_n^L)^2 = 2\int_0^L \lambda \mathbf{1}_{\{Y_n > \lambda\}} d\lambda$, and exactly the same comparisons now give

$$\mathsf{E}[(Y_n^L)^2] \le 2\mathsf{E}[X_n Y_n^L] \le 2\|X_n\|_2 \|Y_n^L\|_2.$$

The truncated norm is finite because $Y_n^L \le L$. If it is nonzero, divide by it to obtain $\|Y_n^L\|_2 \le 2\|X_n\|_2$; if it is zero that conclusion is automatic. Squaring and letting $L \uparrow \infty$, monotone convergence gives

$$\mathsf{E}\left[\max_{0\le j\le n} X_j^2\right] = \mathsf{E}Y_n^2 \le 4\mathsf{E}X_n^2.$$

The truncation supplies a noncircular proof of the finite maximum second moment as well as the inequality.

(k) Assume Z, W are integrable, so their conditional expectations are defined in the ordinary signed sense, and let $D = Z - W \geq 0$. Put $H = \mathsf{E}[D \mid \mathcal{G}]$. If $\mathsf{P}(H < 0) > 0$, the countable union $\{H < 0\} = \bigcup_{m \geq 1} \{H < -1/m\}$ shows that one event A_m has positive probability. It belongs to $\mathcal{G}$ because H is $\mathcal{G}$-measurable. The conditional test identity then gives the contradiction

$$0 \leq \mathsf{E}[D\mathbf{1}_{A_m}] = \mathsf{E}[H\mathbf{1}_{A_m}] \leq -\mathsf{P}(A_m)/m < 0.$$

Hence $H \geq 0$ almost surely. Conditional linearity identifies it as $\mathsf{E}[Z \mid \mathcal{G}] - \mathsf{E}[W \mid \mathcal{G}]$, proving the asserted monotonicity. For nonnegative extended variables, the corresponding ordered conditional expectations are obtained by monotone approximation; arbitrary signed nonintegrable variables would need a separate definition and cannot simply be subtracted.

5 June 2023

Two questions; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space and let $(U_k)_{k \in \mathbb{N}}$ be a sequence of i.i.d. random variables with $U_1 \sim \mathrm{Unif}(0,1)$. Define another sequence $(X_n)_{n \in \mathbb{N}}$ as follows:

$$X_n(\omega) = \begin{cases} 1, & \text{if } U_n(\omega) \in \left[0, \dfrac{1}{n^\alpha}\right], \\ 0, & \text{if } U_n(\omega) \in \left(\dfrac{1}{n^\alpha}, 1\right], \end{cases}$$

where $\alpha > 0$ is a fixed parameter. Set

$$X_\infty := \liminf_{n \to \infty} X_n \text{ and } X^\infty := \limsup_{n \to \infty} X_n.$$

We say that the sequence $(X_n)_{n \in \mathbb{N}}$ converges to zero *in probability* if, for every $\varepsilon \in (0,1)$, $\lim_{n \to \infty} \mathsf{P}(|X_n| > \varepsilon) = 0$. Instead, we say that it converges *almost surely* to zero if

$$\mathsf{P}(X^\infty = X_\infty = 0) = 1.$$

(a) Show that the sequence $(X_n)_{n \in \mathbb{N}}$ converges to zero in probability.
Hint: show that $\mathsf{P}(|X_n| > \varepsilon) = \frac{1}{n^\alpha}$.

(b) Show that $\mathsf{E}[X_\infty] = 0$.
Hint: Fatou's lemma.

(c) Deduce from (b) that $\mathsf{P}(X_\infty = 0) = 1$.

Recall the definition of the "limsup event": $\{X_n = 1, \text{ i.o.}\} = \bigcap_{n \in \mathbb{N}} \bigcup_{m \geq n} \{X_m = 1\}$, where i.o. means *infinitely often*.

(d) Show the equivalence of the events $\{X^\infty = 1\} = \{X_n = 1, \text{ i.o.}\}$.

(e) Use the first Borel–Cantelli lemma to show that, for $\alpha > 1$, $\mathsf{P}(X^\infty = 1) = 0$. Deduce that $\mathsf{P}(X^\infty = 0) = 1$ for $\alpha > 1$.
Hint: you may use that $\sum_{n=1}^\infty \frac{1}{n^\alpha} < \infty$ *for* $\alpha > 1$.

(f) Use the second Borel–Cantelli lemma to show that, for $\alpha \in (0,1]$, $\mathsf{P}(X^\infty = 1) = 1$. *Hint: you may use that* $\sum_{n=1}^\infty \frac{1}{n^\alpha} = \infty$ *for* $\alpha \in (0,1]$.

(g) Comparing the assertions in (c), (e) and (f), what can you conclude about the almost sure convergence of the sequence $(X_n)_{n \in \mathbb{N}}$?

Q.2 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space and let $(\varepsilon_n)_{n \in \mathbb{N}}$, $(\delta_n)_{n \in \mathbb{N}}$ be two independent sequences of i.i.d. random variables with distribution

$$\mathsf{P}(\varepsilon_1 = u) = \frac{1}{2},\ \mathsf{P}(\varepsilon_1 = d) = \frac{1}{2},$$

$$\mathsf{P}(\delta_1 = h) = \frac{1}{2},\ \mathsf{P}(\delta_1 = \ell) = \frac{1}{2},$$

where $d < u$ and $\ell < h$ are fixed constants. Let $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$ be the filtration generated by the two sequences, i.e.,

$$\mathcal{F}_0 = \{\Omega, \emptyset\},\ \mathcal{F}_n = \sigma(\varepsilon_1, \delta_1, \ldots, \varepsilon_n, \delta_n),\ n \in \mathbb{N}.$$

In a financial market there is a stock S and a risk-less asset B. The stock price and the risk-free rate are modelled as stochastic processes $(S_n)_{n\in\mathbb{Z}_+}$ and $(r_n)_{n\in\mathbb{Z}_+}$, described by

$$S_n = S_{n-1}\varepsilon_n,\ S_0 = s > 0,$$

and

$$1 + r_n = \delta_n,\ r_0 = r \in (0, 1).$$

The dynamics $(B_n)_{n\in\mathbb{Z}_+}$ of the risk-less asset depend on the dynamics of the risk-free rate and are given by

$$B_n = (1 + r_n)B_{n-1},\ B_0 = b > 0.$$

(a) Show that $(S_n)_{n\in\mathbb{Z}_+}$, $(r_n)_{n\in\mathbb{Z}_+}$ and $(B_n)_{n\in\mathbb{Z}_+}$ are $(\mathcal{F}_n)$-adapted processes in $L^1(\Omega, \mathcal{F}, \mathsf{P})$.

(b) The discounted stock price $(\widetilde{S}_n)_{n\in\mathbb{Z}_+}$ is defined as $\widetilde{S}_n = \frac{S_n}{B_n}$. Show that it is a martingale if and only if $\frac{\ell+h}{h\ell}(u + d) = 4$.

From now on we assume:

$$\frac{u}{\ell} = \frac{d}{\ell} = \frac{3}{2} \text{ and } \frac{u}{h} = \frac{d}{h} = \frac{1}{2}.$$

It is easy to check that $(\widetilde{S}_n)_{n\in\mathbb{Z}_+}$ is a martingale.

(c) Show that, for every $n \in \mathbb{N}$, the distribution of $\widetilde{S}_n$ conditional on the value of $\widetilde{S}_{n-1}$ (or equivalently on $\mathcal{F}_{n-1}$) is given by

$$\mathsf{P}\left(\widetilde{S}_n = \frac{3}{2}\widetilde{S}_{n-1} \mid \mathcal{F}_{n-1}\right) = \mathsf{P}\left(\widetilde{S}_n = \frac{3}{2}\widetilde{S}_{n-1} \mid \widetilde{S}_{n-1}\right) = \frac{1}{2},$$
$$\mathsf{P}\left(\widetilde{S}_n = \frac{1}{2}\widetilde{S}_{n-1} \mid \mathcal{F}_{n-1}\right) = \mathsf{P}\left(\widetilde{S}_n = \frac{1}{2}\widetilde{S}_{n-1} \mid \widetilde{S}_{n-1}\right) = \frac{1}{2}.$$

Hint: notice that $\widetilde{S}_n/\widetilde{S}_{n-1} = \varepsilon_n/\delta_n$.

Consider a European option that allows one to swap the risk-less asset for the risky asset at maturity at time $n = 2$. We want to calculate the option's price $p_0 = \mathsf{E}\left[\frac{1}{B_2}(S_2 - B_2)^+\right]$.

(d) Assume $s = 4$ and $b = 1$ and show that we can equivalently write the option price as $p_0 = \mathsf{E}\left[\frac{1}{B_2}(S_2 - B_2)\right]$. *Hint: express the payoff as a function of* $\widetilde{S}_2$*, notice that* $\widetilde{S}_0 = s/b = 4$*, and then use the conditional law the conditional-law formula.*

(e) Calculate p_0.
Hint: use the pricing formula in item (d).

AI-Generated Worked Solution

Q.1: rare indicator events and two different modes of convergence

(a) Since X_n takes only zero and one, for a tolerance $0 < \epsilon < 1$ its deviation event is exactly $\{|X_n| > \epsilon\} = \{X_n = 1\} = \{U_n \le n^{-\alpha}\}$. Uniformity gives probability $n^{-\alpha}$, which tends to zero for every $\alpha > 0$. For $\epsilon \ge 1$ the strict deviation event is empty and its probability is zero at every date. These two cases cover all positive tolerances, establishing convergence in probability to zero. It is the growth of the sample index n, rather than independence, which makes this single-date probability vanish.

(b) Nonnegative measurable variables have a nonnegative measurable lower limit X_∞. The mean of an indicator is its probability, so $\mathsf{E}X_n = n^{-\alpha}$. Fatou's lemma therefore gives

$$0 \le \mathsf{E}X_\infty = \mathsf{E}\liminf_n X_n \le \liminf_n \mathsf{E}X_n = \liminf_n n^{-\alpha} = 0.$$

Hence the expected lower limit is zero. Interchanging a pointwise limit with expectation directly would not be justified by (a), which only proves convergence in probability; Fatou provides the appropriate inequality without that interchange.

(c) A nonnegative variable of zero expectation must vanish almost surely. Explicitly, for each integer $m \geq 1$,

$$X_\infty \geq m^{-1}\mathbf{1}_{\{X_\infty > 1/m\}} \implies \mathsf{P}(X_\infty > 1/m) \leq m\mathsf{E}X_\infty = 0.$$

Every positive value exceeds $1/m$ for some m, so $\{X_\infty > 0\}$ is a countable union of these null events. It too has probability zero, proving $X_\infty = 0$ almost surely.

(d) A zero-one sequence has upper limit one exactly when no tail is identically zero, equivalently every tail contains a one. The event expressing that is

$$\{X^\infty = 1\} = \bigcap_{N\geq 1} \bigcup_{n \geq N} \{X_n = 1\} = \{X_n = 1 \text{ infinitely often}\}.$$

If there are only finitely many ones, a cutoff after the last one makes every later term zero and the upper limit is zero. Conversely infinitely many ones force the supremum of every tail to be one, proving the other direction.

(e) For $\alpha > 1$, the series $\sum_n \mathsf{P}(X_n = 1) = \sum_n n^{-\alpha}$ converges. Apply the first Borel–Cantelli lemma to the measurable events $E_n = \{X_n = 1\}$, whose probabilities have just been shown summable. By (d), $\mathsf{P}(X^\infty = 1) = 0$. Since an indicator sequence has upper limit either zero or one, its upper limit is zero almost surely. In fact the paths are eventually identically zero. Independence is not needed for this part.

(f) For $0 < \alpha \leq 1$, the same probability series diverges (including the harmonic case $\alpha = 1$). Now independence is essential for the converse Borel–Cantelli conclusion: each $E_n = \{U_n \leq n^{-\alpha}\}$ depends on a different independent coordinate, so any finite intersection factors and the events are mutually independent. The second lemma gives $\mathsf{P}(\limsup E_n) = 1$. Equation (d) then identifies this with $\mathsf{P}(X^\infty = 1) = 1$.

(g) If $\alpha > 1$, the lower and upper limits both equal zero on a probability-one event, so the sequence converges to zero almost surely. If $0 < \alpha \leq 1$, (c) gives lower limit zero and (f) upper limit one; intersect their two probability-one events to obtain these conclusions on the same paths. There the sequence has infinitely many zeros and infinitely many ones, so has no limit. Nevertheless (a) still gives convergence to zero in probability. The distinction is that small individual-date probabilities need not prevent infinitely many rare occurrences over the whole infinite path.

Q.2: discounted dynamics and contradictory multiplier assumptions

To divide by the bank account we need nonzero bank multipliers; the financial interpretation uses $0 < \ell < h$ and positive stock multipliers. The displayed later equations $u/\ell = d/\ell = 3/2$ force $u = d$, contradicting $d < u$. Moreover retaining the two statements $\mathsf{P}(\varepsilon_n = u) = \mathsf{P}(\varepsilon_n = d) = 1/2$ is impossible when those events coincide. Thus the original complete collection of assumptions has no model. We solve (a)–(b) for the ordinary distinct-branch model, and (c)–(e) for the explicitly stipulated discounted law, clearly separated from that inconsistency.

(a) Iteration gives $S_n = s\prod_{j=1}^n \varepsilon_j$, $B_n = b\prod_{j=1}^n \delta_j$ and $r_n = \delta_n - 1$. Each is a function of the first n observed multipliers, hence is $\mathcal{F}_n$-measurable. At each fixed date those factors take values in a finite set of finite numbers, so their products and r_n are bounded by a date-dependent deterministic bound. Thus all are integrable and adapted. No bound uniform over infinitely many dates is required for this fixed-date assertion; r_0 and the starting account values are constants.

(b) The discounted ratio evolves as $\widetilde{S}_n = \widetilde{S}_{n-1}(\varepsilon_n/\delta_n)$. Given $\mathcal{F}_{n-1}$, the next independent fair stock and bank multipliers generate the four ratios $u/\ell, u/h, d/\ell, d/h$, each with probability $1/4$. Thus

$$\mathsf{E}\left[\frac{\varepsilon_n}{\delta_n} \,\middle|\, \mathcal{F}_{n-1}\right] = \frac{1}{4}\left(\frac{u}{\ell} + \frac{u}{h} + \frac{d}{\ell} + \frac{d}{h}\right) = \frac{(u+d)(\ell+h)}{4h\ell}.$$

Taking out the known $\widetilde{S}_{n-1}$ proves the martingale identity iff this last scalar equals one. Necessity already follows at $n = 1$ because $\widetilde{S}_0 = s/b > 0$; sufficiency follows at every date

by the same formula. The ratio has finite support with nonzero denominators, ensuring integrability. Rearrangement gives exactly $(\ell + h)(u + d)/(h\ell) = 4$.

(c) Algebraically the later displayed equalities force $u = d = 3\ell/2 = h/2$, hence $h = 3\ell$. This proves their incompatibility with the distinct-branch assumptions; there is no derivation of (2) under all of them. A consistent degenerate variant is to replace the stock law by $\varepsilon_n \equiv 3\ell/2$ almost surely and keep $\delta_n = \ell, h = 3\ell$ independently and fairly chosen. Then the ratio equals $3/2$ on the first bank outcome and $1/2$ on the second. Consequently

$$\mathsf{P}(\widetilde{S}_n = \tfrac{3}{2}\widetilde{S}_{n-1} \mid \mathcal{F}_{n-1}) = \mathsf{P}(\widetilde{S}_n = \tfrac{1}{2}\widetilde{S}_{n-1} \mid \mathcal{F}_{n-1}) = \tfrac{1}{2}.$$

Conditioning instead on the smaller sigma-algebra generated by $\widetilde{S}_{n-1}$ gives the same constants by the tower property. One can equivalently take this two-point discounted transition law directly as the intended model for the remaining questions, without pretending the original stock support is consistent.

(d) Under the stipulated discounted law, $\widetilde{S}_0 = s/b = 4$ and each multiplier is at least $1/2$. Therefore $\widetilde{S}_2 \geq 4(1/2)^2 = 1$ on every path. Positivity of B_2 permits the elementary identity

$$\frac{(S_2 - B_2)^+}{B_2} = \left(\frac{S_2}{B_2} - 1\right)^+ = (\widetilde{S}_2 - 1)^+ = \widetilde{S}_2 - 1 = \frac{S_2 - B_2}{B_2}.$$

The positive part is removable because its argument is nonnegative everywhere, not merely because its expectation is positive. Taking expectations gives the equivalent pricing formula requested.

(e) The mean of the stipulated next discounted multiplier is $(3/2 + 1/2)/2 = 1$. Its integrable discounted process is therefore a martingale, giving $\mathsf{E}\widetilde{S}_2 = \widetilde{S}_0 = 4$. The simplified payoff has price $p_0 = \mathsf{E}(\widetilde{S}_2 - 1) = 3$. Direct path enumeration verifies this: the four two-step discounted values are $9, 3, 3, 1$, each with probability $1/4$, and the payoffs are $8, 2, 2, 0$. Their average is $(8 + 2 + 2 + 0)/4 = 3$. This value belongs to the specified transition-law interpretation; it does not repair the incompatible original multiplier statements.

12 September 2023

One question; show all workings.

Q.1 Let (ε_n) be i.i.d. random variables with

$$\mathsf{P}(\varepsilon_1 = 1) = \mathsf{P}(\varepsilon_1 = -1) = \frac{1}{2},$$

and let $X_0 = 0,\ X_n = X_{n-1} + \varepsilon_n$. The stock price is $S_n = e^{X_n}$, and the risk-free rate is zero. Let (Θ_n) be independent of the stock-price sequence, with

$$\mathsf{P}(\Theta_n = 1) = \mathsf{P}(\Theta_n = 0) = \frac{1}{2}.$$

Let $I_0 = 0$ and $I_n = I_{n-1} + \mathbf{1}_{\{S_{n-1}>1\}}\Theta_n$. Let $\mathcal{H}_n := \sigma(\varepsilon_1, \ldots, \varepsilon_n)$. At time $n = 2$ the payoff is $S_2\mathbf{1}_{\{\tau>2\}}$, where $\tau := \inf\{n \in \mathbb{Z}_+ : I_n = 1\}$. The price is $p_0 = \mathsf{E}[S_2\mathbf{1}_{\{\tau>2\}}]$.

(a) Show the equivalence of events $\{\tau > 2\} = \{I_2 = 0\}$. *Use that (I_n) is non-decreasing.*

(b) Use the stock-price formula to show that $\mathsf{P}(I_1 = 1) = 0$.

(c) Use (a) and (b) to show that $\mathsf{P}(\tau > 2 \mid \mathcal{H}_2) = \mathsf{P}\big(\mathbf{1}_{\{S_1>1\}}\Theta_2 = 0 \mid \mathcal{H}_2\big)$, and then show that $\mathsf{P}\big(\mathbf{1}_{\{S_1>1\}}\Theta_2 = 0 \mid \mathcal{H}_2\big) = \mathbf{1}_{\{S_1\leq 1\}} + \frac{1}{2}\mathbf{1}_{\{S_1>1\}}$. *Hint: let $Y := \mathbf{1}_{\{S_1>1\}}\Theta_2$ and use*

$$\{Y = 0\} = \{Y = 0, S_1 \leq 1\} \cup \{Y = 0, S_1 > 1\}.$$

(d) Use (c) to prove $p_0 = \mathsf{E}[S_2\mathbf{1}_{\{S_1\leq 1\}}] + \frac{1}{2}\mathsf{E}[S_2\mathbf{1}_{\{S_1>1\}}]$. *Hint: use the tower property.*

(e) Use (d) and the expressions for S_1 and S_2 to prove that

$$p_0 = \frac{1}{2}(e^{-1} + e)\mathsf{E}[S_1\mathbf{1}_{\{S_1\leq 1\}}] + \frac{1}{4}(e^{-1} + e)\mathsf{E}[S_1\mathbf{1}_{\{S_1>1\}}].$$

Hint: use the tower property.

(f) Calculate p_0.

(g) Explain why the random time τ from the random-time definition is not an $(\mathcal{H}_n)$-stopping time.

(h) Let $\mathcal{G}_n := \sigma(I_k : k \le n)$. Show that τ is a $(\mathcal{G}_n)$-stopping time.

AI-Generated Worked Solution

Q.1 – A stock payoff with an unobserved activation counter

(a) Each increment of I is $\mathbf{1}_{\{S_{n-1}>1\}}\Theta_n$, hence is either zero or one. Starting at zero, the counter is nondecreasing and cannot move from zero to a value larger than one without first taking the value one. If $I_2 = 0$, it has never reached one by date two, so $\tau > 2$. Conversely, if $I_2 \ge 1$, there is a first positive value among the first two dates and that value is one, giving $\tau \le 2$. Since the counter has nonnegative integer values, the only alternative to $I_2 \ge 1$ is $I_2 = 0$. Therefore $\{\tau > 2\} = \{I_2 = 0\}$. Monotonicity alone would not exclude jumping over one; the zero–one increment property is also used here.

(b) The initial stock is $S_0 = e^{X_0} = e^0 = 1$. Because the activation test is strictly $S_0 > 1$, its initial indicator vanishes. Substituting in the counter recursion gives

$$I_1 = I_0 + \mathbf{1}_{\{S_0>1\}}\Theta_1 = 0 + 0 \cdot \Theta_1 = 0$$

on every path. Consequently $\mathsf{P}(I_1 = 1) = 0$, independently of the first Bernoulli outcome.

(c) The previous equality reduces the second-date counter to $I_2 = \mathbf{1}_{\{S_1>1\}}\Theta_2$. The survival event is therefore $\{\mathbf{1}_{\{S_1>1\}}\Theta_2 = 0\}$. Decompose it into the disjoint cases

$$\{S_1 \le 1\} \ \cup \ \big(\{S_1 > 1\} \cap \{\Theta_2 = 0\}\big).$$

The first case always survives; in the second case survival requires the independent fair Bernoulli value to be zero. As S_1 is $\mathcal{H}_2$-measurable and Θ_2 is independent of the entire stock history,

$$\begin{aligned}\mathsf{P}(\tau > 2 \mid \mathcal{H}_2) &= \mathbf{1}_{\{S_1\le 1\}} + \mathbf{1}_{\{S_1>1\}}\mathsf{E}[\mathbf{1}_{\{\Theta_2=0\}} \mid \mathcal{H}_2] \\ &= \mathbf{1}_{\{S_1\le 1\}} + \tfrac{1}{2}\mathbf{1}_{\{S_1>1\}}.\end{aligned}$$

This computation uses conditional expectations of indicators as conditional probabilities; it does not condition on a single point having positive probability.

(d) The payoff is integrable because $0 \le S_2\mathbf{1}_{\{\tau>2\}} \le e^2$. The tower identity $\mathsf{E}W = \mathsf{E}[\mathsf{E}(W \mid \mathcal{H}_2)]$ and bounded taking-out of the known S_2 give

$$\begin{aligned}p_0 &= \mathsf{E}\left[\mathsf{E}(S_2\mathbf{1}_{\{\tau>2\}} \mid \mathcal{H}_2)\right] \\ &= \mathsf{E}\left[S_2\mathsf{P}(\tau > 2 \mid \mathcal{H}_2)\right] \\ &= \mathsf{E}[S_2\mathbf{1}_{\{S_1\le 1\}}] + \tfrac{1}{2}\mathsf{E}[S_2\mathbf{1}_{\{S_1>1\}}].\end{aligned}$$

Thus the counter randomness has been averaged out, while the stock randomness remains.

(e) Write $S_2 = S_1 e^{\varepsilon_2}$. The next sign is independent of $\mathcal{H}_1$ and fair, so

$$\mathsf{E}[S_2 \mid \mathcal{H}_1] = S_1\mathsf{E}e^{\varepsilon_2} = \frac{e + e^{-1}}{2}S_1 =: cS_1.$$

Each event in the expression for p_0 depends on S_1 and is in $\mathcal{H}_1$. For either event A, the conditional test identity therefore gives $\mathsf{E}[S_2\mathbf{1}_A] = \mathsf{E}[\mathbf{1}_A\mathsf{E}(S_2 \mid \mathcal{H}_1)] = c\mathsf{E}[S_1\mathbf{1}_A]$. Substituting separately for the two events proves

$$p_0 = c\mathsf{E}[S_1\mathbf{1}_{\{S_1\le 1\}}] + \frac{c}{2}\mathsf{E}[S_1\mathbf{1}_{\{S_1>1\}}],$$

which has exactly the two coefficients $(e + e^{-1})/2$ and $(e + e^{-1})/4$ requested.

(f) At date one, $S_1 = e^{-1}$ or e, each with probability one half. Hence $\mathsf{E}[S_1\mathbf{1}_{\{S_1\le 1\}}] = e^{-1}/2$ and $\mathsf{E}[S_1\mathbf{1}_{\{S_1>1\}}] = e/2$. It follows that

$$\begin{aligned}p_0 &= \frac{e + e^{-1}}{2}\frac{e^{-1}}{2} + \frac{e + e^{-1}}{4}\frac{e}{2} \\ &= \frac{1 + e^{-2}}{4} + \frac{e^2 + 1}{8} = \frac{e^2 + 3 + 2e^{-2}}{8}.\end{aligned}$$

A path check gives the same result: the two first-up paths have survival factor one half and the two first-down paths factor one, producing $e^2/8 + 1/8 + 1/4 + e^{-2}/4$.

(g) A stopping time for the stock filtration would require $A = \{\tau \leq 2\} \in \mathcal{H}_2$. But from the counter calculation,

$$\mathsf{E}[\mathbf{1}_A \mid \mathcal{H}_2] = \tfrac{1}{2}\mathbf{1}_{\{S_1>1\}}.$$

On $\{S_1 > 1\}$, an event of probability one half, this conditional value is exactly one half. If A belonged to $\mathcal{H}_2$, its conditional expectation would instead equal $\mathbf{1}_A$, which can only take the values zero and one. This contradiction proves $A \notin \mathcal{H}_2$ and τ is not a stock-history stopping time. The missing information is the independent Bernoulli counter increment, not the observed stock value.

(h) The filtration $\mathcal{G}_n = \sigma(I_k : k \leq n)$ observes the counter. Since τ is its first date of taking the value one,

$$\{\tau \leq n\} = \bigcup_{j=0}^{n} \{I_j = 1\}.$$

Each event is in $\mathcal{G}_j \subset \mathcal{G}_n$, and their finite union is in $\mathcal{G}_n$. This verifies the stopping-time definition for every n. If the counter later exceeds one, the union still remembers the earlier first hit, so it is this union, not just the current event $\{I_n = 1\}$, that describes occurrence by date n.

Pricing convention. The answer evaluates the expectation explicitly stipulated in the paper. Under the fair signs, $\mathsf{E}[S_{n+1} \mid \mathcal{H}_n] = \cosh(1)S_n$, so with the stated zero interest the measure is not risk-neutral for this stock. An arbitrage-free risk-neutral valuation would require a different probability or discount convention and is a different calculation from the defined p_0.

14 January 2024

One question; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space, let $(\varepsilon_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d. random variables with

$$\mathsf{P}(\varepsilon_1 = 3/2) = p,\ \mathsf{P}(\varepsilon_1 = 1/2) = 1 - p$$

for some $p \in (0, 1)$, and let $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$ be the filtration generated by $(\varepsilon_n)_{n\in\mathbb{N}}$ with $\mathcal{F}_0 = \{\Omega, \emptyset\}$. In a financial market the risky asset's dynamics are given by $S_n = S_{n-1}\varepsilon_n,\ n \in \mathbb{N}$, with $S_0 > 0$, $S_0 \in m\mathcal{F}_0$ (i.e., S_0 is a constant). The risk-free rate is $r \in \mathbb{R}$.

(a) Show that $(S_n)_{n\in\mathbb{Z}_+}$ is a $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$-adapted process in $L^2(\Omega, \mathcal{F}, \mathsf{P})$.

(b) Show that the measure P is risk-neutral if and only if $r = p - \frac{1}{2}$. *Hint: the discounted asset price must be a martingale. Notice that $r \leq 0$ if $0 \leq p \leq 1/2$ and do not worry about it!*

A European Call option with maturity at time $N = 2$ and strike price $K = 1$ is an option with payoff $\varphi(S_2(\omega)) = (S_2(\omega) - 1)^+$, where $(z)^+ = \max\{0, z\}$ for $z \in \mathbb{R}$. The option's price at time 2 is given by $V_2(S_2(\omega)) = \varphi(S_2(\omega))$. The option prices at time 1 and at time 0 read, respectively, $V_1(S_1(\omega)) = \mathsf{E}\left[(1+r)^{-1}\varphi(S_2) \mid \mathcal{F}_1\right](\omega)$ and $V_0(S_0) = \mathsf{E}\left[(1+r)^{-2}\varphi(S_2)\right]$, for some functions $s \mapsto V_1(s)$ and $s \mapsto V_0(s)$. In the next two questions you will find expressions for the functions V_1 and V_0.

(c) Use the freezing lemma to prove that, for $S_1(\omega) = s$,

$$V_1(s) = \frac{1}{1+r}\left(p\left(\frac{3}{2}s - 1\right)^+ + (1-p)\left(\frac{1}{2}s - 1\right)^+\right).$$

(d) Prove that $V_0(S_0) = \mathsf{E}\left[\frac{1}{1+r}V_1(S_1)\right]$ and deduce the expression for the function V_0 when $S_0 = s$: $V_0(s) = \frac{1}{1+r}\left(pV_1\left(\frac{3}{2}s\right) + (1-p)V_1\left(\frac{1}{2}s\right)\right)$. Calculate $V_0(1)$ with $r = 0$ and $p = 1/2$.

Q.1 (Continued)

An American Call option with maturity at time $N = 2$ and strike price $K = 1$ is an option with the same payoff φ as in the call-payoff formula, but, differently from its European counterpart, the option holder can exercise the option at any time prior to maturity. The option price at time 2 is

$$V_2^{Am}(S_2(\omega)) = \varphi(S_2(\omega)) = V_2(S_2(\omega)).$$

The option prices at time 1 and at time 0 read, respectively,

$$V_1^{Am}(S_1(\omega)) = \max\{\varphi(S_1(\omega)), V_1(S_1(\omega))\},$$

$$V_0^{Am}(S_0) = \max\left\{\varphi(S_0), \mathsf{E}\left[\frac{1}{1+r}V_1^{Am}(S_1)\right]\right\}.$$

The function $V_1^{Am}(s) = \max\{\varphi(s), V_1(s)\}$, $s > 0$, is explicit thanks to (c), and the function $s \mapsto V_0^{Am}(s)$ will be determined in (f) below.

(e) Show that the American option price is larger than its European counterpart. That is, for any $s > 0$ prove $V_1^{Am}(s) \geq V_1(s)$ and use this result along with (d) to prove $V_0^{Am}(s) \geq V_0(s)$.

(f) Prove the formula, for $S_0 = s$,

$$V_0^{Am}(s) = \max\left\{\varphi(s), \frac{1}{1+r}\left(pV_1^{Am}\left(\frac{3}{2}s\right) + (1-p)V_1^{Am}\left(\frac{1}{2}s\right)\right)\right\}.$$

(g) Prove that the functions $s \mapsto V_0^{Am}(s)$ and $s \mapsto V_1^{Am}(s)$ are Borel-measurable.
Hint: start by proving that V_1 is Borel-measurable. Then deduce V_1^{Am} is Borel-measurable etc.

(h) Deduce from (g) that, for $n = 0, 1, 2$, $\{\omega \in \Omega : V_n^{Am}(S_n(\omega)) = \varphi(S_n(\omega))\} \in \mathcal{F}_n$.

It can be proven that the optimal exercise time for the American option is given by

$$\tau_*(\omega) = \inf\{n \in \{0, 1, 2\} : V_n^{Am}(S_n(\omega)) = \varphi(S_n(\omega))\}.$$

That is, the option holder observes the value of the underlying asset S_n at each time $n \in \{0, 1, 2\}$ and exercises the option at the first time when the option price $V_n^{Am}(S_n)$ equals the payoff $\varphi(S_n)$.

(i) Prove that τ_* is a $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$-stopping time.
Hint: notice that $V_n^{Am}(s) \geq \varphi(s)$ for all $n \in \{0, 1, 2\}$ and $s > 0$.

AI-Generated Worked Solution

Q.1 – European and American calls on a two-step binomial stock

(a) Iterating the stock equation gives $S_n = S_0 \prod_{j=1}^{n} \varepsilon_j$. This is a measurable function of the first n increments, so is $\mathcal{F}_n$-measurable and the process is adapted. As each multiplier lies in $\{1/2, 3/2\}$ and S_0 is a positive constant,

$$0 < S_0(1/2)^n \leq S_n \leq S_0(3/2)^n.$$

Consequently $\mathsf{E}S_n^2 \leq S_0^2(3/2)^{2n} < \infty$ at every fixed date. Thus all the required values lie in L^2, without a bound uniform over every date being required.

(b) A financially meaningful discount needs $1+r > 0$, which the resulting condition will guarantee. Let $D_n = (1+r)^{-n}S_n$. The next multiplier is independent of $\mathcal{F}_n$ and has mean $3p/2 + (1-p)/2 = p + 1/2$. Therefore

$$\mathsf{E}[D_{n+1} \mid \mathcal{F}_n] = D_n \frac{p + 1/2}{1+r}.$$

($\Rightarrow$) If D is a martingale, the conditional identity and $D_n > 0$ give $p + 1/2 = 1 + r$, hence $r = p - 1/2$. For $p \in (0, 1)$ the bank growth factor is then $p + 1/2 \in (1/2, 3/2)$ and is positive. A negative r in the allowed range is therefore no obstacle to discounting.
($\Leftarrow$) For $r = p - 1/2$, the established conditional identity reduces to D_n. Adaptedness and integrability follow from the finite-date bounds in (a), so D is a martingale.

(c) At date one the state S_1 is known and $S_2 = S_1\varepsilon_2$. Independence leaves the two multiplier probabilities unchanged under conditioning. Freezing $S_1 = s$, the future call payoff is $(3s/2 -$

$1)^+$ in the up state and $(s/2-1)^+$ in the down state. Thus

$$V_1(s) = \frac{p(3s/2-1)^+ + (1-p)(s/2-1)^+}{1+r}.$$

This averaging is the freezing lemma in its finite-state form: the known argument is held fixed while only the independent future variable is integrated. Both outcomes are finite, so the conditional payoff is integrable.

(d) Apply the tower property to the discounted maturity payoff, then separate one bank factor:

$$\begin{aligned} V_0(S_0) &= \mathsf{E}\left[\mathsf{E}((1+r)^{-2}\varphi(S_2) \mid \mathcal{F}_1)\right] \\ &= \mathsf{E}\left[(1+r)^{-1}V_1(S_1)\right]. \end{aligned}$$

If $S_0 = s$, the first-period states are $3s/2$ and $s/2$ with probabilities $p, 1-p$, giving

$$V_0(s) = \frac{pV_1(3s/2) + (1-p)V_1(s/2)}{1+r}.$$

For $s = 1, p = 1/2, r = 0$, the four terminal prices are $9/4, 3/4, 3/4, 1/4$, each path having probability one fourth. Their call payoffs are $5/4, 0, 0, 0$, so $V_0(1) = 5/16$. Equivalently $V_1(3/2) = 5/8$, $V_1(1/2) = 0$, and their first-period mean is $5/16$.

(e) At date one, $V_1^{Am}(s) = \max\{\varphi(s), V_1(s)\}$ contains $V_1(s)$ as one argument, so is at least it at every state. Multiplication by $(1+r)^{-1} > 0$ and expectation preserve this inequality. Therefore the continuation value entering the American date-zero maximum is at least $\mathsf{E}[(1+r)^{-1}V_1(S_1)] = V_0(s)$. The maximum is at least its continuation argument, proving $V_0^{Am}(s) \geq V_0(s)$. This comparison holds even when $r \leq 0$ in the permitted range, since the discount factor remains positive.

(f) Conditional on $S_0 = s$, waiting until date one yields $V_1^{Am}(3s/2)$ with probability p and $V_1^{Am}(s/2)$ with probability $1-p$. Discounting this one-period value gives the continuation amount. Immediate exercise instead pays $\varphi(s) = (s-1)^+$. Choosing the larger amount therefore gives the explicit backward recursion

$$V_0^{Am}(s) = \max\left\{(s-1)^+, \frac{pV_1^{Am}(3s/2) + (1-p)V_1^{Am}(s/2)}{1+r}\right\}.$$

The date-one optimal value already compares exercise and continuation to maturity, so this single remaining maximum incorporates every permitted exercise date.

(g) The function $x \mapsto x^+$ is continuous, and affine maps and finite weighted sums preserve continuity. The explicit formula for V_1 is consequently continuous and hence Borel-measurable on $(0, \infty)$. The maximum of continuous real functions is continuous, as $\max(f,g) = (f+g+|f-g|)/2$. Hence $V_1^{Am} = \max(\varphi, V_1)$ is continuous. The two affine state substitutions, their weighted discounted sum, and its maximum with φ in the formula for V_0^{Am} preserve continuity once more. Both requested functions are therefore Borel-measurable, with the stronger property of continuity.

(h) For each $n = 0, 1, 2$, S_n is $\mathcal{F}_n$-measurable, and the deterministic maps V_n^{Am} and φ are Borel-measurable; at maturity they coincide. Thus their difference composed with S_n is $\mathcal{F}_n$-measurable. The exercise event is the inverse image of the closed singleton zero:

$$E_n = \{V_n^{Am}(S_n) - \varphi(S_n) = 0\} \in \mathcal{F}_n.$$

This verifies measurability of the decision to exercise using only information available at its decision date.

(i) The first exercise time has occurred by date $n \leq 2$ exactly when at least one of $E_0, \ldots, E_n$ has occurred. Hence $\{\tau_* \leq n\} = \bigcup_{j=0}^{n} E_j \in \mathcal{F}_n$ because the filtration increases. At maturity $V_2^{Am} = \varphi$, so $E_2 = \Omega$ and $\tau_* \leq 2$ on every path. For all $n > 2$, the event $\{\tau_* \leq n\}$ is therefore Ω, also measurable. This proves the stopping-time condition at every date. Its optimality comes from the backward choices defining the value functions, while the argument here verifies that those choices indeed give an admissible information-based stopping rule.

9 December 2024

Two questions; show all workings.

Q.1 Let (ε_n) be a sequence of i.i.d. random variables with

$$\mathsf{P}(\varepsilon_1 = 1) = p,\ \mathsf{P}(\varepsilon_1 = -1) = 1 - p,$$

and let $S_n = S_0 + \sum_{j=1}^{n} \varepsilon_j$. Let $(\mathcal{F}_n)$ be the filtration generated by (ε_n). Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a non-negative Borel-measurable function and, for fixed $N \in \mathbb{N}$, define

$$V_n := \mathsf{E}[\varphi(S_N) \mid \mathcal{F}_n],\ n \in \{0, 1, \ldots, N\}.$$

(a) Show that $V_n = v(n, S_n)$ for a Borel-measurable function $v : \mathbb{Z}_+ \times \mathbb{R} \to [0, \infty]$ such that $v(N, x) = \varphi(x)$ and

$$v(n, x) = \mathsf{E}\left[\varphi\left(x + \sum_{j=n+1}^{N} \varepsilon_j\right)\right],\ n \in \{0, 1, \ldots, N-1\}.$$

Hint: use the freezing lemma to express $v(n, S_n)$.

(b) What guarantees that the function $v(n, x)$ is measurable in x for each $n \in \mathbb{N}$?

(c) Show that $v(n, x) = \mathsf{E}[v(n+1, x + \varepsilon_{n+1})]$. *Hint: use the tower property.*

(d) Assume $N = 2$ and use the backward recursion to obtain

$$v(0, S_0) = p^2\varphi(S_0 + 2) + 2p(1-p)\varphi(S_0) + (1-p)^2\varphi(S_0 - 2).$$

Hint: start with $v(N, x) = \varphi(x)$, then compute $v(n, x) = \mathsf{E}[v(n+1, x + \varepsilon_{n+1})]$.

Q.2 Consider a set function $\rho : \mathbb{R} \to \mathbb{R}$.

(a) Show that $\rho^{-1}((-\infty, c]) \in \mathcal{B}(\mathbb{R})\ \forall c \in \mathbb{R}$ if and only if $\rho^{-1}([c, \infty)) \in \mathcal{B}(\mathbb{R})\ \forall c \in \mathbb{R}$.

(b) Define the σ-algebra $\sigma(\rho)$ generated by ρ.

(c) Let us now consider the family of functions $\{\rho_a : a \in \mathbb{R}\}$ defined as $\rho_a(x) := \mathbf{1}_{(-\infty, a]}(x)$. Calculate $\sigma(\rho_a)$ and prove that $\sigma(\rho_a) \subseteq \mathcal{B}(\mathbb{R})\ \forall a \in \mathbb{R}$.

(d) Is the function ρ_2 measurable with respect to any $\sigma(\rho_a)$ for $a \neq 2$? Motivate your answer.

(e) Let μ be a measure on $\mathcal{B}(\mathbb{R})$. Explain why the limit $\lim_{n\to\infty} \int_{\mathbb{R}} \rho_{1/n}(x)\, \mu(dx)$ exists.

(f) Let μ be a finite measure on $\mathcal{B}(\mathbb{R})$ and let $\mathcal{L}$ be the Lebesgue measure. Show that $\lim_{n\to\infty} \int_{\mathbb{R}} \rho_{1/n}(x)\, \mu(dx) = \mu((-\infty, 0])$ and $\lim_{n\to\infty} \int_{\mathbb{R}} \rho_{1/n}(x)\, \mathcal{L}(dx) = +\infty$.

(g) Calculate $\lim_{n\to\infty} \rho_{1/n}(x)$ and explain why, in the second integral in (f), we cannot use MON and DOM.

AI-Generated Worked Solution

Q.1: pricing by freezing the current state

(a) Separate known and future increments: $S_N = S_n + T_n$, where $T_n = \sum_{j=n+1}^{N} \varepsilon_j$. The first term is $\mathcal{F}_n$-measurable and the second is independent of $\mathcal{F}_n$. Conditioning therefore freezes the current state and averages only the future. If K counts the future up moves, then $T_n = 2K - (N - n)$ and K is binomial $(N - n, p)$. This gives

$$v(n, x) = \sum_{k=0}^{N-n} \binom{N-n}{k} p^k (1-p)^{N-n-k} \varphi(x + 2k - (N-n)),\ V_n = v(n, S_n).$$

To verify the conditional identity, test against $\mathbf{1}_A$, $A \in \mathcal{F}_n$: independence replaces each future-count indicator by its constant probability inside the expectation. At $n = N$ there are no remaining increments and the only shift is zero, so $v(N, x) = \varphi(x)$. Because S_0 is fixed and the horizon is finite, the real-valued payoff is evaluated at finitely many reachable points and is integrable there. Nonnegative extended payoffs can instead be handled by monotone truncation; zero-probability terms are omitted at $p = 0, 1$.

(b) Each shift $x \mapsto x + 2k - (N - n)$ is continuous. Composing it with the Borel payoff gives a Borel function of x. The formula is a finite sum with constant nonnegative coefficients, so $v(n, \cdot)$ is Borel. This checks the measurability required to substitute the random state S_n

without assuming that the payoff is continuous. It is a particular instance of measurability of integrals of nonnegative jointly measurable functions.

(c) Split $T_n = \varepsilon_{n+1} + T_{n+1}$. First condition on the next sign, then average it out. The remaining independent increments have exactly the law used in $v(n+1,\cdot)$, so the tower property gives

$$\begin{aligned} v(n,x) &= \mathsf{E}[\mathsf{E}[\varphi(x+\varepsilon_{n+1}+T_{n+1}) \mid \varepsilon_{n+1}]] \\ &= \mathsf{E}[v(n+1, x+\varepsilon_{n+1})] \\ &= pv(n+1,x+1) + (1-p)v(n+1,x-1). \end{aligned}$$

The recursion averages the two next-state values, not the payoff at an averaged state; these operations differ for a nonlinear payoff.

(d) At maturity, $v(2,x) = \varphi(x)$, hence $v(1,x) = p\varphi(x+1) + (1-p)\varphi(x-1)$. Insert this into the previous step:

$$\begin{aligned} v(0,S_0) &= p[p\varphi(S_0+2) + (1-p)\varphi(S_0)] \\ &\quad + (1-p)[p\varphi(S_0) + (1-p)\varphi(S_0-2)] \\ &= p^2\varphi(S_0+2) + 2p(1-p)\varphi(S_0) + (1-p)^2\varphi(S_0-2). \end{aligned}$$

The central coefficient combines two paths, up–down and down–up. The three weights sum to one, checking the calculation.

Q.2: generated information and a decreasing limit

(a) (⇒) Suppose every $\{\rho \le c\}$ is Borel. Strict lower preimages follow from the countable identity

$$\{\rho < c\} = \bigcup_{m \ge 1} \{\rho \le c - 1/m\}.$$

If $\rho(x) < c$, choose m with $1/m < c - \rho(x)$; conversely every member of the union satisfies the strict inequality. Complementing gives $\{\rho \ge c\}$ Borel.
(⇐) In the reverse direction, use $\{\rho > c\} = \bigcup_{m\ge1}\{\rho \ge c + 1/m\}$ and complement. The proof uses precisely the countable unions and complements permitted in a sigma-algebra.

(b) The smallest sigma-algebra on the domain making ρ measurable is

$$\sigma(\rho) = \{\rho^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}.$$

This preimage collection itself is a sigma-algebra: inverse images preserve the whole set, complements and countable unions. Any other sigma-algebra making ρ measurable must contain all these preimages, proving minimality. Lower-ray preimages suffice as generators because lower rays generate the target Borel sigma-algebra.

(c) Put $A_a = (-\infty, a]$. The indicator is one on A_a and zero on A_a^c. A target Borel set containing neither, one, zero, or both of these values has preimage $\varnothing, A_a, A_a^c$, or $\mathbb{R}$, respectively. All four occur. Therefore

$$\sigma(\rho_a) = \{\varnothing, A_a, A_a^c, \mathbb{R}\} \subset \mathcal{B}(\mathbb{R}).$$

The inclusion follows since both half-lines are Borel; the generated information tells only which side of the threshold a contains the point.

(d) No. If $a < 2$, choose $x \in (a, 2]$ and $y > 2$. Both lie in the same atom A_a^c but $\rho_2(x) = 1$, $\rho_2(y) = 0$. If $a > 2$, choose $x \le 2$ and $y \in (2, a]$, both in A_a, again with different values. Thus ρ_2 is not constant on an information atom. Equivalently $A_2 = \{\rho_2 = 1\}$ is not one of the four available sets, so a necessary measurability test fails.

(e) The thresholds decrease, so $A_{1/n}$ decreases. Consequently $\int \rho_{1/n} d\mu = \mu(A_{1/n})$ is a decreasing sequence in $[0,\infty]$, whose extended limit is its infimum. No finiteness condition is needed merely for existence of this numerical limit. Identifying it with the measure of the intersection is a separate assertion with additional hypotheses.

(f) Every nonpositive point belongs to all $A_{1/n}$; a positive point x is excluded by taking $n > 1/x$.

Thus their intersection is $(-\infty, 0]$. For finite μ, continuity from above yields

$$\lim_n \int \rho_{1/n} d\mu = \mu\left(\bigcap_n A_{1/n}\right) = \mu((-\infty, 0]).$$

One may prove this by applying increasing continuity to $A_1 \setminus A_{1/n}$ and subtracting from the finite $\mu(A_1)$. In contrast every negative half-line has infinite Lebesgue length. Hence each Lebesgue integral, and its limit, equals $+\infty$ by direct calculation, without subtracting infinite masses.

(g) Pointwise the same membership analysis gives $\rho_{1/n} \downarrow \mathbf{1}_{(-\infty,0]}$. The usual MON theorem concerns increasing functions, so it does not apply as stated. DOM would need an integrable dominator. Under Lebesgue measure any dominator of even ρ_1 is at least one on $(-\infty, 1]$ and has infinite integral, so none exists. For finite μ, the constant one is an integrable dominator and DOM is valid. Equality of the infinite Lebesgue integrals does not make either inapplicable theorem a legitimate justification.

4 February 2025

Two questions; show all workings.

Q.1 Let $S = [0, \infty)$, $\Sigma = \mathcal{B}([0, \infty))$ and $\mu(ds) = \lambda e^{-\lambda s} \mathcal{L}eb(ds)$ for a fixed parameter $\lambda > 0$. Here, $\mathcal{L}eb$ denotes the Lebesgue measure on $[0, \infty)$.

(a) Using the definition, prove that (S, Σ, μ) is a probability space.

Let $(\Omega, \mathcal{F}, \mathsf{P})$ be another probability space and let

$$\overline{\Omega} := S \times \Omega,\ \overline{\mathcal{F}} := \Sigma \times \mathcal{F},\ \overline{\mathsf{P}} := \mu \times \mathsf{P}.$$

The space $(\overline{\Omega}, \overline{\mathcal{F}}, \overline{\mathsf{P}})$ is a product measure space.

(b) Write a generic sample point $\overline{\omega} \in \overline{\Omega}$ in terms of sample points in S and Ω.

(c) What sets generate the product σ-algebra $\overline{\mathcal{F}}$?

(d) Show that $(\overline{\Omega}, \overline{\mathcal{F}}, \overline{\mathsf{P}})$ is a probability space.

Let $X : \overline{\Omega} \to [0, \infty)$ be a $\overline{\mathcal{F}}$-measurable function such that

$$\int_\Omega X(s, \omega) \mathsf{P}(d\omega) = \mathsf{E}[X(s, \cdot)] = s$$

for each $s \in S$.

(e) Explain why both equalities hold in the equation below:

$$\overline{\mathsf{E}}[X] = \int_\Omega \int_0^\infty X(s, \omega) \lambda e^{-\lambda s} \mathcal{L}eb(ds) \mathsf{P}(d\omega) = \int_0^\infty \int_\Omega X(s, \omega) \mathsf{P}(d\omega) \lambda e^{-\lambda s} \mathcal{L}eb(ds),$$

where $\overline{\mathsf{E}}[\,\cdot\,]$ is the expectation under the measure $\overline{\mathsf{P}}$.

(f) Prove that $\overline{\mathsf{E}}[X] = \frac{1}{\lambda}$.

Q.2 Let $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n \in \mathbb{Z}_+}, \mathsf{P})$ be a filtered probability space and let $(X_n)_{n \in \mathbb{Z}_+}$ and $(Y_n)_{n \in \mathbb{Z}_+}$ be two martingales in $L^2(\Omega, \mathcal{F}, \mathsf{P})$. Define $Z_n := X_n Y_n,\ n \in \mathbb{Z}_+$.

(a) Show that $(Z_n)_{n \in \mathbb{Z}_+}$ is an adapted process in $L^1(\Omega, \mathcal{F}, \mathsf{P})$.

(b) Show that

$$Z_{n+1} - Z_n = (X_{n+1} - X_n)(Y_{n+1} - Y_n) + X_n(Y_{n+1} - Y_n) + Y_n(X_{n+1} - X_n).$$

(c) Deduce from (b) that $(Z_n)_{n \in \mathbb{Z}_+}$ is a martingale if and only if, for all $n \in \mathbb{Z}_+$,

$$V_n := \mathsf{E}[(X_{n+1} - X_n)(Y_{n+1} - Y_n) \mid \mathcal{F}_n] = 0.$$

(d) State a simple, sufficient condition for the conditional cross-increment condition and write an explicit example that satisfies your sufficient condition.

Assume $V_n = 1$ for all $n \in \mathbb{N}$ and $X_0 = Y_0 = 0$.

(e) What can you conclude about the martingale property of the process $(Z_n)_{n \in \mathbb{Z}_+}$?
Hint: is it martingale, super-martingale, sub-martingale or neither? Why?

(f) We know that the Doob's decomposition $Z_n = Z_0 + M_n + A_n$ holds with $(M_n)_{n\in\mathbb{Z}_+}$ a martingale and $(A_n)_{n\in\mathbb{N}}$ a previsible process such that $M_0 = A_0 = 0$. Determine M_n and A_n and show that

$$Z_0 = 0,\ M_n = X_nY_n - n,\ A_n = n,\ n \in \mathbb{N}.$$

AI-Generated Worked Solution

Q.1: an exponential probability and its product space

(a) Define $\mu(A) = \int \mathbf{1}_A(s)\lambda e^{-\lambda s}ds$ for Borel $A \subset [0,\infty)$. It is nonnegative and vanishes on the empty set. If A_j are disjoint, $\sum_{j=1}^m \mathbf{1}_{A_j} \uparrow \mathbf{1}_{\cup_j A_j}$, so MON proves $\mu(\cup_j A_j) = \sum_j \mu(A_j)$. Thus it is countably additive. Finally

$$\mu([0,\infty)) = \lim_{M\to\infty} \int_0^M \lambda e^{-\lambda s}ds = \lim_{M\to\infty}(1 - e^{-\lambda M}) = 1.$$

This verifies the probability axioms, including measure additivity rather than only normalization.

(b) An outcome is the pair $\overline{\omega} = (s,\omega)$, $s \in [0,\infty)$ and $\omega \in \Omega$. The first coordinate has law μ, the second law P. Holding s fixed and integrating $X(s,\cdot)$ averages only the original experiment; the product expectation averages both coordinates.

(c) The product sigma-algebra is $\Sigma \otimes \mathcal{F} = \sigma\{A \times B : A \in \Sigma, B \in \mathcal{F}\}$. Rectangles generate it but are not generally the whole collection, since a union or complement of rectangles need not be a single rectangle. In particular coordinate projections are measurable because their inverse images are rectangles $A \times \Omega$ and $S \times B$. The question's notation $\Sigma \times \mathcal{F}$ is read in this generated sense.

(d) Both factors are probabilities, hence finite and sigma-finite, so the product measure exists and is countably additive on the product sigma-algebra. On rectangles it satisfies $(\mu\otimes P)(A\times B) = \mu(A)P(B)$. Applying this to $S \times \Omega$ gives total mass $1 \cdot 1 = 1$, proving it is a probability measure.

(e) By definition, $\overline{\mathsf{E}}X = \int_{S\times\Omega} X\, d(\mu \otimes P)$. Joint measurability and nonnegativity allow Tonelli's theorem to identify this with both iterated integrals. Replacing the s-measure by its density gives

$$\overline{\mathsf{E}}X = \int_\Omega \int_0^\infty X(s,\omega)\lambda e^{-\lambda s}ds\, P(d\omega) = \int_0^\infty \int_\Omega X(s,\omega)P(d\omega)\lambda e^{-\lambda s}ds.$$

All expressions are defined in $[0,\infty]$ before finiteness is known. No unproved absolute-integrability assumption is needed for this nonnegative interchange.

(f) Substitute the given inner mean s. On $[0,M]$, integration by parts with $u = s$, $dv = \lambda e^{-\lambda s}ds$ gives

$$\int_0^M s\lambda e^{-\lambda s}ds = -Me^{-\lambda M} + \int_0^M e^{-\lambda s}ds = -Me^{-\lambda M} + \frac{1 - e^{-\lambda M}}{\lambda}.$$

Exponential decay makes both boundary errors vanish as $M \to \infty$, so $\overline{\mathsf{E}}X = 1/\lambda$. The finite result also establishes integrability on the product probability space.

Q.2: products of square-integrable martingales

(a) The pair (X_n, Y_n) is $\mathcal{F}_n$-measurable and multiplication is continuous, so $Z_n = X_nY_n$ is adapted. Cauchy–Schwarz gives

$$\mathsf{E}|Z_n| \le (\mathsf{E}X_n^2)^{1/2}(\mathsf{E}Y_n^2)^{1/2} < \infty.$$

This is the role of the L^2 assumption: products of two merely integrable variables need not be integrable.

(b) Write $\Delta X = X_{n+1} - X_n$, $\Delta Y = Y_{n+1} - Y_n$. Expanding $(X_n + \Delta X)(Y_n + \Delta Y) - X_nY_n$ leaves the three terms $\Delta X\Delta Y + X_n\Delta Y + Y_n\Delta X$. In particular the cross-increment product does not disappear; it is precisely the possible drift of the product process.

(c) Increments are in L^2, hence all products in (b) are integrable by Cauchy–Schwarz. The martingale property gives $\mathsf{E}[\Delta X \mid \mathcal{F}_n] = \mathsf{E}[\Delta Y \mid \mathcal{F}_n] = 0$. Taking out the past factors yields

$$\mathsf{E}[Z_{n+1} - Z_n \mid \mathcal{F}_n] = \mathsf{E}[\Delta X \Delta Y \mid \mathcal{F}_n] = V_n.$$

For an unbounded past factor X_n, justify this by bounded truncation: $\|X_n - X_n^{(k)}\|_2 \|\Delta Y\|_2 \to 0$ bounds the discarded product in L^1, and conditional expectation is an L^1 contraction.
($\Rightarrow$) If Z is a martingale, the computed conditional drift is zero, so $V_n = 0$ a.s. for each n.
($\Leftarrow$) If each $V_n = 0$ a.s., the same drift identity is the martingale identity; the other requirements were verified in (a).

(d) Conditional independence of the increments given $\mathcal{F}_n$ suffices: their conditional product expectation factors into two zero conditional means. For a concrete example use two mutually independent iid fair-sign families ξ_j, η_j, with $X_n = \sum_{j \le n} \xi_j$, $Y_n = \sum_{j \le n} \eta_j$ and the joint natural filtration. The next signs are independent of each other and the entire past, giving $V_n = 0$. The finite sums have finite second moments, verifying all the required assumptions.

(e) Under the intended condition $V_n = 1$ for every $n \ge 0$, $\mathsf{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n + 1$. Thus Z is a strict submartingale, neither a martingale nor a supermartingale. The printed final hypothesis uses $n \in \mathbb{N}$: if that means $\{1, 2, \ldots\}$, V_0 is unspecified, and even the first-step submartingale assertion requires the extra $V_0 \ge 0$. Later positive drifts do not determine the missing first one.

(f) Since $Z_0 = X_0 Y_0 = 0$, the predictable drift is the sum

$$A_n = \sum_{j=0}^{n-1} \mathsf{E}[Z_{j+1} - Z_j \mid \mathcal{F}_j] = \sum_{j=0}^{n-1} V_j.$$

If every $V_j = 1$, including $j = 0$, then $A_n = n$ and $M_n = Z_n - n$. The compensator is deterministic and predictable, while $\mathsf{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = 1 - 1 = 0$, so M is a martingale with $M_0 = A_0 = 0$. If the hypothesis starts only at $j = 1$, the correct formula for $n \ge 1$ is instead $A_n = V_0 + n - 1$, $M_n = Z_n - A_n$. Thus the requested $A_n = n$ needs $V_0 = 1$ too. This explains the exact additional condition rather than silently changing the original indexing.

5 June 2025

INSTRUCTIONS: There are one question to this exam: **Q.1**. The question contains a number of sub-questions. It is possible to answer all sub-questions independently of the previous ones.
Write down all your calculations. Answers provided without clear workings will be penalised.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space. Let $(\varepsilon_n)_{n \in \mathbb{N}}$ be an i.i.d. sequence with

$$\mathsf{P}(\varepsilon_1 = 1) = \frac{2}{3},\ \mathsf{P}(\varepsilon_1 = -1) = \frac{1}{3}.$$

Let $(\mathcal{F}_n)_{n \in \mathbb{Z}_+}$ be a filtration with

$$\mathcal{F}_0 = \{\emptyset, \Omega\} \text{ and } \mathcal{F}_n = \sigma(\varepsilon_j, j \le n),\ n \in \mathbb{N}.$$

Let us introduce a process $(X_n)_{n \in \mathbb{Z}_+}$ with $X_0 = 0$ and $X_n = \sum_{j=1}^n \varepsilon_j,\ n \in \mathbb{N}$.

(a) Show that the process $(X_n)_{n \in \mathbb{Z}_+}$ is a $(\mathcal{F}_n, \mathsf{P})$-submartingale.

(b) Show that the Doob's decomposition $X_n = M_n + A_n$ is given by $M_0 = A_0 = 0$ and

$$M_n = X_n - \frac{1}{3}n,\ A_n = \frac{1}{3}n,\ n \in \mathbb{N}.$$

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) := \frac{3}{4}\mathbf{1}_{\{x=1\}} + \frac{3}{2}\mathbf{1}_{\{x=-1\}}$. Show that f is Borel-measurable. *Hint: show* $\{f > c\} \in \mathcal{B}(\mathbb{R})$ *for all* $c \in \mathbb{R}$.

(d) Let the process $(Z_n)_{n \in \mathbb{Z}_+}$ be defined as $Z_0 = 1$ and $Z_n = \prod_{j=1}^n f(\varepsilon_j),\ n \in \mathbb{N}$. Show that $(Z_n)_{n \in \mathbb{Z}_+}$ is a positive $(\mathcal{F}_n, \mathsf{P})$-martingale with

$$\mathsf{E}[Z_n] = 1 \text{ for all } n \in \mathbb{Z}_+.$$

(e) Fix $N \in \mathbb{N}$ and, for any $E \in \mathcal{F}_N$, let $\mathbb{Q}(E) := \mathsf{E}[Z_N \mathbf{1}_E]$. Show that $\mathbb{Q}$ is a probability measure on $(\Omega, \mathcal{F}_N)$.
Hint: use $\mathsf{E}[Z_N] = 1$ *and* $\mathbf{1}_{\cup_k E_k} = \sum_k \mathbf{1}_{E_k}$ *for any pairwise disjoint sets* $(E_k)_{k\in\mathbb{N}}$.

(f) Let $\mathsf{E}^{\mathbb{Q}}[\,\cdot\,]$ be the expectation under the probability measure $\mathbb{Q}$. Show that, for any $Y \in (m\mathcal{F}_N)^+$, $\mathsf{E}^{\mathbb{Q}}[Y] = \mathsf{E}[Z_N Y]$. *Hint: prove the claim for indicator functions, then for simple functions, etc.*

(g) Show that the result from (f) extends to any $Y \in b\mathcal{F}_N$ (bounded and $\mathcal{F}_N$-measurable).
Hint: if $|Y(\omega)| \le c$, *then* $Y(\omega) + c \ge 0$.

Notice that $(X_n)_{n=0}^N$ is bounded and therefore (g) can be used to compute $\mathsf{E}^{\mathbb{Q}}[X_n]$ for $n = 0, \dots, N$. In the next part of the exam you are going to show that $(X_n)_{n=0}^N$ is a $(\mathcal{F}_n, \mathbb{Q})$-martingale. For that purpose you need to be able to calculate $\mathsf{E}^{\mathbb{Q}}[X_{n+1} \mid \mathcal{F}_n]$ for $n = 0, \dots, N-1$.

(h) Let $Y \in b\mathcal{F}_N$ and, for fixed $n = 0, \dots, N$, denote $W = \mathsf{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_n]$. Using (f), show that for any $E \in \mathcal{F}_n$ it holds $\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_E W] = \mathsf{E}[\mathbf{1}_E W \mathsf{E}[Z_N \mid \mathcal{F}_n]]$ and $\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_E Y] = \mathsf{E}[\mathbf{1}_E \mathsf{E}[Y Z_N \mid \mathcal{F}_n]]$.

(i) By definition of conditional expectation $\mathsf{E}^{\mathbb{Q}}[\mathbf{1}_E W] = \mathsf{E}^{\mathbb{Q}}[\mathbf{1}_E Y]$. Combine the latter with (h) and the martingale property of $(Z_n)_{n\in\mathbb{Z}_+}$ to deduce $\mathsf{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_n] = Z_n^{-1}\mathsf{E}[Z_N Y \mid \mathcal{F}_n]$, P-a.s.

(j) Use (i) to prove that $(X_n)_{n=0}^N$ is a bounded $(\mathcal{F}_n, \mathbb{Q})$-martingale.
Hint: use tower property to simplify $\mathsf{E}[Z_N X_{n+1} \mid \mathcal{F}_n]$ *and then notice that* $Z_{n+1} = Z_n f(\varepsilon_{n+1})$.

(k) Calculate $\mathbb{Q}(\varepsilon_1 = 1)$ and $\mathbb{Q}(\varepsilon_1 = -1)$. *Hint: using (j) you can deduce the result without calculations!*

AI-Generated Worked Solution

A biased walk made fair by a density martingale

(a) Each X_n is a finite sum of the first n signs, hence is adapted and satisfies $|X_n| \le n$. Its next increment has mean $\mathsf{E}\varepsilon_{n+1} = 2/3 - 1/3 = 1/3$ and is independent of $\mathcal{F}_n$. Therefore

$$\mathsf{E}[X_{n+1} \mid \mathcal{F}_n] = X_n + \tfrac{1}{3} \ge X_n.$$

Adaptedness, integrability and the conditional inequality prove the submartingale property. The positive drift is predictable, not a martingale increment.

(b) Remove exactly that drift at each date. Put $A_n = n/3$ and $M_n = X_n - n/3$. The process A is deterministic and therefore predictable, with $A_0 = 0$. Also $M_0 = 0$ and

$$\mathsf{E}[M_{n+1} - M_n \mid \mathcal{F}_n] = \mathsf{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] - \tfrac{1}{3} = 0.$$

The bounded finite-date sums make M integrable and adapted. Thus this is a valid Doob decomposition; its compensator is the sum of the conditional increment means $\sum_{j=0}^{n-1} 1/3$.

(c) The function is zero outside $\{-1, 1\}$, takes value $3/4$ at 1 and $3/2$ at -1. Its strict upper level sets are $\mathbb{R}$ if $c < 0$, $\{-1, 1\}$ if $0 \le c < 3/4$, $\{-1\}$ if $3/4 \le c < 3/2$, and $\varnothing$ if $c \ge 3/2$. Every such set is Borel. Since strict rays generate the target Borel sigma-algebra, this proves measurability, including the threshold endpoint cases.

(d) On the actual sign support, both values of f are strictly positive, so $Z_n > 0$ despite f being zero at other real arguments. Each product is adapted and bounded by $(3/2)^n$, so it is integrable. Its one-step conditional mean is

$$\mathsf{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n \mathsf{E}[f(\varepsilon_{n+1})] = Z_n \left(\tfrac{2}{3}\tfrac{3}{4} + \tfrac{1}{3}\tfrac{3}{2}\right) = Z_n.$$

The expectation of the multiplier is one. Iterating the martingale identity from $Z_0 = 1$ gives $\mathsf{E}Z_n = 1$ for all n, proving both claims.

(e) The definition gives $Q(E) \ge 0$ and $Q(\varnothing) = 0$. For disjoint $E_k \in \mathcal{F}_N$, the partial indicator sums increase to $\mathbf{1}_{\cup_k E_k}$. Because $Z_N \ge 0$, MON yields

$$Q(\cup_k E_k) = \mathsf{E}[Z_N \sum_k \mathbf{1}_{E_k}] = \sum_k Q(E_k).$$

Finally $Q(\Omega) = \mathsf{E}Z_N = 1$. These are the probability axioms. Moreover $Z_N > 0$ everywhere on the sign experiment up to null sets, so Q and P have the same null sets on $\mathcal{F}_N$.

(f) For an indicator $Y = \mathbf{1}_E$, the assertion is the definition of Q. For a nonnegative simple function $Y = \sum_{j=1}^m a_j \mathbf{1}_{E_j}$, finite linearity gives $\mathsf{E}_Q Y = \sum_j a_j Q(E_j) = \mathsf{E}[Z_N Y]$. For arbitrary nonnegative measurable Y, choose simple $Y_k \uparrow Y$. Then $Z_N Y_k \uparrow Z_N Y$ because the density is nonnegative. MON under both probabilities gives

$$\mathsf{E}_Q Y = \lim_k \mathsf{E}_Q Y_k = \lim_k \mathsf{E}[Z_N Y_k] = \mathsf{E}[Z_N Y],$$

with extended values allowed. This construction explains why the formula is not restricted to bounded variables.

(g) For bounded signed Y, take c with $|Y| \le c$. The function $Y + c$ is nonnegative, so the preceding formula applies. Subtracting $c\mathsf{E}Z_N = c$ yields $\mathsf{E}_Q Y = \mathsf{E}[Z_N Y]$. All expectations in this subtraction are finite because $|Z_N Y| \le cZ_N$ and $\mathsf{E}Z_N = 1$; hence there is no difference of infinities.

(h) Let $W = \mathsf{E}_Q[Y \mid \mathcal{F}_n]$. Conditional monotonicity gives $|W| \le \|Y\|_\infty$ and W is $\mathcal{F}_n$-measurable. Applying (g) and then the conditional defining property, for $E \in \mathcal{F}_n$ we obtain

$$\mathsf{E}_Q[\mathbf{1}_E W] = \mathsf{E}[Z_N \mathbf{1}_E W] = \mathsf{E}[\mathbf{1}_E W \mathsf{E}(Z_N \mid \mathcal{F}_n)].$$

Likewise $\mathsf{E}_Q[\mathbf{1}_E Y] = \mathsf{E}[Z_N \mathbf{1}_E Y] = \mathsf{E}[\mathbf{1}_E \mathsf{E}(Z_N Y \mid \mathcal{F}_n)]$. Boundedness of Y, W and integrability of the density justify these tests and taking out the past-measurable factors.

(i) By the definition of W, the two Q-integrals in (h) agree for every $E \in \mathcal{F}_n$. Since Z is a martingale, $\mathsf{E}[Z_N \mid \mathcal{F}_n] = Z_n$. Thus the two integrable $\mathcal{F}_n$-measurable variables WZ_n and $\mathsf{E}[Z_N Y \mid \mathcal{F}_n]$ have the same integral over every $E \in \mathcal{F}_n$. They agree almost surely: if their difference were positive on a set of positive probability, one of its positive-level sets would have a strictly positive integral, contrary to the test identity. Apply the same argument to the negative difference. Strict positivity of Z_n now permits division, giving

$$\mathsf{E}_Q[Y \mid \mathcal{F}_n] = \frac{\mathsf{E}[Z_N Y \mid \mathcal{F}_n]}{Z_n}.$$

The denominator is the current conditional density, not the unconditional mean one. Equivalence of the probabilities makes the equality valid with either null-set convention.

(j) Since $|X_n| \le n$, the walk is bounded on the fixed horizon. Apply the preceding formula with $Y = X_{n+1}$ and first condition on $\mathcal{F}_{n+1}$:

$$\begin{aligned}\mathsf{E}[Z_N X_{n+1} \mid \mathcal{F}_n] &= \mathsf{E}[Z_{n+1} X_{n+1} \mid \mathcal{F}_n] \\ &= Z_n \mathsf{E}[f(\varepsilon_{n+1})(X_n + \varepsilon_{n+1}) \mid \mathcal{F}_n] \\ &= Z_n\{X_n \mathsf{E} f(\varepsilon_1) + \mathsf{E}[\varepsilon_1 f(\varepsilon_1)]\} = Z_n X_n.\end{aligned}$$

Here the first multiplier expectation is one, while the second is $(2/3)(3/4) - (1/3)(3/2) = 0$. Dividing by Z_n proves the Q-martingale identity. The same process can have positive drift under P and zero drift under Q because the probabilities of its future increments have changed.

(k) For $E = \{\varepsilon_1 = 1\} \in \mathcal{F}_1$, the tower property gives $Q(E) = \mathsf{E}[Z_1 \mathbf{1}_E] = (3/4)(2/3) = 1/2$. Similarly $Q(\varepsilon_1 = -1) = (3/2)(1/3) = 1/2$. As a second check, the Q-martingale property at zero implies $\mathsf{E}_Q \varepsilon_1 = 0$. A variable supported on $\{-1, 1\}$ has mean zero exactly when its two probabilities are equal, hence both must be one half.

16 September 2025

One question; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a probability space. Let $(\varepsilon_n)_{n\in\mathbb{N}}$ be an i.i.d. sequence with

$$\mathsf{P}(\varepsilon_1 = 1) = \mathsf{P}(\varepsilon_1 = -1) = \frac{1}{2}.$$

Let $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$ be a filtration with $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma(\varepsilon_j, j \le n),\ n \in \mathbb{N}$. Let us introduce a

process $(X_n)_{n\in\mathbb{Z}_+}$ with $X_0 = 1$ and

$$X_n = \exp\left(\sum_{j=1}^{n} \varepsilon_j - A_n\right),\ n \in \mathbb{N},$$

where $(A_n)_{n\in\mathbb{N}}$ is a $\mathcal{F}_n$-previsible process and $A_0 = 0$. Moreover, denote $(M_n)_{n\in\mathbb{Z}_+}$ the maximum process $M_n = \max\{X_0, X_1, \ldots, X_n\}$.

(a) Show that the process $(X_n)_{n\in\mathbb{Z}_+}$ is a $(\mathcal{F}_n, \mathsf{P})$-martingale if and only if $A_n = \alpha n,\ n \in \mathbb{N}$, where

$$\alpha = \ln\left(\frac{1}{2}e + \frac{1}{2}e^{-1}\right) > 0.$$

(b) Show that the process $(M_n)_{n\in\mathbb{Z}_+}$ is $\mathcal{F}_n$-adapted.

(c) For any $k > 0$, let $\tau_k := \inf\{n \in \mathbb{N} : X_n > k\}$. Show the equivalence of sets

$$\{\tau_k > n\} = \{M_n < k\}$$

and deduce that τ_k is a $\mathcal{F}_n$-stopping time.

In a financial market with zero risk-free rate (i.e., $r = 0$), the process $(X_n)_{n\in\mathbb{Z}_+}$ can be used to model a stock-price. A "look-back knock-in" Call option is an exotic financial derivative with the following features: let a strike price $K > 0$, a knock-in date N_1, a barrier k and a maturity $N_2 > N_1$ be given; at maturity the option pays $(X_{N_2} - K)^+$ if $\tau_k \le N_1$ (i.e., if the stock price exceeds the barrier k before or at time N_1).
The mathematical expression for the option's payoff reads $\mathbf{1}_{\{\tau_k \le N_1\}}(X_{N_2} - K)^+$.

(d) The option's payoff can be written as $\varphi(X_{N_2}, M_{N_1})$ for some measurable function

$$\varphi : [0,\infty)^2 \to [0,\infty).$$

Identify the function φ.
Hint: use (c).

Q.1 (Continued)
Thanks to (d), the price at time zero of the look-back knock-in Call option reads

$$V_0 = \mathsf{E}[\varphi(X_{N_2}, M_{N_1})] = \mathsf{E}[\varphi(X_{N_1} \cdot Z, M_{N_1})],$$

where in the final expression we have set $Z := \frac{X_{N_2}}{X_{N_1}}$.

(e) Write explicitly the expression for Z in terms of the sequence $(\varepsilon_n)_{n\in\mathbb{N}}$. Then show that Z is independent of $\mathcal{F}_{N_1}$.

(f) From the time-zero pricing formula and considerations in (e), combined with tower property, one can deduce that V_0 can be equivalently expressed as $V_0 = \mathsf{E}[h(X_{N_1}, M_{N_1})]$, for some measurable function $h : [0,\infty)^2 \to [0,\infty)$. Explain how to obtain such result.

(g) Show that the function $h(x,m)$ from (f) can be written as $h(x,m) = \mathbf{1}_{\{m\ge k\}}\mathsf{E}[(xZ - K)^+]$.

We now have all the ingredients to calculate V_0.
Assume: $N_1 = 1$, $N_2 = 2$, $k = 1.5$, $K = 0$ **and recall** $X_0 = 1$.

(h) Show that

$$h(x,m) = \mathbf{1}_{\{m\ge 1.5\}}x \text{ for } 0 \le x \le m.$$

Hint: calculations are quicker if you recall (a).

(i) Combine (f) and (h) to obtain $V_0 = \frac{e}{e^1+e^{-1}}$. *Hint: notice that*

$$\frac{2e^{-1}}{e+e^{-1}} < 1.5 < \frac{2e}{e+e^{-1}}.$$

AI-Generated Worked Solution

An exponential martingale and a look-back knock-in payoff

(a) A previsible A_n is $\mathcal{F}_{n-1}$-measurable, so the next compensator increment is known before the next sign. Factor

$$X_n = X_{n-1}\exp(\varepsilon_n - (A_n - A_{n-1})).$$

Independence of the fair sign gives $\mathsf{E}e^{\varepsilon_n} = (e + e^{-1})/2 =: c$. For nonnegative variables the conditional computation is valid even before checking finiteness, and hence

$$\mathsf{E}[X_n \mid \mathcal{F}_{n-1}] = X_{n-1}e^{-(A_n - A_{n-1})}c.$$

If X is a martingale, this must equal X_{n-1}; since the latter is strictly positive, $e^{A_n - A_{n-1}} = c$ almost surely. Thus every increment equals $\alpha = \log c$, and $A_0 = 0$ gives $A_n = n\alpha$. Conversely that deterministic choice makes $X_n \le e^{n(1-\alpha)}$, ensuring integrability, and the displayed computation proves the martingale identity. Since $c = \cosh 1 > 1$, $\alpha > 0$.

(b) X_j is $\mathcal{F}_j$-measurable and $\mathcal{F}_j \subset \mathcal{F}_n$ for $j \le n$. For every real a,

$$\{M_n \le a\} = \bigcap_{j=0}^{n} \{X_j \le a\} \in \mathcal{F}_n.$$

A maximum is at most a exactly when each of its entries is. The lower-ray criterion therefore proves measurability of M_n at time n, which is adaptedness.

(c) The set identity as printed is false: a first passage defined by $X_j > k$ has survival condition $X_j \le k$, not $X_j < k$. Furthermore the infimum in the question begins at $j = 1$, whereas M_n includes X_0. Precisely, for $n \ge 1$,

$$\{\tau_k > n\} = \bigcap_{j=1}^{n} \{X_j \le k\},\ \{\tau_k \le n\} = \bigcup_{j=1}^{n} \{X_j > k\} \in \mathcal{F}_n.$$

The last equality proves the stopping-time property without either faulty convention. If $k \ge X_0 = 1$, the first event also equals $\{M_n \le k\}$. For a counterexample to the strict printed inequality take $k = X_1$ on a reachable node with $X_1 > 1$: no crossing occurs at that node but $M_1 = k$, not $M_1 < k$. For $k < 1$, retaining the initial point in M_n creates an additional mismatch with a passage that ignores time zero.

(d) Under the natural barrier convention $k \ge 1$, the corrected payoff function is

$$\varphi(x, m) = \mathbf{1}_{\{m > k\}}(x - K)^+.$$

It is Borel because $\{m > k\}$ is Borel and the positive-part map is continuous. The barrier tests strict crossing, consistent with τ_k. For arbitrary $k < 1$, M_{N_1} alone can conceal whether the maximum one is only the initial value; to represent the stated contract generally, use the running maximum over dates $1, \dots, N_1$ instead. The exercise's later numerical case has $k = 1.5$, so this initial-time issue does not affect it.

(e) Under the compensator found in (a), the ratio is

$$Z = \frac{X_{N_2}}{X_{N_1}} = \exp\left(\sum_{j=N_1+1}^{N_2} \varepsilon_j - \alpha(N_2 - N_1)\right).$$

It is a measurable function only of signs strictly after N_1. Independence of the sign sequence implies this future block, and therefore Z, is independent of $\mathcal{F}_{N_1}$. Cancelling the factors up to N_1 is what removes dependence on the observed path from the ratio.

(f) The pair (X_{N_1}, M_{N_1}) is known at time N_1, while Z is independent future randomness. Conditional expectation freezes this pair:

$$h(x, m) = \mathsf{E}[\varphi(xZ, m)],\ \mathsf{E}[\varphi(X_{N_1}Z, M_{N_1}) \mid \mathcal{F}_{N_1}] = h(X_{N_1}, M_{N_1}).$$

The function is Borel: the future block has finite support, so the expectation is a finite sum of Borel functions of (x, m). Taking expectations and applying the tower property gives

$V_0 = \mathsf{E}[h(X_{N_1}, M_{N_1})]$. This separates the barrier observation period from the remaining pricing period.

(g) With the strict passage convention, the exact formula is

$$h(x, m) = \mathbf{1}_{\{m>k\}}\mathsf{E}[(xZ - K)^+],$$

since the barrier indicator depends on the frozen m, not on Z. The printed weak condition $m \geq k$ corresponds to a different contract that activates at equality. There is no general reason for equality at a reachable barrier to be null on a binomial tree. The two formulas agree only if that equality is impossible or irrelevant to the payoff; this will hold for the specified barrier 1.5 at the first observation date.

(h) Here $K = 0$ and $x \geq 0$, $Z > 0$, so $(xZ - 0)^+ = xZ$. A single future multiplier has mean $ce^{-\alpha} = 1$, hence $\mathsf{E}Z = 1$ and

$$h(x, m) = \mathbf{1}_{\{m>1.5\}}x$$

for the stated strict contract. The exercise's weak-barrier version is $\mathbf{1}_{\{m\geq 1.5\}}x$; these differ at arbitrary input $m = 1.5$, so its claimed global identity cannot be literally recovered for the strict contract. On the actual reachable states at date one neither maximum equals 1.5, so either expression yields the same price. This distinguishes a correct numerical answer from a false all-state identity.

(i) The first-step values are $x_u = e^{1-\alpha} = 2e/(e + e^{-1})$ and $x_d = e^{-1-\alpha} = 2e^{-1}/(e + e^{-1})$. The first exceeds 1.5 and the second is below one, so the running maxima are x_u on the up branch and 1 on the down branch. Only the up branch activates. With probability 1/2 on that branch and conditional continuation mean equal to the current stock,

$$V_0 = \tfrac{1}{2}x_u = \frac{e}{e + e^{-1}}.$$

There is no discount factor because $r = 0$. The down branch contributes zero, and the equality convention at the barrier has no numerical effect here because neither reachable maximum is 1.5.

23 January 2026

Two questions; show all workings.

Q.1 Let $(\Omega, \mathcal{F}, \mathsf{P})$ be a complete probability space. A sequence $(X_n)_{n\in\mathbb{N}} \subset m\mathcal{F}$ converges in probability to $X \in m\mathcal{F}$ if, for any $\varepsilon > 0$, the limit holds $\lim_{n\to\infty} \mathsf{P}(|X_n - X| > \varepsilon) = 0$. It is a well-known fact that, if the limit above holds, then we can find a subsequence $(X_{n_k})_{k\in\mathbb{N}}$ of the original sequence such that $\lim_{k\to\infty} X_{n_k}(\omega) = X(\omega)$, P-a.s. **Your task in this question is to assume that convergence in probability holds and follow the steps detailed below to prove that almost sure subsequence convergence indeed holds.**

(a) Deduce from convergence in probability that, for any $k \in \mathbb{N}$, there is X_{n_k} such that $\mathsf{P}\big(|X_{n_k} - X| > \frac{1}{2^k}\big) < \frac{1}{2^k}$. This step defines a subsequence $(X_{n_k})_{k\in\mathbb{N}}$.

(b) Set $A_k = \left\{\omega \in \Omega : |X_{n_k}(\omega) - X(\omega)| > \frac{1}{2^k}\right\}$. Use (a) to prove that $\sum_{k\in\mathbb{N}} \mathsf{P}(A_k) < \infty$.

(c) Use (b) to prove that $\mathsf{P}(\limsup_{k\to\infty} A_k) = 0$.

(d) Use (c) to prove that $\mathsf{P}(\liminf_{k\to\infty} A_k^c) = 1$.

(e) Write explicitly the event A_k^c and use the definition of $\liminf_{k\to\infty} A_k^c$ to show that, for $\omega \in \liminf_{k\to\infty} A_k^c$, it holds $\lim_{k\to\infty} |X_{n_k}(\omega) - X(\omega)| = 0$. *Hint: show that* $\exists k_\omega \in \mathbb{N}$ *such that* $|X_{n_j}(\omega) - X(\omega)| < 1/2^j$ *for all* $j \geq k_\omega$.

(f) Combine (d) and (e) to deduce almost sure subsequence convergence.

Q.2 Consider a financial market described by a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, equipped with an i.i.d. sequence $(\varepsilon_j)_{j\in\mathbb{N}}$ such that $\varepsilon_j \in \{1/2, 3/2\}$ with $\mathsf{P}(\varepsilon_j = 1/2) = p$. Let $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$ be the filtration generated by the sequence, i.e.,

$$\mathcal{F}_0 = \{\emptyset, \Omega\} \text{ and } \mathcal{F}_n = \sigma(\varepsilon_j, j \leq n).$$

The risk-free rate is set to zero (i.e., $r = 0$) and the stock price follows the dynamics

$$S_0 = 1,\ S_n = S_{n-1}\varepsilon_n,\ n \in \mathbb{N}.$$

(a) Show that $(S_n)_{n\in\mathbb{Z}_+}$ is adapted to the filtration $(\mathcal{F}_n)_{n\in\mathbb{Z}_+}$, that $S_n \geq 0$, and that

$$\mathsf{E}[|S_n|] < \infty \text{ for all } n \in \mathbb{Z}_+.$$

(b) Show that $(S_n)_{n\in\mathbb{Z}_+}$ is a martingale if and only if $p = 1/2$.

From now on we assume $p = 1/2$. The payoff of an Asian Put option with maturity at time $N = 2$ and strike equal to 1 is given by $AP(\omega) := \big(1 - \frac{1}{2}\big(S_1(\omega) + S_2(\omega)\big)\big)^+$, where $(a)^+ := \max\{0, a\}$ for any $a \in \mathbb{R}$.

(c) Show that $AP \in m\mathcal{F}_2$.

(d) Show that $\mathsf{E}[AP \mid \mathcal{F}_1] = f(S_1)$, with $f(s) = \frac{1}{2}\left(1 - \frac{3}{4}s\right)^+ + \frac{1}{2}\left(1 - \frac{5}{4}s\right)^+$. *Hint: use freezing lemma.*

(e) Deduce from (d) that the option price reads $p_0 = \mathsf{E}[f(S_1)]$ and calculate p_0.
Hint: use tower property.

AI-Generated Worked Solution

Q.1: extracting an almost-surely convergent subsequence

(a) Fix a positive integer k. Convergence in probability at tolerance 2^{-k} supplies an index N_k such that $P(|X_n - X| > 2^{-k}) < 2^{-k}$ for every $n \geq N_k$. Choose recursively $n_k \geq \max(N_k, n_{k-1} + 1)$, starting with $n_0 = 0$. The strict increase is essential to obtain a subsequence rather than an arbitrary list of indices. This deterministic choice gives the required bound at every k.

(b) By definition $P(A_k) < 2^{-k}$. Summing and using nonnegativity,

$$\sum_{k\geq 1} P(A_k) \leq \sum_{k\geq 1} 2^{-k} = 1 < \infty.$$

Finiteness is all that is needed; there is no need to assert a strict bound on the infinite sum from the termwise inequalities.

(c) *We prove that $P(\limsup_k A_k) = 0$, using the first Borel–Cantelli argument.* Put $B_m = \cup_{k\geq m} A_k$. Subadditivity yields $P(B_m) \leq \sum_{k\geq m} P(A_k) \to 0$. The sets B_m decrease to $\limsup A_k$, so continuity from above on the probability space gives zero for that intersection. Independence is not required for this direction.

(d) De Morgan's laws identify the complement event:

$$(\limsup_k A_k)^c = \left(\bigcap_m \bigcup_{k\geq m} A_k\right)^c = \bigcup_m \bigcap_{k\geq m} A_k^c = \liminf_k A_k^c.$$

Its probability is one by the preceding part. This event means that, beyond some finite path-dependent index, every selected error satisfies its bound.

(e) The complement is $A_k^c = \{|X_{n_k} - X| \leq 2^{-k}\}$. If $\omega \in \cup_m \cap_{k\geq m} A_k^c$, there is $m(\omega)$ such that the bound holds for every $k \geq m(\omega)$. Given any $\eta > 0$, choose $K \geq m(\omega)$ with $2^{-K} < \eta$. Then for all $k \geq K$,

$$|X_{n_k}(\omega) - X(\omega)| \leq 2^{-k} \leq 2^{-K} < \eta.$$

This is the definition of pointwise convergence of the selected errors to zero. The hint's strict $< 2^{-k}$ is unnecessary and does not follow from taking the complement of $> 2^{-k}$; the correct weak bound suffices.

(f) Part (d) supplies one event of probability one and part (e) proves convergence at every point of that event. Therefore $X_{n_k} \to X$ almost surely. The construction does not claim convergence of the whole original sequence. Completeness of the probability space is allowed by the question but is not needed for these measurable events and countable operations.

Q.2: an Asian put on a multiplicative binomial stock

(a) Iterating gives $S_n = \prod_{j=1}^{n} \varepsilon_j$. Each factor is measurable at time n, so the product is $\mathcal{F}_n$-measurable. It is strictly positive and bounded by $(3/2)^n$. Consequently $\mathsf{E}|S_n| \le (3/2)^n < \infty$. These are the adaptedness and integrability requirements that must be checked before asserting a martingale property.

(b) Independence of the next multiplier from the past gives

$$\mathsf{E}[S_{n+1} \mid \mathcal{F}_n] = S_n \mathsf{E}\varepsilon_{n+1} = S_n \left(\tfrac{1}{2}p + \tfrac{3}{2}(1-p)\right) = S_n(\tfrac{3}{2} - p).$$

Since $S_n > 0$, equality to S_n holds exactly when $3/2 - p = 1$, or $p = 1/2$. Thus the condition is necessary as well as sufficient. With zero risk-free rate no further discounting changes this martingale equation.

(c) Both S_1 and S_2 are $\mathcal{F}_2$-measurable. Their sum, scalar multiples, and composition with the continuous positive-part map remain measurable. Hence $AP \in m\mathcal{F}_2$. It also satisfies $0 \le AP \le 1$, so every conditional expectation and tower operation used below is integrable.

(d) Substitute $S_2 = S_1\varepsilon_2$ into the average:

$$AP = \left(1 - \tfrac{1}{2}S_1(1+\varepsilon_2)\right)^+.$$

When conditioning on $\mathcal{F}_1$, S_1 is fixed and the independent future multiplier equals $1/2$ or $3/2$, each with probability $1/2$. These give average coefficients $(1 + 1/2)/2 = 3/4$ and $(1 + 3/2)/2 = 5/4$. Therefore

$$\mathsf{E}[AP \mid \mathcal{F}_1] = \tfrac{1}{2}(1 - \tfrac{3}{4}S_1)^+ + \tfrac{1}{2}(1 - \tfrac{5}{4}S_1)^+ = f(S_1).$$

The freezing step can also be checked against any $\mathcal{F}_1$ event by independence. The function f is continuous, hence Borel, since it is a sum of positive parts of affine functions.

(e) By the tower property the undiscounted price is $p_0 = \mathsf{E}AP = \mathsf{E}f(S_1)$. At the down value $S_1 = 1/2$,

$$f(1/2) = \tfrac{1}{2}(1 - 3/8) + \tfrac{1}{2}(1 - 5/8) = \tfrac{5}{16} + \tfrac{3}{16} = \tfrac{1}{2}.$$

At the up value $S_1 = 3/2$, both inner affine expressions are negative, so $f(3/2) = 0$. Therefore

$$p_0 = \tfrac{1}{2}f(1/2) + \tfrac{1}{2}f(3/2) = \tfrac{1}{2}\tfrac{1}{2} + \tfrac{1}{2}(0) = \tfrac{1}{4}.$$

As a path check, the stock pairs and payoffs are

(S_1, S_2)	$(1/2, 1/4)$	$(1/2, 3/4)$	$(3/2, 3/4)$	$(3/2, 9/4)$
AP	$5/8$	$3/8$	0	0

Each path has probability $1/4$. Their weighted payoff sum is $(5/8 + 3/8)/4 = 1/4$, agreeing with the conditional calculation.

AI-Generated Mock Exams

Mock Exam 1: generating classes, random variables and an Asian claim

Questions

Q.1 Work on $\Omega = (0,1)$ with Borel sigma-algebra and Lebesgue probability. Put $U(\omega) = \omega$ and $X = -\log U$.

(a) Prove that X is measurable using preimages of rays, and identify its distribution.
(b) Compute $\mathsf{E}X$ and $\mathsf{E}X^2$ by integration, justifying the improper limits.
(c) Let $\mathcal{H} = \sigma(\{U \le 1/2\})$. Find $\mathsf{E}[X \mid \mathcal{H}]$ directly from the defining integral property.
(d) Show that $X_n = X \wedge n$ converges in L^1 and compute the exact L^1 error.
(e) Explain why verifying equality of two probability measures on all rays $(-\infty, x]$ determines equality on all Borel sets.

Q.2 A two-period binomial market has $S_0 = 4$, $S_{n+1}/S_n \in \{2, 1/2\}$ and bank account $B_n = 1$. The two multipliers are independent under the pricing measure. Let $H = ((S_1 + S_2)/2 - 5)^+$.

(a) Find the risk-neutral up probability and verify the discounted stock martingale condition.
(b) List every path, its probability and its payoff. Does the payoff depend on S_2 alone?
(c) Compute the price at time zero by expectation and by backward induction.
(d) Find the stock and cash holdings at both time-one nodes and at time zero.
(e) Verify the terminal wealth on all four paths and explain uniqueness of the price.

AI-Generated Worked Solution

Q.1

(a) A convenient generating class for the target Borel sigma-algebra consists of rays. For $x < 0$, $\{X \le x\} = \varnothing$, since $-\log U > 0$. For $x \ge 0$, exponentiation preserves the equivalent inequality $U \ge e^{-x}$, so $\{X \le x\} = [e^{-x}, 1) \cap \Omega$ is Borel. Thus X is measurable and

$$P(X \le x) = \begin{cases} 0 & x < 0, \\ 1 - e^{-x} & x \ge 0. \end{cases}$$

Differentiating on $(0, \infty)$ gives density e^{-x}: X is exponential of rate one. Endpoints have zero mass, so weak/strict endpoint choices do not alter its law.

(b) Nonnegative integration permits increasing truncations. Integration by parts on $[0, M]$ yields

$$\int_0^M x e^{-x} dx = 1 - (M+1)e^{-M} \longrightarrow 1.$$

A second integration by parts gives

$$\int_0^M x^2 e^{-x} dx = -M^2 e^{-M} + 2\int_0^M x e^{-x} dx \longrightarrow 2.$$

The boundary terms vanish because exponential decay dominates these powers. Monotone convergence justifies passing to the full expectations. Consequently $X \in L^2$, $\mathsf{E}X = 1$ and $\operatorname{Var} X = 1$.

(c) Write $A = \{U \le 1/2\}$. Any $\mathcal{H}$-measurable variable is constant on A and A^c, so seek $Z = a\mathbf{1}_A + b\mathbf{1}_{A^c}$. The defining property for A requires

$$aP(A) = \mathsf{E}[X\mathbf{1}_A] = \int_0^{1/2} -\log u \, du = \tfrac{1}{2}(1 + \log 2).$$

Since $P(A) = 1/2$, $a = 1 + \log 2$. The full expectation gives $b/2 = 1 - a/2$, hence $b = 1 - \log 2$. These equalities also verify the integral property for A^c, and therefore for every member of $\mathcal{H} = \{\varnothing, A, A^c, \Omega\}$. This proves the conditional expectation, rather than merely guessing a plausible piecewise constant variable.

(d) Pointwise $X_n \uparrow X$. Also $X - X_n = (X-n)^+ \geq 0$, so

$$\mathsf{E}|X - X_n| = \int_n^\infty (x-n)e^{-x}dx = e^{-n}\int_0^\infty ve^{-v}dv = e^{-n}.$$

It tends to zero; hence convergence is in L^1. Monotone convergence alone gives convergence of expectations here, but the exact error also makes the rate explicit.

(e) Rays form a pi-system: the intersection of the rays with endpoints x, y is the ray with endpoint $\min(x, y)$. They generate the Borel sigma-algebra, since complements and countable unions produce open intervals. The two probability measures have the same finite total mass and agree on this generating pi-system. Lemma 1.2 therefore makes them equal on every Borel set; its finite-mass hypothesis is satisfied.

Q.2

(a) If p is the up probability, the martingale equation is $2p+(1/2)(1-p) = 1$, giving $(3/2)p = 1/2$ and $p = 1/3$. Both branches have positive probability, and independence implies $\mathsf{E}[S_{n+1} \mid \mathcal{F}_n] = S_n$ at each node. The stock is adapted and integrable because the tree is finite.

(b) The path table is

Path	S_1	S_2	Probability	H
uu	8	16	1/9	7
ud	8	4	2/9	1
du	2	4	2/9	0
dd	2	1	4/9	0

For example at ud, $(8+4)/2-5 = 1$. The two paths ending at $S_2 = 4$ have different payoffs, so treating this as a function of terminal stock alone would lose essential path information.

(c) No discounting is needed: $V_0 = 7/9+2/9 = 1$. At the up node, $V_1^u = (1/3)7+(2/3)1 = 3$; at the down node, $V_1^d = 0$. Backward induction then gives $V_0 = (1/3)3 + (2/3)0 = 1$, agreeing with the direct path sum.

(d) On the up node solve $16\Delta + C = 7$, $4\Delta + C = 1$. Subtraction gives $12\Delta = 6$, so $\Delta_1^u = 1/2$, $C_1^u = -1$ and cost $(1/2)8 - 1 = 3$. At the down node both payoffs are zero, so take $\Delta_1^d = C_1^d = 0$. At time zero solve $8\Delta_0 + C_0 = 3$, $2\Delta_0 + C_0 = 0$, giving $\Delta_0 = 1/2$, $C_0 = -1$ and initial cost $4/2-1 = 1$. After the first move the old portfolio is worth exactly the cost of the new node portfolio, so rebalancing needs no cash injection: the strategy is self-financing.

(e) At uu the final wealth is $16/2 - 1 = 7$, at ud it is $4/2 - 1 = 1$, and at both down-first paths it is zero. These equal the respective payoffs. Each two-by-two replication system has nonzero stock-price difference, so it has a unique solution at every node. A different claim price from the replication cost would permit buying the cheaper of claim/replica and selling the more expensive, yielding a positive initial surplus and zero terminal liability. Thus the arbitrage-free price is unique.

Mock Exam 2: conditional expectation and a change of probability

Questions

Q.1 On $\Omega = \{1, 2, 3, 4\}$ with equiprobable points, let $X = (0, 2, 4, 6)$ and $\mathcal{G} = \sigma(\{1, 2\}, \{3, 4\})$.

(a) Find $Z = \mathsf{E}[X \mid \mathcal{G}]$ and verify its defining properties.

(b) Prove the tower property for conditioning further on $\{\varnothing, \Omega\}$ in this example.

(c) Compute $\mathsf{E}[X^2 \mid \mathcal{G}]$ and the conditional variance; verify conditional Jensen.

(d) Show directly that Z minimizes $\mathsf{E}[(X - Y)^2]$ over all $\mathcal{G}$-measurable Y.

(e) Is $Z^2 = \mathsf{E}[X^2 \mid \mathcal{G}]$? State exactly when equality would hold in the squared conditional Jensen inequality.

Q.2 Under P, independent signs ξ_1, ξ_2 have probabilities $1/2, 1/2$. Let $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ and $M_n = \prod_{j=1}^n (1 + a\xi_j)$, $M_0 = 1$, where $0 < a < 1$. Define $Q(A) = \mathsf{E}_P[M_2 \mathbf{1}_A]$.

(a) Prove that M is a positive P-martingale.
(b) Verify that Q is a probability equivalent to P and find its four path probabilities.
(c) Show that the signs are independent under Q and find their means.
(d) Derive the formula $\mathsf{E}_Q[Y \mid \mathcal{F}_1] = \mathsf{E}_P[M_2 Y \mid \mathcal{F}_1]/M_1$ for any Y on this tree by testing integrals.
(e) Find a constant b such that $N_n = \sum_{j=1}^{n}(\xi_j - b)$ is a Q-martingale; calculate $\mathsf{E}_Q[N_2^2]$.

AI-Generated Worked Solution

Q.1

(a) A $\mathcal{G}$-measurable variable is constant on each of its two atoms. The average of $0, 2$ is 1, and that of $4, 6$ is 5, so $Z = (1, 1, 5, 5)$. It is measurable and integrable. On the first atom the integral of Z is $2/4 = 1/2$, equalling $(0 + 2)/4$; on the second it is $10/4$, equalling $(4 + 6)/4$. The equalities extend by addition to their union and are trivial for the empty set, verifying every test set in $\mathcal{G}$.

(b) Conditioning on the trivial sigma-algebra gives a constant equal to the expectation. Here $\mathsf{E}Z = (1 + 1 + 5 + 5)/4 = 3$ and $\mathsf{E}X = (0 + 2 + 4 + 6)/4 = 3$. Thus $\mathsf{E}[\mathsf{E}(X \mid \mathcal{G})] = \mathsf{E}X$. This is the defining integral equality tested on the whole space, not a requirement that Z be constant before further conditioning.

(c) Averaging squares on the atoms gives $\mathsf{E}[X^2 \mid \mathcal{G}] = (2, 2, 26, 26)$. Subtracting $Z^2 = (1, 1, 25, 25)$ gives conditional variance equal to 1 everywhere. In particular $Z^2 \le \mathsf{E}[X^2 \mid \mathcal{G}]$, with strict inequality on both atoms. Its expectation yields $\mathsf{E}Z^2 = 13 \le 14 = \mathsf{E}X^2$.

(d) Write $Y = (c, c, d, d)$. Expanding around the atom means gives

$$\mathsf{E}[(X - Y)^2] = \tfrac{1}{4}[c^2 + (2 - c)^2 + (4 - d)^2 + (6 - d)^2] = 1 + \tfrac{1}{2}(c - 1)^2 + \tfrac{1}{2}(d - 5)^2.$$

The minimum is 1, uniquely at $c = 1, d = 5$.

(e) They are unequal because the conditional variance is positive.
($\Rightarrow$) In general, if equality holds in the squared conditional Jensen inequality for $X \in L^2$, the identity $\mathsf{E}[(X - Z)^2 \mid \mathcal{G}] = \mathsf{E}[X^2 \mid \mathcal{G}] - Z^2$ makes the conditional squared residual zero almost surely. Taking expectations makes a nonnegative square have expectation zero, so $X = Z$ almost surely.
($\Leftarrow$) Conversely, if X is already $\mathcal{G}$-measurable, its conditional expectation equals itself and equality holds. Strict convexity must be paired with this measurability condition, not merely with constancy of the unconditional mean.

Q.2

(a) Each factor lies in $(0, 2)$, so $M_n > 0$, and the finite products are integrable and adapted. Independence and $\mathsf{E}_P\xi_{n+1} = 0$ give

$$\mathsf{E}_P[M_{n+1} \mid \mathcal{F}_n] = M_n \mathsf{E}_P[1 + a\xi_{n+1} \mid \mathcal{F}_n] = M_n.$$

Thus $\mathsf{E}_P M_n = M_0 = 1$. All three martingale requirements have been checked: measurability, integrability and the conditional identity.

(b) The weights M_2 are strictly positive, so Q is nonnegative and countably additive as a weighted integral; $Q(\Omega) = \mathsf{E}_P M_2 = 1$. Every singleton has positive Q mass, so P, Q have the same null sets. The masses of $++, +-, -+, --$ are respectively $(1 + a)^2/4, (1 - a^2)/4, (1 - a^2)/4, (1 - a)^2/4$.

(c) These factor as the product of one-step masses $Q(\xi = 1) = (1 + a)/2$ and $Q(\xi = -1) = (1 - a)/2$. Hence the two signs remain independent under Q, but cease to be centered: $\mathsf{E}_Q\xi = (1 + a)/2 - (1 - a)/2 = a$. Changing the probability law can therefore destroy a martingale built from centered increments under the original law.

(d) Let $W = \mathsf{E}_P[M_2 Y \mid \mathcal{F}_1]/M_1$. Its denominator is positive and $\mathcal{F}_1$-measurable. For $A \in \mathcal{F}_1$,

$$\begin{aligned}\mathsf{E}_Q[W\mathbf{1}_A] &= \mathsf{E}_P[M_2 W \mathbf{1}_A] = \mathsf{E}_P[M_1 W \mathbf{1}_A] \\ &= \mathsf{E}_P[\mathsf{E}_P(M_2 Y \mid \mathcal{F}_1)\mathbf{1}_A] = \mathsf{E}_P[M_2 Y \mathbf{1}_A] = \mathsf{E}_Q[Y\mathbf{1}_A].\end{aligned}$$

The second equality uses $\mathsf{E}_P[M_2 \mid \mathcal{F}_1] = M_1$ and the ability to take out a measurable factor. This proves the formula through the defining property; it also explains why the denominator is the density at time one, not the unconditional value one.

(e) Conditional independence under Q gives $\mathsf{E}_Q[\xi_{n+1} - b \mid \mathcal{F}_n] = a - b$, so choose $b = a$. Then N is adapted, integrable, and has centered conditional increments. Since the increments are independent with variance $\mathsf{E}_Q\xi^2 - a^2 = 1 - a^2$, the cross term is zero and $\mathsf{E}_Q N_2^2 = 2(1 - a^2)$. This also follows by expanding $(\xi_1 - a + \xi_2 - a)^2$ and using the factorization of the mixed expectation.

Mock Exam 3: rare events, subsequences and a stopped walk

Questions

Q.1 Let (A_n) be independent events with $P(A_n) = 1/(n+1)$, and set $Y_n = \mathbf{1}_{A_n}$.

(a) Prove that $Y_n \to 0$ in probability.

(b) Show that Y_n does not converge to zero almost surely. State precisely where independence is used.

(c) Construct a deterministic subsequence converging almost surely to zero and prove the claim.

(d) Find $\|Y_n\|_p$ for every finite $p \geq 1$. Is this in conflict with the almost-sure behavior?

(e) For an arbitrary sequence $Z_n \to Z$ in probability, construct an almost-surely convergent subsequence using quantitative probability bounds.

Q.2 Let ξ_n be iid fair signs, $S_0 = 0$, $S_n = \sum_{j=1}^n \xi_j$ and $\tau = \inf\{n \geq 0 : |S_n| = 2\}$. Let $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$.

(a) Show that τ is a stopping time and that S and $S_n^2 - n$ are martingales.

(b) Without invoking an unbounded stopping theorem, show that $S_{n\wedge\tau}$ is a bounded martingale.

(c) Prove $P(\tau > 2k) = 2^{-k}$ and deduce that $\tau < \infty$ almost surely.

(d) Find $P(S_\tau = 2)$ and $\mathsf{E}\tau$, justifying all limits.

(e) Explain why the same method cannot simply be applied to a first hitting time of $+2$ with no lower barrier to conclude finite expected hitting time.

AI-Generated Worked Solution

Q.1

(a) For $0 < \epsilon < 1$, $P(|Y_n| > \epsilon) = P(A_n) = 1/(n+1) \to 0$; for $\epsilon \geq 1$ it is zero. The two tolerance ranges exhaust $\epsilon > 0$, proving convergence in probability.

(b) The sum $\sum_n P(A_n)$ diverges. By the second Borel–Cantelli lemma, independence implies $P(A_n \text{ infinitely often}) = 1$. On this event $Y_n = 1$ infinitely often and hence cannot converge to zero. The first lemma alone would not yield this conclusion: its implication requires a convergent sum. Independence is essential to the divergent-sum implication used here.

(c) Take $n_k = 2^k$. Then $\sum_k P(A_{n_k}) \leq \sum_k 2^{-k} < \infty$. The first Borel–Cantelli lemma, which needs no independence, says that only finitely many selected events occur almost surely. Thus eventually $Y_{2^k} = 0$, proving almost-sure convergence along that subsequence.

(d) Since $Y_n^p = Y_n$, $\|Y_n\|_p = (\mathsf{E}|Y_n|^p)^{1/p} = (n+1)^{-1/p} \to 0$. This is convergence in every finite L^p and hence in probability. Neither implies almost-sure convergence of the entire sequence. Small individual event probabilities and infinitely many occurrences are compatible when their sum diverges.

(e) For each k, convergence in probability permits choosing $n_k > n_{k-1}$ so that $P(|Z_{n_k} - Z| > 2^{-k}) < 2^{-k}$. The sum of these probabilities is finite. Almost surely, therefore, there is $k_0(\omega)$ such that $|Z_{n_k}(\omega) - Z(\omega)| \leq 2^{-k}$ for all $k \geq k_0$. The upper bound tends to zero, giving pointwise convergence outside one null set. The indices are chosen deterministically from the probability bounds; they are not random path-dependent choices.

Q.2

(a) $\{\tau \le n\} = \bigcup_{j=0}^{n} \{|S_j| = 2\} \in \mathcal{F}_n$. The sums are adapted and integrable, and $\mathsf{E}[S_{n+1} \mid \mathcal{F}_n] = S_n$ by centering and independence of ξ_{n+1}. Also

$$S_{n+1}^2 - (n+1) = S_n^2 - n + 2S_n\xi_{n+1} + \xi_{n+1}^2 - 1.$$

Conditional expectation removes the mixed term and the last two terms cancel since $\xi^2 = 1$. Thus the second process is also a martingale.

(b) Directly,

$$S_{(n+1)\wedge\tau} - S_{n\wedge\tau} = \mathbf{1}_{\{\tau>n\}}\xi_{n+1}.$$

The indicator is $\mathcal{F}_n$-measurable and the future sign is centered independently, so this increment has conditional mean zero. Before hitting either boundary the walk lies in $\{-1, 0, 1\}$; at the boundary it stays at ± 2. Hence $|S_{n\wedge\tau}| \le 2$, verifying bounded integrability without any unproved stopping assertion.

(c) Starting at zero, after two moves equal signs hit a boundary, with probability $1/2$, while opposite signs return to zero, also with probability $1/2$. Survival through $2k$ moves requires an opposite-sign pair at each of the k independent pairs. Thus its probability is $(1/2)^k$. These decreasing tail events have intersection $\{\tau = \infty\}$, so continuity from above gives probability zero for infinite survival.

(d) Since $S_{n\wedge\tau} \to S_\tau$ almost surely and is bounded by 2, dominated convergence gives $\mathsf{E}S_\tau = 0$. Its only possible values are ± 2, so $2p - 2(1-p) = 0$ and $p = 1/2$. For the mean hitting time use the nonnegative tail-sum identity, justified by Tonelli:

$$\mathsf{E}\tau = \sum_{n\ge 0} P(\tau > n).$$

Hitting can occur only at even times, so $P(\tau > 2k+1) = P(\tau > 2k) = 2^{-k}$. Therefore $\mathsf{E}\tau = 2\sum_{k\ge 0} 2^{-k} = 4$. One can check this from $\mathsf{E}S_{n\wedge\tau}^2 = \mathsf{E}(n \wedge \tau)$, then dominated convergence for the bounded square and monotone convergence for the stopping time give $4 = \mathsf{E}\tau$.

(e) With only an upper barrier the stopped walk can make arbitrarily large negative excursions; the uniform bound 2 is lost. Almost-sure convergence of stopped values alone does not justify convergence of expectations. To see why finite expectation is impossible, suppose the one-sided hitting time τ_+ had $\mathsf{E}\tau_+ < \infty$. Then it would be finite almost surely, and $|S_{n\wedge\tau_+}| \le \tau_+$ would be an integrable dominating bound. The bounded-time increment calculation gives $\mathsf{E}S_{n\wedge\tau_+} = 0$ for every n, while dominated convergence would give $\mathsf{E}S_{\tau_+} = 2$, a contradiction. Thus its expectation cannot be finite. This argument identifies the exact limiting hypothesis that was available with two barriers and is now missing.

Mock Exam 4: convergence, domination and a digital hedge

Questions

Q.1 On $(0, 1)$ with Lebesgue probability let $f_n = n\mathbf{1}_{(0,1/n)}$ and $g_n = n^{-1}\mathbf{1}_{(0,1)}$.

(a) Determine their pointwise limits and their integrals.

(b) Apply Fatou's lemma to f_n and identify whether its inequality is strict.

(c) Prove that no integrable function dominates every f_n almost everywhere.

(d) Determine the L^p convergence behavior of both sequences for finite $p \ge 1$.

(e) Let $h_n = f_n + g_n$. Does convergence in probability together with bounded L^1 norms imply L^1 convergence? Use this example and calculate a uniform-integrability tail test.

Q.2 A three-period binomial market has $S_0 = 1$, $u = 2$, $d = 1/2$, $r = 0$. The claim pays $H = \mathbf{1}_{\{S_3>1\}}$.

(a) Find the pricing probability and reduce the payoff to a statement about the number of up moves.

(b) Compute the time-zero price by the binomial distribution.

(c) Compute the claim values at all distinct time-two stock nodes.

(d) Find the time-two hedges and then the time-zero stock holding by backward induction.

(e) Derive the one-step replication formula and explain why it remains valid for a path-dependent claim if the whole node history is retained.

AI-Generated Worked Solution

Q.1

(a) For fixed $x > 0$, choose $n > 1/x$; then $x \notin (0, 1/n)$ and $f_n(x) = 0$. Also $g_n(x) = 1/n \to 0$. Nevertheless $\int f_n = n(1/n) = 1$, whereas $\int g_n = 1/n \to 0$. The mass in f_n is compressed towards the omitted endpoint zero while its height grows.

(b) Every f_n is measurable and nonnegative, so Fatou applies without domination:

$$\int \liminf_n f_n = 0 \leq \liminf_n \int f_n = 1.$$

The inequality is strict. Fatou is an inequality theorem and does not promise equality merely because pointwise convergence holds.

(c) Suppose $G \in L^1$ and $f_n \leq G$ almost everywhere for every n. The union of the countably many exceptional null sets is still null, so the inequalities hold simultaneously outside one null set. Since $f_n \to 0$, dominated convergence applied with $|f_n| \leq |G|$ would force $\int f_n \to 0$, contradicting the value one.

(d) For finite $p \geq 1$, $\int |f_n|^p = n^p/n = n^{p-1}$, so $\|f_n\|_p = n^{1-1/p}$. This stays at one for $p = 1$ and diverges for $p > 1$; no L^p convergence to zero occurs. By contrast $\|g_n\|_p = 1/n$ for every such p, so $g_n \to 0$ in every finite L^p. These explicit norms are stronger evidence than inspecting the pointwise limit.

(e) $h_n \to 0$ pointwise, hence in probability on this probability space, while $\|h_n\|_1 = 1 + 1/n \to 1$, so it does not converge in L^1. For any $K > 0$, choose an integer $n > K + 1$. On $(0, 1/n)$, $h_n = n + 1/n > K$ and

$$\mathsf{E}[|h_n| \mathbf{1}_{\{|h_n| > K\}}] \geq (n + 1/n)/n = 1 + 1/n^2 \geq 1.$$

Thus the supremum over n of the tail expectation never tends to zero: the family is not uniformly integrable. Bounded expectations alone do not control concentration of mass on rare events.

Q.2

(a) The equation $2p + (1 - p)/2 = 1$ gives $p = 1/3$. If K is the number of up moves, $S_3 = 2^K (1/2)^{3-K} = 2^{2K-3}$. It is greater than one exactly when $2K - 3 > 0$, or $K \geq 2$.

(b) Under the risk-neutral law the independent branch count is binomial $(3, 1/3)$. With no discounting,

$$V_0 = P(K \geq 2) = \binom{3}{2}(1/3)^2(2/3) + (1/3)^3 = 6/27 + 1/27 = 7/27.$$

(c) The time-two prices are $4, 1, 1/4$. From 4, either last move produces a stock above one (8 or 2), so $V_2(4) = 1$. From 1, the successors are 2 and $1/2$, so $V_2(1) = 1/3$. From $1/4$, both successors are at most one, so $V_2(1/4) = 0$.

(d) At stock s, the last-step hedge solves $\Delta us + C = H_u$, $\Delta ds + C = H_d$. At $s = 4$, the payoffs are $1, 1$, hence $\Delta = 0, C = 1$. At $s = 1$, they are $1, 0$, hence $\Delta = 2/3, C = -1/3$ and cost $1/3$. At $s = 1/4$, both holdings are zero. At time one the values are

$$V_1(2) = \tfrac{1}{3}(1) + \tfrac{2}{3}(1/3) = 5/9,\ V_1(1/2) = \tfrac{1}{3}(1/3) + \tfrac{2}{3}(0) = 1/9.$$

The time-zero hedge then has $\Delta_0 = (5/9 - 1/9)/(2 - 1/2) = 8/27$ and cash $C_0 = 1/9 - (8/27)(1/2) = -1/27$. Its cost is $8/27 - 1/27 = 7/27$, verifying the expectation price. Rebalancing at each node uses the current portfolio value, establishing self-financing.

(e) In general with bank growth $1 + r$, solve

$$\Delta us + (1 + r)C = H_u,\ \Delta ds + (1 + r)C = H_d.$$

Subtraction gives $\Delta = (H_u - H_d)/(s(u - d))$, then $C = (uH_d - dH_u)/((1+r)(u-d))$. Adding the current stock cost gives $\Delta s + C = [pH_u + (1-p)H_d]/(1+r)$ with $p = ((1+r) - d)/(u - d)$.

The system is local to a node, so it does not require the claim to be a function of stock alone. For a path-dependent claim one must keep separate histories that happen to meet at the same stock price; their continuation payoffs and hedges may differ.

Mock Exam 5: product measures and predictable transforms

Questions

Q.1 Let U, V be independent uniform variables on $(0,1)$, and $W = \max(U,V)$, $J = \mathbf{1}_{\{V \le U\}}$.

(a) Establish measurability of W, J from Borel preimages.
(b) Find the distribution and mean of W using the product law.
(c) Compute $\mathsf{E}[J \mid \sigma(U)]$ by verifying the conditional integral property for every Borel test set.
(d) Find $\mathsf{E}[W \mid \sigma(U)]$ and verify its expectation agrees with part (b).
(e) Is J independent of U? Exhibit a concrete event calculation.

Q.2 Let ξ_n be iid centered square-integrable variables of variance $v > 0$, $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$, and H_n a bounded $\mathcal{F}_{n-1}$-measurable variable. Define $M_0 = 0$, $M_n = \sum_{j=1}^n H_j \xi_j$.

(a) Prove that M is a square-integrable martingale.
(b) Prove orthogonality of its distinct increments and derive $\mathsf{E}M_n^2$.
(c) Show that $M_n^2 - v\sum_{j=1}^n H_j^2$ is a martingale.
(d) Take fair signs, $H_j = 2^{-j}\mathbf{1}_{\{\xi_1=1\}}$ for $j \ge 2$, and $H_1 = 0$. Calculate $\mathsf{E}M_n^2$ and show M_n converges in L^2.
(e) Explain why replacing H_j by ξ_j for fair signs invalidates the martingale conclusion.

AI-Generated Worked Solution

Q.1

(a) The pair (U,V) is measurable into the product Borel space. The maximum map is continuous, hence Borel, so its composition W is measurable. Also $\{(u,v) : v \le u\}$ is closed in $\mathbb{R}^2$, being the preimage of $(-\infty, 0]$ under the continuous map $(u,v) \mapsto v - u$. Its indicator composed with (U,V) is a measurable random variable J.

(b) Independence means the joint law on the unit square is product Lebesgue measure. For $0 \le w \le 1$, $P(W \le w) = P(U \le w, V \le w) = w^2$. The distribution is zero below zero and one above one, with density $2w$ on $(0,1)$. Consequently $\mathsf{E}W = \int_0^1 2w^2 dw = 2/3$. All integrands are bounded/nonnegative, so Tonelli and ordinary finite integration apply without integrability concerns.

(c) Candidate $Z = U$ is $\sigma(U)$-measurable and bounded. For any Borel $A \subset (0,1)$, Tonelli gives

$$\mathsf{E}[J\mathbf{1}_{\{U\in A\}}] = \int_A \int_0^1 \mathbf{1}_{\{v\le u\}} dv\, du = \int_A u\, du = \mathsf{E}[U\mathbf{1}_{\{U\in A\}}].$$

Sets of this form constitute $\sigma(U)$ by definition. The defining property is therefore verified on all required sets, so $\mathsf{E}[J \mid \sigma(U)] = U$. In particular $P(J=1) = \mathsf{E}U = 1/2$.

(d) Hold $U = u$ fixed and integrate out the independent V:

$$g(u) = \int_0^1 \max(u,v) dv = \int_0^u u\, dv + \int_u^1 v\, dv = u^2 + \frac{1-u^2}{2} = \frac{1+u^2}{2}.$$

As above, testing against $\mathbf{1}_{\{U\in A\}}$ and using product integration proves $\mathsf{E}[W \mid \sigma(U)] = g(U)$. Then $\mathsf{E}g(U) = \frac{1}{2}(1 + \int_0^1 u^2 du) = \frac{1}{2}(1 + 1/3) = 2/3$, as required by the tower property. The product-law calculation is what justifies integrating out V while freezing U.

(e) If they were independent, $P(J = 1, U \le 1/2)$ would equal $(1/2)(1/2) = 1/4$. In fact the conditional calculation yields $\int_0^{1/2} u\, du = 1/8$. Thus they are dependent, even though the two underlying uniform variables are independent. A function involving both coordinates need not be independent of either one.

Q.2

(a) Each product is adapted because H_j is known at $j-1$ and ξ_j at j. Boundedness of H_j and square integrability of ξ_j give a finite second moment for every finite sum. Independence of the next increment from $\mathcal{F}_n$ gives

$$\mathsf{E}[M_{n+1} \mid \mathcal{F}_n] = M_n + H_{n+1}\mathsf{E}[\xi_{n+1} \mid \mathcal{F}_n] = M_n.$$

Taking out H_{n+1} is valid precisely because it is predictable, meaning measurable before the next shock.

(b) For $i < j$, the product $H_i\xi_i H_j$ is $\mathcal{F}_{j-1}$-measurable. Conditioning on that sigma-algebra,

$$\mathsf{E}[H_i\xi_i H_j\xi_j] = \mathsf{E}[H_i\xi_i H_j\mathsf{E}(\xi_j \mid \mathcal{F}_{j-1})] = 0.$$

Also $\mathsf{E}[H_j^2\xi_j^2] = v\mathsf{E}H_j^2$. Expanding the square of the sum eliminates all cross terms, yielding the isometry $\mathsf{E}M_n^2 = v\sum_{j=1}^n \mathsf{E}H_j^2$.

(c) Expand one step:

$$M_{n+1}^2 - M_n^2 = 2M_nH_{n+1}\xi_{n+1} + H_{n+1}^2\xi_{n+1}^2.$$

Its conditional expectation is vH_{n+1}^2. Subtract the increment of the predictable compensator $v\sum H_j^2$ to obtain conditional mean zero. All terms are integrable, proving the claimed martingale.

(d) Here $v = 1$ and for $j \geq 2$, $\mathsf{E}H_j^2 = 4^{-j}P(\xi_1 = 1) = 4^{-j}/2$. Therefore

$$\mathsf{E}M_n^2 = \frac{1}{2}\sum_{j=2}^{n} 4^{-j} = \frac{1}{24}(1 - 4^{-(n-1)}),\ n \geq 2.$$

For $n = 0, 1$, the sum is empty and $M_n = 0$. For $m > n \geq 1$, the orthogonality established in (b) gives $\mathsf{E}|M_m - M_n|^2 = \frac{1}{2}\sum_{j=n+1}^m 4^{-j} \to 0$ uniformly in $m > n$. Hence (M_n) is Cauchy in the complete space L^2 and converges there to some M_∞. This establishes convergence directly, rather than claiming it from bounded expectations without a theorem.

(e) If $H_j = \xi_j$, it is generally not measurable at time $j-1$. For fair signs $H_j\xi_j = \xi_j^2 = 1$, so the alleged transform is $M_n = n$, whose increments have conditional expectation one. It is not a martingale. Anticipating the current shock destroys the centering calculation.

Mock Exam 6: independence of sigma-algebras and exponential martingales

Questions

Q.1 On a probability space, let $\mathcal{I}, \mathcal{J}$ be pi-systems containing Ω, and assume $P(A \cap B) = P(A)P(B)$ for $A \in \mathcal{I}, B \in \mathcal{J}$.

(a) For fixed $A \in \mathcal{I}$, show that the sets B satisfying this identity form a lambda-system.

(b) Extend the identity to $A \in \sigma(\mathcal{I}), B \in \sigma(\mathcal{J})$ using two successive extensions.

(c) Deduce factorization of $\mathsf{E}[f(X)g(Y)]$ for bounded Borel functions of independent variables.

(d) Explain how to extend this to nonnegative functions without assuming finite expectations, avoiding $0 \cdot \infty$ ambiguities.

(e) Give an example of two dependent variables X, Y with zero covariance.

Q.2 Under P, iid signs satisfy $P(\xi_j = 1) = p \in (0,1)$, $S_n = \sum_{j=1}^n \xi_j$. For $t \in \mathbb{R}$, set $c(t) = pe^t + (1-p)e^{-t}$ and $M_n = e^{tS_n}/c(t)^n$.

(a) Prove that M is a positive martingale for its natural filtration.

(b) On a fixed horizon N, define $Q_t(A) = \mathsf{E}_P[M_N\mathbf{1}_A]$. Find the law of each sign under Q_t.

(c) Find t for which the signs are fair under Q_t.

(d) Show that $S_n - n(2p-1)$ is a P-martingale and compute its variance.

(e) For $p = 1/2$, $t \neq 0$, show $M_n \to 0$ almost surely but $\mathsf{E}M_n = 1$ for every n. Is the convergence in L^1?

AI-Generated Worked Solution

Q.1

(a) Fix A and define $\mathcal{D}_A = \{B : P(A\cap B) = P(A)P(B)\}$. It contains Ω since $P(A\cap\Omega) = P(A)$. If $B \in \mathcal{D}_A$, subtraction of finite probabilities gives $P(A\cap B^c) = P(A) - P(A\cap B) = P(A)(1 - P(B))$. For disjoint $B_n \in \mathcal{D}_A$, countable additivity yields $P(A \cap \bigcup B_n) = \sum P(A \cap B_n) = P(A)\sum P(B_n)$. These prove the lambda-system axioms.

(b) The assumption puts $\mathcal{J} \subset \mathcal{D}_A$; the pi–lambda theorem gives $\sigma(\mathcal{J}) \subset \mathcal{D}_A$ for each original $A \in \mathcal{I}$. Now fix an arbitrary $B \in \sigma(\mathcal{J})$ and define $\mathcal{C}_B = \{A : P(A \cap B) = P(A)P(B)\}$. The complement and disjoint-union arguments in (a), with the roles of the coordinates exchanged, make this a lambda-system, and the first extension shows it contains $\mathcal{I}$. A second application gives $\sigma(\mathcal{I}) \subset \mathcal{C}_B$. Both sigma-algebras have therefore been extended, not just one.

(c) For indicators the identity is the event independence just proved. For simple $f = \sum_i a_i \mathbf{1}_{A_i}$ and $g = \sum_j b_j \mathbf{1}_{B_j}$, distributivity and finite linearity give

$$\mathsf{E}[f(X)g(Y)] = \sum_{i,j} a_i b_j P(X \in A_i, Y \in B_j) = \mathsf{E}f(X)\mathsf{E}g(Y).$$

Approximate bounded nonnegative Borel functions by increasing simple functions, then use monotone convergence. For bounded signed functions split into positive and negative parts; every resulting term is finite, so subtraction is legitimate.

(d) For nonnegative f, g, apply the bounded result to $f \wedge n, g \wedge n$ and let $n \uparrow \infty$ by monotone convergence on both sides. This defines the limiting product through the increasing products of finite expectations. If $\mathsf{E}f = 0$, then $f(X) = 0$ almost surely, so the product variable is zero almost surely (for finite-valued g), even if $\mathsf{E}g = \infty$; treat this case separately. Otherwise a positive finite or infinite expectation multiplied by another positive expectation has its usual extended meaning. This avoids cancelling or multiplying ambiguous extended values without explanation.

(e) Take X uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Then $\mathsf{E}X = 0$, $\mathsf{E}XY = \mathsf{E}X^3 = 0$, so $\mathrm{Cov}(X, Y) = 0$. They are dependent because $P(Y = 0, X = 0) = 1/3$, whereas $P(Y = 0)P(X = 0) = 1/9$. Zero covariance is only a moment restriction, much weaker than event independence.

Q.2

(a) $c(t) > 0$, and for each finite n, $|S_n| \le n$, so M_n is adapted, positive and bounded by a finite deterministic constant. Independence gives

$$\mathsf{E}[M_{n+1} \mid \mathcal{F}_n] = \frac{e^{tS_n}}{c(t)^{n+1}}\mathsf{E}[e^{t\xi_{n+1}}] = M_n.$$

In particular $\mathsf{E}M_n = 1$. Normalizing by $c(t)^n$ removes exactly the average growth of the exponential product.

(b) A specified sign path $x_1, \ldots, x_N$ has Q_t mass

$$\prod_{j=1}^{N} \frac{e^{tx_j} P(\xi_j = x_j)}{c(t)}.$$

This factorization proves independence under the new law with up probability $p_t = pe^t/c(t)$ and down probability $(1-p)e^{-t}/c(t)$. Positivity and expectation one of the density verify that it defines an equivalent probability on the finite horizon.

(c) Fairness requires $pe^t = (1-p)e^{-t}$. Multiplying by e^t and dividing by p gives $e^{2t} = (1-p)/p$, hence $t = \frac{1}{2}\log((1-p)/p)$. At this value the two weights coincide and each normalized probability is $1/2$.

(d) Each centered increment has conditional mean $\mathsf{E}\xi_j - (2p-1) = 0$, proving the martingale identity. Its variance is $1 - (2p-1)^2 = 4p(1-p)$. Independence eliminates cross covariances, so the variance at time n is $4np(1-p)$.

(e) For fair signs the strong law gives $S_n/n \to 0$ almost surely. Since $c(t) = \cosh t > 1$ when $t \neq 0$,

$$\frac{1}{n}\log M_n = t\frac{S_n}{n} - \log(\cosh t) \longrightarrow -\log(\cosh t) < 0.$$

Thus $M_n \to 0$ almost surely at an exponential logarithmic rate. Yet its expectation remains one. Since $M_n \geq 0$, $\mathsf{E}|M_n - 0| = 1$, ruling out L^1 convergence. The nonnegative martingale convergence theorem permits this loss of expectation; uniform integrability would prevent it and hence fails in this example.

Mock Exam 7: conditional Jensen and a barrier claim

Questions

Q.1 Let $X \in L^2$, $\mathcal{G} \subset \mathcal{F}$ and $Z = \mathsf{E}[X \mid \mathcal{G}]$.

(a) Prove $|Z| \leq \mathsf{E}[|X| \mid \mathcal{G}]$ by monotonicity of conditional expectation.

(b) Prove $Z \in L^2$ and $\mathsf{E}Z^2 \leq \mathsf{E}X^2$ without assuming in advance that Z has a second moment.

(c) Show $\mathsf{E}[(X - Z)Y] = 0$ for every $\mathcal{G}$-measurable $Y \in L^2$.

(d) Deduce the Pythagorean identity for the squared error of any such Y and characterize its minimizer.

(e) If X, Y are independent iid integrable variables, find $\mathsf{E}[X \mid \sigma(X + Y)]$. State whether a second moment is needed for this answer.

Q.2 In the three-period binomial market $S_0 = 1, u = 2, d = 1/2, r = 0$, a claim pays one if the stock has reached 4 by time three, otherwise zero. Reaching means $S_j \geq 4$ for at least one $j \in \{0, 1, 2, 3\}$.

(a) Find the pricing measure and identify exactly which paths trigger the barrier.

(b) Calculate the price at time zero and compare with the terminal-only digital $\mathbf{1}_{\{S_3 \geq 4\}}$.

(c) Determine time-two continuation values, keeping track of barrier status.

(d) Find the hedge at the two time-one nodes and at time zero.

(e) Verify why a node where the barrier has already been reached must have value one even if the future stock falls.

AI-Generated Worked Solution

Q.1

(a) From $-|X| \leq X \leq |X|$ and conditional monotonicity, $-\mathsf{E}[|X| \mid \mathcal{G}] \leq Z \leq \mathsf{E}[|X| \mid \mathcal{G}]$. The right-hand variable is nonnegative, giving the absolute-value inequality. Integrability of X follows from L^2 on a probability space by Cauchy–Schwarz, so all these conditional expectations exist.

(b) We avoid multiplying by an unbounded Z prematurely. Set $Z_n = Z\mathbf{1}_{\{|Z| \leq n\}}$, a bounded $\mathcal{G}$-measurable variable. The defining integral property, extended from indicators to bounded measurable test variables, gives

$$a_n := \mathsf{E}[Z^2 \mathbf{1}_{\{|Z| \leq n\}}] = \mathsf{E}[X Z_n] \leq (\mathsf{E}X^2)^{1/2} (\mathsf{E}Z_n^2)^{1/2} = (\mathsf{E}X^2)^{1/2} a_n^{1/2}.$$

If $a_n = 0$ the bound is trivial; otherwise division yields $a_n \leq \mathsf{E}X^2$. Monotone convergence now gives $\mathsf{E}Z^2 \leq \mathsf{E}X^2 < \infty$. This proves the square-integrability needed for subsequent products.

(c) First take Y bounded and $\mathcal{G}$-measurable: the conditional defining property gives $\mathsf{E}[XY] = \mathsf{E}[ZY]$. For general $Y \in L^2$, truncate it to $Y_n = (-n) \vee (Y \wedge n)$. Since $Y_n \to Y$ in L^2, Cauchy–Schwarz gives

$$|\mathsf{E}[(X - Z)(Y - Y_n)]| \leq \|X - Z\|_2 \|Y - Y_n\|_2 \to 0.$$

The bounded-test identity proved above therefore passes to the limit and gives the required orthogonality. All products are integrable by the same inequality.

(d) Decompose $X - Y = (X - Z) + (Z - Y)$ and expand its square. Since $Z - Y$ is $\mathcal{G}$-measurable in L^2, the cross term is zero by part (c), yielding

$$\mathsf{E}(X - Y)^2 = \mathsf{E}(X - Z)^2 + \mathsf{E}(Z - Y)^2.$$

The second term is nonnegative and vanishes precisely if $Y = Z$ almost surely. Thus conditional expectation is the unique almost-sure minimizer in the $L^2(\mathcal{G})$ space.

(e) Exchangeability gives $\mathsf{E}[X\mathbf{1}_{\{X+Y\in A\}}] = \mathsf{E}[Y\mathbf{1}_{\{X+Y\in A\}}]$ for every Borel A. Therefore the two conditional expectations given $\sigma(X+Y)$ coincide. Their sum is $\mathsf{E}[X+Y \mid \sigma(X+Y)] = X+Y$, so each equals $(X+Y)/2$. Integrability suffices; no L^2 assumption is needed. Equality in distribution of the pair under exchange is used, rather than a false claim that X and its sum are independent.

Q.2

(a) The up probability is $p = (1-1/2)/(2-1/2) = 1/3$. At time one the stock is at most 2. At time two only the path uu reaches 4. At time three, a value at least 4 requires three up moves, which have already reached the barrier at time two. Thus the triggering paths are exactly uuu, uud, equivalently the first two moves are up.

(b) Their combined probability is $p^2 = 1/9$, so $V_0 = 1/9$. The terminal-only claim pays only on uuu, since terminal prices are $8, 2, 1/2, 1/8$. Its price is $p^3 = 1/27$. The difference $2/27$ is exactly the probability of uud, where the barrier was hit but the final stock is below it.

(c) At the time-two history uu, the payoff has become certain: $V_2 = 1$ for either final branch. At ud, du, dd, the highest possible terminal price is respectively $2, 2, 1/2$, below the barrier, so all three values are zero. Keeping the history is the safe general method; in this particular tree the stock value 4 also identifies the only already-triggered node.

(d) At time one after an up move the stock is 2 and successors are $4, 1$ with continuation values $1, 0$. Thus $\Delta_1^u = (1-0)/(4-1) = 1/3$ and $C_1^u = -1/3$, with value $2/3 - 1/3 = 1/3$. After a down move the continuation values are zero, so the holdings and value are zero. At time zero solve $2\Delta_0 + C_0 = 1/3$, $(1/2)\Delta_0 + C_0 = 0$. Hence $\Delta_0 = 2/9$, $C_0 = -1/9$, cost $1/9$. These node portfolios exactly finance the next rebalance.

(e) Once triggered, $H = 1$ on every continuation of that history. A unit of cash replicates this sure payoff, and with $r = 0$ its price is exactly one. Using a fresh terminal barrier test at the next node would incorrectly erase the contractual memory and fail to replicate uud.

Mock Exam 8: a conditional distribution and martingale decomposition

Questions

Q.1 Let X, Y be independent Poisson variables of means $\lambda, \mu > 0$ and let $T = X + Y$.

(a) Derive the law of T by summing the joint masses.

(b) Derive the conditional law of X given $T = n$, including its support.

(c) Find $\mathsf{E}[X \mid \sigma(T)]$ and verify the defining property for arbitrary events in $\sigma(T)$.

(d) Find $\mathsf{E}[X(X-1) \mid \sigma(T)]$ and the conditional variance.

(e) Use the total-variance identity to recover $\operatorname{Var} X = \lambda$, showing the calculations.

Q.2 Let Z_n be an integrable adapted process, $Z_0 = 0$, on a finite horizon $0 \le n \le N$. Set $a_j = \mathsf{E}[Z_j - Z_{j-1} \mid \mathcal{F}_{j-1}]$, $A_n = \sum_{j=1}^{n} a_j$, $M_n = Z_n - A_n$.

(a) Prove that M is a martingale and that A is predictable.

(b) If Z is a submartingale, prove that A has nonnegative increments.

(c) Prove uniqueness of this decomposition when $M_0 = A_0 = 0$ and A is predictable with integrable increments.

(d) Apply the construction to $Z_n = S_n^2$ for a simple symmetric random walk; identify A and M.

(e) If $\sigma \le N$ is a stopping time, prove $\mathsf{E}M_\sigma = 0$ directly by writing a sum of stopped increments.

AI-Generated Worked Solution

Q.1

(a) For integer $n \ge 0$, independence and convolution give

$$P(T=n) = \sum_{k=0}^{n} e^{-\lambda}\frac{\lambda^k}{k!}e^{-\mu}\frac{\mu^{n-k}}{(n-k)!} = \frac{e^{-(\lambda+\mu)}}{n!}\sum_{k=0}^{n}\binom{n}{k}\lambda^k\mu^{n-k}.$$

The binomial theorem makes the sum $(\lambda+\mu)^n$, so T is Poisson with mean $\lambda+\mu$.

(b) Since $P(T=n)>0$, for $k=0,\dots,n$ the conditional mass is the joint mass divided by the preceding marginal:

$$P(X=k\mid T=n)=\binom{n}{k}\left(\frac{\lambda}{\lambda+\mu}\right)^k\left(\frac{\mu}{\lambda+\mu}\right)^{n-k}.$$

It is zero outside this support. Thus it is binomial (n,q) with $q=\lambda/(\lambda+\mu)$. The two independent counts become dependent after their sum is fixed.

(c) The binomial mean gives candidate qT. Every event of $\sigma(T)$ is $\{T\in B\}$ for some subset B of the nonnegative integers. Summing over its atoms,

$$\mathsf{E}[X\mathbf{1}_{\{T\in B\}}]=\sum_{n\in B}P(T=n)\sum_{k=0}^{n}kP(X=k\mid T=n)=\sum_{n\in B}qnP(T=n)=\mathsf{E}[qT\mathbf{1}_{\{T\in B\}}].$$

Nonnegative sums justify the exchanges and integrability follows from $\mathsf{E}X=\lambda$. Measurability and the integral property therefore prove $\mathsf{E}[X\mid\sigma(T)]=qT$.

(d) For a binomial (n,q), use $k(k-1)\binom{n}{k}=n(n-1)\binom{n-2}{k-2}$ to obtain factorial second moment $n(n-1)q^2$. Hence $\mathsf{E}[X(X-1)\mid T]=T(T-1)q^2$ and $\mathsf{E}[X^2\mid T]=T(T-1)q^2+Tq$. Subtracting $(qT)^2$ gives $\operatorname{Var}(X\mid T)=Tq(1-q)$.

(e) Expanding $X-\mathsf{E}X=(X-\mathsf{E}[X\mid T])+(\mathsf{E}[X\mid T]-\mathsf{E}X)$ and conditioning shows the cross term has expectation zero. Therefore

$$\operatorname{Var}X=\mathsf{E}[Tq(1-q)]+\operatorname{Var}(qT)=(\lambda+\mu)q(1-q)+q^2(\lambda+\mu)=(\lambda+\mu)q=\lambda.$$

The first term is residual uncertainty after observing the total, and the second the variation in the conditional mean.

Q.2

(a) Conditional expectations of integrable increments are integrable. Also a_j is $\mathcal{F}_{j-1}$-measurable, so A_n is $\mathcal{F}_{n-1}$-measurable for $n\geq 1$, the discrete predictable property. The increment of M is $Z_j-Z_{j-1}-a_j$, with conditional expectation zero given $\mathcal{F}_{j-1}$. Since M is adapted and integrable, this proves the martingale identity.

(b) Submartingality says $\mathsf{E}[Z_j\mid\mathcal{F}_{j-1}]\geq Z_{j-1}$. Subtracting the already measurable Z_{j-1} gives $a_j\geq 0$ almost surely. The compensator A is consequently increasing, with $A_0=0$.

(c) If $Z=M+A=\widetilde{M}+\widetilde{A}$, then $D=M-\widetilde{M}=\widetilde{A}-A$ is both a martingale and predictable, starting at zero. Predictability makes both D_j and D_{j-1} measurable with respect to $\mathcal{F}_{j-1}$, hence so is their difference. Martingality gives conditional mean zero, but conditioning a measurable integrable increment leaves it unchanged. Every increment is thus zero almost surely. A finite union of exceptional null sets gives equality of the whole processes on the fixed horizon.

(d) With $S_j=S_{j-1}+\xi_j$, the squared increment is $2S_{j-1}\xi_j+1$. Its conditional expectation is one. Hence $a_j=1$, $A_n=n$, and $M_n=S_n^2-n$. This obtains the familiar square martingale from the general predictable-drift subtraction method.

(e) Because $\sigma\leq N$,

$$M_\sigma=\sum_{j=1}^{N}\mathbf{1}_{\{\sigma\geq j\}}(M_j-M_{j-1}).$$

The event $\{\sigma\geq j\}=\{\sigma\leq j-1\}^c$ belongs to $\mathcal{F}_{j-1}$. Each summand is integrable and has zero expectation by conditioning on that sigma-algebra. Summing finitely many zero expectations proves $\mathsf{E}M_\sigma=0$, with no unbounded-stopping limit or hidden optional-stopping hypothesis.

Mock Exam 9: null sets, measures and discounted replication

Questions

Q.1 Let $(\Omega,\mathcal{F},P)$ be a probability space and $N\in\mathcal{F}$ with $P(N)=0$. A subset $A\subset N$ need not belong to $\mathcal{F}$. Let $X=0$ and $Y=\mathbf{1}_A$ as pointwise functions.

(a) Prove that Y is measurable exactly when $A \in \mathcal{F}$.

(b) Explain why equality outside a measurable null set does not, on an incomplete space, guarantee measurability.

(c) Define the completed sigma-algebra and its probability. Explain why the extension is well defined.

(d) Show that the completed space makes Y measurable with expectation zero.

(e) If measurable X_n vanish almost surely for each n, show that their sum is zero almost surely wherever the series is defined, using one common null set.

Q.2 A two-period market has $S_0 = 10$, up multiplier $u = 3/2$, down multiplier $d = 3/4$, and risk-free growth $1 + r = 9/8$. The claim pays $H = (S_2 - 10)^+$.

(a) Find the risk-neutral probability and verify the no-arbitrage inequalities.

(b) Compute the three distinct terminal stock values, payoffs and time-one claim values.

(c) Compute the time-zero price from discounted expectation.

(d) Find all stock holdings and cash values needed for a self-financing replicating portfolio.

(e) Verify explicitly that the discounted price process is a martingale under the pricing measure.

AI-Generated Worked Solution

Q.1

(a) **(⇒)** If Y is measurable, $A = Y^{-1}(\{1\})$ is measurable because $\{1\}$ is Borel.
(⇐) Conversely, if $A \in \mathcal{F}$, its indicator is measurable: the preimage of a Borel set is one of $\varnothing, A, A^c, \Omega$, all in $\mathcal{F}$. This verifies both directions without assuming the null-subset property.

(b) Outside N, both functions are zero. Yet if A is a nonmeasurable subset of N, the first part shows Y is not measurable, although X is. Such a subset exists by the definition of an incomplete measure space. Therefore an almost-everywhere modification preserves measurability automatically only in a complete space, or when the modifying sets/functions are separately known measurable. The exceptional containing set must itself be a measurable null set when using this formulation.

(c) Define $\overline{\mathcal{F}} = \{B \subset \Omega : \exists E, F \in \mathcal{F},\ E \subset B \subset F,\ P(F \setminus E) = 0\}$. It is closed under complements by reversing the bounding sets, and under countable unions by taking unions of the lower and upper sets: the new gap is contained in the union of their null gaps. Define $\overline{P}(B) = P(E) = P(F)$. If another pair E', F' represents B, the bounds give $E \subseteq F'$ and $E' \subseteq F$. Monotonicity and the null gaps give $P(E) \leq P(F') = P(E')$; exchanging the two pairs gives the reverse inequality. Thus the assigned mass is independent of representation. Disjoint countable additivity follows by choosing disjoint measurable lower sets $E_n \subset B_n$; their union is a lower representative of the union, with a null upper gap. It extends P by taking $E = F = B$ for original measurable sets.

(d) Take lower set $\varnothing$ and upper set N to get $A \in \overline{\mathcal{F}}$ with $\overline{P}(A) = 0$. Its indicator is now a measurable nonnegative variable, with expectation equal to the mass of its support, namely zero. The completion changed the allowed measurable sets, not the probabilities of any original measurable event.

(e) For each n, choose a measurable null set N_n outside which $X_n = 0$. Countable subadditivity gives $P(\bigcup_n N_n) = 0$. Outside this single union all terms are simultaneously zero, so every partial sum and its limit are zero. The countability is essential: an uncountable union of null sets need not be null.

Q.2

(a) We have $d = 3/4 < 9/8 < 3/2 = u$, so a strictly positive branch pricing measure exists. Its up probability solves $pu + (1-p)d = 9/8$, hence

$$p = \frac{9/8 - 3/4}{3/2 - 3/4} = \frac{3/8}{3/4} = \frac{1}{2}.$$

Independent branching with this probability makes S_n/B_n a martingale, where $B_n = (9/8)^n$.

(b) Terminal stock values are $S_{uu} = 45/2$, $S_{ud} = S_{du} = 45/4$, $S_{dd} = 45/8$, with payoffs $25/2, 5/4, 0$. At time one,

$$V_1^u = \frac{(25/2 + 5/4)/2}{9/8} = \frac{55}{9},\ V_1^d = \frac{(5/4 + 0)/2}{9/8} = \frac{5}{9}.$$

The discount factor is applied at each backward step because these are current values, not future cash amounts.

(c) The four path probabilities are $1/4$ each. Thus

$$V_0 = (9/8)^{-2}\left[\tfrac{1}{4}\tfrac{25}{2} + \tfrac{1}{2}\tfrac{5}{4}\right] = \frac{64}{81}\frac{15}{4} = \frac{80}{27}.$$

Backward induction also gives $(8/9)(V_1^u + V_1^d)/2 = (8/9)(10/3) = 80/27$.

(d) At time one after an up move, the stock is 15; solve $\Delta(45/2) + (9/8)C = 25/2$ and $\Delta(45/4) + (9/8)C = 5/4$. Their difference gives $\Delta_1^u = 1$, then $C_1^u = -80/9$, with cost $15 - 80/9 = 55/9$. After a down move the stock is $15/2$; subtracting the terminal equations gives $\Delta_1^d = (5/4)/(45/8) = 2/9$. Then $(9/8)C_1^d = -(2/9)(45/8) = -5/4$, so $C_1^d = -10/9$ and cost $5/9$. At time zero,

$$\Delta_0 = \frac{55/9 - 5/9}{15 - 15/2} = \frac{20}{27},\ \frac{9}{8}C_0 = \frac{5}{9} - \frac{20}{27}\frac{15}{2} = -5,$$

giving $C_0 = -40/9$. Its cost is $10(20/27) - 40/9 = 80/27$. Here C denotes the cash value currently invested in the bank, whose next-period value is $(9/8)C$; the number of bank units would be C/B_n. The portfolio value after the first move equals the node's rebalancing cost, making the strategy self-financing.

(e) At either time-one node the equation $(9/8)V_1 = \mathsf{E}[H \mid \mathcal{F}_1]$ yields $\mathsf{E}[V_2/B_2 \mid \mathcal{F}_1] = V_1/B_1$. At time zero, $\mathsf{E}[V_1/B_1] = (8/9)(55/9 + 5/9)/2 = 80/27 = V_0$. Adaptedness and integrability hold on the finite tree. These equalities verify the discounted martingale property directly and show why omitting discount factors would fail.

Mock Exam 10: revealing binary information and a lookback claim

Questions

Q.1 Let U be uniform on $[0, 1)$ and let $\mathcal{F}_n$ be generated by the dyadic intervals $I_{n,k} = [k2^{-n}, (k + 1)2^{-n})$, $0 \le k < 2^n$. Set $M_n = \mathsf{E}[U \mid \mathcal{F}_n]$.

(a) Find an explicit formula for M_n on each atom and verify it by integration.

(b) Prove that $(\mathcal{F}_n)$ is increasing and (M_n) is a bounded martingale.

(c) Prove convergence to U almost surely and in every finite L^p using a deterministic error bound.

(d) Compute the exact squared L^2 error and $\mathsf{E}M_n^2$.

(e) Compute $\mathsf{E}[(M_{n+1} - M_n)^2]$ and explain the orthogonality of successive information increments.

Q.2 In a three-step binomial model $S_0 = 1, u = 2, d = 1/2, r = 0$, let $L_3 = \max(S_0, S_1, S_2, S_3)$ and $H = L_3 - S_3$.

(a) List all eight paths, their maxima and payoffs.

(b) Compute the arbitrage-free time-zero price by path enumeration.

(c) Show that stock price alone does not determine the time-two claim value, using the histories ud and du.

(d) Calculate the four time-two claim values and the two time-one values.

(e) Compute the initial stock holding and cash balance and explain how the state can be enlarged to support replication.

AI-Generated Worked Solution

Q.1

(a) On an atom $[a, a+h)$, $h = 2^{-n}$, the conditional mean is its midpoint $a + h/2$. Indeed its mass is h, and $\int_a^{a+h} u\,du = ah + h^2/2 = h(a + h/2)$. Thus

$$M_n = \frac{\lfloor 2^n U \rfloor + 1/2}{2^n}.$$

It is constant on the atoms, measurable and bounded. The integral equality on each atom extends to every union of atoms, so the conditional defining property holds on $\mathcal{F}_n$.

(b) Each dyadic atom at level n splits into two level-$n+1$ atoms, hence $\mathcal{F}_n \subset \mathcal{F}_{n+1}$. Taking conditional expectations twice gives $\mathsf{E}[M_{n+1} \mid \mathcal{F}_n] = \mathsf{E}[\mathsf{E}(U \mid \mathcal{F}_{n+1}) \mid \mathcal{F}_n] = \mathsf{E}[U \mid \mathcal{F}_n] = M_n$. Alternatively, the two child midpoints are equally likely and average to the parent midpoint. The variables lie in $[0, 1]$, so adaptation and integrability are automatic.

(c) Every U and its cell midpoint differ by at most half a cell: $|M_n - U| \le 2^{-n-1}$. This proves convergence at every point of $[0, 1)$, not just almost surely. For finite $p \ge 1$, $\mathsf{E}|M_n - U|^p \le 2^{-p(n+1)}$, giving L^p convergence. This quantitative argument supplies the limiting variable explicitly, beyond merely invoking a martingale convergence theorem.

(d) In one cell of width h, shift to its midpoint to obtain $\int_{-h/2}^{h/2} v^2 dv = h^3/12$. There are $2^n = 1/h$ cells, so

$$\mathsf{E}(U - M_n)^2 = \frac{h^2}{12} = \frac{4^{-n}}{12}.$$

Because M_n is bounded and $\mathcal{F}_n$-measurable, the conditional test identity gives $\mathsf{E}[(U - M_n)M_n] = 0$. Expanding the square therefore yields $\mathsf{E}U^2 = \mathsf{E}M_n^2 + \mathsf{E}(U - M_n)^2$. Since $\mathsf{E}U^2 = 1/3$, $\mathsf{E}M_n^2 = 1/3 - 4^{-n}/12$. At $n = 0$, this is $1/4$, consistent with $M_0 = 1/2$.

(e) A child midpoint differs from its parent by $\pm h/4$, so $(M_{n+1} - M_n)^2 = h^2/16 = 4^{-n}/16$ everywhere. Its conditional mean increment is zero. For an earlier increment $D_j = M_j - M_{j-1}$ and later D_k, $j < k$, D_j is $\mathcal{F}_{k-1}$-measurable, so $\mathsf{E}[D_j D_k] = \mathsf{E}[D_j \mathsf{E}(D_k \mid \mathcal{F}_{k-1})] = 0$. The exact increment variance also equals $\mathsf{E}M_{n+1}^2 - \mathsf{E}M_n^2$, providing a check on the geometric factors.

Q.2

(a) The up probability is $1/3$ and the down probability $2/3$. Remember that the initial stock is included in the running maximum:

Path	S_3	L_3	H
uuu	8	8	0
uud	2	4	2
udu	2	2	0
udd	1/2	2	3/2
duu	2	2	0
dud	1/2	1	1/2
ddu	1/2	1	1/2
ddd	1/8	1	7/8

For example, after udd the path is $1, 2, 1, 1/2$, so the payoff is $2 - 1/2 = 3/2$.

(b) The nonzero contributions to the expectation are

$$V_0 = \frac{2}{27}(2) + \frac{4}{27}\frac{3}{2} + \frac{4}{27}\frac{1}{2} + \frac{4}{27}\frac{1}{2} + \frac{8}{27}\frac{7}{8} = \frac{4+6+2+2+7}{27} = \frac{7}{9}.$$

The probability weights use the full branch path, rather than assigning equal mass to distinct final stock values.

(c) At ud and du, $S_2 = 1$, but the maxima so far are 2 and 1. From ud the continuation payoffs are $0, 3/2$, giving value 1. From du they are $0, 1/2$, giving value $1/3$. Thus merging these two histories solely because their stock prices match gives a wrong claim value.

(d) At uu, continuation payoffs are $0, 2$, so $V_2^{uu} = 4/3$. We already have $V_2^{ud} = 1$, $V_2^{du} = 1/3$. At dd, payoffs are $1/2, 7/8$, so $V_2^{dd} = (1/3)(1/2) + (2/3)(7/8) = 3/4$. Then

$$V_1^u = \tfrac{1}{3}\tfrac{4}{3} + \tfrac{2}{3}(1) = \frac{10}{9},\ V_1^d = \tfrac{1}{3}\tfrac{1}{3} + \tfrac{2}{3}\tfrac{3}{4} = \frac{11}{18}.$$

Their risk-neutral mean is $(1/3)(10/9) + (2/3)(11/18) = 7/9$, agreeing with part (b).

(e) At time zero solve $2\Delta_0 + C_0 = 10/9$, $(1/2)\Delta_0 + C_0 = 11/18$. Subtraction yields $\Delta_0 = (1/2)/(3/2) = 1/3$ and then $C_0 = 11/18 - 1/6 = 4/9$, with total cost $7/9$. At later dates use the enlarged state (S_n, L_n), where $L_{n+1} = \max(L_n, S_{n+1})$. Each such node still has two successors and a nonzero stock difference, so its two continuation values determine a unique local stock/cash hedge. Holding the selected portfolio until the next node and then rebalancing at the continuation price gives a self-financing terminal replication even though stock alone is insufficient to label the state.